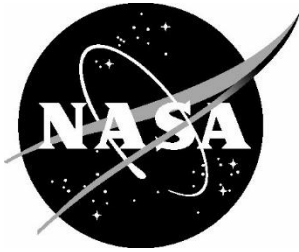


NASA/TM–20240009010



**EXTRAVEHICULAR ACTIVITY AND HUMAN SURFACE
MOBILITY PROGRAM (EHP)
EXPLORATION EVA (XEVA) SYSTEM COMPATIBILITY
STANDARDS**

Christine N. Kovich

The Aerospace Corporation

Johnson Space Center (JSC) Houston

August 2024

NASA STI Program Report Series

Since its founding, NASA has been dedicated to the advancement of aeronautics and space science. The NASA scientific and technical information (STI) program plays a key part in helping NASA maintain this important role.

The NASA STI program operates under the auspices of the Agency Chief Information Officer. It collects, organizes, provides for archiving, and disseminates NASA's STI. The NASA STI program provides access to the NTRS Registered and its public interface, the NASA Technical Reports Server, thus providing one of the largest collections of aeronautical and space science STI in the world. Results are published in both non-NASA channels and by NASA in the NASA STI Report Series, which includes the following report types:

- **TECHNICAL PUBLICATION.** Reports of completed research or a major significant phase of research that present the results of NASA Programs and include extensive data or theoretical analysis. Includes compilations of significant scientific and technical data and information deemed to be of continuing reference value. NASA counterpart of peer-reviewed formal professional papers but has less stringent limitations on manuscript length and extent of graphic presentations.
- **TECHNICAL MEMORANDUM.** Scientific and technical findings that are preliminary or of specialized interest, e.g., quick release reports, working papers, and bibliographies that contain minimal annotation. Does not contain extensive analysis.
- **CONTRACTOR REPORT.** Scientific and technical findings by NASA-sponsored contractors and grantees.

- **CONFERENCE PUBLICATION.** Collected papers from scientific and technical conferences, symposia, seminars, or other meetings sponsored or co-sponsored by NASA.
- **SPECIAL PUBLICATION.** Scientific, technical, or historical information from NASA programs, projects, and missions, often concerned with subjects having substantial public interest.
- **TECHNICAL TRANSLATION.** English-language translations of foreign scientific and technical material pertinent to NASA's mission.

Specialized services also include organizing and publishing research results, distributing specialized research announcements and feeds, providing information desk and personal search support, and enabling data exchange services.

For more information about the NASA STI program, see the following:

- Access the NASA STI program home page at <http://www.sti.nasa.gov>



National Aeronautics and
Space Administration

EHP-10028

BASELINE

EFFECTIVE DATE: MAY 9, 2024

EXTRAVEHICULAR ACTIVITY AND HUMAN SURFACE MOBILITY PROGRAM (EHP)

EXPLORATION EVA (XEVA) SYSTEM COMPATIBILITY STANDARDS

Verify that this is the correct version before use

This document has been reviewed for Proprietary, CUI, and Export Control (ITAR/EAR) and has been determined to be non-sensitive. It has been approved for public release via the NASA Scientific and Technical Information (STI) Process.

DAA #20240009010

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 2 of 280
Title: EHP xEVA System Compatibility Standards	

REVISION AND HISTORY PAGE

Revision No.	Change No.	Description	Release Date
Baseline	EHP-C0570	Initial Release- Approval and release authorized by signed Directive EHP-D0570	05/09/24
Admin Chg 1	EHP-D0758	Straight-to-Directive (S2D) acknowledges publicly available marking DAA #20240009010, and authorizes release of the document.	8/9/24

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 3 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE OF CONTENTS

SECTION	PAGE
1.0 INTRODUCTION	12
1.1 Purpose	12
1.2 Scope	12
1.3 Applicability/Allocations	15
1.4 Conventions and Notations.....	16
1.4.1 Shall Statements (Standards/Requirements).....	16
1.4.2 Requirement Identification	16
1.4.3 Rationale.....	16
1.4.4 Units.	16
1.5 Change Authority/Responsibility	17
2.0 DOCUMENTS.....	18
2.1 Applicable Documents	18
2.2 Reference Documents	18
3.0 REQUIREMENTS.....	21
3.1 Clearances	21
3.1.1 Microgravity Translation Path Hardware Protrusion.....	21
3.1.2 Partial Gravity Translation Path Hardware Protrusion	22
3.1.3 Tool Interface Clearance.....	24
3.1.4 Bolt Clearance	25
3.1.5 Gloved-Hand Clearance	26
3.1.6 Handrail/Handhold Mounting Clearance	26
3.1.7 Handrail/Handhold Side Clearance.....	27
3.1.8 APFR EVA Installation/Removal Clearance.....	28
3.2 Translation.....	30
3.2.1 Contiguous EVA Translation Path	30
3.2.2 Translation Path.....	30
3.2.3 Incapacitated Crewmember Interfaces	30
3.2.4 EVA Contact Hazards Preclusion Beyond Translation Corridor.....	31
3.2.5 Partial Gravity EVA Contact Hazards Preclusion.....	32
3.3 EVA Handling	33

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 4 of 280
Title: EHP xEVA System Compatibility Standards	

3.3.1	Microgravity EVA Equipment Handling Features	33
3.3.2	Partial Gravity EVA Handling Capability <TBR-EHP-10028-003>	34
3.3.3	Partial Gravity Step Design Parameter	38
3.3.4	Airlock and Hatch Transfer Volumetric Constraints	39
3.3.5	Pressurized xEVA Suit Transfer through Airlock Hatch	39
3.3.6	EVA Lunar Hatch Dimensions	39
3.3.7	Handling Alignment Criteria	40
3.3.8	Manual Locking Features.....	41
3.3.9	Unique Labeling.....	43
3.3.10	Hand Tools	43
3.3.11	EVA Actuated Controls	45
3.3.12	EVA Handrails/Handholds	46
3.4	Worksite Outfitting	58
3.4.1	Worksite Mobility.....	58
3.4.2	Worksites.....	58
3.4.3	Dedicated EVA Worksite.....	60
3.4.4	Robotic Assisted EVA Worksites	61
3.4.5	Free-Float Worksites.....	61
3.4.6	Installation of Passive Worksite Interface (WIF) Sockets.....	61
3.4.7	Foot Restraint Ingress/Egress Aids	62
3.4.8	Visual Contrast at Worksite.....	63
3.5	Loads.....	64
3.5.1	Equipment Tether Attach Points	64
3.5.2	Microgravity EVA Crew-Induced Loads	66
3.5.3	Partial Gravity EVA Crew-Induced Loads	70
3.5.4	EVA Actuated Loads.....	73
3.5.5	EVA Mechanical Feedback.....	75
3.5.6	WIF Structural Loading	75
3.6	Connectors	77

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 5 of 280
Title: EHP xEVA System Compatibility Standards	

3.6.1	One-Handed EVA Operable Connectors	77
3.6.2	Connector Accessibility	78
3.6.3	Gloved-Hand Connector Accessibility	78
3.6.4	Incorrect Connector Mating Prevention.....	78
3.6.5	Connector Coding	79
3.6.6	Connector Protective Caps	79
3.6.7	Connector Rotational Travel	80
3.6.8	Connector Locking Feature.....	80
3.6.9	Visible Connector Alignment Markings	80
3.6.10	Connector Locking Ring.....	80
3.6.11	Connector Mating Status	81
3.6.12	Electrical Connectors	81
3.6.13	Fluid Connectors.....	83
3.7	EVA Operable Fasteners	84
3.7.1	Captive Fastener Design	84
3.7.2	Locking Features	84
3.7.3	EVA Hand-Actuated Fasteners Operation	85
3.7.4	EVA Push-in-Pull (PIP) Pins	87
3.7.5	Tool-Actuated Fasteners.....	88
3.8	Hardware Design	94
3.8.1	Battery Powered Devices.....	94
3.8.2	EVA Tools.....	95
3.8.3	Cable Routing/Restraint.....	96
3.8.4	Cable Routing Spacing	96
3.8.5	Paint.	96
3.9	Safety	97

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 6 of 280
Title: EHP xEVA System Compatibility Standards	

3.9.1	Hazardous Material Control	97
3.9.2	Venting of Hazardous Materials.....	97
3.9.3	Electrical and Safety Design	98
3.9.4	Shock Protection.....	98
3.9.5	Operation During Pressure Change.....	99
3.9.6	Component Hazardous Energy Provision	99
3.9.7	Detachable Parts	100
3.9.8	Touch Temperatures.....	100
3.9.9	xEVA Suit Glove Palm External Touch-Temperature Compliance for Small, Handheld Hardware	101
3.9.10	Incidental and Unlimited Contact Heat Transfer Rates	103
3.9.11	Sharp Edge and Protrusions Protection.....	104
3.9.12	Levers, Cranks, Hooks, and Controls	107
3.9.13	Exposed Holes or Slots.....	107
3.9.14	Pinch Hazard Protection	107
3.9.15	Equipment Clearance for Entrapment Hazard	108
3.9.16	Protection from Moving or Rotating Equipment	108
3.9.17	Emergency Ingress of Pressurized Habitable Environment.....	109
3.9.18	Emergency Ingress/Egress of Unpressurized Mobile Asset	110
3.9.19	External Surface Cleanliness Level	110
3.9.20	Tethering / Restraint	111
3.9.21	Crew Safety Tether Attachment.....	111
3.9.22	Crew Safety Restraint.....	112
3.9.23	Color for Crew Load-Rated Handhold/Handrails, Crew Safety Tethers and Safety Restraint Attachment Points.....	112
3.9.24	Manual Overrides	114
3.9.25	Entanglement, Cable Clamps, Ducts, or Retractors	114
3.10	Environments.....	115
3.10.1	Magnetic Field Limit.....	115
3.10.2	Voltage Potential Discharge.....	116
3.10.3	Lunar Surface Dust Tolerance	116
3.10.4	Lunar Surface Dust Mitigation.....	117
4.0	VERIFICATION	119
4.1	Responsibilities.....	119
4.2	Verification approach.....	119
4.3	Verification methods	121

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 7 of 280
Title: EHP xEVA System Compatibility Standards	

4.4	Verification Matrix	121
4.5	Artemis System Verifications	134
4.6	EHP Verifications.....	189
APPENDIX A	ACRONYMS AND ABBREVIATIONS AND GLOSSARY OF TERMS.	
		244
Appendix A-1	Acronyms and Abbreviations.....	244
Appendix A-2	Glossary of Terms	249
APPENDIX B	EVA BEST PRACTICES (BP)	254
APPENDIX C	NON-EVA BPS FOR CONSIDERATIONS.....	264
APPENDIX D	OPEN WORK.....	266
Appendix D-1	To Be Determined	266
Appendix D-2	To Be Resolved.....	266
Appendix D-3	Forward Work.....	268
APPENDIX E	ARTEMIS SYSTEM ALLOCATIONS.....	269

TABLES

Table 1.3.1-1	Convention and Notations Descriptions	16
Table 2.1-1	Applicable Documents List.....	18
Table 2.2-1	Reference Documents List.....	19
Table 3.3.1-1	Constraints for EVA Transport of Objects	34
Table 3.3.2 -1	Constraint for Partial Gravity Large Mass Items – Lifting	36
Table 3.3.2-2	Constraint for Partial Gravity Large Mass Items – Carrying	37
Table 3.3.2 -3	Partial Gravity Volume Limit Surface Carry	37
Table 3.3.3-1	Partial Gravity Step Design Parameter.....	38
Table 3.3.8.5-1	Soft Dock Force	43
Table 3.3.12.6.1-1	Partial Gravity Work Envelope	53
Table 3.4.3-1	Worksite Forces	60
Table 3.5.1.1-1	EVA Handling and EVA Equipment Tether Point Colors <TBR-EHP-10028-012>	64
Table 3.5.2-1	Microgravity EVA Crew-Induced Loads.....	67
Table 3.5.3-1	Partial Gravity EVA Crew-Induced Loads <TBR-EHP-10028-002>	71
Table 3.5.4-1	Miscellaneous Crew Induced Limit Loads	74
Table 3.7.3.4-1	EVA External Hexagonal Drive Interface.....	87
Table 3.7.5.4-1	Fastener Torque Data	91
Table 3.9.5-1	Depress and Repress Rates	99
Table 3.9.8-1	Touch Temperatures <TBR-EHP-10028-011>.....	101

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 8 of 280
Title: EHP xEVA System Compatibility Standards	

Table 3.9.9-1 30-Minute Touch Temperature Compliance for Small, Handheld Hardware <TBR-EHP-10028-011>	102
Table 3.9.9-2 30-Minute Touch Temperature Heat Rate Compliance <TBR-EHP-10028-011>	102
Table 3.9.10-1 Heat Transfer Rates <TBR-EHP-10028-011>	103
Table 3.9.11-1 Edge, Corner, and Protrusion Criteria – Edge and In-Plane Corner Radii	105
Table 3.9.11-2 Edge, Corner, and Protrusions Criteria – Protrusions and Outside Corners	106
Table 3.9.17-1 Emergency Ingress Capability.....	109
Table 3.9.19-1 External Surface Cleanliness Level.....	111
Table 3.9.23-1 Crew Load-Rated Colors <TBR-EHP-10028-012>.....	114
Table 3.10.1-1 Magnetic Field Limit	115
Table 4.4-1 Artemis Systems Verification Matrix.....	122
Table 4.5-1 Artemis System Verifications.....	134
Table 4.6-1 EHP Verifications	189
Table B-1 PIP Pin Acceptance Vibration Test (AVT) Levels	255
Table C-1 Standard Interface Temperature Differential.....	264
Table D1-1 To Be Determined	266
Table D2-1 To Be Resolved	266
Table D3-1 Forward Work	268
Table E-1: Artemis System Allocation Matrix	269

FIGURES

Figure 3.1.1-1 Translation Path Hardware Protrusion Envelope	22
Figure 3.1.1-2 Translation Path Minimum Diameter	22
Figure 3.1.2-1 Top View Partial Gravity Crew Envelope Diameter	23
Figure 3.1.2-2 Side View Partial Gravity Crew Envelope Height	24
Figure 3.1.3-1 Tool Interface Clearance.....	25
Figure 3.1.5-1 Gloved-Hand Clearance.....	26
Figure 3.1.6-1 EVA Handrail/Handhold Mounting Clearance Envelope	27
Figure 3.1.7-1 Handrail/Handhold Side Clearance	28
Figure 3.1.8-1 APFR Installation/Removal EVA Clearance Envelope – Top Mounted WIF	29
Figure 3.2.4-1 EVA Contact Hazard Preclusions Envelope.....	32
Figure 3.2.5-1 Side View Partial Gravity EVA Contact Hazards Height.....	33
Figure 3.3.12.1-1 Handrail/Handhold Profile Dimensions.....	47

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 9 of 280
Title: EHP xEVA System Compatibility Standards	

Figure 3.3.12.1-2 Partial Gravity Handrail/Handhold Profile Dimensions	47
Figure 3.3.12.4-1 Handrail/Handhold Gaps	49
Figure 3.3.12.6.1-1 Work Envelope – Microgravity	51
Figure 3.3.12.6.1-2 Partial Gravity Work Envelope	52
Figure 3.3.12.6.1-3 Partial Gravity Work Envelope – Reach Envelope	54
Figure 3.3.12.6.1-4 Two-Handed Reach Envelope Width	55
Figure 3.3.12.6.4-1 Crew Aids for Stability	57
Figure 3.4.2.2-1 Microgravity Working Volume for Manipulative EVA Tasks.....	59
Figure 3.4.2.2-2 Partial Gravity Working Volume	60
Figure 3.4.7-1 Ingress and Egress Aids for Foot Restraint.....	63
Figure 3.5.1.2-1 EVA Equipment Tether Point Dimensions.....	65
Figure 3.5.2-1 Opposing Forces on Handrail/Handhold	70
Figure 3.5.6-1 Top Mounted WIF Worksite Half Loading Criteria.....	77
Figure 3.7.3.3-1 Flight Knob Examples	86
Figure 3.7.5.2-1 11 mm Hexagonal EVA and Robotic Compatible Bolt	89
Figure 3.7.5.2-2 7/16-Inch Hexagonal Eva and Robotic Compatible Bolt.....	90
Figure 3.9.11-1 Exposed Corner and Edges	105
Figure B-1 EVABP-27 Connector Spacing	260
Figure B-2 EVABP-30 Minimum Don/Doff Volume Dimensions for a Single xEVA Suit	262

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 10 of 280
Title: EHP xEVA System Compatibility Standards	

Preface

Over the life of the Extravehicular Mobility Unit (EMU), series of products detailing “how to build Extravehicular Activity (EVA) System hardware”, “how to interface with EVA System hardware” and “how to design hardware EVA will access and utilize” were created to provide consolidated locations of the best practices and lessons learned of the EVA Community to projects unfamiliar with EVA. There were multiple documents describing these “how to” about EVA, from three different eras of human spaceflight. While these documents may be associated with specific programs or address specific needs such as payload requirements, there is an overlap of the type of information contained in each of them. For instance, three documents refer to the needs for “handling features to interface with other mass handling systems” such as the Body Restraint Tether (BRT), with some providing outdated values, no values at all or the correct values. Since the inception of each of these documents, some have continued to receive updates due to the necessity of an on-going program while others have remained unchanged for 20+ years and have led to discrepancies between the current accepted values and these outdated documents.

To prevent conflicts between multiple sources of information and to assess what is truly applicable to Exploration EVA System (xEVA) activities at Exploration Destinations (e.g., cislunar space and lunar surface), EVA-EXP-0035, *Exploration EVA System Compatibility* document was generated to produce a set of EVA design standards that apply to all hardware and systems that interface with the xEVA Suit and EVA activities at Exploration Destinations. While prior Space Shuttle Program, International Space Station (ISS) and Constellation Program (CxP) documents were reviewed and in part incorporated into this document, no document from those specific programs will be directly levied on any future Exploration Mission. This does not supersede documents for current and inactive programs, as documentation applicable to any on-going program will continue until the conclusion of that program.

Efforts in the past, such as the creation of JSC 28918, *EVA Design Requirements and Considerations* were attempted to provide a common source, that included ISS lessons learned, for all EVA requirements for future programs, but due to the on-going churn of ISS-EVA documentation and a lack of new programs going into development, resources were spent on maintaining and updating only the active program documentation. JSC 28918 was never officially levied as an ISS program document. Thus, JSC 28918 has not been updated. Instead, the focus has continued to keep ISS Program-levied EVA requirements documents up to date.

After generation of tailored versions for Artemis Systems such as Gateway and Human Landing System (HLS) Programs, and the existing EVA-EXP-0035 document, the newly formed Extravehicular Activity and Human Surface Mobility Program (EHP) determined that there was a need to consolidate into one document that would apply to all Artemis Systems. This led to the creation of this document EHP-10028, *EHP Exploration EVA (xEVA) System Compatibility Standards*.

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 11 of 280
Title: EHP xEVA System Compatibility Standards	

This document establishes design standards that minimize the performance risks to EVA flight crewmembers in human space flight programs and applies to future space vehicles, habitats, facilities, or any other hardware that will interface with the suited crewmember during space flight in various missions (e.g., cislunar space and lunar surface). As Design Reference Missions (DRM) evolve, this document will be updated to reflect utilizing the xEVA System at that destination.

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 12 of 280
Title: EHP xEVA System Compatibility Standards	

1.0 INTRODUCTION

Extravehicular Activity (EVA) is a significant capability for Artemis Systems which includes Gateway Program (modules), Human Landing System (HLS) Program (Integrated Lander and Human Class Delivery Lander), Lunar Terrain Vehicle (LTV), Pressurized Rover (PR), Payloads, and any other future lunar surface assets that will be utilized for exploration or science. The EVA and Human Surface Mobility Program (EHP) will provide the following: Exploration EVA (xEVA) suits, xEVA tools and crew aids along with vehicle integration/support hardware, LTV, and PR. Being a distributed and dynamic capability (moving components), xEVA System hardware will interface with almost all elements of the Artemis architecture. As such, it is important to document a singular set of EVA compatibility standards at the programmatic level to articulate “how to interface with xEVA suit hardware” and “how to design hardware that EVA crewmembers will access and manipulate.”

1.1 PURPOSE

The EHP xEVA Compatibility Standards document establishes a set of EVA compatibility design standards for hardware and systems that have an interface with the xEVA suit and/or must facilitate EVA activities. This document will be used by all Artemis Systems and contains the fundamental necessary information required for building and verifying that Artemis Systems hardware are compatible with suited EVA flight crewmembers to perform successful EVAs in microgravity/lunar orbit and on the lunar surface. It is to be used for the development of lower-level program/project-specific requirements. Compatibility standards are in section 3.0 with gravity field (microgravity and partial gravity) associated with each requirement.

As an interface standard, EHP-10028 levies requirements on the xEVA System (managed by EVA Development Project), interfacing EHP Projects (LTV and PR) and cross-program partners (Human Landing System, Gateway, and Payloads). It contains verification guidance for Hardware Providers as well as EHP.

The Gateway Program is maintained and inspected both by the Gateway External Robotics Systems (GERS) capability (reference GP 10027, Gateway Concept of Operations) and by EVA capability. GERS is the Extravehicular Robotics (EVR) capability on Gateway. EVA capability is used for certain pre-defined maintenance tasks as well as for contingencies such as failure of the GERS, inability of the GERS to complete a task that was intended for GERS compatibility, or unforeseen tasks. When both accommodations of EVA and GERS operations are required, which are not explicitly defined in this document, an integrated strategy which will be certified or evaluated on a mission-specific basis to determine how best to account for both GERS and EVA design constraints and requirements necessary to support mission objectives.

1.2 SCOPE

This document defines standards controlled by the EHP ensuring that all hardware and Artemis System’s design are compatible with the xEVA suits and EVA activities in any

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 13 of 280
Title: EHP xEVA System Compatibility Standards	

given mission destination (e.g., cislunar space, and lunar surface). The standards herein are applicable to any hardware internally or externally that will be near, come in contact with or has features that can present hazards in any way to the xEVA crewmember. Projects developed for the International Space Station (ISS) will continue to reference SSP 30256 *EVA Interface Control Document (ICD)*, other applicable Interface Requirements Documents (IRDs), and JSC 26626 *EVA Hardware Generic design Requirements Document* as appropriate.

This document does not include information on the interfaces for space suit servicing, performance, and checkout equipment (e.g., items that directly interface with the space suit, such as fluid and electrical connectors), or non-mechanical interfaces (such as wireless communication, data, video, and audio). All these items will be defined in EHP-10029, *xEVA System to Artemis Systems Interface Requirements Document (IRD)* **<TBR-EHP-10028-016>**.

As part of the construction of this document, active EVA documents from the Shuttle Program, ISS Program and Constellation Program (CxP) eras were assessed for requirements pertaining to hardware compatibility with the EVA System and operations. These documents were cross compared, and the relevant information was used to form the basis of the standards included in EHP-10028 and updates were made to correspond with the latest mission destinations. The initial documents reviewed with the specified revisions included the following:

- JSC 26626 Revision A, *Extravehicular Activity (EVA) Hardware Generic Design Requirements Document (GDRD)*
- JSC 26626 Draft Revision B, *GDRD (unreleased)*
- JSC 28918 Baseline, *EVA Design Requirements and Considerations*
- SSP 30256 Revision J, *Extravehicular Activity (EVA) Standard ICD*
- SSP 57003 Revision L, *ISS External Payload IRD*
- SSP 50005 Revision F, *International Space Station (ISS) Flight Crew Integration Standard (NASA-STD-3001/T) – intent of this document for ISS is satisfied by NASA-STD-3001*
- SSP 50021 Baseline Document Change Notice (DCN) 005, *Safety Requirements Document* (Note – this document was superseded by Baseline of SSP 51721, *Safety Requirements Document*)
- CxP 70130 Revision A, *CxP EVA Design and Construction Specification* (document no longer active)
- NASA–STD-3001 V2 Revision A, *Human Factors, Habitability, and Environmental Health*.

**Evaluation of revisions to the still active documents are assessed for impacts to standards contained in this document.

In many cases the values used for defining microgravity compatibility with a space suit are based on heritage United States (US) EVA capabilities as carried over from the Space

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 14 of 280
Title: EHP xEVA System Compatibility Standards	

Shuttle Program and modified for the International Space Station (ISS) Program. It is expected that the xEVA Suit will either meet or exceed the capabilities of the original EVA suit capabilities. The xEVA System is provided to National Aeronautics and Space Administration (NASA) on a competitive service contract in which vendors are awarded Task Orders for future missions, tasks, and activities in an ongoing basis. Due to the uncertainty of the specific suit design that will be used for each individual mission/task/activity as a result of the ongoing Task Order award schedule, the terminology and noted dimensions/values in this document will be based on the government reference next generation EVA Suit design, also known as the Exploration Extravehicular Mobility Unit (xEMU). In general, an xEVA System consists of the EVA Suit (interchangeable with “xEVA Suit” or “xEMU” for the purposes of this document), EVA tools, and ancillary hardware that support EVA functions.

Each standard was assessed for application in a partial gravity environment and captures partial gravity considerations in the standard. As the xEVA suit design matures and/or as other suppliers of EVA hardware are identified, updates to the standards will be proposed to the appropriate change authority to reflect the improvement/changes in EVA capabilities. This document covers both microgravity and partial gravity EVA requirements. Partial gravity references in this document are specific to the lunar gravity field (1/6 gravity). As missions beyond lunar surface are developed, consideration of additional gravity fields (e.g., Martian gravity) will be addressed. References to weight in this document take into account the applicable gravity field (e.g., approximately 1/6th of Earth gravity for Lunar conditions).

For the purposes of this document, the term payload applies to any utilization science/research specific hardware that will interact with EVA equipment or tools. Appendix A-2 provides glossary terms associated with payloads. There are external payloads on Gateway, which are not expected to be EVA serviceable but will be designed to protect EVA crewmembers. Gateway Payloads, Maintenance Items (MI), etc. designed for EVA cannot preclude GERS operations.

GP 10031, Gateway Visiting Vehicle Interface Requirements Document (IRD) defines the subset of EVA compatibility requirements for those Visiting Vehicles that may protrude into Gateway EVA translation paths. Any hardware that protrudes into the EVA contact hazard preclusion envelope must meet EVA kickloads, touch temperatures, and EVA safety requirements to protect the crewmember while performing an EVA on Gateway as well as protect the docked Visiting Vehicle hardware, even though the Visiting Vehicle may not have EVA requirements imposed on it. *GP 10031, Gateway Visiting Vehicle IRD Annex 1: HLS Requirements* requirement INFO-VV-HLS-0112 specifies that when HLS is docked to Gateway that both vehicles will meet EVA compatibility requirements.

Crew health and safety standards that describe the human-to-suit interfaces (like anthropometric requirements), or Intravehicular Activity (IVA) suit-to-vehicle interfaces are not included in this document. Those standards and their requirement guidance are

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 15 of 280
Title: EHP xEVA System Compatibility Standards	

included in NASA-STD-3001, Volume 1: Crew Health and Volume 2: Human Factors, Habitability, and Environmental Health, or tailored through respective Program/Project System Requirements Documents (SRDs). Specific crew health and safety related standards are found in NASA-STD-3001 V2. A supplementary NASA document, NASA/SP-2010-3407, *Human Integration Design Handbook (HIDH)* is a compilation of human space flight history and provides a resource for deriving program-specific requirements in accordance with NASA-STD-3001, Volume 2. EHP Projects, Gateway and HLS tailor NASA-STD-3001 through respective Project/Program documents.

1.3 APPLICABILITY/ALLOCATIONS

All requirements in this standard are applicable to the development of all Artemis program/projects that perform EVA operations or interact with the xEVA System. Each requirement contains the Gravity Field Applicability along with Allocation to the Artemis Systems. For the purposes of this document, “Artemis System” is considered to include the following:

1. External Program Spacecraft with EVA Capabilities – This document is applicable to vehicles that require EVA operations performing nominal and contingency: maintenance, repair, and replacement, payload, and science utilization. Currently two external Programs (Gateway and HLS) have been identified, but any future EHP external Exploration Programs will utilize this document for EVA compatibility standards. HLS Assumptions for microgravity application: Microgravity requirements would only apply in cases where translation paths are required from HLS airlock to HLS EVA worksites (if vehicle had any), and that being docked to the staging vehicle cannot impede the HLS translation paths with no identified hazards. If no microgravity worksites required, then N/A, but requirement is applicable to drive review of scenarios and vehicle design.
2. Internal EHP Project Spacecrafts (e.g., LTV and PR) with EVA Capabilities – This document is applicable to vehicles that require EVA operations performing nominal and contingency: maintenance, repair and replacement, and payload operations.
3. EVA System (managed by EVA Development Project) – This document is applicable to the xEVA System and decomposition of allocated requirements will be performed by EVA Development Project. This document applies to the xEVA System, and specifically to portions of the xEVA System that will be operated by the suited crewmember.
4. Payloads – This document is applicable to science and research instrument hardware that will be near to, come in contact with, or present hazards in any way to xEVA suits or suited EVA flight crewmembers.

All hardware providers need to make provisions for protecting the EVA crewmember from anything that presents a danger or safety hazard. EVA hazards that are not controlled on hardware that are outside the translation paths or associated zones will be declared a keep out zone or no touch area and this information provided to the Flight Operations Directorate (FOD) as operational controls.

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 16 of 280
Title: EHP xEVA System Compatibility Standards	

1.4 CONVENTIONS AND NOTATIONS

1.4.1 Shall Statements (Standards/Requirements)

The convention used in this document which indicates requirements, standards, goals, and statements of facts is as follows:

TABLE 1.3.1-1 CONVENTION AND NOTATIONS DESCRIPTIONS

Designation	Description
“Shall”	used to indicate a requirement that is binding, which must be implemented, and its implementation verified in the design.
“Should”	used to indicate good practice or a goal which is desirable, but not mandatory and does not require a formal verification.
“Will”	used to indicate a statement of fact or declaration of purpose on the part of the government that is reflective of decisions or realities that exist and are to be taken as a given and not open to debate or discussion.

Those statements that fall more into a design consideration/best practice are in APPENDIX B and APPENDIX C. While not considered requirements, these best practices are highly desirable for the design of hardware compatibility with EVA operations.

1.4.2 Requirement Identification

As documents are revised, there are cases where sections will move; therefore, each requirement in this document is uniquely numbered, and identified by the presence of a tag. The requirements tag “[EVASC.XXXX]” provides a trace to historical information across multiple Programs and Projects.

1.4.3 Rationale

A rationale statement is included for each requirement only when necessary. The purpose of the rationale statement is to indicate why the requirement is needed, the basis for its inclusion in a requirements document, and to provide context and examples to stakeholders. It is important to note that a rationale is not binding, and it only provides supporting information. In the event there is an inconsistency between the requirement and the rationale, the requirement is binding and takes precedence.

1.4.4 Units

Artemis Systems utilize NIST SP 811, *Guide for the Use of International System of Units (SI)*, for standardization and conversions of the units of measure. For the purposes of this document, the unit of measure should be considered as metric as primary with English units in parenthesis unless otherwise noted. Microgravity requirements with associated figures and tables were derived from heritage ISS EVA requirements documents that contained English units. SI units in this document for those microgravity

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 17 of 280
Title: EHP xEVA System Compatibility Standards	

applications were converted from the heritage ISS English values. Therefore, the English units are considered to be the driver for metric conversion.

NOTE: Some referenced drawings and documents may use English units as primary and may or may not have metric units as secondary.

1.5 CHANGE AUTHORITY/RESPONSIBILITY

The Extravehicular Activity and Human Surface Mobility Program (EHP) Board Authority for Approval of this product is the EHP Control Board (EHPCB). Board Authority for future revisions of this product is not delegated. Board Authority for changes is the EHPCB.

The administration of future revisions will be managed by the Office of Primary Responsibility (OPR). Proposed changes to this product shall be submitted via Change Management Workflow (CMW). All such requests will adhere to the EHP-20000, *Extravehicular Activity and Human Surface Mobility Program (EHP) Configuration and Data Management Plan*.

The appropriate NASA Office of Primary Responsibility (OPR) identified for this product is the EHP Systems Engineering and Integration (SE&I) Office (DI). The EHP Requirements, Interfaces, Verification, and Analysis (RIVA) Panel will be used to bring updates for community and stakeholder review prior to CR release. Approval of the technical accuracy of this document resides within the EHP SE&I Control Board (CB), with Programs/Projects providing agreement/concurrence at the EHPCB.

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 18 of 280
Title: EHP xEVA System Compatibility Standards	

2.0 DOCUMENTS

2.1 APPLICABLE DOCUMENTS

The following documents include specifications, models, standards, guidelines, handbooks, and other special publications. The documents listed in Table 2.1-1 are applicable to the extent specified herein. The EHP specific applicable documents and standards revisions listed are in accordance with EHP-10012, *Extravehicular Activity and Human Surface Mobility Program (EHP) Systems Requirement Document*, section 2.2.1. Any notation of NASA approved alternate standard needs to have Cross-Program Chief Engineer approval. Gateway Program applicable document revisions listed in Table 2.1-1 are in accordance with GP 10000, *Gateway Program System Requirements* section 2.2.1. HLS Program applicable document revisions listed in Table 2.1-1 are in accordance with the respective HLS SRD (reference Table 2.2-1).

TABLE 2.1-1 APPLICABLE DOCUMENTS LIST

Document Number	Document Title
EHP-10012	Extravehicular Activity and Human Surface Mobility Program (EHP) Systems Requirements Document
EHP-10013	Extravehicular Activity and Human Surface Mobility Program (EHP) Verification and Validation Plan
EHP-10029 <TBR-EHP-10028-016>	Exploration Extravehicular Activity (xEVA) System to Artemis System Interface Requirements Document (IRD)
EHP-10074 <TBR-EHP-10028-008>	EHP EVA Tool List and Process for EVA Operations
EHP-20000	Extravehicular Activity and Human Surface Mobility Program (EHP) Configuration and Data Management Plan
EVA-EXP-0049	Extravehicular Activity (EVA) Office Exploration EVA (xEVA) System Labeling Guidelines
GP 10000	Gateway Program System Requirements
GP 11461	Gateway Requirements for the Control of Electromagnetic Interference (EMI) Characteristics of Subsystems and Equipment
GSDO-RQMT-1080	Cross-Program Contamination Control Requirements
HLS-STD-007	Human Landing System (HLS) Requirements for the Control of Electromagnetic Interference (EMI) Characteristics of Subsystems and Equipment
JPR 5322.1	Contamination Control Requirements Manual
NASA-STD-4003	Electrical Bonding for NASA Launch Vehicles, Spacecraft, Payloads, and Flight Equipment
SDG33107728	Tube – Handrail and Handgrip (via SEG33106347, Top Mounted Handrail Assy)

2.2 REFERENCE DOCUMENTS

The following documents contain supplemental information to guide the user in the application of this document and can include Type 3 Technical Authority (TA) Standards. Reference Documents may be called out in any section or as a part of a requirement

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 19 of 280
Title: EHP xEVA System Compatibility Standards	

rationale and are not binding. Table 2-2.1 only lists the reference documents and revisions for this product.

TABLE 2.2-1 REFERENCE DOCUMENTS LIST

Document Number	Document Revision	Document Title
2024-ABF-001	January 2024	Suited Capability Requirements for Partial Gravity – Anthropometry and Biomechanical Perspective
ACD-50012	Current	Artemis Campaign Development (ACD) Enterprise Verification and Validation Plan
AML-01-02-R1	March 22, 2001	Advanced Materials Laboratory Test Report: Hot D-Ring / Space Suit Assembly (SSA) Lay-up Incidental Contact Test
EHP-10004	Current	Extravehicular Activity and Human Surface Mobility Program (EHP) Systems Engineering Management Plan (SEMP)
ES6-23-044	September 18, 2023	Lunar EVA Crew Induced Loads on Restraint Attachment Points from Crew Fall
ES6-24-001	February 22, 2024	Lunar EVA Crew Induced Loads
EX-24-008	April 23, 2024	EX-24-005, Artemis System Lunar Dust Tolerance – Amount of Dust Justification and Calculation
EVA-EXP-0031	Baseline	EVA Office Extravehicular Activity (EVA) Airlocks and Alternative Ingress/Egress Methods Document
EVA-EXP-0032	Baseline	EVA Office EVA-ISS Interface Definition Document
EVA-EXP-0034	Revision C	Extravehicular Activity (EVA) Office Exploration EVA (xEVA) System Technical Standards
EVA-EXP-0035	Baseline w/Admin Change 1	EVA Office xEVA System Compatibility
EVA-EXP-0067	Revision C	Human Landing System (HLS) Program Extravehicular Activity (EVA) Interface Requirements Control Document (IRCD)
EVA-EXP-0070	Revision D with Change 2	HLS Program Extravehicular Activity (EVA) Compatibility Interface Requirements Document (IRD)
GP 10027	Current	Gateway Concept of Operations
GP 10031	Current	Gateway Visiting Vehicle IRD
GP 10031-01	Current	Gateway Visiting Vehicle IRD Annex 1: HLS Requirements
HLS-RQMT-002	Current	HLS Partner System Requirements Document (PaSRD)
HLS-RQMT-006	Current	HLS Integrated Lander Requirements Document – Sustained Phase
HLS-RQMT-007	Current	HLS Human-Class Delivery Lander (HDL) Requirements Document – Sustained Phase
HLS-RQMT-009	Current	HLS Provider Integrated Lander Systems Requirements Document -Sustained - SpaceX
HLS-RQMT-109	Current	HLS Provider Integrated Lander Systems Requirements Document -Sustained – Blue Origin
JSC 28528	A	Preparation and Revision of Cx Mockup Development Requirements Document (MDRD)

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 20 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 2.2-1 REFERENCE DOCUMENTS LIST

Document Number	Document Revision	Document Title
JSC 39117	C	Extravehicular Mobility Unit (EMU) International Space Station (ISS) Extravehicular Activity (EVA) Thermal Environment Requirements for Certification
JSC 39198	3/14/2018	Incidental Contact Temperature Test on Extravehicular Mobility Unit (EMU) Softgoods Materials Lay-Up
JSC 65828	B w/ Change 1	Structural Design Requirements and Factors of Safety for Spaceflight Hardware
MIL-PRF-19500	R	Semiconductor Devices, General Specification
MIL-DTL-81381	C Amendment 1	Wire, Electric, Polyimide-Insulated, Copper, or Copper Alloy, Detail Specification
NASA/SP-2010-3407	Rev 1	Human Integration Design Handbook (HIDH)
NASA-STD-3001 Volume 2	D	NASA Spaceflight Human-System Standard Volume 2: Human Factors, Habitability, and Environmental Health
NASA-STD-5020	B	Requirements for Threaded Fastening Systems in Spaceflight Hardware
NASA-STD-6016	C	Standard Materials and Processes Requirements for Spacecraft
NIST SP 811	3/2008	Guide for the Use of the International System of Units (SI)
NPR 8705.4	A	Risk Classification for NASA Payloads
SDG33107506	Latest	Knob, Push, Pitch Joint Assembly Articulating Portable Foot Restraint (APFR), ISS
SED33107181	Latest	EVA Knob Assembly, Bridge Clamp
SEG33114987	Latest	1.0 Inch Quick Disconnect (QD) Button Depress Tool Assy
SLS-SPEC-159	I	Cross-Program Design Specification for Natural Environments (DSNE)
SSP 30256	Revision K	Extravehicular Activity (EVA) Standard Interface Control Document (ICD)
SSP 30423	L	Space Station Approved Electrical, Electronic and Electromechanical (EEE) Part List
SSP 57003	M	ISS Payloads Interface Requirements Document (IRD)

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 21 of 280
Title: EHP xEVA System Compatibility Standards	

3.0 REQUIREMENTS

This section defines the minimum requirements and constraints that apply to performance, design, interoperability, maintainability, and safety of hardware that may come in contact or be operated by crew in pressurized EVA suits. The gravity field applicability (microgravity and partial gravity) is included after each requirement along with allocation to Artemis Systems. Variances to these requirements may be proposed to EHP, considered by the EHP RIVA Panel in accordance with EHP-10004, *Extravehicular Activity and Human Surface Mobility Program (EHP) Systems Engineering Management Plan (SEMP)*, and approved via the EHP Variance CR process described in EHP-20000, *Extravehicular Activity and Human Surface Mobility Program (EHP) Configuration and Data Management Plan*.

3.1 CLEARANCES

3.1.1 Microgravity Translation Path Hardware Protrusion

[EVASC.0001] The Artemis System shall design or configure equipment to not protrude into the minimum 1.1 m (43 in) diameter translation path envelope, excluding translation aids (handrails/handholds) in microgravity as outlined in Figure 3.1.1-1 *Translation Path Hardware Protrusion Envelope* and Figure 3.1.1-2 *Translation Path Minimum Diameter*.

Rationale: In microgravity, the minimum diameter of the translation corridor of 1.1 m (43 in) is to accommodate hand-over-hand EVA crewmember translation. Interference in translation paths may pose hazards to EVA crewmembers or impact the success of EVA activities by precluding translation to worksites. This includes any obstructions or intrusions in the envelope. This microgravity minimum diameter is based on the heritage ISS EMU suit and is associated with microgravity application.

Gravity Field Applicability: Microgravity

Allocation: EVA Dev, Gateway, HLS

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 22 of 280
Title: EHP xEVA System Compatibility Standards	

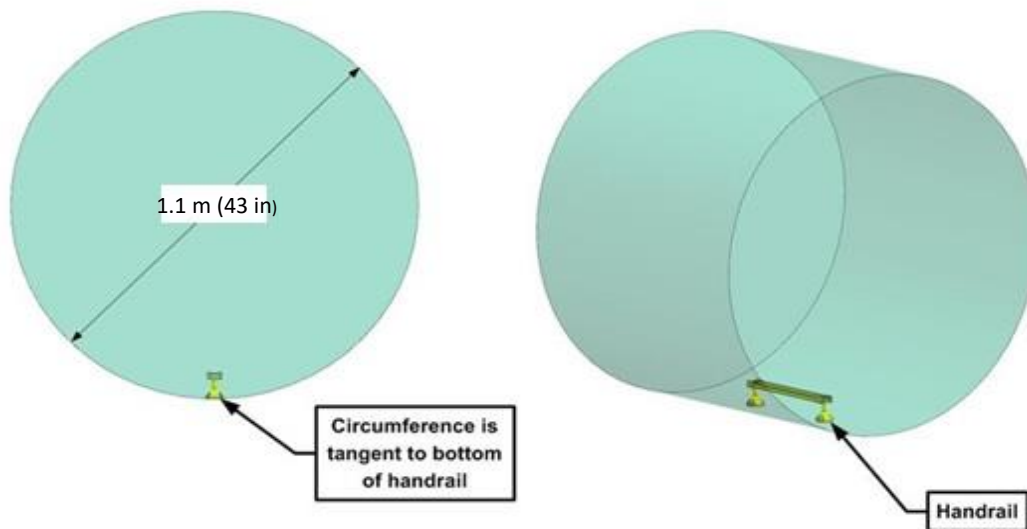


FIGURE 3.1.1-1 TRANSLATION PATH HARDWARE PROTRUSION ENVELOPE

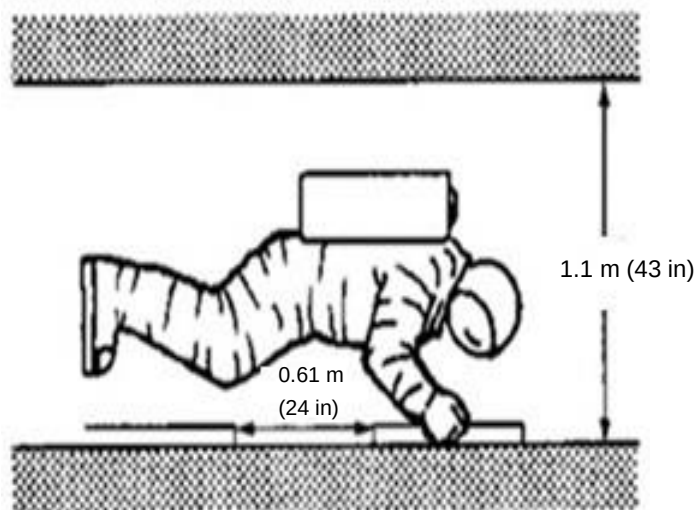


FIGURE 3.1.1-2 TRANSLATION PATH MINIMUM DIAMETER

3.1.2 Partial Gravity Translation Path Hardware Protrusion

[EVASC.0171] The Artemis System shall design or configure equipment to not protrude into the minimum 1.1 m (43 in) diameter translation path envelope, excluding translation aids (handrails/handholds) and provide a translation path free of equipment and protrusions that accommodates the volume of the suited crewmember in partial gravity as outlined in Figure 3.1.2-1 *Top View Partial Gravity Crew Envelope Diameter* and Figure 3.1.2-2 *Side View Partial Gravity Crew Envelope Height*.

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 23 of 280
Title: EHP xEVA System Compatibility Standards	

Rationale: In partial gravity, the breadth and height of the translation path must accommodate the volume of the suited crewmember in any direction which the crew is expected to move. Interference in translation paths may pose hazards to EVA crewmembers or impact the success of EVA activities by precluding translation to worksites. This includes any obstructions or intrusions in the envelope. Partial gravity application ensures crewmembers can traverse around, to and from vehicle worksite areas (e.g., catwalk around a cargo lander, walking under the lander, ingress/egress of vehicle to the surface, areas around unpressurized and PRs).

Gravity Field Applicability: Partial Gravity

Allocation: EVA Dev, LTV, PR, HLS

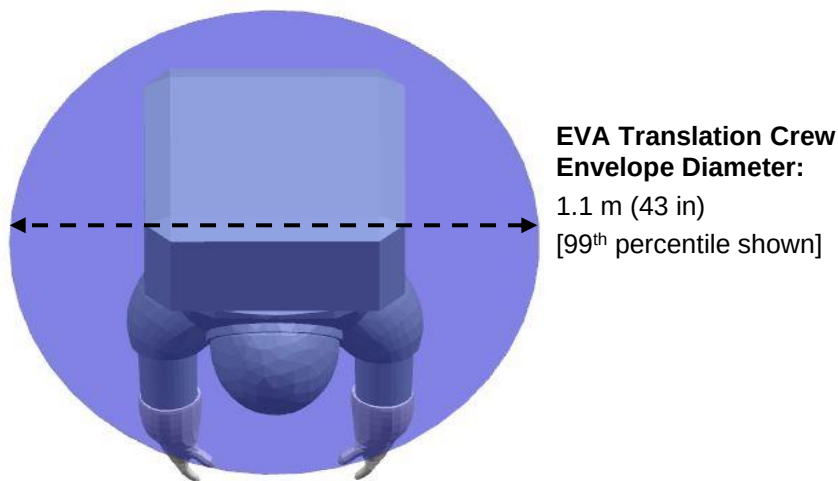


FIGURE 3.1.2-1 TOP VIEW PARTIAL GRAVITY CREW ENVELOPE DIAMETER

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 24 of 280
Title: EHP xEVA System Compatibility Standards	

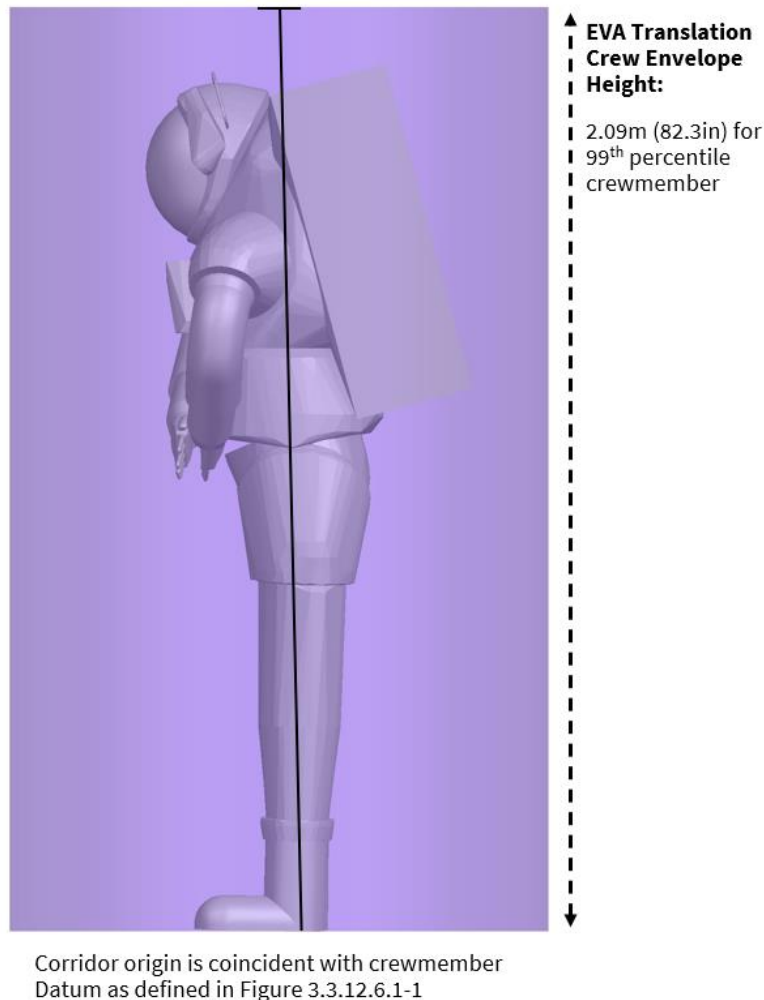


FIGURE 3.1.2-2 SIDE VIEW PARTIAL GRAVITY CREW ENVELOPE HEIGHT

3.1.3 Tool Interface Clearance

[EVASC.0002] The Artemis System shall maintain 7.6 cm (3.0 in) of clearance between the gripping portions of a tool interface and any surrounding hardware or obstructions with a tool handle offset maximum of 35.6 cm (14.0 in) as defined in Figure 3.1.3-1 *Tool Interface Clearance*.

Rationale: Tools to be used during EVAs are to be identified by the Artemis System, and clearances for efficient application are to be accommodated to ensure that the EVA task can be accomplished. The intent is that this requirement applies to EVA crew operated manual tools interfacing with EVA fasteners. To ensure the task can be accomplished, the clearance between the work surface and the tool design needs to include operation by a pressurized gloved hand. This includes

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 25 of 280
Title: EHP xEVA System Compatibility Standards	

protecting for the 7.6 cm (3.0 in) clearance to the nearest obstruction(s) in addition to the grasp length of the tool handle. Reference [EVASC.0004] which defines the handrail mounting clearance distance that accommodates grasp of the pressurized gloved hand. The bolt head tool clearance is defined in requirement [EVASC.0007]. Minimum grip length is assumed to be length of handrail/handhold/handle requirement [EVASC.0030].

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

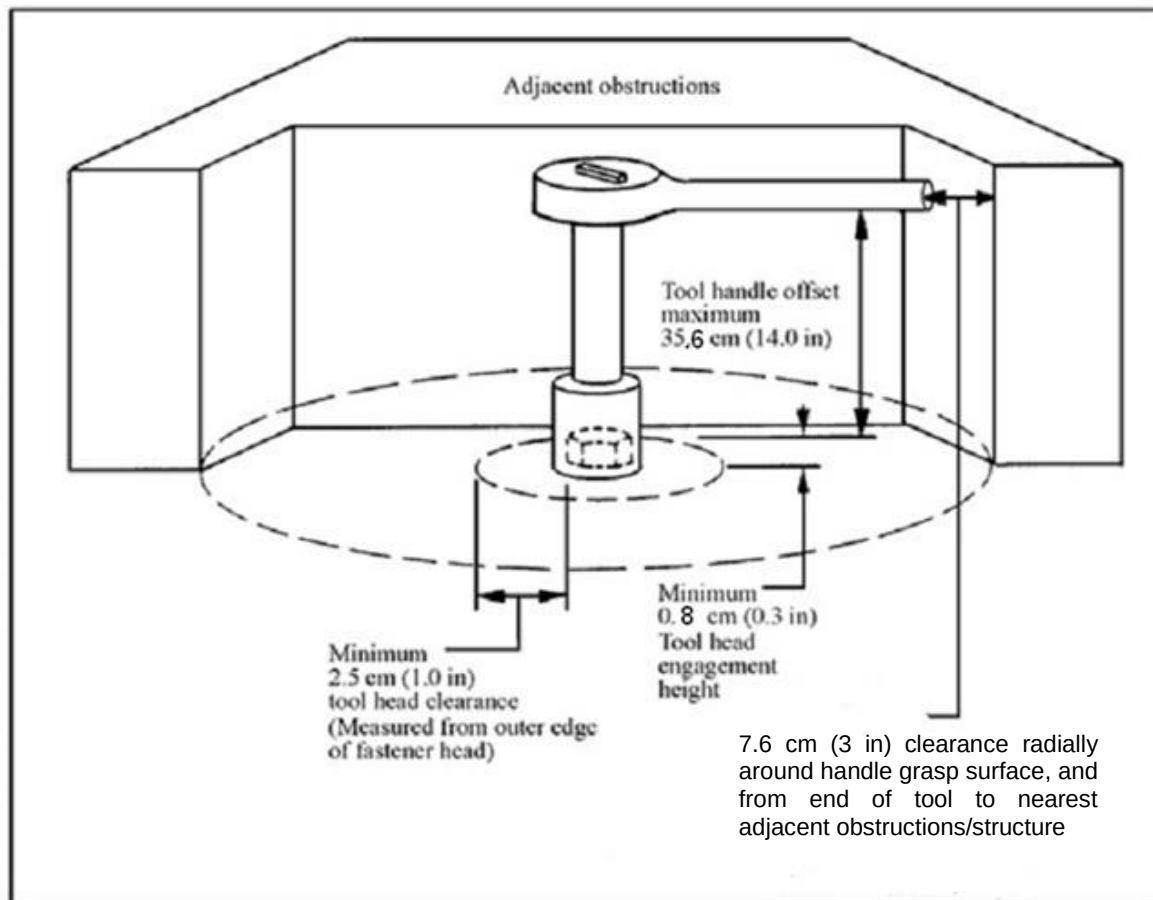


FIGURE 3.1.3-1 TOOL INTERFACE CLEARANCE

3.1.4 Bolt Clearance

[EVASC.0007] The Artemis System shall maintain a 2.5 cm (1.0 in) clearance around the bolt head outer edge with a minimum tool head engagement height of 0.8 cm (0.3 in) to

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 26 of 280
Title: EHP xEVA System Compatibility Standards	

ensure access as defined in requirement [EVASC.0002] Figure 3.1.3-1 *Tool Interface Clearance*.

Rationale: The intent of this requirement is to ensure tool access with the fastener head.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

3.1.5 Gloved-Hand Clearance

[EVASC.0003] The Artemis System shall maintain minimum clearance envelope for a gloved hand access as specified in Figure 3.1.5-1 *Gloved-Hand Clearance*.

Rationale: Pressurized EVA gloves have their own thickness in addition to the human hands they protect. Interfaces to be operated by the EVA crewmember must accommodate the dimensions of the hand/glove combination.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

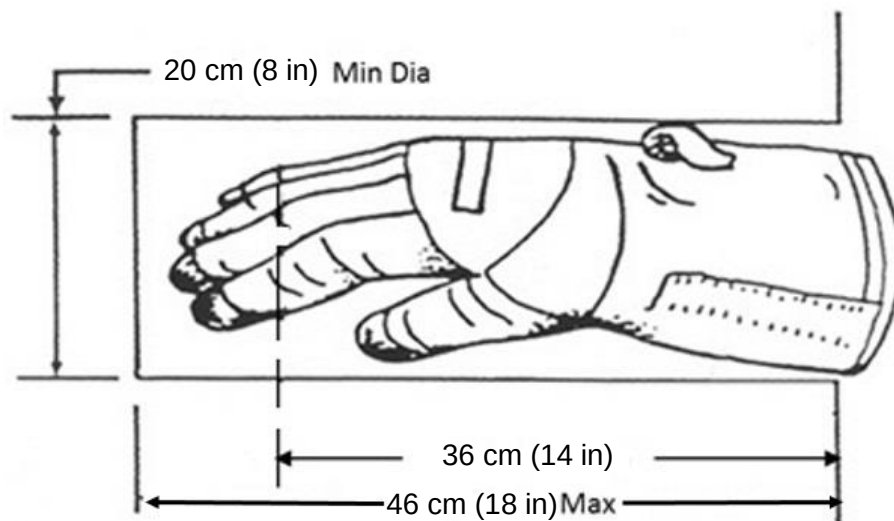


FIGURE 3.1.5-1 GLOVED-HAND CLEARANCE

3.1.6 Handrail/Handhold Mounting Clearance

[EVASC.0004] The Artemis System shall maintain a minimum clearance distance between the lower surface of the handrail/handhold and the mounting surface of 5.72 cm

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 27 of 280
Title: EHP xEVA System Compatibility Standards	

(2.25 in) as shown in Figure 3.1.6-1 *EVA Handrail/Handhold Mounting Clearance Envelope*.

Rationale: EVA handrail cross sections themselves have been designed to facilitate grasping by the gloved EVA crewmember's hand. However, it is theoretically possible to mount a handrail in such a way (such as immediately next to structure or over an exposed structural rib) that the EVA crewmember's gloved hand cannot access the EVA handrail. To ensure EVA handrails are mounted in a manner that preserves/facilitates their use, the area around and underneath handrails/handholds must be free of obstructions within the specified clearance dimensions. Adequate clearance is required to facilitate use by a pressurized gloved hand. This also supports clearance for tether hook attachment to the stanchions.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

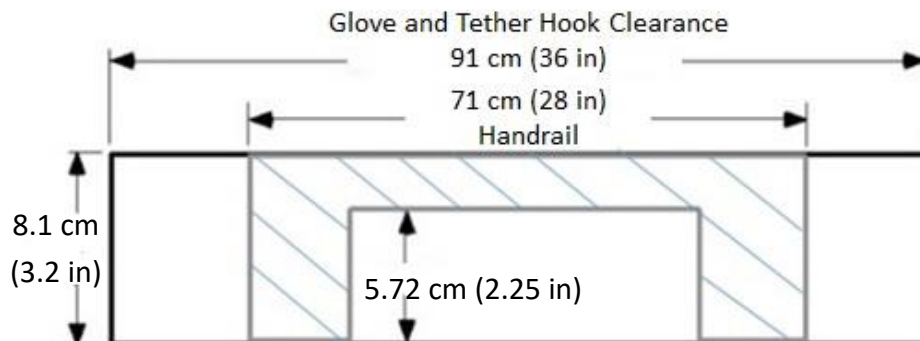


FIGURE 3.1.6-1 EVA HANDRAIL/HANDHOLD MOUNTING CLEARANCE ENVELOPE

Note: The 71 cm (28 in) handrail is an example.

3.1.7 Handrail/Handhold Side Clearance

[EVASC.0005] The Artemis System shall maintain a minimum 10 cm (4 in) clearance to both sides of handrail/handholds as defined in Figure 3.1.7-1 *Handrail/Handhold Side Clearance*.

Rationale: To ensure EVA handrails are mounted in a manner that preserves/facilitates their use, the area around and underneath handrails/handholds must be free of obstructions within the specified clearance

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 28 of 280
Title: EHP xEVA System Compatibility Standards	

dimensions. Adequate clearance is required to facilitate use by a pressurized gloved hand.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

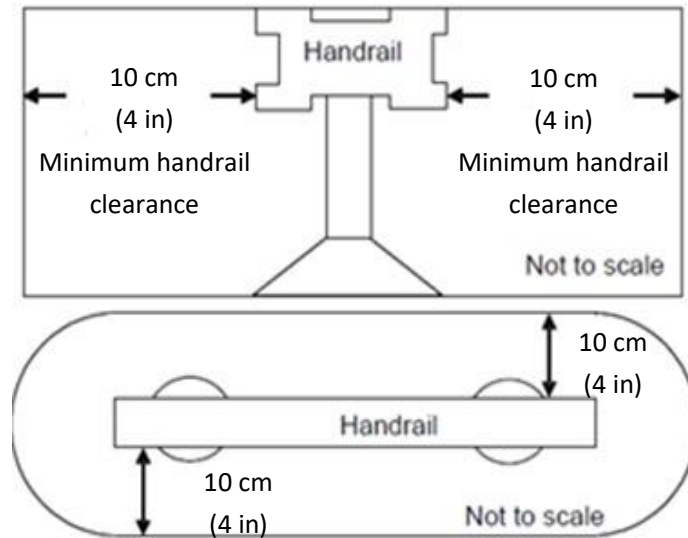


FIGURE 3.1.7-1 HANDRAIL/HANDHOLD SIDE CLEARANCE

3.1.8 APFR EVA Installation/Removal Clearance

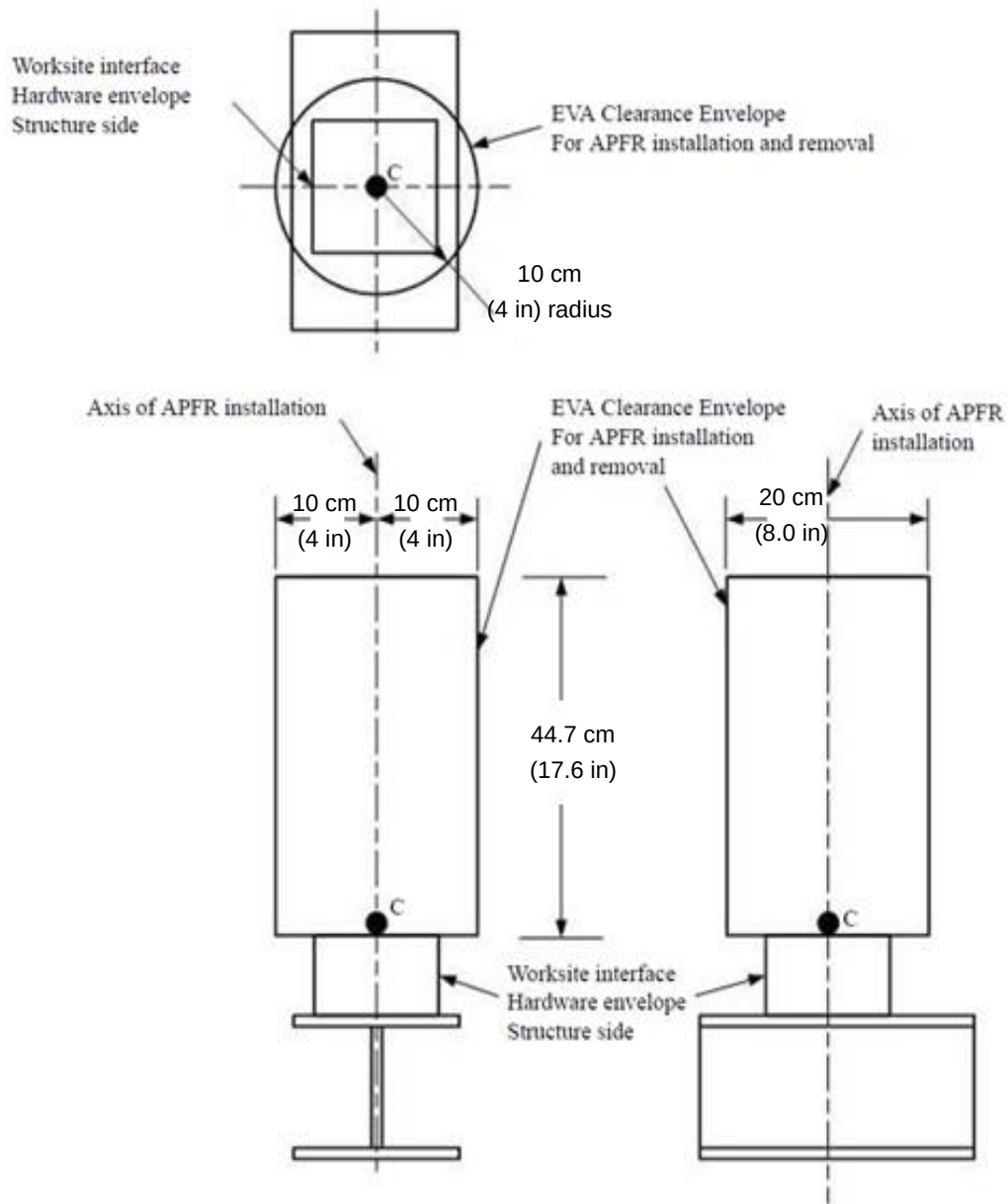
[EVASC.0006] The Artemis System shall maintain a clearance envelope around the Passive Worksite Interface (WIF) to support EVA installation/removal of the APFR as defined in Figure 3.1.8-1 *APFR Installation/Removal EVA Clearance Envelope – Top Mounted WIF*.

Rationale: The intent is to ensure adequate clearance is provided to support installation/removal of hardware designed to interface with the passive WIF. Contact EHP for consideration and potential use of other EVA WIF configurations. Nominally used in microgravity environment.

Gravity Field Applicability: Microgravity

Allocation: Gateway, HLS

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 29 of 280
Title: EHP xEVA System Compatibility Standards	



Notes:

1. Drawing not to scale
2. Point C is the centroid of the bolt pattern

FIGURE 3.1.8-1 APFR INSTALLATION/REMOVAL EVA CLEARANCE ENVELOPE – TOP MOUNTED WIF

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 30 of 280
Title: EHP xEVA System Compatibility Standards	

3.2 TRANSLATION

3.2.1 Contiguous EVA Translation Path

[EVASC.0152] The Artemis System shall provide EVA compatible accommodations to support contiguous translation paths to adjacent berthed/docked vehicles.

Rationale: A contiguous translation path is defined as the path from/to the airlock to/from any vehicle worksite that a crewmember needs to access for EVA operations. For vehicles that are berthed/docked to other vehicles, these contiguous translation paths need to provide for translation from one vehicle to another. Intentionally designed translation paths include illumination and features such as handrails and tether points free from contact hazards that promote safe and efficient EVA translation. These translation paths allow the EVA crewmember to safely translate across multiple vehicles which may include a safe and quick emergency return to the airlock. Translation paths to EVA worksites are addressed in [EVASC.0008].

Gravity Field Applicability: Microgravity

Allocation: Gateway, HLS

3.2.2 Translation Path

[EVASC.0008] The Artemis System shall design a translation path to all EVA worksites.

Rationale: These translation paths within the design constraints of the host vehicle. Intentionally designed translation paths include illumination and features such as handrails and tether points free from contact hazards that promote safe and efficient EVA translation. These translation paths allow the EVA crewmember to safely access worksites as well as to return to the airlock safely and quickly in case of an emergency for microgravity application. Partial gravity application ensures crewmembers can traverse around, to and from vehicle worksite areas (e.g., catwalk around a cargo lander, walking under the lander, ingress/egress of vehicle to the surface).

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: Gateway, HLS

3.2.3 Incapacitated Crewmember Interfaces

[EVASC.0154] The Artemis System shall provide hardware that aids a suited crewmember in the rescue of an incapacitated suited crewmember from the surface or microgravity location and return for ingressing the vehicle through the EVA hatch, culminating in positioning the suited incapacitated crewmember in an xEVA System for extraction which may require a 1st percentile female to rescue a 99th percentile male.

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 31 of 280
Title: EHP xEVA System Compatibility Standards	

Rationale: Incapacitated pressurized-suited crewmembers may be unable to ingress the vehicle (e.g., Tool cart, LTV, PR, HLS) on their own and may require assistance from a second crewmember. Depending on architecture and hatch height from the lunar surface, this can include items such as winch and rescue harness, deck space/porch, hatch seal cover, etc. Devices may need to be operated by a suited crewmember both internal and external to the vehicle. Microgravity considerations include increased stability and aids for the incapacitated crew handling and/or restraint. NASA will assist in the design definition by identifying credible scenarios and the degree by which a crewmember is incapacitated.

This includes the volume to perform the assisted task, hatch opening size and operation, as well as the external translation path and mobility/stabilization aids. Removing an incapacitated crewmember from a rear entry suit could be challenging and could require extra support structure, or possible tools to assist the crewmember inside as well as removal from the suit. This is an integrated risk response regarding inputs from EHP and External Programs and clarifications for provision of specific hardware will be realized based on architectural designs.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS

3.2.4 EVA Contact Hazards Preclusion Beyond Translation Corridor

[EVASC.0010] The Artemis System shall preclude EVA contact hazards (e.g., EVA Kick loads as defined in [EVASC.0050] and EVA safety requirements defined in section 3.9) within 0.61 m (24 in) beyond the translation corridor dimension as outlined in Figure 3.2.4-1 *EVA Contact Hazard Preclusions Envelope*.

Rationale: The additional 0.61 m (24 in) radius around the translation corridor protects for inadvertent contact hazards outside of the nominal translation corridor (1.1 m (43 in) + 1.2 m (48 in) = total diameter of 2.3 m (91 in) referenced in Figure 3.2.4-1 EVA Contact Hazard Preclusions Envelope for EVA kickloads and sharp edges). This additional envelope is based on the heritage ISS EMU suit.

Gravity Field Applicability: Microgravity

Allocation: Gateway, HLS

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 32 of 280
Title: EHP xEVA System Compatibility Standards	

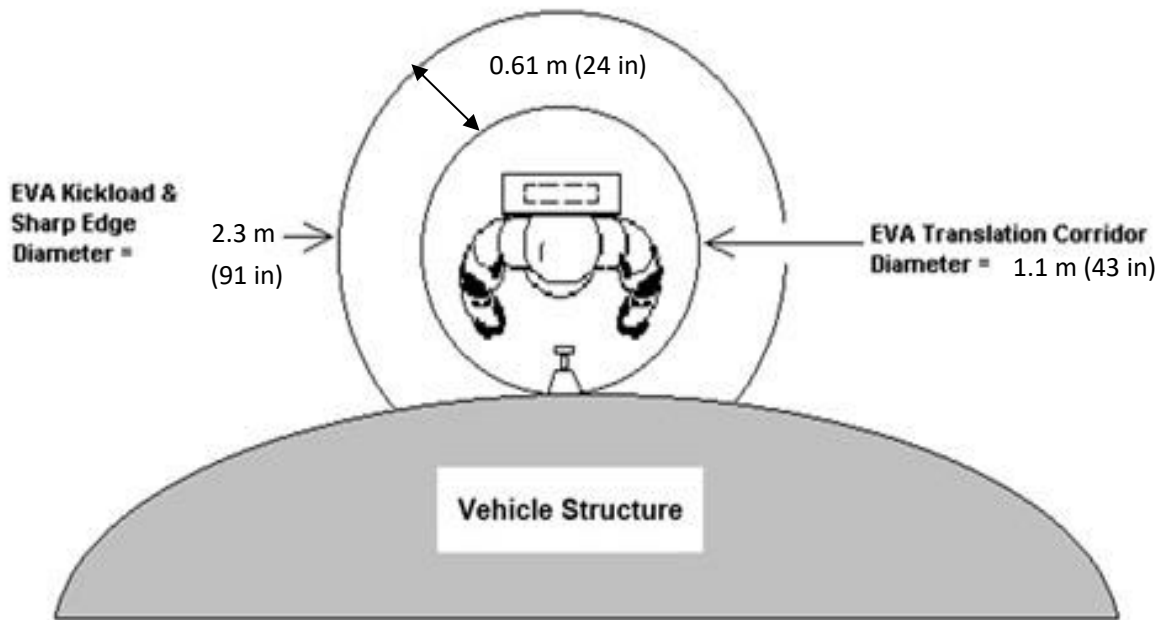


FIGURE 3.2.4-1 EVA CONTACT HAZARD PRECLUSIONS ENVELOPE

3.2.5 Partial Gravity EVA Contact Hazards Preclusion

[EVASC.0159] The Artemis System shall preclude EVA contact hazards (e.g., loads as defined in [EVASC.0155] and EVA safety requirements defined in section 3.9) within the specified constraints outlined in Figure 3.2.5-1 *Side View Partial Gravity EVA Contact Hazards Height*.

Rationale: Intent is to outline constraints that can cause a safety concern for the EVA crewmember while working on the lunar surface. This applies for external hardware, IVA hardware accessible by the suited crewmember, which is why there is no width, or horizontal dimensions specified. Crew will have capability to walk/traverse to all Interior Airlock surfaces and have similar accessibility to all External Vehicle features in all directions at the Lunar/Ground Level. EVA Contact Hazards will be precluded or controlled up to 2.5 meters high (98 inches).

Gravity Field Applicability: Partial Gravity

Allocation: LTV, PR, HLS, Payloads

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 33 of 280
Title: EHP xEVA System Compatibility Standards	

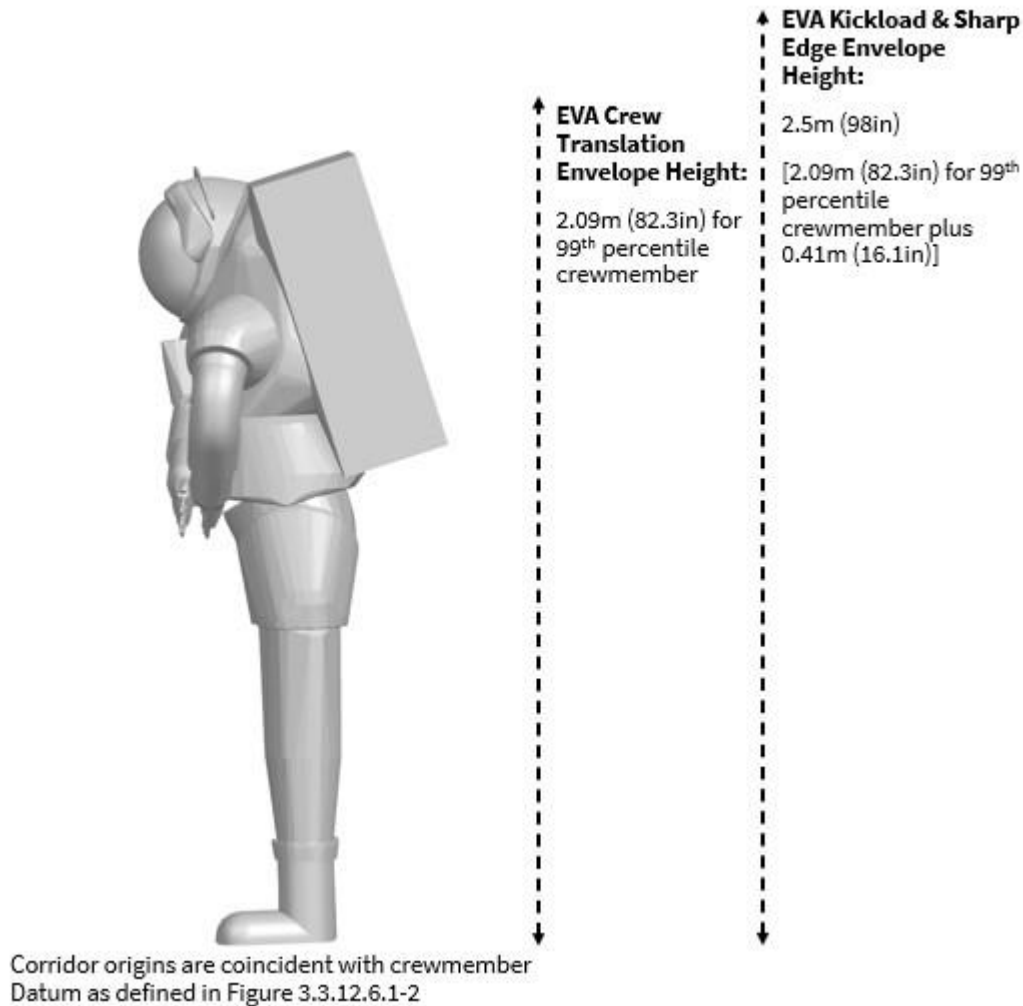


FIGURE 3.2.5-1 SIDE VIEW PARTIAL GRAVITY EVA CONTACT HAZARDS HEIGHT

3.3 EVA HANDLING

3.3.1 Microgravity EVA Equipment Handling Features

[EVASC.0012] The Artemis System shall include EVA handling features that are commensurate with the size of the hardware and the expected EVA operations as defined in Table 3.3.1-1 Constraints for EVA Transport of Objects.

Rationale: This is to ensure that there will be provisions for proper EVA handling and transport. The handling features required progress from a simple tether point, to handrails, and fixture for robotic arm translation. Similarly, the translation mode of the crewmember progresses from hand-over-hand to Remote Manipulator System (RMS) with object on a tether. For shorter distances, Body Restraint Tether (BRT) or foot restraints can be used for transfer of Mis. The larger or more massive the object, the more extensively the object must be tethered. Items that

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 34 of 280
Title: EHP xEVA System Compatibility Standards	

an EVA crewmember needs to manipulate that are over 340 kg (750 lbm) require robotic arm transport through a support fixture (e.g., grapple, micro square, etc.) for robotic arm translation. Variances to this requirement, based on mission objectives or contingency operations, may be proposed to EHP via the EHP RIVA Panel.

Gravity Field Applicability: Microgravity

Allocation: EVA Dev, Gateway, HLS, Payloads

TABLE 3.3.1-1 CONSTRAINTS FOR EVA TRANSPORT OF OBJECTS

Object/Size	Object Translation Mode	Object Handling Method	Design Features Required for Translation
0 – 9 kg (0 – 20 lbm), 0 – 0.056 m ³ (0 – 2 ft ³) max dim: 0.61 m (24 in)	Hand-over-hand crew translation with object on a tether	Tether	Tether point
0 – 136 kg (0 – 300 lbm), 0.056 – 0.14 m ³ (2 – 5 ft ³) max dim: 1.52 m (60 in)	Hand-over-hand crew translation with object on a BRT	BRT	Handrail section ¹
0 – 340 kg (0 – 750 lbm), 0.14 – 2.83 m ³ (5 – 100 ft ³) max dim: 2.44 m (96 in)	Object on a transfer device (Crewmember on robotic arm with an APFR)	Transfer device interface	Two handrails are required ¹

¹ Handrail requirements can also be met using handling aids like scoops interfacing with fixtures on the object (e.g., ISS used square or round scoops). The spacecraft/Program must supply the fixture. A universal fixture is highly desired over multiple types of fixtures.

3.3.2 Partial Gravity EVA Handling Capability <TBR-EHP-10028-003>

[EVASC.0160] The Artemis System shall limit the properties of items to be handled by suited crewmember to the values outlined in the following Tables: Table 3.3.2-1 Constraint for Partial Gravity Large Mass Items – Lifting, Table 3.3.2-2 Constraint for Partial Gravity Large Mass Items – Carrying, and Table 3.3.2-3 Partial Gravity Volume Limit Surface Carry.

Rationale: The limitations levied by this requirement apply to items intended for handling (i.e., relocation by carrying) by a single, suited crewmember, in one or two-handed operations. Suited EVA crewmembers perform tasks to meet the mission goals and operate human-system interfaces required for use during suited EVA operations. Pressurized suits can limit the crew’s mobility, dexterity, and tactility capabilities below that of an unsuited crewmember. It is anticipated that the crew will be required to relocate items on the surface, including payloads and cargo, between a surface worksite or rover and the originating vehicle. The intent of this requirement is to provide physical attributes (i.e., mass and volume) of an item that is carried by a suited crewmember during surface operations. Payload mass limits for lifting, payload mass limits for carrying, and payload volume limits

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 35 of 280
Title: EHP xEVA System Compatibility Standards	

for carrying are provided in Tables 3.3.2-1, 3.3.2-2, and 3.3.2-3. Reference 2024-ABF-001, Suited Capability Requirements for Partial Gravity – Anthropometry and Biomechanical Perspective which defines methodology and analysis approach for mass and volume capabilities.

The provided suited lifting limit was determined by evaluating the maximum load a single suited crewmember could lift without experiencing instability issues that can lead to mishaps or injury. Limits of stability were assessed through center of pressure comparisons to the feet's base of support using suit-human mass models in different static lifting conditions. At the most conservatively safe lifting position, the maximum suited lifting limit was then calculated for a payload placed at 28 cm (11 in) in front of feet center, and for the center of mass of the payload at 51 cm (20 in) above the ground. The conservative lift position is based on ergonomic guidelines such as the Washington State Hazard Checklist's that provides recommendations on an ergonomic "power zone" that maximizes lifting performance. A safety margin factor of 10% was applied to the base of support boundary to account for potential postural variations and inadvertent impulse forces. Suited Human-in-the-Loop (HITL) testing was performed to confirm and supplement the suit-human mass modeling results. The HITL testing findings do support that this payload mass value could be lifted in a stable manner and that ensuing suited lifting dynamics did not inhibit the lifting motion. Suited crewmember strength was not considered in the modeling analysis as it was assumed to be a minimal factor when compared to center of gravity (CG) management difficulties of lifting a payload in front of the body.

The suited carrying limit provided here was determined by evaluating the maximum load a suited crewmember could hold/maintain over various carrying distance durations. Specifically, force-endurance curve modeling efforts were performed to estimate carrying task capabilities as a function of time. Similar models have been used in ergonomics literature for assessing static manual materials handling capabilities. Thus, maximal load during a sustained suited carry was calculated for various time durations as a percentage of the maximum two-handed suited grasp hold strength. The maximum two-handed suited grasp capability was then derived for the 1st percentile individual from the HITL maximal grasp evaluations.

The suited volume limits for one-handed and two-handed carry were determined based on the reach envelope of a person with a 1st percentile arm-hand length. The specific dimensions were identified by the limiting anthropometric measurement for each dimension while avoiding interferences with the suit. For one-handed carry, the box dimensions were determined by minimizing the collision of the payload's width and depth with suit components including the thigh and Portable Life Support Subsystem (PLSS). Height was calculated from a person with a 1st percentile stature, minimizing collision with the 20° slope terrain. For two-handed carry, the box dimensions were determined from 1st percentile

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 36 of 280
Title: EHP xEVA System Compatibility Standards	

shoulder-to-shoulder breadth measurement for width. Depth and height were calculated to minimize the collision of the container with the suit components, including the Display and Control Unit (DCU), helmet and thigh, while maximizing the container volume. Suited HITL testing was performed with one suited subject using a simulated payload volume, the HITL testing findings supported that the payload volume defined could be carried by a suited crewmember and did not inhibit field of view or interferences with the suit.

Variances to this requirement, based on mission objectives or contingency operations, may be proposed to EHP via the EHP RIVA Panel.

Gravity Field Applicability: Partial Gravity

Allocation: EVA Dev, LTV, PR, HLS, Payloads

TABLE 3.3.2 -1 CONSTRAINT FOR PARTIAL GRAVITY LARGE MASS ITEMS – LIFTING

Task	Mass Limit for Suited Lifting kg (lbm)	Assumptions
Two-handed lifting	67.5 (149)	1 through 14

Assumptions:

- 1 This requirement is applicable to transfer of payloads or other equipment on the surface and is not applicable to small hand-carried or body-worn items (e.g., tools) or sample collections.
- 2 Applicable items that are intended for relocation.
- 3 The analysis was constrained to worse-case, heavy mass handling scenarios. It is recommended that all heavy objects be designed for 2-handed carries. For lighter 1-handed objects, it is recommended that crewmember discretion be used on when the object should be carried one-handed or two-handed.
- 4 An ergonomic squat position is assumed to be the starting pose for lifting.
- 5 The payload size does not inhibit the suited crewmember from maintaining an ergonomic squat position.
- 6 The challenges associated with managing the high-inertial properties of a payload (e.g., high mass, low weight) could not be analyzed and assumed to induce minimal difficulties but cannot be confirmed without testing.
- 7 The weight of the payload is evenly distributed with the center of mass closely aligned to the geometric center of the payload.
- 8 Both modeling and HITL testing were performed under the assumption that the starting point for lifting has the center of mass of the payload at 50.8 cm (20 in) above ground. It is possible that lifts performed lower to the floor can result in compromised proper suited lifting postures.
- 9 The subject is assumed to have good handle coupling on both sides of the payload and operating in a non-restricted workspace.
- 10 Item is lifted from the ground with handles at a height of 22.9 cm (12.3 in) or greater along the vertical axis.
- 11 Load is less than 40.6 cm (16.0 in) forward of the feet centerline.
- 12 Assumes lift is symmetrical without back twisting.

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 37 of 280
Title: EHP xEVA System Compatibility Standards	

- 13 Container is a box with opposing handles on the lateral sides and the handles are positioned even with the center of mass/load of the loaded container.
- 14 Based on static CGs

TABLE 3.3.2-2 CONSTRAINT FOR PARTIAL GRAVITY LARGE MASS ITEMS – CARRYING

Task	Mass Limit for Suited Carrying kg (lbm)	Distance m (ft)	Duration seconds	Assumptions
Two-handed carrying	75.7 (167)	8.8 (29)	10	1,2,3,4,5,6,7,8,9,10,12
	44.6 (98)	35.7 (117)	40	1,2,3,4,5,6,7,8,9,10,12
	26.5 (58)	107.3 (352)	120	1 through 12

Assumptions:

1. This requirement is applicable to transfer of payloads or other equipment on the surface and is not applicable to small hand-carried or body-worn items (e.g., tools).
2. Applicable to items that are intended for relocation.
3. The challenges associated with managing the high-inertial properties of a payload (e.g., high mass, low weight) could not be analyzed and assumed to induce minimal difficulties but cannot be confirmed without testing.
4. HITL testing was performed in a static offloaded posture; the effect of synergistic muscles and varied payload resting positions were not considered.
5. The payload size is designed to keep the payload CG close to the body.
6. An average suited walking speed of 3.2 kph (2mph) was used to find transversal distances.
7. Distance refers to the physical distance over which a suited crewmember is expected to comfortably handle and relocate the item.
8. Suited crewmembers are walking on level ground with a forward tilt posture.
9. Two-handed carry occurs with arms outstretched with slightly bent elbows at waist level.
10. Container is a box with opposing handles on the lateral sides and the handles are positioned even with the center of mass/load of the loaded container.
11. Based on static CGs.
12. The analysis considers that there is ample rest between payload carry tasks. It is recommended that crewmember discretion be used to determine how often rest breaks are taken.

TABLE 3.3.2 -3 PARTIAL GRAVITY VOLUME LIMIT SURFACE CARRY

Task	Maximum Width cm (in)	Maximum Depth cm (in)	Maximum Height cm (in)	Assumptions
Two-Handed Carry	33 (13.0)	44 (17.3)	61 (24.0)	1,2,4

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 38 of 280
Title: EHP xEVA System Compatibility Standards	

Task	Maximum Width cm (in)	Maximum Depth cm (in)	Maximum Height cm (in)	Assumptions
One-Handed Carry	45 (17.7)	69 (27.2)	68 (26.8)	1,3,4

Assumptions:

1. Crew are walking on ground with a forward tilt posture (Figure 3.3.12.6.1-2)
2. In the two-handed condition, the container is a box with opposing handles on the lateral sides at the center of the surface.
3. In the one-handed condition, the container is a box with a handle on the top surface.
4. Does not apply to small, hand-carried tools or to portable, well-balanced items (for example, a narrow mast of antenna with a much greater length than diameter)

3.3.3 Partial Gravity Step Design Parameter

[EVASC.0162] The Artemis System shall limit the height of a crewmember's step to the values outlined in Table 3.3.3.-1 Partial Gravity Step Design Parameter.

Rationale: Pressurized suits can limit the crew's mobility, dexterity, and tactility below that of an unsuited crewmember. Step height needs to be considered for scenarios such as moving through a hatch, up into a vehicle or rover, and getting down off a vehicle or rover. An overhead restriction refers to an overhead obstruction that requires crew to simultaneously reduce their height while elevating their foot to move through an area (e.g., crew bending over/crunching while lifting their leg to move through a hatch of the dimensions in [EVASC.0153]). Taller crew may not have the capability to duck low enough to translate through a given hatch size. Variances to this requirement, based on mission objectives or contingency operations, may be proposed to EHP via the EHP RIVA Panel. Reference 2024-ABF-001, Suited Capability Requirements for Partial Gravity – Anthropometry and Biomechanical Perspective which defines methodology and analysis approach for values in Table 3.3.3-1.

Gravity Field Applicability: Partial Gravity

Allocation: EVA Dev, LTV, PR, HLS

TABLE 3.3.3-1 PARTIAL GRAVITY STEP DESIGN PARAMETER

Type of Step / Threshold	Overhead Restriction cm (in)	No Overhead Restriction cm (in)	Assumptions
Front Step Up/Down	Maximum of 30.5 (12)	Maximum of 30.5 (12)	1,2,3

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 39 of 280
Title: EHP xEVA System Compatibility Standards	

Type of Step / Threshold	Overhead Restriction cm (in)	No Overhead Restriction cm (in)	Assumptions
Sidestep Over	Maximum of 30.5 (12)	Maximum of 30.5 (12)	1,2,3

Assumptions:

- 1 Deltas in lunar boot thickness, mobility or suit overall size may impact step heights.
- 2 Available data set is not comprehensive across all ranges of anthropometry; the step threshold height was selected so that smaller suited subjects could step over the threshold while still allowing taller suited subjects to crouch below a set overhead restriction. The specific threshold height and overhead restriction height was determined from a compromise between HITL testing results and from static suit modeling efforts that integrated 1st and 99th percentile stature measurements.
- 3 Stability aids such as handles should be made available for crewmember's stability.

3.3.4 Airlock and Hatch Transfer Volumetric Constraints

For best practices relating to airlock and hatch transfer volumetric constraints for crew and hardware between pressurized and unpressurized environments, refer to Appendix B, EVABP-30.

3.3.5 Pressurized xEVA Suit Transfer through Airlock Hatch

[EVASC.0135] The Artemis System shall design hatches to comply with xEVA suited pressurized crewmember transfer between pressurized and unpressurized environments with minimum hatch diameter of 1000 mm (39.37 in).

Rationale: Suited pressurized crewmembers cannot fit through NASA Docking System (NDS). Analysis and experience have shown that a 1000 mm (39.37 in) hatch (e.g., circular, square, "D" hatch) allows a pressurized suited crewmember to transfer through the hatch in microgravity. The intent is to prevent the vehicle hatch hinge from intruding into the transfer corridor. Requirement also accounts for incapacitated crew scenarios.

Gravity Field Applicability: Microgravity

Allocation: EVA Dev, Gateway, HLS

3.3.6 EVA Lunar Hatch Dimensions

[EVASC.0153] The Artemis System shall provide a minimum EVA Lunar hatch opening of 1.02 m (horizontal dimension) X 1.52 m (vertical dimension) (40 in X 60 in) for EVA egress/ingress.

Rationale: A minimum EVA hatch opening is needed to allow a pressurized crewmember to enter and exit a structure in partial gravity without the xEVA suit coming in contact with the surrounding hatch structure. A sufficiently sized hatch opening allows a crewmember to ingress/egress without crawling, without

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 40 of 280
Title: EHP xEVA System Compatibility Standards	

contorting their body such as to cause undue strain (due to size, shape, transition frame), and without spending time aligning themselves with direct center of the opening. These dimensions assume that the EVA crewmember translates horizontally through the hatch and does not account for other potential hatch configurations or dimensions. Alternative configurations and dimensions can be approved by EHP as a part of a waiver to this requirement. EVA Hatch sizes must allow for the incapacitated crewmembers or crewmember requiring assistance to pass through the EVA hatch. Apollo-sized EVA hatches caused undue strain to crewmembers and caused their metabolic rate to increase during ingress/egress and did not allow for incapacitated crewmember operations. Requirement also accounts for incapacitated crew scenarios.

Gravity Field Applicability: Partial Gravity

Allocation: EVA Dev, PR, HLS

3.3.7 Handling Alignment Criteria

3.3.7.1 Mounting Alignment Devices

[EVASC.0014] Alignment devices shall be provided for hardware that requires mounting to preclude incorrect installations.

Rationale: Alignment devices are preferred over markings alone. Alignment devices are often used to preclude incorrect installation and help provide gross alignment.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

3.3.7.2 Handling Alignment Marks

[EVASC.0015] Alignment marks shall be incorporated to aid in proper interface orientation, installation, and mechanism operation.

Rationale: Alignment marks can also be used to help indicate complete installation or actuation. This could include alignment marks, color coding, appropriate visual cues, and/or orientation arrows. A method for visual alignment (e.g., straight, or curved lines of a width and length to allow for accurate alignment) can also be applied to both mating parts of the hardware to assist in alignment in the operational position. Alignment devices are preferred over markings alone.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 41 of 280
Title: EHP xEVA System Compatibility Standards	

3.3.8 Manual Locking Features

3.3.8.1 Locking Device

[EVASC.0016] Locking devices (such as latches and latch locks) shall travel at least 30% of their full travel before changing state (locked to unlocked or vice versa).

Rationale: The intent is to travel beyond 30% before changing state. Designing a locking device mechanism that requires a large range of travel before changing state ensures that the EVA crewmember is aware what position the device is in at all times. Locking devices that have safety-critical functions may require additional locking features like Push-in-Pull (PIP) Pins to maintain sufficient failure tolerance as determined during the safety review process.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

3.3.8.2 Over Center Device Force

[EVASC.0124] Over center devices shall have a minimum peak force of twice the force required to initiate travel, not to exceed 44 N (10 lbf).

Rationale: Dexterity and tactility are degraded when wearing a pressurized xEVA suit, thus requiring increased levels of tactile feedback. Maximum force does not exceed 44 N (10 lbf) to comply with the finger actuation force requirement [EVASC.0018]. See Appendix A-2 Glossary of Terms for definition and examples of over the center devices.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

3.3.8.3 Soft Dock or Soft Capture Mechanisms

[EVASC.0017] A soft dock or soft capture mechanism shall be provided for hardware that is not held in place while being permanently installed.

Rationale: Soft dock or soft capture is utilized to hold hardware in place during installation in microgravity. A soft dock mechanism holds the item in place, but the crewmember can remove it by applying a slight force to overcome the mechanism (i.e., magnets or ball detents, reference [EVASC.0126] for minimum soft dock forces). Soft capture release requires an additional operation of a mechanism for release (i.e., mechanical latch). The intent of this requirement is from the perspective of what works best for EVA. This may or may not be acceptable for EVR Systems. Depending on the Program this requirement is applied to and their plans for EVA and EVR, tailoring may be required. While docked to staging vehicle (Gateway), EVR is the primary mode of mating/demating external payloads. Soft

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 42 of 280
Title: EHP xEVA System Compatibility Standards	

docks should be avoided as much as possible, especially those requiring high loads.

Gravity Field Applicability: Microgravity

Allocation: EVA Dev, Gateway, HLS, Payloads

3.3.8.4 Soft Capture

[EVASC.0125] Soft capture, as opposed to soft dock, shall be used in the following cases to increase safety for hardware that is not held in place during the process of being permanently installed:

- 1) Hardware more massive than 272 kg (600 lbm)
- 2) Large hardware that presents crewmember visibility problems when verifying position for mating
- 3) Complex hardware with unusual shapes or centers of gravity which preclude a stable or positive soft capture
- 4) Hardware installations where partial gravity vectors can be shown to maintain a stable or positive configuration before a hard capture is engaged in.

Rationale: Soft capture helps restrain hardware items, while final alignment/attachment can be made. Hardware that does not require a soft dock/soft capture may still employ these devices to improve EVA operability. The intent of this requirement is from the perspective of what works best for EVA. This may or may not be acceptable for EVR systems. Depending on the Program this requirement is applied to and their plans for EVA and EVR, tailoring may be required. While docked to staging vehicle (Gateway), EVR is the primary mode of mating/demating external payloads. Soft docks should be avoided as much as possible, especially those requiring high loads. Soft capture can be the first step in the process for a very large object. Soft capture can be used instead of soft dock for these cases in microgravity. However, once captured, additional soft dock may also be used to establish fine alignment prior to final mechanical attachment, such as bolt torques or latch throws.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: LTV, PR, Gateway, HLS, Payloads

3.3.8.5 Soft Dock Force

[EVASC.0126] The force required to initially and temporarily secure an object (soft dock) shall be sized proportionally to the mass of the object per Table 3.3.8.5-1 Soft Dock Force.

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 43 of 280
Title: EHP xEVA System Compatibility Standards	

Rationale: The larger the mass, the more difficult it may be to perceive the tactile feedback that soft dock has been achieved. Existing rigidizable tether configurations only support applying forces of up to 22 N (5 lbf). Imparting forces beyond 22 N (5 lbf) will require alternate methods of restraining the crewmember. The intent of this requirement is from the perspective of what works best for EVA. This may or may not be acceptable for EVR systems. Depending on the Program this requirement is applied to and their plans for EVA and EVR, tailoring may be required. While docked to staging vehicle (Gateway), EVR is the primary mode of mating/demating external payloads. Soft docks should be avoided as much as possible, especially those requiring high loads.

Gravity Field Applicability: Microgravity

Allocation: EVA Dev, Gateway, HLS, Payloads

TABLE 3.3.8.5-1 SOFT DOCK FORCE

Hardware Mass kg (lbm)	Soft dock Force N (lbf)	Soft Dock Force Tolerance N (lbf)
0–90.7 (0–200)	22 (5)	2.2 (0.5)
91.2–181.4 (201–400)	44 (10)	4.4 (1.0)
181.9–272.2 (401–600)	67 (15)	6.7 (1.5)
272.6–340.2 (601–750)	89 (20)	8.9 (2.0)

3.3.9 Unique Labeling

[EVASC.0024] The Artemis System shall code and/or mark translation paths, worksites, and any other callouts for the EVA crewmember with appropriate cues/indicators/labels for intended use as defined in EVA-EXP-0049, *EVA Office Exploration EVA (xEVA) System Labeling Guidelines*.

Rationale: Labeling can include alignment markings, operating instructions (bolt turns/torque/direction), Connector identification, MI part number and serial number (to prevent confusion during change out), caution and warning labels, location identification, etc.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

3.3.10 Hand Tools

3.3.10.1 Hand Tool Operations

[EVASC.0025] Hand tools shall be capable of being operated one-handed, ambidextrously for engagement/disengagement.

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 44 of 280
Title: EHP xEVA System Compatibility Standards	

Rationale: EVA operations require the crewmember to actively use both hands to fully complete a task. In microgravity, this can mean that one hand is used to stabilize the crewmember while the other is used to operate the tool.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

3.3.10.2 Ratchet Wrenches

3.3.10.2.1 Hand Ratcheting

[EVASC.0135a] Worksites, equipment, and tools shall accommodate a throw angle for manual EVA ratcheting of at least 90° on all EVA operable fasteners.

Rationale: This requirement applies to all EVA operable fasteners (bolts, screw, nuts, etc.) and hex-head drive shafts. Ratchet wrenches can be used where at least 90° of handle movement is possible.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

3.3.10.2.2 Throw Angle

[EVASC.0026] A ratcheting handle shall be capable of providing torque with a minimum throw angle of 45°.

Rationale: Efficient tool usage requires adequate access space.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

3.3.10.2.3 Ratchet Mechanism

[EVASC.0127] Hardware requiring use of a ratchet wrench shall incorporate a fine-tooth mechanism to allow engagement every 10° maximum.

Rationale: Ratchet increments should be as small as possible. The 10° increments provides 18 ratchet cycles in 180° of handle movement.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 45 of 280
Title: EHP xEVA System Compatibility Standards	

3.3.10.2.4 Back Drive Torque

[EVASC.0128] EVA wrenches shall have a maximum back drive torque of 0.05 Nm (7 in-oz) while operated by a pressurized glove.

Rationale: Incorporating this will ensure one-handed operation by the crewmember.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

3.3.10.2.5 Running Torque

[EVASC.0129] Fasteners to be used with an EVA ratchet wrench shall have a running torque that is greater than the maximum back drive torque of 0.05 Nm (7 in-oz).

Rationale: Incorporating this will ensure one-handed operation by the crewmember.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

3.3.11 EVA Actuated Controls

3.3.11.1 Pressurized Glove Compatibility

[EVASC.0027] Controls operated by an EVA suited crewmember shall be compatible with a pressurized glove.

Rationale: Controls that necessitate actuation by a pressurized glove are to have the appropriate features for pressurized suited operation (controls can include the following and are not all inclusive: touch screens, toggle switches, fasteners, knobs, latches, etc.). The controls may need to be adjusted to accommodate the appropriate tactile feedback when used by the pressurized-glove hand to ensure accurate operation of the control compared to unsuited performance. While dexterity and tactility are degraded when using a pressurized EVA suit, fatigue is also an issue. Fatigue happens when repeatedly applying a significant load to achieve a task.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

3.3.11.2 Inadvertent Actuation of EVA-Actuated Controls

[EVASC.0087] EVA-actuated controls shall prevent inadvertent actuation.

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 46 of 280
Title: EHP xEVA System Compatibility Standards	

Rationale: This can be accomplished via switch guards, recessing, etc. EVA worksites are often congested, and inadvertent contact is often hard to avoid.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

3.3.12 EVA Handrails/Handholds

3.3.12.1 Handrail/Handhold Cross-Section

[EVASC.0029] The Artemis System shall design cross-sectional profiles for handholds/handrails to meet Figure 3.3.12.1-1 *Handrail/Handhold Profile Dimensions*, for microgravity/dual gravity field application with tolerances per SDG33107728, via Top Mounted Handrail drawing SEG33106347 latest revision or Figure 3.3.12.1-2 *Partial Gravity Handrail/Handhold Profile Dimensions*.

Rationale: The intent of this requirement is to ensure that all handholds/handrails are designed to accommodate the various EVA interfaces (e.g., pressurized glove, tools, tethers, etc.). Artemis Systems are encouraged to contact the Johnson Space Center (JSC) EHP before manufacturing handrails/handholds to ensure that the handrail/handhold meets interface requirements, or if another configuration of handrail/handhold is necessary. Handrail cross-section is intended for use in microgravity environment with heritage ISS EVA tools. The microgravity environment application of handrail cross-section design needs to meet the loads outlined in [EVASC.0050] for use of existing heritage ISS tools (e.g., BRT). If HLS is to dock with Gateway and has a potential for performing an EVA, then the preferred handrail/handhold profile is outlined in Figure 3.3.12.1-1. In partial gravity, there is an additional envelope profile that can be utilized and is outlined in Figure 3.3.12.1-2. SDG33107728, via Top Mounted Handrail drawing SEG33106347 provides tolerances for both Figures 3.3.12.1-1 and 3.3.12.1-2. Sharp Edge and minimum Radius 1mm (0.04 inches) per Table 3.9.11-1 remain applicable. This provides common hand-grasp profile for suited glove and potential Tool-interface applications.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 47 of 280
Title: EHP xEVA System Compatibility Standards	

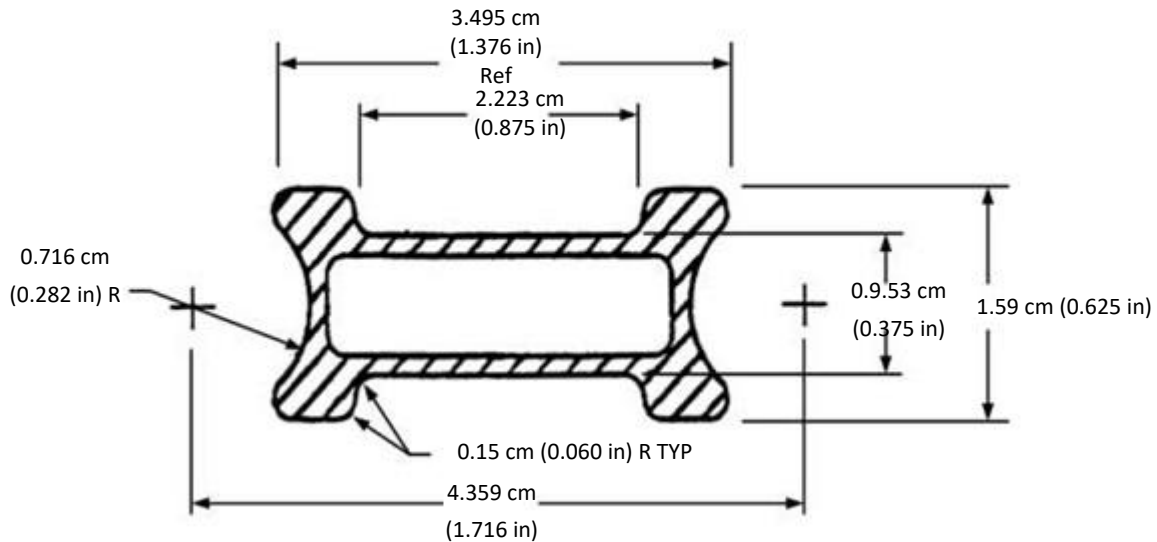
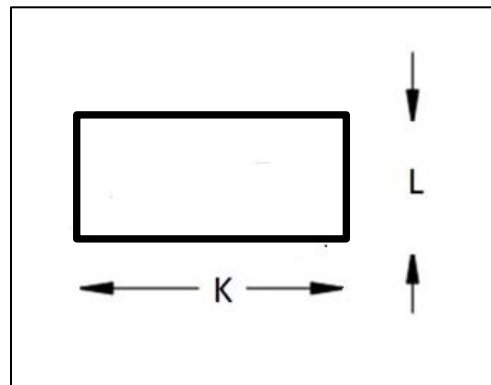


FIGURE 3.3.12.1-1 HANDRAIL/HANDHOLD PROFILE DIMENSIONS



	cm	in
K	3.50	1.376
L	1.59	0.625

FIGURE 3.3.12.1-2 PARTIAL GRAVITY HANDRAIL/HANDHOLD PROFILE DIMENSIONS

3.3.12.2 Handrail/Handhold/Handle Length

[EVASC.0030] The Artemis System shall design handholds/handrails/handles with a minimum grip length of 0.15 m (6 in) that can be operated by a pressurized glove.

Rationale: Grip length is essential in designing translation aids for EVA crewmembers. There needs to be adequate length for the EVA pressurized glove to access for EVA activities. In microgravity, this also will accommodate attachment of a BRT if it meets the torsional loading on the handrail/handhold from the interface connection point.

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 48 of 280
Title: EHP xEVA System Compatibility Standards	

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

3.3.12.3 Handrail/Handhold Accessibility

[EVASC.0031] The Artemis System shall design handrail/handholds and other EVA interfaces located in the defined crew translation path or worksite to be left exposed and usable to the crew as a translation/stability aid.

Rationale: Obscured or blocked handholds/handrails can impede the accomplishment of EVA tasks. Thermal covers on spacecraft and hardware should be designed such that handholds/handrails remain accessible. For handrails which are used only as handling aids (and not translation aids), the handrail may be covered if it is accessible via an EVA-operable access cover or flap.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: LTV, PR, Gateway, HLS

3.3.12.4 Continuous Handrail/Handhold Spacing

[EVASC.0032] The Artemis System shall space continuous handholds/handrails and the integral safety tether attach points to not exceed 0.61 m (24 in) between the standoff and handrail/handhold as defined in Figure 3.3.12.4-1 *Handrail/Handhold Gaps*.

Rationale: The 0.61 m (24 in) spacing is the comfortable and acceptable arm reach for EVA crewmembers. In microgravity, this spacing enables hand-over-hand translation by a pressurized suited crewmember. In partial gravity, this spacing enables ascent or descent from items such as a ladder but may include other features where continuous handholds are needed for crew stability.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: LTV, PR, Gateway, HLS

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 49 of 280
Title: EHP xEVA System Compatibility Standards	

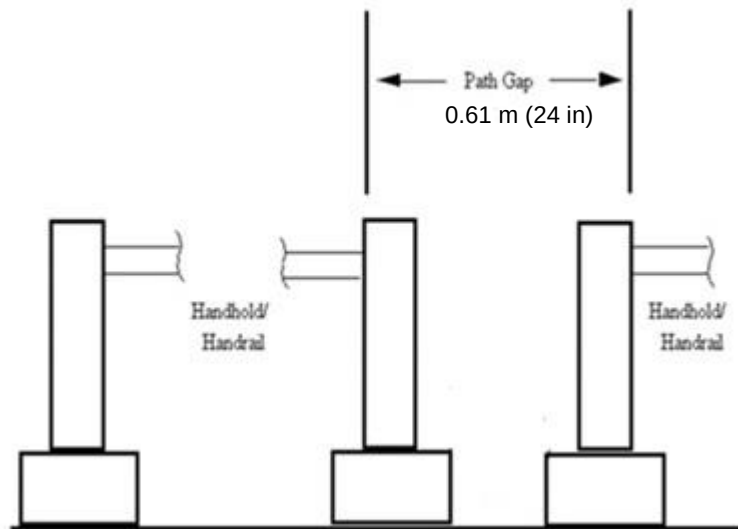


FIGURE 3.3.12.4-1 HANDRAIL/HANDHOLD GAPS

3.3.12.5 EVA Egress Handrail Spacing

[EVASC.0144] The Artemis System shall space handrail/handholds required for EVA egress to be within 30.5 cm (12 in) of the EVA egress location.

Rationale: Need to ensure that there is adequate aid for EVA ingress/egress operations. This is the necessary at the transition in/out of the Airlock in which typically, the crew must reach around a hatch bulkhead, and has reduced reach during that transition around corners. This provides the necessary control and stability needed by the crew for both Microgravity and Partial Gravity ingress/egress at the Airlock. The nominal continuous handrail/handhold spacing is defined in [EVASC.0032].

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: PR, Gateway, HLS

3.3.12.6 EVA Worksites

3.3.12.6.1 Work Envelope

[EVASC.0035] The Artemis System shall ensure EVA worksites with either one- or two-handed EVA glove operations reside within the work envelopes as defined in the following: Figure 3.3.12.6.1-1 *Work Envelope – Microgravity*, Figure 3.3.12.6.1-2 *Partial Gravity Work Envelope*, Table 3.3.12.6.1-1 *Partial Gravity Work Envelope*, Figure 3.3.12.6.1-3 *Partial Gravity Work Volume – Reach Envelope*, and Figure 3.3.12.6.1-4 *Two-Handed Reach Envelope Width*.

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 50 of 280
Title: EHP xEVA System Compatibility Standards	

Rationale: If the microgravity EVA task requires two-handed operations, the two-handed work envelope should be used. If the EVA task can be done one-handed, the one-handed work envelope should be used. Microgravity work volume is based on the heritage ISS EMU suit associated with the APFR. Partial gravity application work envelope is based on the Exploration EMU (xEMU) reference design, and the reach envelopes were represented for the fixed Hard Upper Torso (HUT), without considering the lower body motions. Thus, work envelope may not be interpreted to be smaller in partial gravity than microgravity. Crewmembers can move the lower body and traverse in partial gravity, substantially increasing the effective work volume. HUT or lower body mobility was excluded from reach envelope calculation due to the large number of degrees of freedom. Reference 2024-ABF-001, Suited Capability Requirements for Partial Gravity – Anthropometry and Biomechanical Perspective which defines methodology and analysis approach for one and two-handed work envelope.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 51 of 280
Title: EHP xEVA System Compatibility Standards	

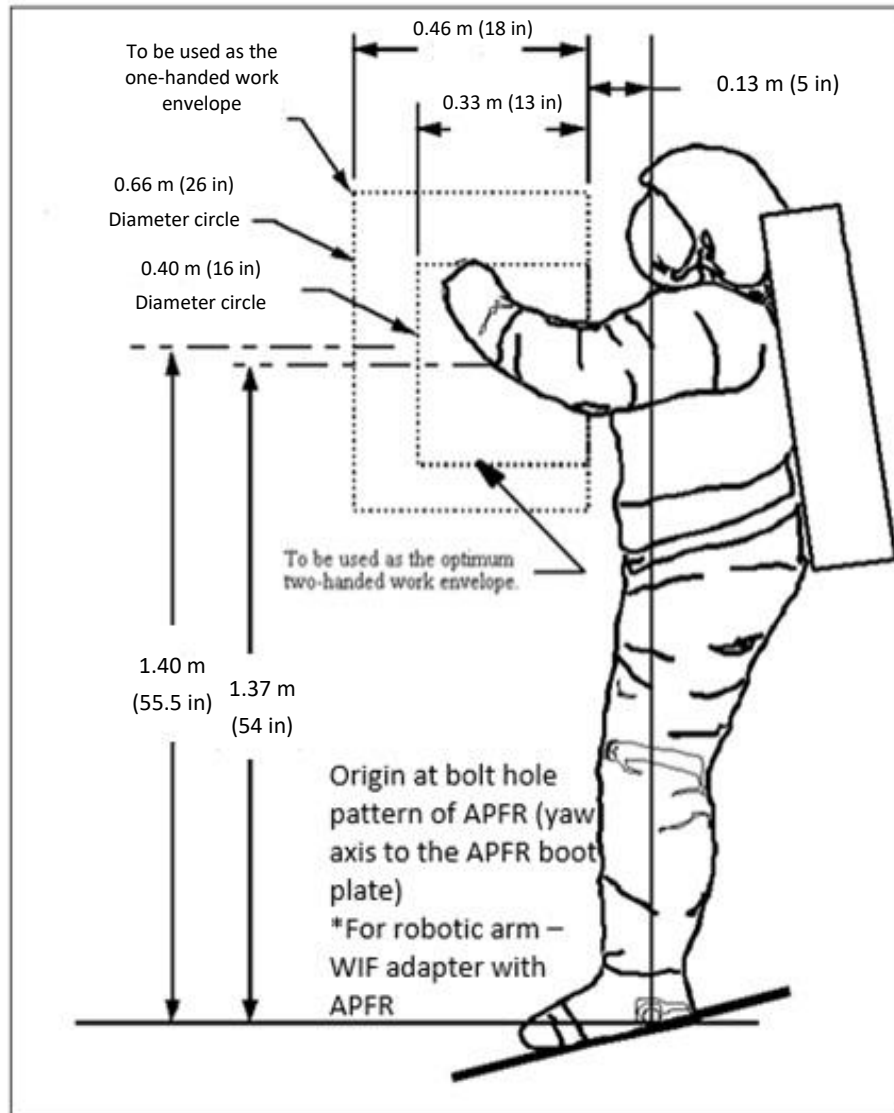


FIGURE 3.3.12.6.1-1 WORK ENVELOPE – MICROGRAVITY

Note:

1. Reach/work envelope values are from a fixed foot restraint platform and include torso motion.

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 52 of 280
Title: EHP xEVA System Compatibility Standards	

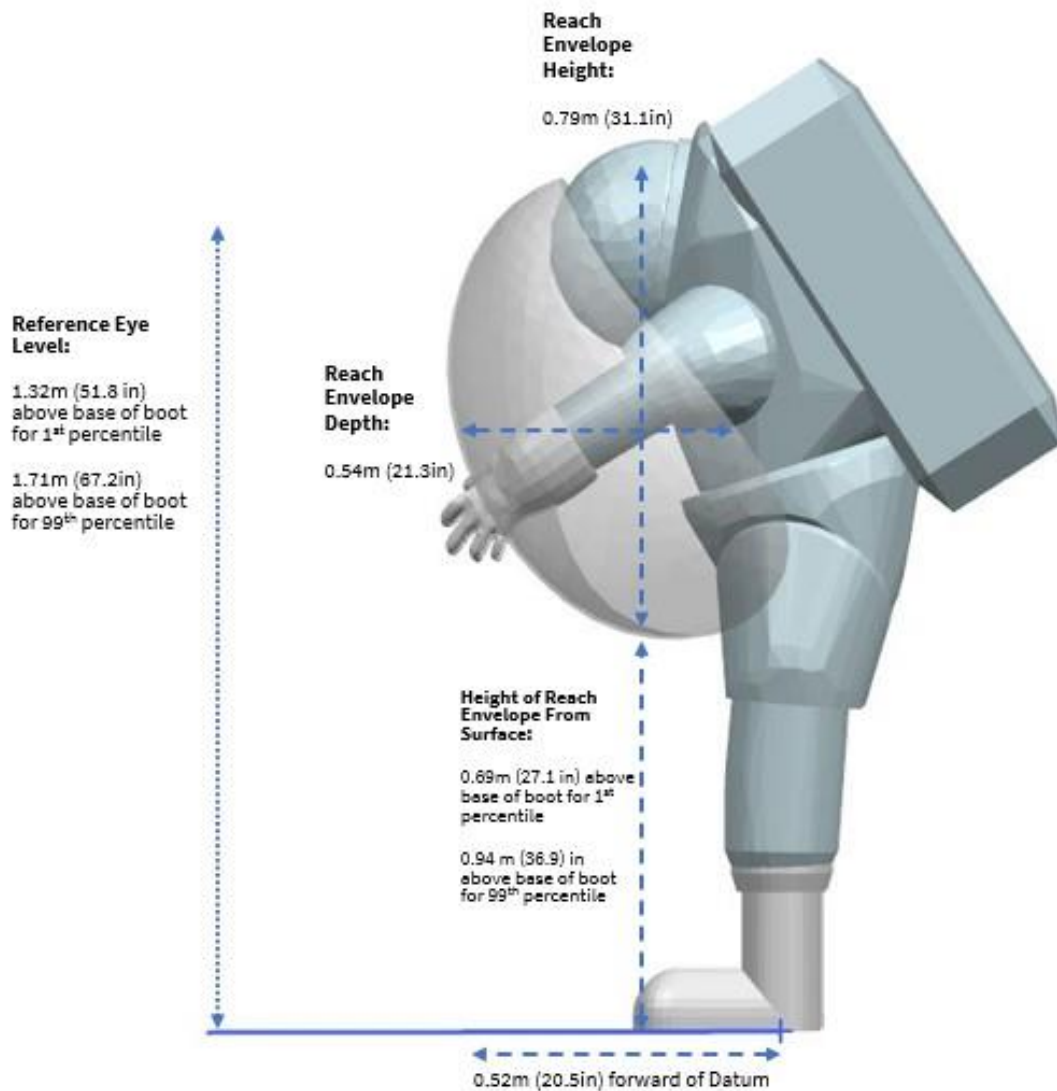


FIGURE 3.3.12.6.1-2 PARTIAL GRAVITY WORK ENVELOPE

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 53 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 3.3.12.6.1-1 PARTIAL GRAVITY WORK ENVELOPE

	Width (1-Handed) m (in)	Width (2-Handed) m (in)	Width (Total) m (in)	Height (1-Handed) m (in)	Height (2-Handed) m (in)	Depth (1-Handed) m (in)	Depth (2-Handed) m (in)	Assumptions
Maximum	0.77 (30.3)	0.18 (6.9)	1.35 (53.1)	0.79 (31.1)	0.54 (21.2)	0.54 (21.3)	0.14 (5.3)	1,2,3,5,6,7,8,9,10,11,12
Preferred	0.31 (12.1)	0.07 (2.8)	0.54 (21.3)	0.32 (12.4)	0.22 (8.5)	0.22 (8.5)	0.06 (2.2)	1,2,3,4,5,6,7,8,9,10,11,12

Assumptions:

1. Unlike EMU work envelope (Figure 3.3.12.6.1-1), the reach/work envelope dimensions are fixed at the HUT and do not include torso motion (Conservative approach). Crew can reposition feet to optimize the work envelope.
2. Person-to-person variability is substantial for reach envelope shapes and dimensions. The listed envelopes represent the worst-case capability of a 1st percentile hand-arm length crew in an xEMU thus correspond to a "worst-case" condition.
3. Partial gravity analogue testing demonstrated a forward tilt posture. This is assumed to maintain proper balance due to the weight of the suit and PLSS, of which the combined CG is located in the aft of the body. The specific posture angle of the suit can vary across individual crewmembers.
4. It is assumed that the work objects should be placed inside of the envelope to allow small persons to reach. However, due to the mechanical constraints and pressurization of the suit, large persons may need to exert a higher force for objects close to the body.
5. Preferred reach envelope is approximately 40% of the maximum envelope size, of which measurements are from a prolonged duration EVA test. However, the specific location within the maximum reach envelope may differ.
6. The reach envelope was measured from the simulated 1/6-g posture, with the upper body slightly bent forward. The subject was asked to keep the elbow straight, but elbow flexion was not strictly enforced thus naturally occurred during testing especially at the peripheral regions.
7. The reach envelopes are based on the government reference design xEMU suit. Thus, the dimensions and geometries are different from the previous suits, including the heritage EMU.
8. For the xEMU, hand positions were represented in the fixed HUT coordinate system, while the EMU reach volume incorporated the movements of the whole body, including the HUT, Lower Torso Assembly, legs, and APFR.
9. Thus, the reach envelope may not be directly compared between the xEMU and EMU. The shown differences are due to the different measurements techniques and may not be interpreted for the reduced capabilities of the xEMU. If the HUT and lower body mobility were included the xEMU reach volume can be substantially larger, for example, if the HUT or Lower Torso Assembly (LTA) bending was incorporated.
10. HUT or lower body mobility was excluded from reach volume calculation due to the large number of degrees of freedom of the xEMU. HITL tests were not able to sufficiently evaluate the virtually infinite permutations of joint variables.
11. The DCU, shoulder and arm stiffness also play a significant role in reach volume. Reach kinematics also vary with individual Anthropometry, strength capability, and experience level. Human-suit interactions are more complex and difficult to measure in some regions of the reach envelopes, especially near the boundaries.
12. Due to these factors, some of which are difficult to measure, reach envelopes may not be synthetically permuted (e.g., artificially adjusting elbow flexion angle) to calculate for hypothetical reach volume.

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 54 of 280
Title: EHP xEVA System Compatibility Standards	

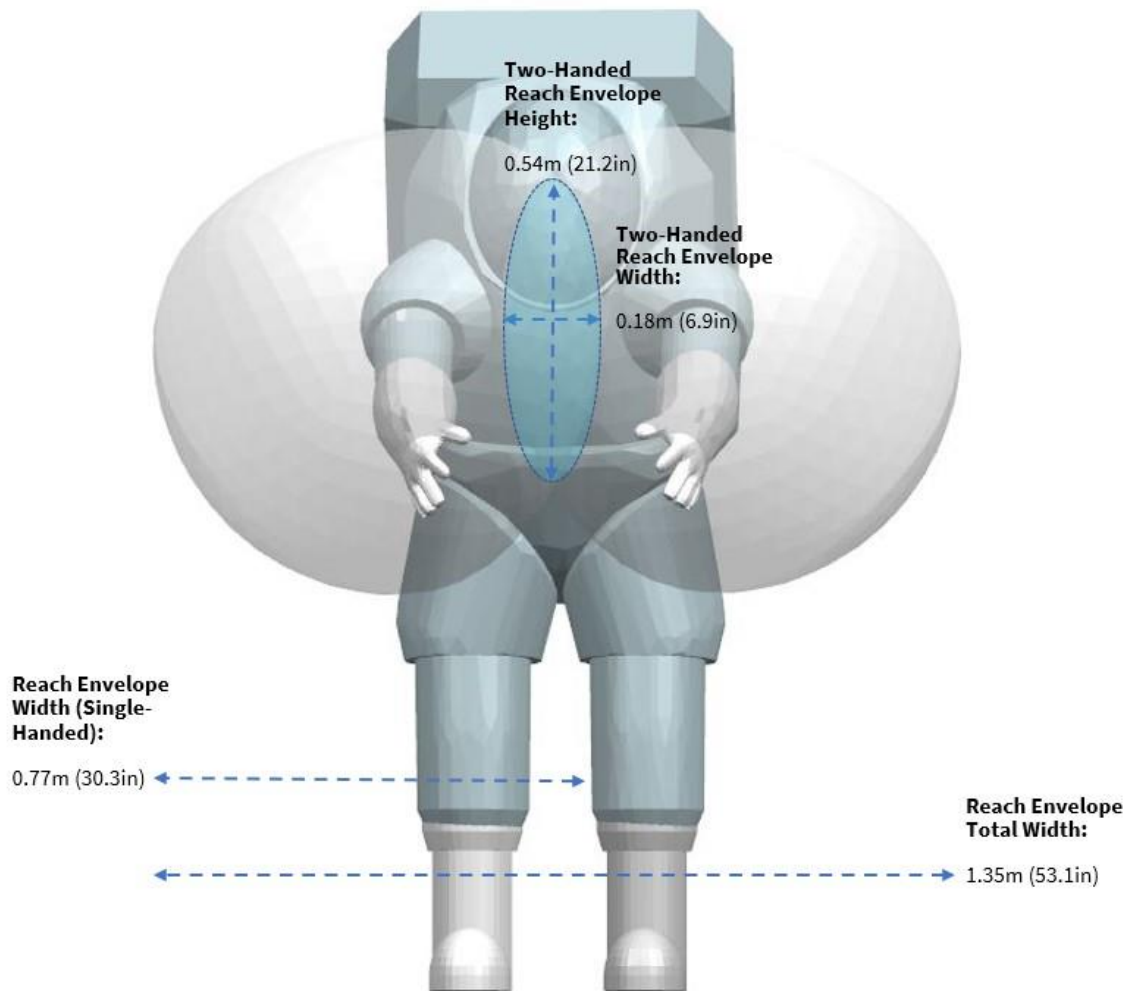


FIGURE 3.3.12.6.1-3 PARTIAL GRAVITY WORK ENVELOPE – REACH ENVELOPE

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 55 of 280
Title: EHP xEVA System Compatibility Standards	

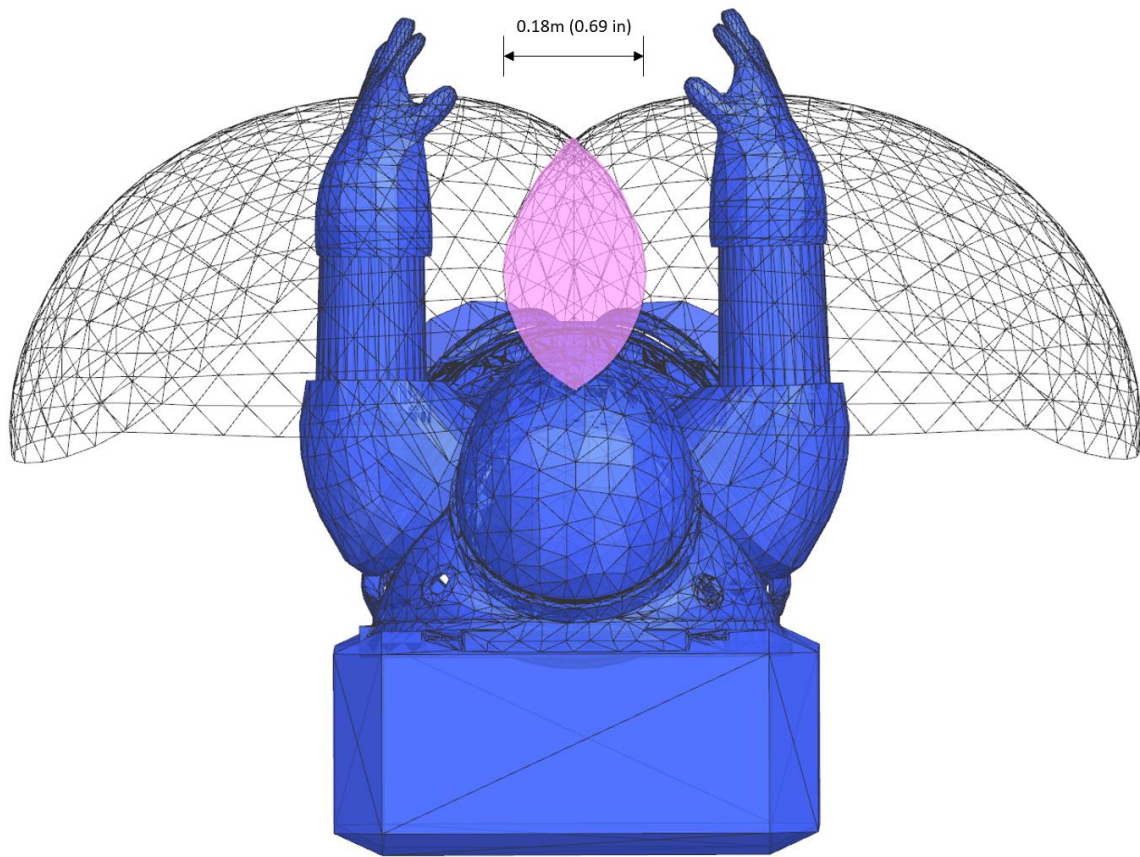


FIGURE 3.3.12.6.1-4 TWO-HANDED REACH ENVELOPE WIDTH

3.3.12.6.2 Working Volume Operation of Vehicle

[EVASC.0170] The Artemis System seated crewmembers working volume for operation of vehicle while on the lunar surface shall be within **<TBD-EHP-10028-019>**.

Rationale: The vehicle control and features need to be accessible by seated crewmembers. Due to restrained or limited waist/torso movement while seated, working volume (one and two handed) are expected to be reduced than those while standing.

Gravity field: Partial gravity

Allocation: LTV

3.3.12.6.3 EVA Egress/Ingress Methods

[EVASC.0157] The Artemis System shall provide EVA egress/ingress methods from/to the vehicle that accommodates the pressurized crewmember's mobility.

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 56 of 280
Title: EHP xEVA System Compatibility Standards	

Rationale: The intent of this requirement is to ensure that the pressurized crewmember has a compatible egress/ingress method from the vehicle to the lunar surface, and back to the vehicle after the EVA. If the vehicle design includes a method, such as a ladder, gate, step, etc., then the design has to accommodate the pressurized crewmember mobility with regards to pressurized glove interface, handhold spacing and width, as well as lower torso mobility to climb. In addition, safety of the crewmember will also need to be taken into consideration with regards to fall protection. Vehicles will need to jointly develop and demonstrate the egress/ingress methods with the EVA suit provider to ensure compliance with this requirement.

Gravity Field Applicability: Partial Gravity

Allocation: EVA Dev, LTV, PR, HLS

3.3.12.6.4 Crew Aids for Stability

[EVASC.0036] The Artemis System shall place crew stability aids or grasp points for a dedicated, robotic, or free-float worksite within 0.46 m (18 in) above or below the center of the crewmember's two-handed envelope, 0.61 m (24 in) to the left or right of the body centerline, and between 0.13 m (5.0 in) and 0.58 m (23 in) from the bolt hole pattern of the foot restraint (same depth as the one-handed work envelope) as defined in Figure 3.3.12.6.4-1 *Crew Aids for Stability*.

Rationale: Stability aids in the specified envelope provide usable and reachable interfaces to enable performance of EVA tasks. Stability aids might include handrails, handholds, or equivalent structure. The one and two-handed envelopes are defined in requirement [EVASC.0035] and are depicted in Figure 3.3.12.6.4-1 below, pink for the two-handed and yellow for the one-handed envelope. This is based on the heritage ISS EMU suit. The intent is to preserve the envelopes for microgravity EVAs.

Gravity Field Applicability: Microgravity

Allocation: Gateway, HLS

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 57 of 280
Title: EHP xEVA System Compatibility Standards	

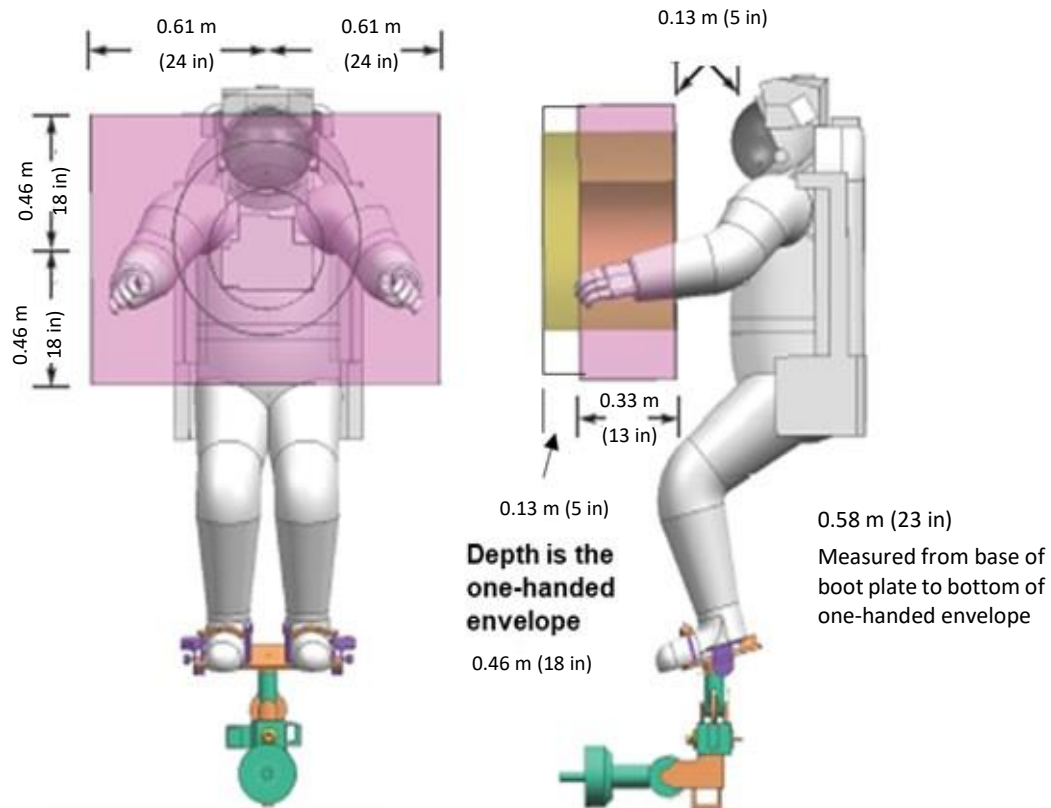


FIGURE 3.3.12.6.4-1 CREW AIDS FOR STABILITY

NOTE: For definitions of the one- and two-handed envelope, reference Figure 3.3.12.6.1-1 Work Volume – Microgravity, in requirement [EVASC.0035].

3.3.12.6.5 EVA Worksites Using Robotic Work Platform

[EVASC.0037] The Artemis System shall design worksites requiring a robotic work platform to include handholds/handrails within the EVA crewmember worksite area to react forces or moments and provide stability.

Rationale: Intent is to ensure that the forces or moments created by the crewmember do not back-drive the robotic work platform. It is intended that the handrails/handholds meet the Crew Aids for Stability requirement [EVASC.0036].

Gravity Field Applicability: Microgravity,

Allocation: Gateway, HLS

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 58 of 280
Title: EHP xEVA System Compatibility Standards	

3.3.12.7 EVA Hardware

3.3.12.7.1 Lubricant Use

[EVASC.0038] The Artemis System shall select and utilize hardware lubricants in a manner that precludes the external use of or exposure to oil or grease, which in turn can transfer onto the xEVA glove/suit or must accept that the lubricant area will be designated as a “no touch” area.

Rationale: It has been shown that dry films are the best for EVA cleanliness purposes but may not be appropriate for all tool performance applications. Other lubricants may be acceptable but must not transfer to the xEVA glove/suit. This requirement applies only to vehicle areas that will be interacted with by the suited crewmembers.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

3.4 WORKSITE OUTFITTING

Worksites are defined in Appendix A-2 Glossary of Terms.

3.4.1 Worksite Mobility

[EVASC.0040] The Artemis System shall utilize foot restraints that accommodate the crewmember’s range of motion within the constraints of the xEVA suit while at the EVA worksite defined in Figure 3.3.12.6.1-1 *Work Volume – Microgravity*, in requirement [EVASC.0035].

Rationale: The work volume is based on the heritage ISS EMU suit. The intent is to preserve the envelopes for microgravity EVAs.

Gravity Field Applicability: Microgravity

Allocation: Gateway, HLS

3.4.2 Worksites

3.4.2.1 Worksite Field of View

[EVASC.0041] The Artemis System shall configure worksites such that the EVA crewmember has an unobstructed line of sight to the worksite.

Rationale: The intent is to ensure that EVA interfaces are not obstructed by surrounding hardware and includes time periods a crewmember is driving or at worksites during surface operations. EVA suits limit the field of vision and range of motion of EVA crewmembers when compared with non-suited activities. The location of equipment to be operated during EVA must accommodate these

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 59 of 280
Title: EHP xEVA System Compatibility Standards	

limitations. Vehicle needs to provide field of view for crewmembers utilizing the vehicle while seated.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: LTV, PR, Gateway, HLS, Payloads

3.4.2.2 Working Volume

[EVASC.0042] The Artemis System shall design equipment to be EVA-manipulated to provide the working volume as specified in Figure 3.4.2.2-1 *Microgravity Working Volume for Manipulative EVA Tasks*, and Figure 3.4.2.2-2 *Partial Gravity Working Volume*.

Rationale: The work volume (internal or external to the vehicle) allows the crewmember to work comfortably without entrapment or confinement concerns and preserves a free area around the crewmember to reduce the risk of inadvertent EVA contact hazards. In Microgravity, this volume is centered at the footplate/restraint location as necessary by the task/worksites objective. In Partial Gravity, this is centered at the standing foot location. The work volume is based on the heritage ISS EMU, but still intend to apply the 1.2 m (48 in) cylinder around the base of the surface vehicle at known EVA worksites for the vehicle. This requirement volume does not apply once the crewmembers are away from the vehicle.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: LTV, PR, Gateway, HLS, Payloads

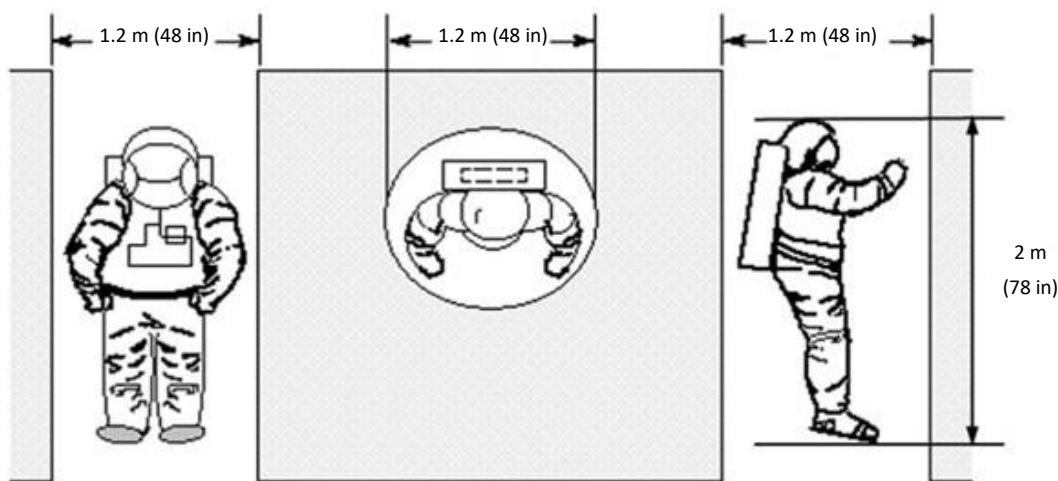


FIGURE 3.4.2.2-1 MICROGRAVITY WORKING VOLUME FOR MANIPULATIVE EVA TASKS

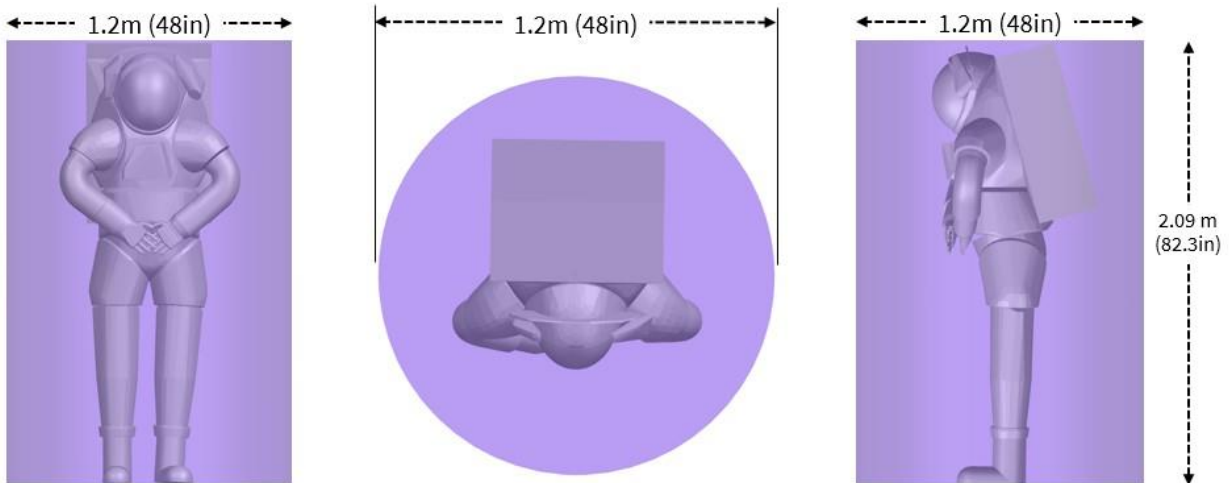


FIGURE 3.4.2.2-2 PARTIAL GRAVITY WORKING VOLUME

3.4.3 Dedicated EVA Worksite

[EVASC.0148] Dedicated EVA worksites shall provide restraint accommodations and capability for the crewmember at the worksite with either an APFR or BRT where any of the following conditions apply as specified in Table 3.4.3-1 Worksite Forces, for contingency cases where robotic crew transport and restraint capability cannot be used:

- (1) Two-handed operations are required to complete the task
- (2) Forces are required which exceed the following:

TABLE 3.4.3-1 WORKSITE FORCES

Linear Force	Duration
4.4 N (1.0 lbf)	4.5 sec
22.2 N (5.0 lbf)	2.1 sec
44.5 N (10.0 lbf)	1.4 sec

- (3) Tools are required to complete the task
- (4) Objects for manipulation do not have a soft dock/soft capture mechanism to aid in installation

Rationale: Restraint provisions at a worksite vary, depending on the complexity and specific requirements of the task. Reference Figure 3.3.12.6.4-1 Crew Aids for Stability [EVASC.0036] for relative placement of stability aids. Typically, the remote manipulator arm (with crew attached), requires the use of a foot restraint

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 61 of 280
Title: EHP xEVA System Compatibility Standards	

*and would limit crew translation – the BRT does not require foot restraint, and as such does **not** limit translation.*

Gravity Field Applicability: Microgravity

Allocation: Gateway, HLS

3.4.4 Robotic Assisted EVA Worksites

[EVASC.0149] EVA worksites requiring crew positioning when the robotic arm platform is stationary shall provide a force reaction mechanism (e.g., crew stability aid or grasp point) as defined in requirement [EVASC.0036], independent of the robotic stabilization platform for EVA worksite operations requiring crew-generated forces greater than 10 lbf (44 N).

Rationale: The intent is to ensure that if a robotic assisted worksite is needed, that the hardware provider considers definition of the worksite to be within the kinematic and reach envelope of the manipulator utilized. In addition, that the robotic arm platform satisfies the Crew Aids for Stability requirement [EVASC.0036].

Gravity Field Applicability: Microgravity

Allocation: Gateway, HLS

3.4.5 Free-Float Worksites

[EVASC.0150] All tasks to be performed by a free-float crewmember shall be located within the one-handed envelope defined in Figure 3.3.12.6.1-1 *Work Envelope – Microgravity*, in requirement [EVASC.0035].

Rationale: Free-float worksites must not be a dedicated worksite as defined in requirement [EVASC.0148] or a Robotic Assisted EVA Worksite as defined in [EVASC.0149].

Gravity Field Applicability: Microgravity

Allocation: Gateway, HLS

3.4.6 Installation of Passive Worksite Interface (WIF) Sockets

[EVASC.0045] Any EVA worksites which include passive WIF sockets shall provide handholds/handrails located within 0.61 m (24 in) of the passive WIF to support installation of the foot restraint by the EVA crewmember.

Rationale: Based on previous ISS experience, the intent of this requirement is to ensure that interfaces necessary to interface with the foot restraint socket are

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 62 of 280
Title: EHP xEVA System Compatibility Standards	

considered part of the EVA worksite and include the appropriate stability aids for installation.

Gravity Field Applicability: Microgravity

Allocation: Gateway, HLS

3.4.7 Foot Restraint Ingress/Egress Aids

[EVASC.0046] EVA worksites with WIFs shall include crew mobility aids located within 0.91 to 1.2 m (36 to 48 in) above the foot restraint/boot interface, 0.61 m (24 in) to the left or right of the foot restraint/boot interface centerline, and 0.46 m (18 in) in front of the centerline axis of the foot restraint/boot interface to aid in ingress and egress of the foot restraint as specified in Figure 3.4.7-1 *Ingress and Egress Aids for Foot Restraint*.

Rationale: Ingress and egress aids should be placed at appropriate positions to accommodate the limitations of the xEVA Suit design. This requirement only applies to those EVA worksites that require the use of a WIF. These dimensions are based on the heritage ISS EMU suit. The intent is to preserve the envelopes for microgravity EVAs.

Gravity Field Applicability: Microgravity

Allocation: Gateway, HLS

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 63 of 280
Title: EHP xEVA System Compatibility Standards	

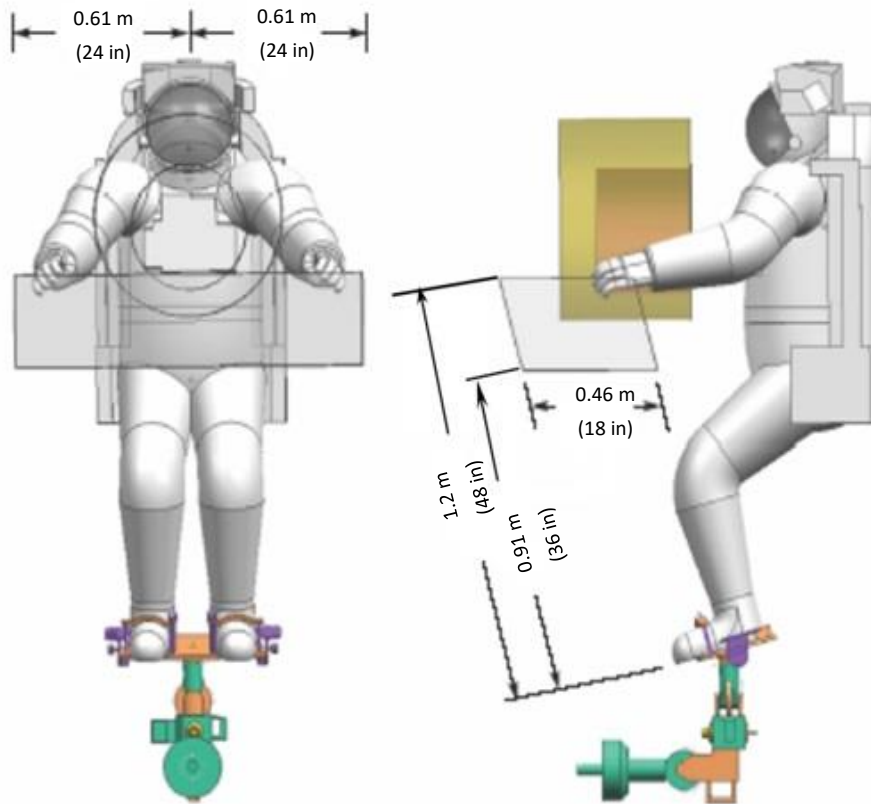


FIGURE 3.4.7-1 INGRESS AND EGRESS AIDS FOR FOOT RESTRAINT

3.4.8 Visual Contrast at Worksite

[EVASC.0047] EVA worksites and all associated equipment, such as labeling, markings, displays, and installation cues shall be clearly readable by crewmembers in extreme lighting conditions with a contrast ratio of 5:1.

Rationale: EVA crewmembers require adequate lighting to perform EVA tasks safely and efficiently, including reading of hardware labels while EVA. Extreme lighting conditions range from both maximum solar irradiance (1370 W/m² at 1 Astronomical Unit (AU)) and darkness with widely varying angles of incidence; these conditions can challenge the crewmember's ability to interpret EVA labels. Thus, all labels to be read by the EVA crewmember while EVA must all be designed with extreme lighting conditions in mind. For instance, EVA crewmembers utilize EVA Helmet Visors that reduce glare and direct sunlight incidence on the EVA crewmembers' eyes. Similarly, the xEVA suit provides its own EVA suit-mounted lights (or supplied by the xEVA System) that ensure minimum required lighting is immediately available at the EVA worksite during eclipse or shaded operations. Vehicle and hardware builders must demonstrate designs are legible in these extreme conditions and that the design, while

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 64 of 280
Title: EHP xEVA System Compatibility Standards	

acceptable in one condition (such as maximum solar irradiance), does not render it illegible in the other (such as darkness). For any EVA interface that has an installed and removed position, the vehicles must also clearly visually indicate successful installation and removal. EHP-10012, EHP Systems Requirement Document requirement [EHP-R-013] provides additional context for surface lighting environmental conditions. The recommended contrast ratio and color coding can be found in the NASA/SP-2010-3407/REV1, NASA Human Integration Design Handbook (HIDH), sections 9.7.2.2.2 and 10.4.7.3.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

3.5 LOADS

3.5.1 Equipment Tether Attach Points

3.5.1.1 Color for EVA Handling and EVA Equipment Tether Points

[EVASC.0033] Handholds or tether points intended to be used only for equipment handling and not crew translation, crew safety tethering or restraint, shall be colored according to Table 3.5.1.1-1 EVA Handling and EVA Equipment Tether Point Colors. **<TBR-EHP-10028-012>**

Rationale: The intent is to ensure that if the crewmember only needs to handle the MI versus translate over or across it, the additional loading requirements on the handle are not necessary. Similarly, EVA Equipment Tether Point requirement [EVASC.0048] is available for tethering equipment at loads less than those required for Crew Safety Tethers [EVASC.0116] or Crew Safety Restraints [EVASC.0161]. Reference requirement [EVASC.0117] for handhold/handrails that can withstand the loads.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

TABLE 3.5.1.1-1 EVA HANDLING AND EVA EQUIPMENT TETHER POINT COLORS <TBR-EHP-10028-012>

Item	Gravity Field Applicability	Color	Notes
EVA Handles on removable equipment intended to be used only for equipment handling	Microgravity, Partial Gravity	Silver or white	EVA Handles on hardware for handling (i.e., MI handles) need not be designed for translation or safety loads and therefore should not be colored the same.
EVA Equipment Tether Points	Microgravity, Partial Gravity	Silver or white	EVA Equipment Tether Points are defined in [EVASC.0048].

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 65 of 280
Title: EHP xEVA System Compatibility Standards	

3.5.1.2 EVA Equipment Tether Point

[EVASC.0048] Non-soft goods tether point shall have a minimum load of 334 N (75 lbf) in any direction per Figure 3.5.1.2-1 *EVA Equipment Tether Point Dimensions*, in order to accommodate EVA Tether hooks.

Rationale: This requirement applies to the equipment tether attach points, and not to crewmember safety attach points (reference [EVASC.0050] Table 3.5.2-1 Microgravity EVA Crew-Induced Loads, Design Limit Load Type: "Tether Point"). The tether attach point is designed such that two equipment hooks can attach to it for transfer of hardware from one tether to another. Other types of tether attach points, like tether points integral to existing structure, are also acceptable, but shall meet the intent of Figure 3.5.1.2-1 EVA Equipment Tether Point Dimensions, in minimum hole size and maximum cross-sectional dimensions. EVA equipment tether point design features should allow the tether point to be accessed with a hook without manipulation. This is a crewmember one-handed operational task. For example, softgoods tether loops will be stiff, and keyring-type tether loops will be rigidly secured to prevent movement during tethering operations. EVA equipment tether points are not required for partial gravity application, but if used, must meet the load requirement in the gravity vector direction at 334 N (75 lbf) to account for inertial loads if item is dropped.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

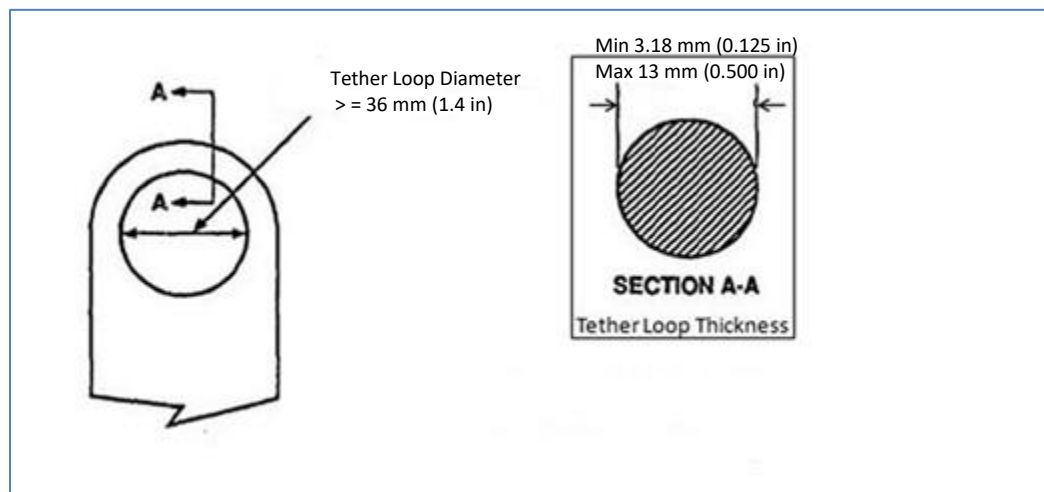


FIGURE 3.5.1.2-1 EVA EQUIPMENT TETHER POINT DIMENSIONS

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 66 of 280
Title: EHP xEVA System Compatibility Standards	

3.5.1.3 Soft Goods Tether Point Loads

[EVASC.0134] Soft goods tether point attached to soft good hardware shall be within the tether loop thickness limits defined in requirement [EVASC.0048] Figure 3.5.1.2-1 *EVA Tether Point Dimensions*, across its entire length for a minimum load of 133 N (30 lbf) in any direction in order to accommodate EVA Equipment Tether hooks.

Rationale: This requirement is applicable to any hardware that includes an integral soft good tether point attached to soft good hardware (reference [EVASC.0050] Table 3.5.2-1 Microgravity EVA Crew-Induced Loads, Design Limit Load Type: “Soft Good Tether Point for Handling”). [EVASC.0048] Figure 3.5.1.2-1 EVA Equipment Tether Point Dimensions, outlines the cross-section diameter (tether loop thickness) maximum of 13 mm (0.500 in). The cross-section needs to be maintained to ensure that the EVA crew hook does not slide off. EVA Tether Points are not required for partial gravity application, but if used must meet the load requirement in the gravity vector direction at 133 N (30 lbf) to account for inertial loads if item is dropped.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

3.5.2 Microgravity EVA Crew-Induced Loads

[EVASC.0050] The Artemis System shall design external hardware to withstand specified EVA crew-induced loads as defined in Table 3.5.2-1 *Microgravity EVA Crew-Induced Loads*.

Rationale: External hardware is defined to be anything that is in the EVA crewmember’s pressurized, suited translation path. This includes anything (e.g., structures) inside the vehicle cabin/airlock that is in the vicinity of the pressurized, suited crewmember, as well as outside the vehicle within the translation path corridor. The hardware, when exposed to Microgravity EVA-crew-induced loads should not create a hazardous condition, suffer loss of structural integrity, or suffer a loss of functionality. These loads apply to any hardware that is susceptible to EVA crew contact. Reference EVA-EXP-0032, EVA Office EVA-ISS Interface Definition Document, for loads that are induced into the xEVA suit by historically used ISS EVA tools.

Gravity Field Applicability: Microgravity

Allocation: EVA Dev, Gateway, HLS, Payloads

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 67 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 3.5.2-1 MICROGRAVITY EVA CREW-INDUCED LOADS

Design Limit Load Type	Limit Load	Type of Loading	Direction	Category of Structure	Application Comments
Loads for Handrails/Handholds and Structure Supporting Handrails/Handholds for Translation					
EVA Handrail/ Handhold– Primary Translation Path	979 N (220 lbf)	Quasi–static load applied over 7.6 cm (3.0 in) length of handrail or handhold at worst location	Any direction	Handrails, handholds, and structure to which handrails and handholds are mounted	This load applied to the translation path used by the crewmember to return to the Airlock. Can also apply to payloads that are in the EVA translation path. This load is only good for use in conjunction with a crew safety tether that provides rip-away stitching for load limiting well below 890 N (200 lbf) and energy absorption for a prescribed length of stroke.
Crew Safety Tether Attachment	890 N (200 lbf)	Quasi–static load applied to crew tether loop attachment	Any direction	Crew tether loops/handrail tether point, attach hardware and support structure	Primary path handrail loads envelope this load. Can also apply to payloads that are in the EVA translation path.
Other EVA-Induced Loads					
EVA Handling Load	200 N (45 lbf) 3.95 Nm (35 lbf-in) for connector paths for mate/de-mate of connector)	Quasi-static concentrated load over a 3.18 cm-radius (1.25-in) circular area	Any direction	Mis and nonstructural closures and covers (including shields, cables, cable connector brackets, cable connector panels, cable clamps)	All hardware shall be designed to this load as a minimum. This load can be applied anytime to any hardware by the EVA crewmember. This force would be applied by a crewmember manipulating hardware. Payloads that are handled by the crew shall meet this load.
Soft-good Tether Point for Handling	133 N (30 lbf) or weight of the item (whichever is greater)	Concentrated load-pull (tension)	Any direction	Tether point for handling a soft-good item	This load is applied to the tether point on any soft good. A tether point shall exist on any soft good that could be handled by EVA. This tether point is not to be used for local tethering after the item is installed.

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 68 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 3.5.2-1 MICROGRAVITY EVA CREW-INDUCED LOADS

Design Limit Load Type	Limit Load	Type of Loading	Direction	Category of Structure	Application Comments
					This shall apply to a payload if it includes soft goods that would be handled by EVA.
Inadvertent Kick, Bump	556 N (125 lbf)	Quasi-static, concentrated load over a 13 mm (0.5 in) diameter circular area	Any direction	Secondary structure near (within 0.61 m (24 in)) a translation path or worksite Reference [EVASC.0010]	This is an accidental impact. It should be applied to hardware near (within 0.61m (24 in)) translation paths and/or worksites. Reference [EVASC.0010]) This applies to any payloads in the vicinity of EVA translations paths and worksites.
EVA Load for Design of APFR ¹ Supporting Structure	1220 N (274 lbf) force; 474 Nm (4200 lbf-in) moment	Quasi-static loads applied at APFR socket to structure interface	Force in any direction; moment about any axis	All structures on which a foot restraint is attached	Force and moment applied simultaneously. Reference requirement [EVASC.0143] This load is only good for use in conjunction with an APFR, which is designed to provide energy absorption to limit loads to the 1220 N (274 lbf) / 474 Nm (4200 lbf-in) value for a prescribed length of stroke.
Handhold/handrail loads during/to accommodate APFR Ingress/Egress only	267 N (60 lbf)	Quasi-static load applied over 7.6 cm (3.0 in) length of handrail or handhold at worst location	Any direction	Handrails, handholds, and structure to which handrails and handholds are mounted	This load applied to any handrail used by crew to assist in APFR ingress and egress
Local Maneuvering	267 N (60 lbf)	Quasi-static load applied over 7.6 cm (3.0 in) length of handrail or handhold at worst location	Any direction	Handrails, handholds, and structure to which handrails and handholds are mounted	This load is applied to any handrail/handhold used to position the crewmember body at a worksite. This maneuver is defined to occur within 1.2 m (48 in) after translation has been completed. These handrails/handholds are not primary translation path aids.

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 69 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 3.5.2-1 MICROGRAVITY EVA CREW-INDUCED LOADS

Design Limit Load Type	Limit Load	Type of Loading	Direction	Category of Structure	Application Comments
					Not applicable to external payloads.
Tool impact	556 N (125 lbf)	Concentrated load on a 0.15 cm-radius (0.06-in) circular area.	Any direction	Shatterable materials such as windows and exposed glass surfaces	Intended to protect shatterable materials from an EVA tool impact. This load applies to all payloads.
Tethering Loads					
Tether Point	334 N (75 lbf)	Quasi-static Concentrated load applied at worst location	Any direction	Any interface to which a tether is attached and the supporting structure of that interface	This is a local tethering load that is applied to any interface for waist tether/and or Retracting End Effector Assy (REEA) attachment. It is also applicable to both tool and tether attach points where tools are attached by a tether. This tether is not for safety tethering. Applies to payloads that contain EVA tether points.
BRT (loose or rigidized)	F = 334 N (75 lbf) M = 34 Nm (300 lbf-in)	Quasi-static Concentrated load applied at worst location	Any direction	EVA Handrail	This is a local tethering load that is applied to an EVA handrail for BRT attachment. EVA crewmembers use the BRT to provide body stabilization at a worksite. EVA crewmembers may need to BRT to handrails after the payload is in its installed location.

Table Footnotes:

¹APFR – EVA load for design of APFR supporting structure incorporates the evaluation for kick load, layback, quick grab, handling, and oscillatory loads.

Overall General Notes:

- a. Pry loads, as defined as the load imparted to a handrail by a U.S. EVA crew hook, have been determined to have a sufficiently low probability of occurring such as to render the load case non-credible. Therefore, pry loads are not to be considered as an EVA Induced Load.

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 70 of 280
Title: EHP xEVA System Compatibility Standards	

- b. EVA Microgravity EVA Induced Loads for inadvertent kick do not apply to hardware or worksites, which are assembled or maintained using only robotic systems (or only a crewmember restrained on robotic system).

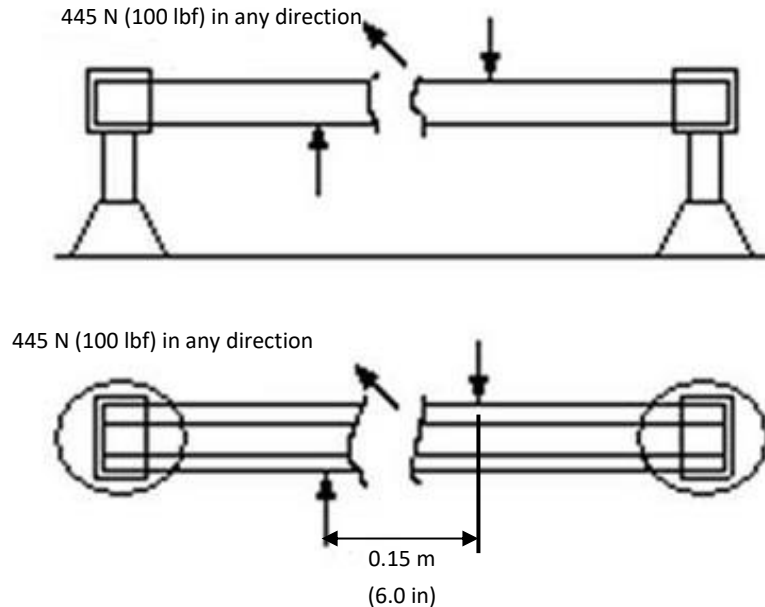


FIGURE 3.5.2-1 OPPOSING FORCES ON HANDRAIL/HANDHOLD

3.5.3 Partial Gravity EVA Crew-Induced Loads

[EVASC.0155] The Artemis System shall design external hardware, or IVA hardware expected to interface with the suited crewmember to withstand specified EVA crew-induced loads as defined in Table 3.5.3-1 *Partial Gravity EVA Crew-Induced Loads*.

Rationale: The hardware, when exposed to Partial Gravity EVA-crew-induced loads should not create a hazardous condition, suffer loss of structural integrity, or suffer a loss of functionality. These loads apply to any hardware that is susceptible to EVA crew contact. Loads provided are Limit loads. Emergency/contingency events were also taken into consideration when appropriate; the emergency/contingency load cases are used to assess structural capability and are enveloped by Ultimate loads as defined in JSC 65828, Structural Design Requirements and Factors of Safety for Spaceflight Hardware. Emergency/contingency load cases are severe with low probability of occurring events. The intent of specifying energy is to not be overly prescriptive or overly conservative on quasi-static or static force/load definitions that do not consider the local compliance of the impacted structure. The energy approach allows the structural designers some freedom to design localized structures and support structures to various levels of force magnitude but are still compatible with the impact energy. Force in L12 assumes safety tether and crew restraint, these assumptions need to be confirmed based on the exact design of crew restraint or

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 71 of 280
Title: EHP xEVA System Compatibility Standards	

safety tether. Reference ES6-24-001, Lunar EVA Crew Induced Loads Memo for further information on the values in Table 3.5.3-1.

Gravity Field Applicability: Partial Gravity

Allocation: EVA Dev, LTV, PR, HLS, Payloads

TABLE 3.5.3-1 PARTIAL GRAVITY EVA CREW-INDUCED LOADS <TBR-EHP-10028-002>

LOAD/ENERGY CATEGORIES		Maximum Magnitude Metric Units	Notes#
L1*	Bump Energy @ Impact	21 J	
L2	Kick Energy @ Impact	68 J	
L3*	Crew mass support load	288 N	
L4*	Drop Energy @ Impact	12*H J	H = height from which object is dropped, in meters
L5*	Fall Energy @ Impact	296 J	Encompasses EVA crew ground level falls only
L6*	Crew Handling Loads	769 N	
L7*	Tools Handling Loads	729 N	
L8*	Stomp/Jump Energy @ Impact	272 J	
L9*	Suited crew mass load	757 N	
L10*	Walking or stepping load	792 N	
L11* +	Handrails Grasp/Pull load	861 N	
L12*	Swinging or Falling	1370 N	Reference ES6-23-044, Lunar EVA Crew Induced Loads on Restraint Attachment Point from Crew Fall xEMU interface Restraint loads back to the suit were assumed to be a static 334 kg for purposes of the applicable Loads Calculations in Table 3.5.3-1.

*Magnitude value is maximum of subcategories. See individual tables below for additional options, including lower magnitude loads for specific load cases.

+Emergency/Contingency load considered.

All values assume xEMU government reference design mass and dimensions of EVA suit and tools. Only metric units are provided so that no rounding offsets would be introduced in the English conversions.

L1	Bump Energy	
	Crew contact point	Energy

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 72 of 280
Title: EHP xEVA System Compatibility Standards	

L1A	Helmet	3 J
L1B	PLSS/Upper Torso	21 J

L3	Crew Mass Support Load	
	Crew Action	Load
L3A	Using crew aid or walking stick to stand from falling	149 N +
L3B	Leaning against a wall	99 N +
L3C	Leaning on crew aid or walking stick	288 N

L4	Drop Energy	
	Mass of item dropped	Drop Load
L4A	Hammer Tool	3*H J
L4B	Loaded Tool Belt	12*H J
L4C	Chisel/Scoop Tool	2*H J
L4D	Sample Marker Tool	1*H J
L5	Fall Energy	
	Fall Type	Energy
L5A	Face plant /whole body	146 J
L5B	Fall to knees	296 J
L5C	Catch with hands	47 J

Crew Handling Loads			
L6A	Push Load Max Load		436 N
	Structure	Direction	Load
L6A1	Top Panels	Upwards Push	436 N
L6A2	Side Panels	Horizontal Push	271 N +
L6A3	Floor Panels	Downward Push	281 N
L6B	Pull Load Max Load		769 N
	Structure	Direction	Load
L6B1	Floor Panels	Upwards Pull	769 N
L6B2	Side Panels	Horizontal Pull	478 N +
L6B3	Top Panels	Downward Pull	315 N

L7	Generic Tool Handling Loads		
	Direction	Non-critical	Critical
L7A	Concentric (Grip)	51 N	330 N
L7B	Generic	124 N	729 N

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 73 of 280
Title: EHP xEVA System Compatibility Standards	

L8	Stomp/Jump Energy	
	Crew Action	Energy
L8A	Stomping	43 J
L8B	Jumping	272 J

L9	Suited Crew Mass Load		
	Crew position	Crew contact surface(s)	Load
L9A	Standing	Load at boots	563 N
L9B	Seated	Load at waist brief	600 N
L9C	Single Kneel	Load at single knee contact point	399 N
L9D	Double Kneel	Load at the knees	757 N

L10	Walking Load	
	Crew Action	Load
L10A	Walking on ground level	623 N
L10B	Going down/up ramp	792 N
L10C	Stepping down/up stairs	659 N

L11	Handrails Grasp/Pull Load	
	Highly dynamic short duration crew action	Load
L11A	Push	516N
L11B	Pull	861 N

Footnote for L5 and L8:

- Total weight = 3272 N earth | 545.3 N moon
- Human weight = 1081 N earth | 180 N moon
- Suit (xEMU government reference) = 1966 N earth | 327.8 N moon
- Tools = 225.1 N earth | 37.4 N moon

3.5.4 EVA Actuated Loads

[EVASC.0018] The Artemis System shall design items to be actuated by EVA crewmembers to remain within the loads defined in Table 3.5.4-1 *Miscellaneous Crew Induced Limit Loads*.

Rationale: Dexterity and tactility are degraded when using a pressurized suit which necessitates increased levels of tactile feedback. This does not include the EVA crew-induced loads (reference requirement [EVASC.0050] for EVA induced loads or those outlined in [EVASC.0155] for partial gravity). The intent of this requirement is from the perspective of what works best for EVA, by aiding in reducing fatigue and repetition while working with pressurized gloves. It is anticipated that there will be issues with getting proper reaction on lunar surface, e.g., as illustrated in analog testing for drilling applications. This may or may not

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 74 of 280
Title: EHP xEVA System Compatibility Standards	

be acceptable for EVR systems. Depending on the Program this requirement is applied to and their plans for EVA and EVR, tailoring may be required.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

TABLE 3.5.4-1 MISCELLANEOUS CREW INDUCED LIMIT LOADS

Crew System	Type of Load	Limit Load	Direction of Load
Repetitive Actuation Force	Concentrated Load	Actuation forces to not exceed 89 N (20 lbf) for hand tools and other hardware items which require repetitive/continuous operation.	Any Direction
Finger Actuation Force	Concentrated Load	9 to 44 N (2 to 10 lbf)	Any Direction
Actuation Torque	Concentrated Load	Maximum actuation torque of 3.4 Nm (30 lbf-in) for knobs, fasteners, latches, and other hardware requiring a twisting/turning hand/wrist motion	Clockwise (CW) or Counterclockwise (CCW)
		Actuation forces for hardware that requires fine motor activity by a pressurized gloved hand, such as knobs and devices, to not exceed 0.2 Nm (30 in. oz).	Any Direction
Non-repetitive Actuation Force	Concentrated Load	Actuation forces of hardware to not exceed 111 N (25 lbf) for momentary, non-repetitive gross motor skill operations in foot restraints or BRT, such as operating a latching mechanism or opening/closing a door.	Any Direction
EVA driven bolts	Concentrated Load	<89 N (<20 lbf) for engagement or disengagement of the ratchet type tools onto the bolt. * This force is not required to be maintained while driving the bolt. 0.17 Nm (24 in-oz) minimum torque for installation of bolt.**	Axial
		0.17 Nm (24 in-oz) minimum torque for installation of bolt**	CW or CCW
EVA Hatches, Doors, and Drawers	Concentrated Load	<111 N (<25 lbf)	Any Direction
EVA Connectors	Concentrated Load	3.95 Nm (<35 lbf-in) (mate/demate) for twisting-type connectors (i.e., NATC)	CW or CCW

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 75 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 3.5.4-1 MISCELLANEOUS CREW INDUCED LIMIT LOADS

Crew System	Type of Load	Limit Load	Direction of Load
		<111 N (<25 lbf) for bail and pull connectors (i.e., NASA Zero Gravity Lever (NZGL))	Any direction
Mounting Hardware (install/remove) (EVA)	Concentrated Load	<111 N (<25 lbf)	Any Direction
Fastener Torque (without torque reaction interface)	Concentrated Load	*** <34 Nm (<25 ft-lb)	CW or CCW
High Torque Fasteners (with torque reaction interface)	Concentrated Load	<136 Nm (<100 ft-lb)	CW or CCW
Push and Turn Fasteners	Concentrated Load	<13 N (3 lbf)	Any direction

*Note: Example – A tool must overcome the anti-rotation device inside a modified micro fixture.

**Note: The rationale for this is 0.05 Nm (7 in-oz) maximum torque for back drive of the ratchet wrench.

***Note: The Pistol Grip Tool (PGT) with associated sockets has lower limits that are based on tool stack-up and (Revolutions per Minute (RPMs)). The details are shown in SSP 30256:001 section 3.2.1.4.

3.5.5 EVA Mechanical Feedback

[EVASC.0164] EVA mechanical feedback actuation force shall be greater than 15.5 N (3.5 lbf).

Rationale: Mechanical control feedback needs to be sufficient such as to override the EVA glove attenuation and allow the EVA crewmember to receive positive indication that the control function has been completed.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

3.5.6 WIF Structural Loading

[EVASC.0143] The Artemis System shall maintain positive margins of safety for loads at the WIF to structure interface of 1220 N (274 lbf) in any direction and a moment of 474 Nm (4200 lbf-in) in any direction, simultaneously, as defined in Figure 3.5.6-1 *Top Mounted WIF Worksite Half Loading Criteria*, and to be used only in conjunction with a limiting APFR.

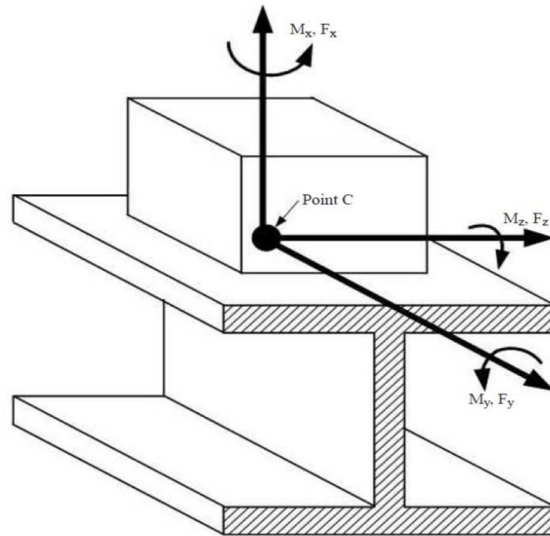
Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 76 of 280
Title: EHP xEVA System Compatibility Standards	

Rationale: Ensures that the structural loading for the hardware where the WIF is attached has been defined. This provides additional clarification to Table 3.5.2-1 Microgravity EVA Crew Induced Loads, Requirement [EVASC.0050] table line item for “EVA Load for Design of APFR Supporting Structure.” The interface loads values here, of 1220 N (274 lbf) and 474 Nm (4200 lbf-in), are only appropriate in conjunction with use of the load-limiting APFR which is designed with an energy absorber to limit the crew-induced loads over a distance of stroke that accommodates EVA crew range of motion at the foot restraint (without bottoming out on a hard stop). If the load limiting APFR is not used for crew foot restraint, the foot restraint loads at the interface can be much higher or different than these as represented.

Gravity Field Applicability: Microgravity

Allocation: EVA Dev, Gateway, HLS

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 77 of 280
Title: EHP xEVA System Compatibility Standards	



Maximum Interface Loads	
Moment	Force
Any Direction 474 Nm (4200 lbf-in)	Any Direction 1220 N (274 lbf)
<u>Notes:</u> 1) The above loads should be applied simultaneously. 2) All loads are to be applied at Point C. 3) Point C, the force and moment origin, is located at the center of the bolt pattern at the mating surface.	

FIGURE 3.5.6-1 TOP MOUNTED WIF WORKSITE HALF LOADING CRITERIA

3.6 CONNECTORS

Unless otherwise specified, the requirements in this section apply to all types of EVA operable connectors, including electrical, fluid, fiber optic, structural, etc. It is important to note that in this context when the word “design” is used, it does not preclude hardware developers from using previously existing products that may or may not have been uniquely created for this application. Instead, the intent is to communicate that regardless of how the parts are sourced, the resulting assembly must be operable by the EVA crewmember.

3.6.1 One-Handed EVA Operable Connectors

[EVASC.0053] The Artemis System shall design EVA connectors to be aligned and mated/demated one-handed and ambidextrously, while wearing a pressurized glove, and without the use of a tool.

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 78 of 280
Title: EHP xEVA System Compatibility Standards	

Rationale: The intent for microgravity application is to not require the use of a foot restraint for stabilization to operate the connectors. One-handed alignment and mating are necessary so that the other hand can be used for stabilization if foot restraints are not provided. If a foot restraint is provided, connectors can be held in two hands and reacted against each other.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

3.6.2 Connector Accessibility

[EVASC.0054] The Artemis System shall design individual EVA connectors capable of being mated, demated, or replaced without having to remove or replace other connectors.

Rationale: Experience has shown that proper spacing connectors can be utilized to provide easier accessibility without having to remove or replace other connectors and provides more efficient packaging of the EVA worksite (reference EVABP-27).

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

3.6.3 Gloved-Hand Connector Accessibility

[EVASC.0138] The Artemis System shall provide gloved hand access per Figure 3.1.5-1 *Gloved-Hand Clearance*, in requirement [EVASC.0003], around any point of the EVA connector that the crew has to actuate, including sweep of the actuating mechanism if applicable.

Rationale: It does not preclude the use of protective hardware such as covers, caps, etc. Experience has shown that closer spacing for small, simple connectors can be utilized to provide more efficient packaging of the EVA worksite (reference EVABP-27).

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

3.6.4 Incorrect Connector Mating Prevention

[EVASC.0055] The Artemis System shall design EVA connectors to prevent incorrect mating or cross-connection.

Rationale: Intent is to provide only one possible mechanical combination for connector mating to ensure that all system connections are manufactured as

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 79 of 280
Title: EHP xEVA System Compatibility Standards	

designed and installed so they cannot be mated/demated incorrectly or in the wrong orientation. Identification alone is not sufficient to preclude incorrect orientation.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

3.6.5 Connector Coding

[EVASC.0056] The Artemis System shall locate unique identifier labels or codes on both sides of mating EVA connectors and associated items that are visible when both connected and disconnected.

Rationale: Mechanical methods of ensuring correct mating of connectors are not sufficient alone. To ensure efficient EVA operations, labels are also needed to mate the correct connectors. Matching EVA connector pair coding simplifies training, eliminates confusion, and speeds operations. Reference EVA-EXP-0049, EVA Office Exploration EVA (xEVA) System Labeling Guidelines, for details on connector labeling.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

3.6.6 Connector Protective Caps

[EVASC.0057] The Artemis System shall design tether capable (or integral tethered) protective caps for plugs and jacks whenever they are not connected to the mating part to prevent physical damage and contamination that account for the following:

- 1) Protection from moisture, lunar dust, and other foreign materials.
- 2) Protection against damage to sealing surfaces, threads, or pins.
- 3) Resistance to abrasion, chipping, or flaking.
- 4) Same finish and material as the connector is protecting (ensuring compatibility with the connector material).
- 5) Brightly colored red (for those not used in flight); ensure that is removed prior to flight.
- 6) Maintained at cleanliness level equivalent to plug or jack that is being protected.

Rationale: The intent is to protect the plugs and receptacles from moisture, lunar dust, and other foreign materials and damage to sealing surfaces, threads, and pins. Also, to ensure compatibility with the connector that is being protected.

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 80 of 280
Title: EHP xEVA System Compatibility Standards	

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

3.6.7 Connector Rotational Travel

[EVASC.0058] The Artemis System shall ensure that the rotational travel of the EVA connector is limited to a maximum of 180° for installation or removal of twist-on connectors.

Rationale: Performing a twist on/off motion is considered awkward and difficult in a pressurized xEVA suit.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

3.6.8 Connector Locking Feature

[EVASC.0059] The Artemis System shall provide EVA connector plugs that include a locking feature to prevent inadvertent demate.

Rationale: Experience has shown that many of the EVA worksites are congested. Design of the worksite should include features which aid in precluding inadvertent demate that could result in a hazardous situation.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

3.6.9 Visible Connector Alignment Markings

[EVASC.0060] The Artemis System shall design visible alignment markings or other features on EVA connectors to ensure proper alignment during mating/demating.

Rationale: Connector alignment should not require so much precision as to be difficult to operate one-handed.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

3.6.10 Connector Locking Ring

[EVASC.0061] The Artemis System shall use EVA connectors with a locking ring that cannot turn independently from the connector body, such that the crewmember does not have to align both the ring and the body of the connector separately.

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 81 of 280
Title: EHP xEVA System Compatibility Standards	

Rationale: The crew should not have to align both the ring and the body of the connector separately, which would otherwise make this a two-handed task. Experience has shown that to ensure successful completion of EVA tasks, hardware operations must be kept as simple as possible. Utilizing a connector, which requires the alignment of both the ring and the body of the connector separately, is an unnecessary complication.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

3.6.11 Connector Mating Status

[EVASC.0062] The Artemis System shall design EVA connectors that indicate full mate/demate status.

Rationale: Full connector mating includes pin engagement. Alignment markings, pin detents, or an over-center feature are examples of features that can indicate connector mating status. It is especially important that EVA crewmembers know if connectors have been successfully mated/demated during periods of Loss of Signal (LOS).

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

3.6.12 Electrical Connectors

3.6.12.1 Mating/Demating of Powered Connectors

[EVASC.0052] The Artemis System shall design EVA connectors mated/demated during an EVA that prevent shock hazards to the EVA crewmember.

Rationale: Intent of requirement is to ensure that EVA safety hazards are considered when designing connectors that need to be operated by a suited pressurized crewmember.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

3.6.12.2 Connectors – Electrical Bonding

[EVASC.0065] The Artemis System shall provide electrical bonding of EVA electrical connector assemblies and conductive outer surfaces of electrical and electronic assemblies in accordance with NASA-STD-4003, *Electrical Bonding for NASA Launch Vehicles, Spacecraft, Payloads, and Flight Equipment*, Class H (or NASA approved

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 82 of 280
Title: EHP xEVA System Compatibility Standards	

alternate electrical bonding standard with Cross-Program Chief Engineer approval), as a minimum to prevent electrical shock hazards to EVA crewmembers.

Rationale: Intent of this requirement is to address floating connectors and electrical assembly housing issues. Proper bonding ensures the majority of harmful electrical current under hazardous fault conditions is shunted away from the EVA crewmember, preventing a shock hazard. Reference NASA-STD-4003, Electrical Bonding for NASA Launch Vehicles, Spacecraft, Payloads, and Flight Equipment.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

3.6.12.3 Scoop-Proof Electrical Connector Design

[EVASC.0066] The Artemis System shall design EVA electrical connectors to prevent electrical pins from inadvertently contacting the opposing connector half during mating/demating.

Rationale: Using scoop-proof connectors is one way to meet this requirement. The use of mechanical design features are necessary to fully enclose or shroud the power connector pins and sockets during mate/demate activities. This minimizes the potential for molten metal caused by FOD or bent pins. Scoop Proof designs provide a longer shell design on the pin half of a connector. Scoop-proof connectors minimize creation of potential hazards (electrical shock, sparks, arcs, or molten metal) as well as potential pin damage leading to the inability to connect the connector.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

3.6.12.4 Connector Grip Point

[EVASC.0063] The Artemis System shall design EVA connector backshells with grip point(s) to provide for engagement/disengagement with the pressurized gloved hand.

Rationale: A grip point provides access for the pressurized gloved hand to operate the connector.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 83 of 280
Title: EHP xEVA System Compatibility Standards	

3.6.12.5 Arc Containment

[EVASC.0131] The Artemis System shall design EVA electrical connectors that confine and isolate electrical arcs and sparks during mating/demating.

Rationale: Certain provisions are required when electrical connectors must be mated or demated during EVA to protect the crew from generation of molten metal, damage to safety critical circuits, and, potentially, shock, all of which are considered catastrophic hazards.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

3.6.13 Fluid Connectors

3.6.13.1 Uncontrolled Motion of Flex Hoses

[EVASC.0132] The Artemis System shall restrain or otherwise capture EVA serviceable fluid flex hoses and lines with delta pressure or under pressure.

Rationale: Delta pressure in a line could cause the line to move/flex, and pressurized lines can rigidize in unwanted configurations, so both should be accounted for. Hose whip can occur as a result of release of pressure in a line or from hose memory, and captive features are required to prevent hose whip that could result in crew injury or damage to adjacent hardware. This requirement does not apply to an xEVA suit umbilical.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

3.6.13.2 Venting Feature

[EVASC.0133] The Artemis System shall ensure that EVA serviceable fluid connectors and lines contain a method for removing pressure before demating.

Rationale: Excess fluid and pressure in a connector during demating may expose the crewmember to hazardous materials (either to the suit or to the IVA environment if brought back inside the vehicle), or to hazardous forces when the pressure is released during separation. This requirement does not apply to an xEVA suit umbilical.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 84 of 280
Title: EHP xEVA System Compatibility Standards	

3.6.13.3 Indication of Pressure Flow – Nonbrazed or Nonwelded

[EVASC.0145] All nonbrazed or nonwelded gas and liquid lines shall be provided with a positive indication of the gas pressure/fluid flow to verify that the line is passive before disconnection of non-self-sealing connectors.

Rationale: Need to ensure flow and pressure is zero via gauge indication prior to maintenance on fluid connectors. The intent is to protect the crewmember from pressurized lines in operations on non-QD connections. These types of connections should be avoided in anticipated EVA operations as QDs are typically used to facilitate operations with a pressurized EVA glove possible.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

3.7 EVA OPERABLE FASTENERS

The following section outlines those fasteners that are to be operated by an EVA crewmember in a pressurized xEVA suit. It is important to note that in this context when the word “design” is used, it does not preclude hardware developers from using previously existing products that may or may not have been uniquely created for this application. Instead, the intent is to communicate that regardless of how the parts are sourced, the resulting assembly must be operable by the EVA crewmember.

3.7.1 Captive Fastener Design

[EVASC.0068] The Artemis System shall design all EVA operable fasteners to be captive without the use of tethers or lanyards as the captive method.

Rationale: In microgravity, loose parts or hardware poses an impact hazard to the EVA crewmember or the vehicle. Loss of parts/hardware required for critical functions is also considered a hazard and an impact to mission success.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

3.7.2 Locking Features

3.7.2.1 Prohibited Locking Methods for Fasteners

[EVASC.0069] The Artemis System shall provide a method for locking EVA operable fasteners that does not involve the use of safety wire, cotter pins, or jam nuts.

Rationale: Safety wire and cotter pins may be a source for sharp edges and are, therefore, prohibited. Safety wire and cotter pins may be used on internal fasteners only. The use of anaerobic locking compounds as the sole secondary

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 85 of 280
Title: EHP xEVA System Compatibility Standards	

locking feature on structural or critical fasteners has been shown to be ineffective in the vacuum environment.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

3.7.3 EVA Hand-Actuated Fasteners Operation

3.7.3.1 One-Handed Fastener Operations

[EVASC.0067] The Artemis System shall design fasteners intended to be manipulated without a tool to be operable one-handed using a pressurized xEVA glove.

Rationale: The hand-operable fastener should not require a high degree of dexterity such that it necessitates two-handed operation. Hand-operated fasteners do not require a tool.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

3.7.3.2 Scalloped Knobs

[EVASC.0072] The Artemis System shall design scalloped knobs for actuation by the pressurized EVA gloved hand with no fewer than four (4) and no more than eight (8) scallops.

Rationale: Knob design specifications will improve performance when used with a pressurized, gloved hand. Three examples of knobs designed for EVA hand actuation are Knob, Push, Pitch Joint Assembly APFR, ISS (SDG33107506), EVA Knob Assembly – Bridge Clamp (SED33107181), and 1.0 Inch QD Button Depress Tool Assy (SEG33114987). Reference Figure 3.7.3.3-1 Flight Knob Examples in requirement [EVASC.0073]; the minimum scallop value allows for deeper scallops as needed.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

3.7.3.3 Knobs Head Diameter and Height

[EVASC.0073] The Artemis System shall design knobs for xEVA suited, pressurized, gloved hand operation to have head diameters between the ranges of 3.8 cm – 5.1 cm (1.5 in – 2.0 in) with a minimum head height of 1.9 cm (0.75 in).

Rationale: Knob design specifications will improve performance when used with a gloved hand. For reference only, Figure 3.7.3.3-1 Flight Knob Examples,

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 86 of 280
Title: EHP xEVA System Compatibility Standards	

provides example shapes that are known to be compatible with Phase VI gloves; implementation of mounting shaft interface will be design-dependent and vary.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

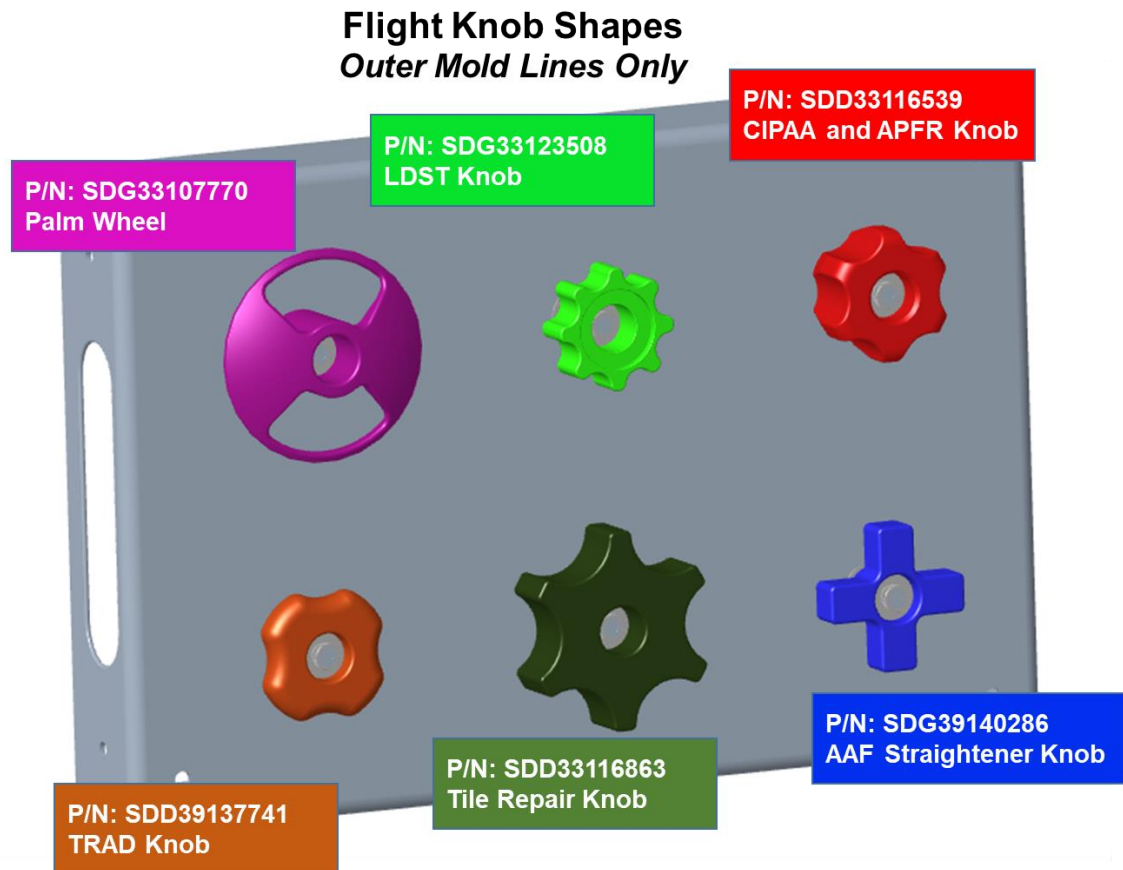


FIGURE 3.7.3.3-1 FLIGHT KNOB EXAMPLES

3.7.3.4 EVA Hand-Actuated Fasteners Contingency Operation

[EVASC.0074] The Artemis System shall provide an external hexagonal drive interface as outlined in Table 3.7.3.4-1 EVA External Hexagonal Drive Interface on rotational fasteners per section 3.7.5 for contingency operation with a hand tool.

Rationale: Same size fastener leads to a single tool that works for all cases.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 87 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 3.7.3.4-1 EVA EXTERNAL HEXAGONAL DRIVE INTERFACE

Hexagonal Drive Interface	Artemis System
11mm	HLS xEVA Tools LTV / PR Payloads
7/16-in.	Gateway

3.7.4 EVA Push-in-Pull (PIP) Pins

Refer to APPENDIX B EVA Best Practices (BPs) for PIP Pins (EVABP-2). Use of PIP Pins has been closely scrutinized by the JSC Structures and Mechanisms System, and these pins are designed so reliably that the item on its own is considered one fault tolerant; however, the Artemis System must receive appropriate risk acceptance from the customer Program. Please verify acceptability of specific part numbers or design with EHP for the NASA Structures and Mechanisms System Team review when using PIP pins.

3.7.4.1 EVA PIP Pin Grasping Interface

[EVASC.0075] The Artemis System shall mount a 3.8 cm (1.5 in) minimum diameter ring or T-handle to the pin that is compatible with a pressurized glove.

Rationale: T-handles or rings large enough to use with a pressurized, gloved hand are required to simplify use with a gloved hand and to minimize the risk of lost hardware. It is preferred that the ring is mounted to the actuation shaft.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

3.7.4.2 PIP Pin Restraint Tether

[EVASC.0076] The Artemis System shall attach a restraint tether to the body (handle) of the PIP pin and not to the activation mechanism.

Rationale: PIP pins may also be captive to the structure, which is preferable to a separate tether.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 88 of 280
Title: EHP xEVA System Compatibility Standards	

3.7.5 Tool-Actuated Fasteners

3.7.5.1 Driver-Type Push Force Tools

[EVASC.0078] The Artemis System shall design fasteners that do not require a driver-type hand tool that requires a push force greater than 9N (2 lbf) to maintain tool engagement while providing torque.

Rationale: Slotted and Phillips screws and screwdrivers are prohibited for EVA interface use. The dexterity that is lost when using pressurized EVA gloves makes using fasteners, which require a constant force to remain engaged, impractical.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

3.7.5.2 EVA Fastener Head

[EVASC.0079] All EVA-actuated bolts shall conform to the following configurations from the minimum to maximum operational temperatures to enable interface with the standard EVA hand and power tools:

- 1) Figure 3.7.5.2-1 11 mm Hexagonal EVA and Robotic Compatible Bolt
- 2) Figure 3.7.5.2-2 7/16-inch Hexagonal EVA and Robotic Compatible Bolt (for Gateway applications only)

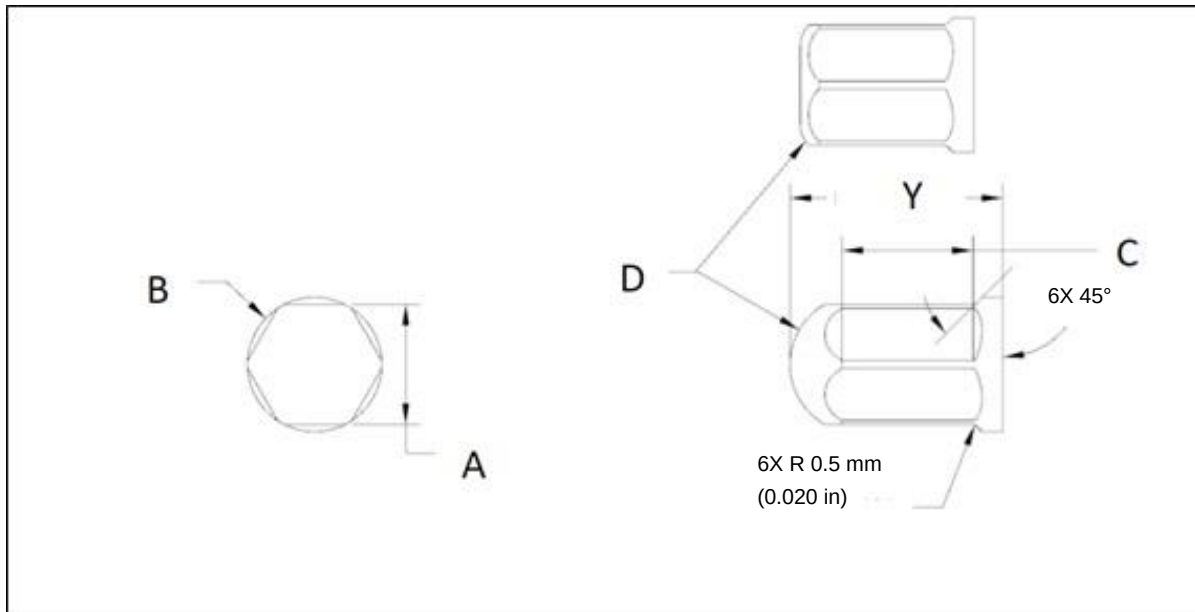
Rationale: This requirement ensures compatibility with future xEVA tools and provides for common EVA tool interface between vehicles. During the 7/28/22 Gateway Program CB, it was decided that Gateway will only utilize the 7/16-in EVA Bolt head outlined in Figure 3.7.5.2-2.

7/16-inch EVA Bolt heads with 12-point design have not been utilized (to date) for ISS operations. The 12-point bolt head design also presents additional manufacturing and gauge challenges versus those experienced with a 6-point bolt-head design, thereby reducing risk to cost and schedule. The existing suite of ISS 7/16-inch sockets can accommodate a 12-point bolt head; however, the socket is limited to 120 ft-lb (163 Nm) rating. If required by hardware design, alternate EVA fasteners may drive design and certification of different sizes of EVA tools such as socket extensions and should be worked with EHP personnel.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 89 of 280
Title: EHP xEVA System Compatibility Standards	



11 mm Nominal bolt head size	Max/Min dimensions (mm)
(A) Flat to Flat	10.96 10.83
(B) Peak to Peak	12.35 12.22
(C) Effective height	9.55 9.30
(D) Wrench guide	Full R 2.75
(Y) Overall height	22.35 10.90

FIGURE 3.7.5.2-1 11 MM HEXAGONAL EVA AND ROBOTIC COMPATIBLE BOLT

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 90 of 280
Title: EHP xEVA System Compatibility Standards	

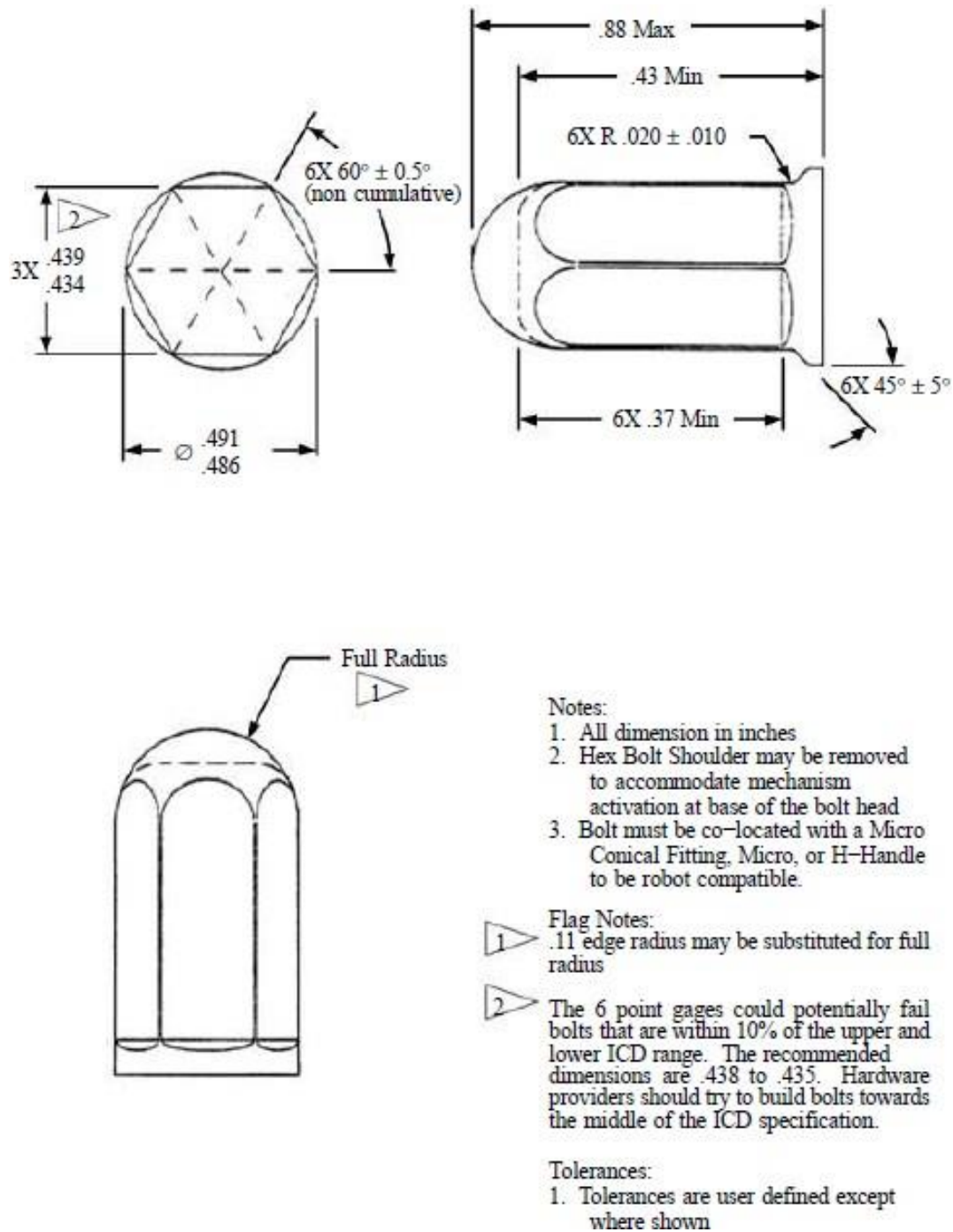


FIGURE 3.7.5.2-2 7/16-INCH HEXAGONAL EVA AND ROBOTIC COMPATIBLE BOLT

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 91 of 280
Title: EHP xEVA System Compatibility Standards	

3.7.5.3 Fastener Deep Socket

[EVASC.0084] EVA fasteners, which are planned to be operated by a hand or power tool, shall have an extra deep well on the socket that is at minimum equivalent to the double hex-head height.

Rationale: An extra-deep well in the socket and/or a wobble feature helps mitigate stripping concerns. The double bolt height is recommended for ease of use and versatility (reference requirement [EVASC.0079]). A wobble feature may assist with fastener alignment and off-angle drive applications in microgravity environment.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

3.7.5.4 Fastener Torque Data

[EVASC.0151] Data Deliverable: When fastener torquing to specifications is required, provide per Table 3.7.5.4-1 Fastener Torque Data.

Rationale: The intent is to ensure that the Artemis System delivers the torque instructions for on-orbit actuation. Additional materials in contact with threads (such as lubricants, locking compounds or mechanical locking features), need to be evaluated for effects within the environmental conditions, and are taken into consideration for torque calculations during EVA installation/ removal. Depending on the state of lubrication, the effects of temperature and pressure (i.e., vacuum) can significantly affect the friction characteristics of the sliding interfaces. Recorded tightening, running, and release torque values are to be provided upon request from Program Management.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

TABLE 3.7.5.4-1 FASTENER TORQUE DATA

Row #	Description	Additional Information
1	Fastener Part Number	If applicable, state additional cautions, in the form of training and updates to EVA procedures, should be taken.
2	Next Assembly Part Number of Fastener	
3	Insert or Part Number	
4	Next Assembly Insert of Nut Part Number	
5	EVA Label for Bolt or Nut Next Higher Assembly	

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 92 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 3.7.5.4-1 FASTENER TORQUE DATA

Row #	Description	Additional Information
6	Name/Description	
7	Bolt is used to secure MI (Yes/No)	
8	MI Part Number	
9	Ground Installation Torque	
10	Maximum On-Orbit Installation Torque	
11	Minimum On-Orbit Installation Torque	
12	No. of bolt turns for clamp-up/Removal	
13	More important: Torque OR Turns OR Both	During on-orbit installation, state if it is more important for the crewmember to achieve the torque value specified or the number of turns specified. If both torque and turns are required, enter "Both".
14	Maximum Expected Breakaway Torque	This should be a value for breakaway that is the highest you would expect to see, whether the bolt was installed on the ground or on-orbit. For non-microconical interfaces, if value is greater than 35 Nm (310 lbf-in), fill in row 15.
15	Maximum Expected Breakaway Torque when install torque is per Row 11. (This row is filled in only if value in Row 14 is greater than 35 Nm (310 lbf-in) for non-microconical interfaces)	This value should also account for on-orbit temperature effects.
16a	Recommended Torque to use for Removal	This is the recommended torque value to set the EVA tool for removal of the bolt. If Row 15 has been filled in, enter the value in Row 16 based on Row 15.
16b	Recommended Torque to use for Installation	This is the recommended torque value to set the EVA tool to for installation of the bolt.
19	Absolute Max. Torque not to be exceeded in order to prevent failure. Factor of Safety (FS) = 1.	Torque values specified are the torque applied at the head of the fastener independent of the tool or extension used. (This differs from all other rows where torque calculations are done with the approved Structural Integrity Factor for Safety Requirements).
20	Failure Mode if torque exceeded.	That is, what item will fail first? What are the expected failure results: bolt thread failure, nut thread failure, bolt shank failure, secondary structure yield, secondary structure failure, primary structural yield, and primary structural failure, etc.? Again, the failure predictions are based on a FS of 1 (Row 19).
21	Failure data from Analysis (A) or Test (T)	Specify whether data has been obtained from testing or from analysis.

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 93 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 3.7.5.4-1 FASTENER TORQUE DATA

Row #	Description	Additional Information
22	Special Operations or constraints in order to minimize binding	If there are any special operations or constraints in order to minimize binding, please specify. If none, leave blank. For example, describe any steps taken to prevent galling; however, galling is a very specific failure mode. Bolt binding does not always equal galling. Side loads, misalignment, thermal mismatch are all sources of binding that present similar to galling, and if the design is vulnerable to these or any others, they should be described in this table.
23	Potential Recovery/Repair Schemes (including captive fastener replacement methods, if applicable)	Potential Recovery/Repair Schemes (including captive fastener replacement methods, if applicable). Expand on Row 23 in note form: e.g., Note 2. Recovery scheme for this fastener is...etc.
24	Has this bolt been analyzed to be out/absent?, while all others remain installed to meet a Fail-Safe Condition? Yes or No Note: This Row (24) is for contingency reference only and must not be used for nominal or work-ahead EVA planning.	This row should be filled in "Yes" or "No" based on existing strength analysis.

General Notes:

- 1) Rows 17 and 18 are intentionally not used.
- 2) Format for torque values: torque Nm, (lbf-in)

3.7.5.5 Power and Hand Tool Operability

[EVASC.0080] EVA fasteners installed with power tools shall be removable and/or installable with a hand-operated tool.

Rationale: Hardware should be designed to be operated/torqued by a power tool and a hand operated tool for contingency/clearance issues. There is an option to use the ratchets or wrenches with sockets for the fasteners.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 94 of 280
Title: EHP xEVA System Compatibility Standards	

3.7.5.6 Fastener Gauge/Measurement

[EVASC.0141] EVA bolt heads shall be gauged/measured ensuring they meet dimensional requirements at the minimum to maximum operational temperatures prior to installation into assembly to verify all bolt head features can be inspected.

Rationale: Go/No gauges allow thermal and manufacturing tolerances to be assessed. Hardware should be designed to accept torque values with as wide a tolerance as possible. Thermal extremes can adversely affect the ability for a tool to achieve a specific torque. Provide verification of bolts prior to installation into assembly to prevent future issues with undersized/oversized bolt heads.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

3.7.5.7 Fastener Alignment Features

[EVASC.0083] The Artemis System shall provide EVA fastener alignment features to ensure initial engagement and preclude cross-threading.

Rationale: Examples include chamfered threaded ends and/or bolt guides.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

3.8 HARDWARE DESIGN

3.8.1 Battery Powered Devices

3.8.1.1 Battery Replacement

[EVASC.0085] Any batteries that necessitate changeout via EVA shall be replaceable at the worksite.

Rationale: The intent is to ensure that if the battery must be changed during EVA, it is done at the worksite without the overhead of bringing it back inside the vehicle/pressurized environment. This provides flexibility for the development of vehicle design operational concepts. Time is better managed with replacement of batteries at the worksite, e.g., if they do not last as long as the intended task. Battery removal and replacement must also be environmentally tolerant so as not to have any impacts removing or installing old or new batteries and that the batteries can tolerate debris.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 95 of 280
Title: EHP xEVA System Compatibility Standards	

3.8.1.2 Battery Voltage Monitoring

[EVASC.0086] Battery-powered devices shall include battery monitoring capability before/during EVA usage to protect against catastrophic hazards and insufficient capacity to support mission time.

Rationale: Battery power indication should be incorporated as part of a power device, not the battery or battery pack itself. The function of battery power indication is intended to inform the EVA crewmembers of battery charge before and during the EVA, so crewmembers can ensure safety critical hardware is sufficiently charged for the duration of the EVA. Battery voltage monitoring is necessary for safe function and mission success of those operable devices.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

3.8.1.3 Auxiliary Controls

[EVASC.0028] Auxiliary controls shall be positioned such that an EVA-gloved hand can activate the control without disturbing the tool/fastener position.

Rationale: The intent is to ensure that if an auxiliary control on the device must be operated while holding it, that there is sufficient clearance to not perturb the tool/fastener position.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

3.8.2 EVA Tools

[EVASC.0088] The Artemis System shall utilize EVA tools listed in EHP-10074, *EHP EVA Tool List and Process for EVA Operations* <**TBR-EHP-10028-008**>.

*Rationale: The EHP configuration-managed list contains tools intended to enable EVA crew to perform operations externally either in microgravity or partial gravity. Vehicle designs should consider these available tools when planning for operations that are needed for maintenance or contingent vehicle activity. EVA tools designated to be used for maintenance on any hardware outside of the habitable volume, such as an external antenna, solar arrays, a rover, or an external science payload, are included in EHP-10074, *EHP EVA Tool List and Process for EVA Operations* <**TBR-EHP-10028-008**>. Tools should be selected from the tool list while considered for destination environment; EHP can assist with selection of tools. Any designs that create the need for an additional EVA tool outside of this set of tools should be routed through the appropriate Program CBs for approval.*

Gravity Field Applicability: Microgravity, Partial Gravity

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 96 of 280
Title: EHP xEVA System Compatibility Standards	

Allocation: LTV, PR, Gateway, HLS, Payloads

3.8.3 Cable Routing/Restraint

[EVASC.0089] Cable routing/restraint for hardware shall preclude use of EVA handling and translation aids as those routing paths or restraints.

Rationale: EVA handling and translation aids are not to be used as routing or restraints for cabling.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: LTV, PR, Gateway, HLS, Payloads

3.8.4 Cable Routing Spacing

[EVASC.0090] EVA-installed cable routing shall include restraints spaced at a maximum of 0.61m (24 in).

Rationale: Designers are to define areas of activity and route fixed lines and cables so that they are both protected from and do not interfere with these activities. Also, system designers are to focus on non-fixed lines and cables that may be unstowed or moved for a specific task or temporary rearrangement. While the rerouted cable or line may accommodate a specific need, the routing path may interfere with other, non-related activities. Designers are to identify potential uses for lines and cables and ensure the start points, end points, and cable and line routes in between will accommodate all crew activities. For ISS, EVA-installed cables are typically secured every 0.61 m (24 in).

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: LTV, PR, Gateway, HLS

3.8.5 Paint

[EVASC.0091] The Artemis System shall provide evidence that painted surfaces or features are able to withstand environmental conditions and wear-and-tear due to EVA contact (planned and inadvertent) on EVA worksites and within EVA translation paths/traverses, or areas on the vehicle which the EVA crewmember will access without flaking or contaminating the EVA crewmember.

Rationale: Paint (often considered the “simple” solution to improve passive thermal properties or provide corrosion control) needs to be minimized on EVA equipment, on EVA worksites, and near EVA translation paths for microgravity application. Paint is difficult to apply properly, difficult to maintain without damage preflight and during processing, and may flake off in a vacuum environment after thermal cycles. Additionally, paint is easily damaged by any contact from an EVA

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 97 of 280
Title: EHP xEVA System Compatibility Standards	

crewmember. Damaged paint will change the surface optical properties and can become Foreign Object Debris (FOD) which clogs mechanisms and increases Micrometeoroid Orbital Debris (MMOD) risk. Partial gravity application includes any hardware or portions of the vehicle that the EVA crewmember will need to access.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: LTV, PR, Gateway, HLS, Payloads

3.9 SAFETY

3.9.1 Hazardous Material Control

[EVASC.0092] The Artemis System shall provide means of containment commensurate with the hazard level for toxic or hazardous chemicals/materials to preclude contaminating an EVA crewmember or xEVA suit (hazardous to EVA crewmember, xEVA suit performance, or to IVA environment if brought back inside habitable volume).

Rationale: Hazardous materials (e.g., toxic or corrosive) may need to be intentionally present for mission operations. However, to protect the safety of the EVA crewmember during the EVA and of IVA crewmembers following EVAs, those materials should be contained in accordance with the particular perspective on fault tolerance of each Artemis System. Exposure to toxic or corrosive materials can cause damage to the xEVA suit or contaminate the airlock and/or associated IVA environment. Therefore, toxic, or corrosive materials need to be completely contained whenever possible. Materials that could cause discoloration of the xEVA suit should not be considered since they can affect the thermal properties of the suit.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

3.9.2 Venting of Hazardous Materials

3.9.2.1 Venting or Exposure Compatibility

[EVASC.0093] The Artemis System shall design hardware to preclude venting or exposure of hazardous materials to an EVA crewmember (hazardous to EVA crewmember or to IVA environment if brought back inside habitable volume).

Rationale: Exposure of hazardous materials may impact/damage xEVA suit materials. Examples of IVA-contaminated environment that can impact the xEVA suit include consumables such as water, oxygen, etc. Exposure to hazardous material contamination could induce a very serious health condition to an EVA crewmember or IVA crew.

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 98 of 280
Title: EHP xEVA System Compatibility Standards	

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

3.9.2.2 Deliberate Hazardous Venting

[EVASC.0094] The Artemis System shall direct deliberate hazardous venting away from the EVA crewmember and translation path/worksites.

Rationale: Hazardous materials may need to be intentionally vented for successful spacecraft operation. However, to protect the safety of crewmembers during and after the EVA, the orientation and operation of those vents need to be designed and operated so that crewmembers and translation paths/worksites are not directly impinged upon by hazardous materials during the EVA. Likewise, such venting operations should not deposit hazardous materials on any surface prior to an EVA, with which the crewmember might intentionally or inadvertently come into contact. These steps preclude contact/transference with vented substances and the possibility of bringing such materials inside the habitable volume following EVA activities.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

3.9.3 Electrical and Safety Design

[EVASC.0097] The Artemis System shall design equipment to preclude exposure of hazardous electrical potential to the EVA crewmember.

Rationale: Electrical hazards could include shock, corona discharge, and high electromagnetic fields. Electrical hazards may not be visibly obvious and so should be controlled by means other than the awareness of EVA crewmembers. Ways to meet this are through inhibits/safing systems when EVA is present and lock out/tag out. For Voltage Potential Discharge, reference requirement [EVASC.0123].

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

3.9.4 Shock Protection

[EVASC.0098] Any exposed conducting surfaces that are accessible to EVA crewmember contact (e.g., translation corridors, work areas, and incidental contact zones) shall prevent shock hazards.

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 99 of 280
Title: EHP xEVA System Compatibility Standards	

Rationale: Reference NASA-STD-4003, Electrical Bonding for NASA Launch Vehicles, Spacecraft, Payloads, and Flight Equipment, for electrical bonding procedures.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

3.9.5 Operation During Pressure Change

[EVASC.0099] The Artemis System shall design or use equipment expected to function during depressurization and repressurization per EHP-10029, *Exploration EVA (xEVA) System to Artemis Systems Interface Requirements Document (IRD) <TBR-EHP-10028-016>*, requirements outlined in Table 3.9.5-1 Depress and Repress Rates, to operate without producing hazards to the EVA crewmember.

Rationale: EVA hardware needs to withstand the differential pressure of depressurization, repressurization, and the depressurized condition without resulting in a hazard.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

TABLE 3.9.5-1 DEPRESS AND REPRESS RATES

Artemis System	Requirement	Rate Condition
xEVA	xEVA.0187 xEVA.0189	Rapid Cabin Depress/Emergency Repress
LTV, PR, Payloads, Gateway, HLS	xEVA.0186 xEVA.0188	Rapid Cabin Depress/Emergency Repress <u>Note:</u> Depress/repress operations do not apply to LTV, rather applies to any LTV hardware that is to be brought back into the pressurized vehicle

3.9.6 Component Hazardous Energy Provision

[EVASC.0100] The Artemis System shall preclude exposing the EVA crewmember to the release of hazardous energy.

Rationale: This applies to mechanical, chemical, and electrical sources of stored energy. Stored energy can exist in mechanical systems and pressurized flex hoses, hence, tether hoses to prevent hose whip when pressurized. Any item with the potential to release stored energy could inadvertently release this energy in the presence of the EVA crew and cause injury to the crew or damage the EVA suit

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 100 of 280
Title: EHP xEVA System Compatibility Standards	

which may result in loss of a crewmember. Examples include springs that can release or propel a mass, or cold propellants that can be released creating a pushing force, or chemicals mixing and creating explosive forces or high temperatures (e.g., hot firing propellants). Implementing vehicle/hardware design features that allow for safing of potential energy, along with confirmation that safing is successful, aids in ensuring that the hazard is controlled.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

3.9.7 Detachable Parts

[EVASC.0101] The Artemis System shall design provisions for securing detachable parts.

Rationale: Detachable parts could include safety pins, caps, etc. All detachable parts must be secured to prevent inadvertent release. This is necessary to prevent inadvertent interference with mechanisms or unintentional re-contact with spacecraft elements. For Partial Gravity applications, this minimizes contamination or loss of cap if dropped on Lunar Surface. Securing detachable parts should not rely on the continuous grasp of an EVA crewmember.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

3.9.8 Touch Temperatures

[EVASC.0103] The Artemis System shall design the hardware to protect the xEVA suit from specified high- and low-touch-temperature extremes as defined in Table 3.9.8-1 *Touch Temperatures <TBR-EHP-10028-011>*, in the translation corridors, work areas, and incidental contact zones.

Rationale: For ISS, the continuous touch temperature requirement is -43°C to 63°C (-45°F to +145°F), and incidental touch temperature requirement is -118°C to 113°C (-180°F to +235°F). The touch temperature requirement shall be verified using Beginning of Life (BOL) optical property data for cold temperature design, and End of Life (EOL) optical property data for hot temperature design. The requirement can be verified based on the absolute temperature range, or by heat transfer analysis in accordance with the performance of the xEVA suit. These incidental and extended brush limits on the Thermal Micrometeoroid Garment (TMG) have been determined by test and analysis (JSC 39198, "Incidental Contact Temperature Test on Extravehicular Mobility Unit (EMU) Softgoods Materials Lay-Up," AML-01-02-R1, "Advanced Materials Laboratory (AML) Test Report: Hot D-Ring / SSA Lay-Up Incidental Contact Test", and JSC 39117, "Extravehicular Mobility Unit (EMU) International Space Station (ISS) Extravehicular Activity (EVA)

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 101 of 280
Title: EHP xEVA System Compatibility Standards	

Thermal Environment Requirements for Certification”). In rare cases, EVA operations and/or hardware design require long-duration compression of the TMG orthofabric, thereby, thermally shorting the multilayer insulation and making TMG layers vulnerable to thermal damage. Examples of these are tools with gauntlets or grips that extend around the backside of the glove, or hardware in or containing confined space that requires any portion of the EMU to be compressed during EVA operations. These extended EMU TMG contact limits have been determined by analysis (JSC 39117).

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

TABLE 3.9.8-1 TOUCH TEMPERATURES <TBR-EHP-10028-011>

Temperature Range °C (°F)	Type of Contact
-43°C to 63°C (-45°F to +145°F)	Unlimited touch
-118°C to 113°C (-180°F to +235°F)	Incidental glove palm contact
-153°C to 160°C (-244°F to 320°F)	Incidental TMG brush & bump*
-126°C to 116°C (-195°F to 240°F)	Extended compression of TMG for anything longer than a brush or bump*~
-122°C to 97°C (-187°F to 206°F)	Extended compression of TMG for back of glove for anything longer than a brush or bump*

* Brush or bump is defined as 3 seconds for bump at 7k Pa (1 psi) or 30 seconds for brush at 0.7 kPa (0.1 psi). These may be invoked if or when the first order thermal analysis for items EVA contacts does not meet the values for unlimited glove palm touch or incidental glove palm contact as appropriate.

~ EVA hardware that causes glove or xEVA suit compression during nominal operation shall be evaluated via a hazard/safety analysis.

3.9.9 xEVA Suit Glove Palm External Touch-Temperature Compliance for Small, Handheld Hardware

[EVASC.0011] The Artemis System shall design hardware that complies with the 30-minute xEVA suit glove palm touch temperature limits for crewmember protection from high and low skin touch temperatures as defined in Table 3.9.9-1 *30-Minute Touch Temperature Compliance For Small, Handheld Hardware <TBR-EHP-10028-011>*, and Table 3.9.9-2 *30-Minute Touch Temperature Heat Rate Compliance <TBR-EHP-10028-011>*.

Rationale: This requirement applies for hardware that should meet the unlimited glove palm temperature limits. As an alternate, this Table 3.9.9-1 30-Minute Touch Temperature Compliance for Small, Handheld Hardware, can be used to demonstrate 30-minute touch-temperature compliance of hardware, subject to the following limitations: small, handheld hardware with no more than 320 cm² (50 in²) surface area, and materials predominantly of aluminum, stainless steel, titanium,

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 102 of 280
Title: EHP xEVA System Compatibility Standards	

and/or soft goods. These limits do not apply to the backside of the glove which is made from different material that meets other standards.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

TABLE 3.9.9-1 30-MINUTE TOUCH TEMPERATURE COMPLIANCE FOR SMALL, HANDHELD HARDWARE <TBR-EHP-10028-011>

Hardware Mass kg (lbm)	Allowable Touch-Temperature Range °C (°F)
0 to 0.09 (0.0 to 0.2)	-129 to 121 (-200 to +250)
>0.09 to 0.10 (0.2 to 0.23)	-129 to 116 (-200 to +240)
>0.10 to 0.11 (0.23 to 0.25)	-129 to 110 (-200 to +230)
>0.11 to 0.13 (0.25 to 0.28)	-129 to 104 (-200 to +220)
>0.13 to 0.14 (0.28 to 0.30)	-129 to 99 (-200 to +210)
>0.14 to 0.16 (0.30 to 0.35)	-129 to 93 (-200 to +200)
>0.16 to 0.18 (0.35 to 0.40)	-129 to 88 (-200 to +190)
>0.18 to 0.27 (0.40 to 0.60)	-118 to 82 (-180 to +180)
>0.27 to 0.36 (0.60 to 0.80)	-101 to 79 (-150 to +175)
>0.36 to 0.54 (0.80 to 1.2)	-90 to 77 (-130 to +170)
>0.54 to 0.91 (1.2 to 2.0)	-82 to 74 (-115 to +165)
>0.91 to 1.8 (2.0 to 4.0)	-71 to 71 (-95 to +160)
>1.8 (4.0)	-62 to 66 (-80 to +150)

TABLE 3.9.9-2 30-MINUTE TOUCH TEMPERATURE HEAT RATE COMPLIANCE <TBR-EHP-10028-011>

Object Temperature °C [°F] <i>Before Contact</i>	Continuous Contact Duration (minutes)	Contact Boundary Temperature °C [°F]	Contact Conductance W/m ² -°C [BTU/hour square inch-° F]	Maximum Average Heat Rate (1,2) W/m ² [BTU/hour-square inch]
> 66 [150]	30	38 [101]	2.698 [0.0033]	93.12 [0.205]
< -62 [-80]	30	24 [75]	2.698 [0.0033]	-264.8 [-0.583]

Notes:

- (1) Positive denotes heat out of the object.
- (2) Averaged over 30 minutes of simulated contact.

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 103 of 280
Title: EHP xEVA System Compatibility Standards	

3.9.10 Incidental and Unlimited Contact Heat Transfer Rates

[EVASC.0104] The Artemis System shall limit heat transfer rates for incidental and unlimited contact to the values defined in Table 3.9.10-1 *Heat Transfer Rates <TBR-EHP-10028-011>*.

Rationale: Limiting heat transfer is an alternative method to maintaining the touch temperature values for incidental and unlimited contact as defined in the requirement [EVASC.0103], Table 3.9.9-1 30-Minute Touch Temperature Compliance for Small, Handheld Hardware <TBR-EHP-10028-011> and Table 3.9.9-2 30-Minute Touch Temperature Heat Rate Compliance <TBR-EHP-10028-011>. End item temperature can be controlled passively or actively. Passive controls include the choice of material, coatings, or insulation, while active controls include items such as heaters or cold plates. The incidental contact temperatures in Table 3.9.10-1 Heat Transfer Rates <TBR-EHP-10028-011>, assumes the skin temperature as the boundary (from medical limits and testing), with a linear conductor to account for resistances and material thermal resistances of the glove (based on testing), and the maximum allowable heat rate (based on testing). The end item provider must apply these values to the object to be contacted by the skin boundary through the prescribed linear conductor. The object initial temperature and material properties must be considered. These values are based on the heritage ISS EMU suit. This requirement applies to designated EVA interfaces that include the following: EVA tools and support equipment, EVA translation aids (e.g., handrails, handholds, etc.), EVA restraints (e.g., foot restraints, tethers, tether points, etc.), all EVA translation paths (e.g., handrails or structure identified for use as a translation path), all surfaces identified for handling, transfer, or manipulation of hardware), EVA stowage, EVA worksite accommodations (e.g., handrails, foot restraint ingress aid, EVA lights, etc.), and any EVA MI handling and transfer equipment.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

TABLE 3.9.10-1 HEAT TRANSFER RATES <TBR-EHP-10028-011>

Object Temperature	Contact Duration (minutes)	Boundary Node Temperature ° C [° F]	Linear Conductor K/W [BTU/hr ° F]	Maximum Average Heat Rate ¹ W [BTU/hr]
Hot Object	Unlimited	45 [113]	2.179 [1.149]	13.34 ² [42.52]
	Incidental (0.5 max)	45 [113]	2.739 [1.444]	51.64 ³ [176.2]
Cold Object	Unlimited	4 [40]	2.015 [1.062]	-38.89 ² [-132.7]

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 104 of 280
Title: EHP xEVA System Compatibility Standards	

Object Temperature	Contact Duration (minutes)	Boundary Node Temperature ° C [° F]	Linear Conductor K/W [BTU/hr ° F]	Maximum Average Heat Rate ¹ W [BTU/hr]
	Incidental (0.5 max)	4 [40]	2.804 [1.4782]	-95.31 ³ [-325.2]

Notes:

- ¹ Positive denotes heat out of the object, negative denoted heat into the object.
- ² Averaged over 30 minutes of simulated contact (excursions up to 1.5 times this rate for 3-minute intervals is allowable).
- ³ Averaged over 2 minutes of simulated contact (excursions up to 2.5 times this rate for 12-second intervals is allowable)

3.9.11 Sharp Edge and Protrusions Protection

[EVASC.0105] The Artemis System shall protect the EVA crewmember from end item edges and protrusions on all hardware located within 0.61 m (24 in) of a translation path (reference requirements [EVASC.0010] and [EVASC.0159]), within a worksite envelope, and on each equipment item needing an EVA interface through compliance with Table 3.9.11-1 *Edge, Corner, and Protrusion Criteria – Edge and In-Plane Corner Radii*, Figure 3.9.11-1 *Exposed Corner and Edges*, and Table 3.9.11-2 *Edge, Corner, and Protrusions Criteria – Protrusions and Outside Corners*.

Rationale: This requirement applies to any protrusion, burr, or exposed sharp edge that has the potential to cut an EVA glove or puncture the spacesuit. There are instances where hardware can comply with this requirement, but burrs (ragged edges can occur during manufacturing/assembly and must be screened and removed. Sharp edges could damage the softgoods of the xEVA suit and potentially create a catastrophic hazard due to pressure loss. It is critical that hardware designers address both sharp edges that may be considered "on purpose" as well as those that can be generated during the manufacturing process as "accidental". A swatch test can also be used to evaluate potential sharp edges. Only trained personnel can perform a swatch test (reference EHP-10013, EHP Verification and Validation Plan Section 6.2.1 for additional information).

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 105 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 3.9.11-1 EDGE, CORNER, AND PROTRUSION CRITERIA – EDGE AND IN-PLANE CORNER RADII

Application	Radius				Remarks	Figure
	Outer mm	in	Inner mm	in		
(a) Openings, panels, covers (corner radii in plane of panel)	6.4 3.0	0.25 0.12	3.0 1.5	0.12 0.06	Preferred Minimum	
(b) Exposed corners:	13	0.5	–	–	Minimum	(a)
(c) Exposed edges: (1) Thicker than 2.0 mm (0.08 in)	1.0	0.04	–	–		(b)
(2) 0.5-2.0 mm (0.02 to 0.08 in) thick	Full Radius		–	–		(c)
(3) less than 0.5 mm (0.02 in) thick	Rolled or Curled					(d)
(d) Flanges, latches, controls, hinges, and other small hardware operated by the pressurized-gloved hand	1.0	0.04	–	–	Minimum required to prevent glove snagging	–
(e) Small protrusions (less than approximately 4.8 mm (3/16 in)) on toggle switches, circuit breakers, connectors, latches, and other manipulative devices	1.0	0.04	–	–	Absolute minimum unless protruding corner is greater than 120°	

Edge and In-Plane Corner Radii - A 45° chamfer by 1.5 mm (0.06 in) (minimum) with smooth broken edges is also acceptable in place of a corner radius. The width of chamfer should be selected to approximate the radius corner described above.

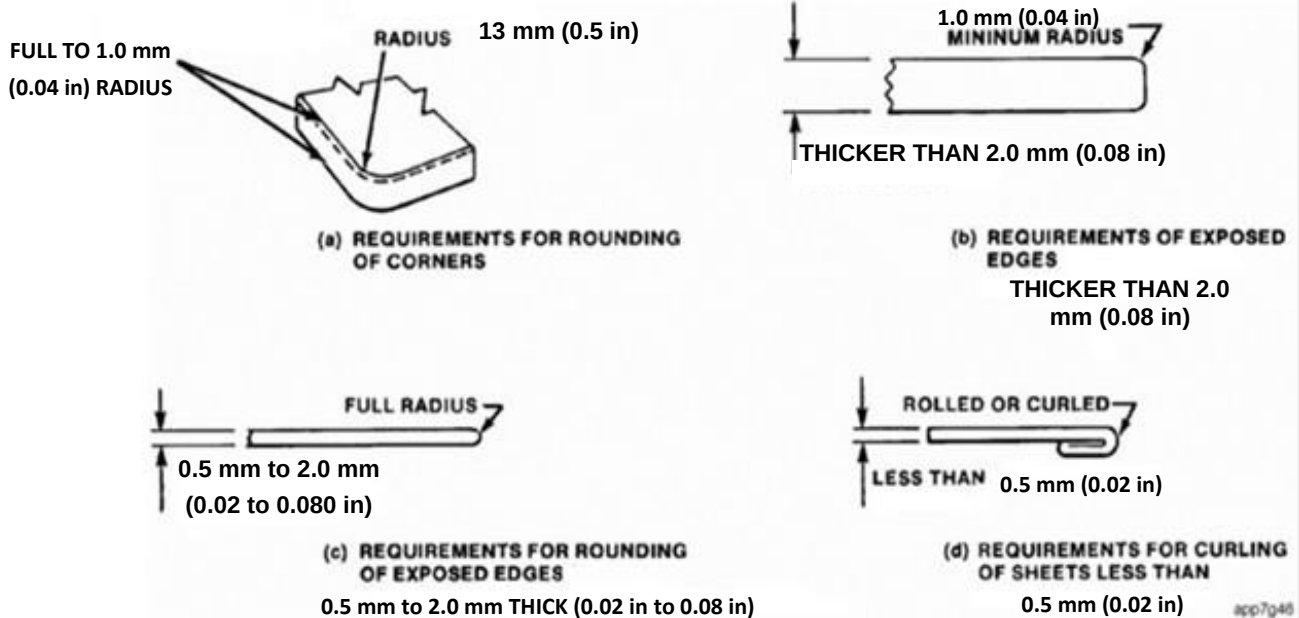


FIGURE 3.9.11-1 EXPOSED CORNER AND EDGES

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 106 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 3.9.11-2 EDGE, CORNER, AND PROTRUSIONS CRITERIA – PROTRUSIONS AND OUTSIDE CORNERS

APPLICATION	CRITERIA/REMARKS
Latching devices	Cover or design all latching devices to preclude gaps or overhangs that can catch fabrics or pressure xEVA suit appendages. All surfaces and edges must be smooth, rounded, and free of burrs.
Lap joints in sheet metal and mismatching of adjacent surfaces	All surfaces must be mated within 0.8 mm (0.03 in) of flat surface at edges or must be butted or recessed. All exposed edges must be smooth and radiused 1.5 mm (0.06 in) minimum, chamfered 45°, or must be covered with an appropriate material to protect EVA gloves.
Safety wire (lock wire) and cotter pins	Ensure that no safety wire is used on external surfaces. Reference requirement [EVASC.0069] for prohibited locking fasteners.
Sheet metal structure, box and cabinet three-plane intersecting corners	Use spherical welded or formed radii unless corners are protected with covers.
Thin Materials	Materials less than 2.0 mm (0.08 in) thick, with exposed edges that are uniformly spaced, not to exceed 13 mm (0.5 in) gaps, flush at the exposed surface plane and shielded from direct EVA interaction, must have edge radii greater than 0.08 mm (0.003 in). Note - this is for radiator-type applications.
Screw heads, bolts, nuts, and nut plates, excess threads and rivets that can be contacted by crewmember	Design all screw heads and bolt heads to face the outside of the structure, if possible. Where nuts, nut plates, and threads are exposed, cover the nuts, nut plates, and threads in a secure manner that does not preclude removal of the fastener. Recessed heads or the use of recessed washers is recommended. Overall height of heads shall be within 3.18 mm (0.125 in) or covered unless more than 7 head diameters apart from center to center. Height of roundhead or oval head screws is not limited. Screw heads or bolt heads more than 6.4 mm (0.25 in) deep must be recessed or be covered with a fairing, except those intended to be EVA crew interfaces.
	Design rivet heads to face out on all areas accessible to crewmembers and to protrude no more than 1.5 mm (0.06 in) unless spaced more than 3.5 head diameters from center to center. In all exposed areas where unset ends of rivets extend more than 3.1 mm (0.12 in), or 13 mm (0.50 in) of unset and diameter if more than 3.1 mm (0.12 in), install a fairing over them. This applies to explosive, blind, or pull rivets, etc. Unset ends of rivets must have edges chamfered 45° or ground off to a minimum radius of 1.5 mm (0.06 in).
	Allow a maximum gap of 0.5 mm (0.02 in) only between one side of a fastener head and its mating surface.
	Prevent or eliminate burrs. Use of Allen heads is preferred. For torque-set, slotted, or Phillips head screws, cover with tape or other protective materials or individually deburr before flight.
	Screws or bolts with exposed threads protruding greater than 3.1 mm (0.12 in) in length, must have protective features that do not prevent installation or removal of the fastener.

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 107 of 280
Title: EHP xEVA System Compatibility Standards	

3.9.12 Levers, Cranks, Hooks, and Controls

[EVASC.0107] The Artemis System shall locate levers, cranks, hooks, and controls where they cannot pinch, snag, cut, abrade, or provide kickback to the EVA crewmember appendages or equipment such as tethers.

Rationale: Levers, cranks, hooks, and controls can generate snag/safety hazards that may pose a threat to EVA crewmembers during translation.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

3.9.13 Exposed Holes or Slots

[EVASC.0109] The Artemis System shall ensure that exposed holes that are round or slotted, within the range of 13 mm (0.5 in) to 36 mm (1.4 in) in diameter are covered to preclude entrapment of an xEVA-gloved finger.

Rationale: Holes with diameter in the range of 13 mm (0.5 in) to 36 mm (1.4 in) could entrap the crewmember's finger. Irregularly shaped holes will be evaluated individually during the safety review process. EVA crewmembers are trained to recognize tether points, so although they are specified to be in the entrapment range, they are exempt from this requirement. This requirement should also be applied to loose equipment and to holes, even if the back side is accessible, because the hazard is not only that the crew will entrap the finger, but that damage may occur to the finger in the effort to remove it from the entrapped condition.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

3.9.14 Pinch Hazard Protection

[EVASC.0110] The Artemis System shall protect the EVA crewmember from pinch hazards.

Rationale: This applies to, but is not limited to, any hardware that pivots, retracts, flexes, has the potential to pinch any appendage, (including legs or other body parts), or has a configuration such that a gap in the range of 13 mm (0.5 in) to 36 mm (1.4 in) (glove pinching), exists between the equipment and adjacent structure (reference requirement [EVASC.0109]).

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 108 of 280
Title: EHP xEVA System Compatibility Standards	

3.9.15 Equipment Clearance for Entrapment Hazard

[EVASC.0111] The Artemis System shall provide clearance for equipment removal and replacement to prevent the creating of a crew entrapment hazard that can be caused by insufficient clearance that results in not meeting any or all of the following requirements:

- 1) Exposed Holes or Slots (entrapment of crew fingers) – reference [EVASC.0109]
- 2) Gloved Operation (entrapment of crew hand) – reference [EVASC.0003]
- 3) Translation Paths (entrapment of crew body) – reference [EVASC.0001] and [EVASC.0171].

Rationale: Other hardware-specific entrapment hazards can also exist that are outside of the three listed above; hazard analyses conducted as part of the safety review process can determine other potential entrapment concerns to be addressed. This requirement applies to hardware in translation paths and worksites. It assumes that hardware (e.g., all latches, fasteners, etc.) are designed to preclude entrapment of the equipment or crewmember. Worksites for equipment removal/replacement tasks must prevent crewmember entrapment hazard.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

3.9.16 Protection from Moving or Rotating Equipment

[EVASC.0112] The Artemis System shall protect EVA crewmembers from impacts from moving, rotating equipment, or inadvertent deployment, separation, or jettison of hardware.

Rationale: Moving or rotating equipment can present a hazard to the EVA crewmember. Equipment that gets separated can present a contact hazard for the EVA crewmember and for the spacecraft. The vehicle needs to include design features to protect EVA crewmembers from hazards due to movable or rotating equipment that could come into contact with an EVA crewmember, snagging or cutting the xEVA suit or umbilical, resulting in catastrophic leak or injury to the EVA crewmember. These design features could include the use of a two-fault tolerant design, use of physical barriers, use of two inhibits to prevent accidental actuation of these types of hardware, or a combination of these design features. Components that contain stored energy (springs, pyro activated devices, pressure vessels, etc.) could constitute a hazard to an EVA crewmember if not properly controlled.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 109 of 280
Title: EHP xEVA System Compatibility Standards	

3.9.17 Emergency Ingress of Pressurized Habitable Environment

[EVASC.0113] The Artemis System shall provide EVA translation path that aids the suited crewmembers in navigating the shortest paths to an airlock, including hardware, aids, and pathway markings or indications for vehicle ingress and egress so that an EVA crewmember can complete an emergency ingress as specified in Table 3.9.17-1 *Emergency Ingress Capability*.

Rationale: xEVA suit design includes 1 hour of emergency life support to protect for most contingencies.

It is assumed for lunar surface operations that tools/samples would be discarded prior to touching the vehicle and there would be no lunar dust mitigation prior to ingress. After ingress and hatch closure, an emergency repress to > 13.3 kPa (1.92 psia) Inspired Partial Pressure of Oxygen (PIO2) (minimum survivable inspired O₂ pressure) would occur such that the crewmember can begin doffing the suit. PIO2 is a function of cabin pressure, oxygen partial pressure and the vapor pressure in psia of water at 37° C (98.6° F) body temperature. The 1 hour-contingency life support duration must account for identifying the failure, physically returning to the vehicle (walking or traversing in a rover) and ingress. Accordingly, an emergency ingress needs to be as quick as possible, since it is only one component of a larger contingency timeline which depends on the type of failure, distance from the vehicle and terrain. A 10 min ingress/repress would force an aggressive walkback with no margin while a 5 min ingress/repress would allow conservatism in walkback speed and/or a small margin. While hardware limitations and crew safety considerations make the problem challenging, alternate solutions to allow additional ingress/hatch/repress time may be considered.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, PR, Gateway, HLS

TABLE 3.9.17-1 EMERGENCY INGRESS CAPABILITY

Artemis System	Time	Conditions
Gateway	30 minutes	Based on suit secondary oxygen design and current operation on ISS but allows for margin with a smaller vehicle stack for the Gateway design. For situations where EVA is on the end of the EVR robotic manipulator at the time of the EVA emergency, the EVR is assumed to bring the EVA crewmember to structure within 7 minutes (of the 30 minutes). This will allow for the crewmember to egress the foot restraint and return to the Airlock.

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 110 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 3.9.17-1 EMERGENCY INGRESS CAPABILITY

Artemis System	Time	Conditions
HLS	10 minutes (threshold) 5 minutes (goal)	Time begins from first contact with the vehicle and ends when the vehicle cabin reaches >13.3 kPa (1.92 psia) PIO2.
PR	10 minutes (threshold) 5 minutes (goal)	Time begins from first contact with the vehicle and ends when the vehicle cabin reaches >13.3 kPa (1.92 psia) PIO2.

3.9.18 Emergency Ingress/Egress of Unpressurized Mobile Asset

[EVASC.0169] The Artemis System shall design EVA crew translation paths, aids, accommodations, and restraints so that an EVA crewmember can complete an emergency ingress/egress within a total time of:

- Five minutes (Threshold)
- Three minutes (Goal)

Duration includes the total summation of time allowable for both ingress and egress. Duration is the time period from first contact with the vehicle to when the vehicle begins driving for ingress, and from vehicle stopping to fully exiting vehicle for egress.

Rationale: Rapid vehicle ingress and egress during a contingency protects for crew being able to drive back to a habitable environment within the time available while on suit emergency life support. It is assumed that tools/samples/ payloads will be discarded or abandoned in place and Manipulator manually stowed or jettisoned with priority on crew transportation only. Ingress and egress times do not include driving time when vehicle is in motion to destination. For example, three minutes to ingress and two minutes to egress sums to a total of five minutes to meet the requirement. This time does not include the time to load an incapacitated crew.

Gravity Field Applicability: Partial Gravity

Allocation: LTV

3.9.19 External Surface Cleanliness Level

[EVASC.0140] The hardware that interfaces with external surfaces of xEVA System hardware shall meet a Visibly Clean-Sensitive (VC-S) level upon delivery as identified in Table 3.9.19-1 *External Surface Cleanliness Level* (or NASA approved cleanliness alternate standard with Cross-Program Chief Engineer approval), or higher.

Rationale: This is intended for any hardware that interfaces with EVA. At the completion of Pre-Delivery Acceptance/Pre-Installation Acceptance (PDA/PIA),

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 111 of 280
Title: EHP xEVA System Compatibility Standards	

hardware is expected to be verified as VC-S or higher and that cleanliness level maintained through the receive-in process at NASA or NASA's Cargo Integrator controlled storage. It is assumed that contamination sources in flight will be controlled by higher-level Program mechanisms. In general, xEVA System hardware delivered for launch as VC-S is not expected to be maintained to VC-S once launched unless specific documentation such as Hazard Controls, Flight Rules, and Operational Control Agreement Database (OCADs) specifically request such precautions.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

TABLE 3.9.19-1 EXTERNAL SURFACE CLEANLINESS LEVEL

Artemis System	Contamination Control Specification
EVA Dev (suit and tools), LTV, PR, payloads	JPR 5322.1, <i>Contamination Control Requirements Manual</i>
Gateway	GSDO-RQMT-1080, <i>Cross-Program Contamination Control Requirements</i> (tailored for Gateway)
HLS	JPR 5322.1, <i>Contamination Control Requirements Manual</i>

3.9.20 Tethering / Restraint

[EVASC.0115] Hardware items shall provide at least one means of positive restraint independent of EVA grasp.

Rationale: Grasping is not considered an acceptable form of positive restraint for a translating EVA crewmember, so tethers and tether points are required on hardware with handholds. This requirement applies for microgravity EVA operations. Surface EVAs may require the use of tethers for hand tools (reference EVABP-32).

Gravity Field Applicability: Microgravity

Allocation: EVA Dev, Gateway, HLS, Payloads

3.9.21 Crew Safety Tether Attachment

[EVASC.0116] Handholds/handrails on microgravity translation paths intended for crewmember restraint shall incorporate a safety tether attachment point into its support structure to withstand the specified EVA crewmember safety tether loads identified in [EVASC.0050] Table 3.5.2-1 Microgravity EVA Crew-Induced Loads, Design Limit Load

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 112 of 280
Title: EHP xEVA System Compatibility Standards	

Type “Crew Safety Tether Attachment”, and only to be used in conjunction with the load-limiting crew safety tethers.

Rationale: The 890 N (200 lbf) Crew Safety Tether load protects for crew in microgravity conditions should they lose physical grip or structural restraint but must be used in conjunction with the load-limiting crew safety tethers (without which, can experience much higher or different tether loading conditions). Crew safety tethers not only have a load limiting rip-away stitching feature, but also have a hook design which preclude a "pry load" on the handrail. This load is encompassed by the Primary Translation path load requirement of 979 N (220 lbf) for General worksite loads outlined in [EVASC.0050].

Gravity Field Applicability: Microgravity

Allocation: EVA Dev, Gateway, HLS, Payloads

3.9.22 Crew Safety Restraint

[EVASC.0161] Attachment point for Crew Safety Restraints, employed in areas of EVA operations where reaction to Crew Slips or Falls is necessary for safety of crewmember, shall be incorporated into support structure that withstands the specified EVA crewmember loads identified in requirement [EVASC.0155], Table 3.5.3-1 Partial Gravity EVA Crew-Induced Loads, under Load/Energy Category “L12, Swinging/ or Falling”.

Rationale: Anticipating that crew may utilize a fall arrest system or restraint system during high-risk operations identified during the safety review process, the restraint attachment point needs to withstand emergency loads. In one potential application, the load in [EVASC.0161] corresponds to the anticipated load transferred into the vehicle structure if a rip-stitch tether is employed, to arrest the fall of a suited crewmember to 1 m (39.4 in). Reference ES6-23-044, Lunar EVA Crew Induced Loads on Restraint Attachment Point from Crew Fall for further information regarding load identified in requirement [EVASC.0155], Table 3.5.3-1 Partial Gravity EVA Crew-Induced Loads, under Load/Energy Category “L12, Swinging/ or Falling”.

Gravity Field Applicability: Partial Gravity

Allocation: EVA Dev, LTV, PR, HLS

3.9.23 Color for Crew Load-Rated Handhold/Handrails, Crew Safety Tethers and Safety Restraint Attachment Points

[EVASC.0117] Handholds/handrail, Crew Safety Tethers and Crew Safety Restraint attachment points intended for use by the crew shall be colored according to Table 3.9.23-1 *Crew Load-Rated Colors*, where the color value has a contrast ratio of 5:1 or greater than the background. **<TBR-EHP-10028-012>**

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 113 of 280
Title: EHP xEVA System Compatibility Standards	

Rationale: The EVA crewmember needs to be able to readily identify handholds/handrails that, when attached to the vehicle structure, are designed to withstand loads to enable crew translation and safety. Designs and materials selected for color must meet NASA-STD-6016, Standard Materials and Processes Requirements for Spacecraft, a Type 2 standard levied in EHP-10012, EHP Systems Requirements Document, and accordingly maintain their unique, colored identification for the intended life of the vehicle. Human-rated handrails/handholds in the primary translation path are defined as those that meet microgravity EVA crew-induced loads (requirement [EVASC.0050], Table 3.5.2-1, Microgravity EVA Crew-Induced Loads, Design Limit Load Type “EVA Handrail Handhold – Primary Translation Path” loads), and sharp edges. For partial gravity handholds/handrails that enable surface mobility (i.e., ascent, descent, ingress, and egress), partial gravity EVA crew-induced loads (requirement [EVASC.0155], Table 3.5.3-1, Partial Gravity EVA Crew-Induced Loads, under Load/Energy Category L11 Handrails Grasp/Pull Load Limit Load%) are enveloped by the microgravity EVA-induced load. Accordingly, all handrails/handholds will be colored gold or yellow. The Crew Safety Restraint is a restraint attachment point for partial gravity that is intended to meet an emergency operational load ([EVASC.0155], Table 3.5.3-1, L11). This corresponds to the load anticipated on the restraint attachment point if a rip-stitch tether is employed to arrest the fall of a suited crewmember to 1m (39.4 in). In order for crew to differentiate between attachment points and identify those that are rated for partial gravity fall arrest, those Crew Safety Restraint attachment points will be colored blue. Loads for the design of the handrails/handholds in the primary translation path, Crew Safety Tether and Crew Safety Restraint are captured in the following requirements: [EVASC.0050], [EVASC.0155], [EVASC.0116], and [EVASC.0161]. Handrails that are not intended for enabling crew attachment or as handholds/handrails attached to vehicle structure will be colored differently as outlined in [EVASC.0033] so as not to misinform crew. The recommended contrast ratio and color coding can be found in the NASA/SP-2010-3407/REV1, NASA Human Integration Design Handbook (HIDH), sections 9.7.2.2.2 and 10.4.7.3.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 114 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 3.9.23-1 CREW LOAD-RATED COLORS <TBR-EHP-10028-012>

Item	Gravity Field Applicability	Color	Notes
Handrail/ Handhold– Primary Translation Path	Microgravity, Partial Gravity	Gold or yellow	Handrail/handholds are defined in [EVASC.0050] for microgravity application. Handholds/Handrails are defined in [EVASC.0155] for partial gravity application under Load/Energy Category L11 Handrails Grasp/Pull Load Limit Load %.
Crew Safety Tether	Microgravity	Gold or yellow	Crew Safety Tethers are defined in [EVASC.0116].
Crew Safety Restraint	Partial Gravity	Blue	Crew Safety Restraints are defined in [EVASC.0161].

3.9.24 Manual Overrides

[EVASC.0118] The Artemis System shall include a manual override that accommodates EVA operation with a hand tool for remotely commanded mechanisms that require EVA as a backup.

Rationale: The intent is to ensure that the hardware design for the remotely commanded mechanism (e.g., robotic work platform) provides access to an EVA override or equivalent drive out/back drive feature. This is like the tasks associated with release of a potentially stuck robotic arm end effector.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: LTV, PR, Gateway, HLS, Payloads

3.9.25 Entanglement, Cable Clamps, Ducts, or Retractors

[EVASC.0137] Cables, conductors, bundles and hoses within an EVA worksite or translation path shall be secured by means of restraints (e.g., clamps, wire ties, etc.) unless they are contained in a cable management system (wiring ducts or cable retractors).

Rationale: Intent is to ensure that the connector worksites and translation paths are free from obstructions.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: LTV, PR, Gateway, HLS, Payloads

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 115 of 280
Title: EHP xEVA System Compatibility Standards	

3.10 ENVIRONMENTS

3.10.1 Magnetic Field Limit

[EVASC.0121] The Artemis System shall not generate static or time varying magnetic fields that exceed values outlined in Table 3.10.1-1 *Magnetic Field Limit* in crew translation corridors, work areas, and incidental contact zones during EVA operations internal and external to the pressurized volume.

Rationale: The intent of this requirement was to prevent magnetic interference with the xEVA suit and xEVA tools. Magnetic field strengths inside the vehicle pressurized volumes are far lower and would pose no threat to any portion of the xEVA suit.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

TABLE 3.10.1-1 MAGNETIC FIELD LIMIT

Artemis System	Requirement	Description	Source Document
xEVA Suit, LTV, PR, Payloads (microgravity transit and surface deployed)	RE 101 A	Radiated Emissions, Time-Varying Magnetic Field	GP 11461, Gateway Requirements for the Control of Electromagnetic Interference (EMI) Characteristics of Subsystems and Equipment reference <TBR-EHP-10012-027> in EHP 10012 Revision A, EHP-C0619 that indicates that EHP will be Baselineing a tailored version and intends to replace GP 11461, Gateway Requirements for the Control of Electromagnetic Interference (EMI) Characteristics of Subsystems and Equipment with new EHP document. (Or NASA approved alternate EMI standard - with Cross-Program Chief Engineer approval)
	RE 101 D	Radiated Emissions, Static Magnetic Field	
Gateway (xEVA System on Gateway)	RE 101 A	Radiated Emissions, Time-Varying Magnetic Field	GP 11461, Gateway Requirements for the Control of Electromagnetic Interference (EMI) Characteristics of Subsystems and Equipment
	RE 101 D	Radiated Emissions, Static Magnetic Field	

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 116 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 3.10.1-1 MAGNETIC FIELD LIMIT

Artemis System	Requirement	Description	Source Document
			(Or NASA approved alternate EMI standard with Cross-Program Chief Engineer approval)
HLS, Payloads (surface ops on HLS)	RE 101 A	Radiated Emissions, Time-Varying Magnetic Field	HLS-STD-007, Human Landing System (HLS) Requirements for the Control of EMI Characteristics of Subsystems and Equipment (Or NASA approved alternate EMI standard with Cross-Program Chief Engineer approval)
	RE 101D	Radiated Emissions, Static Magnetic Field	
	RE 101D	Radiated Emissions, Static Magnetic Field	

3.10.2 Voltage Potential Discharge

[EVASC.0123] The Artemis System shall meet performance requirements during and after exposure to voltage/current transients of **<TBD-EHP-10028-014>** that occur during contact with xEVA operations when the vehicle and xEVA suit or payloads come to equilibrium.

Rationale: Prior to contact, xEVA hardware and Exploration vehicles/payloads/hardware could be at extremely different potential discharge due to interactions with the natural environment. Exploration vehicles/hardware (e.g., Gateway, HLS, LTV, PR, payloads, etc.) should be immune to the resulting voltage and transient currents that occur when the xEVA/Exploration vehicles/hardware potentials come to equilibrium.

Gravity Field Applicability: Microgravity, Partial Gravity

Allocation: EVA Dev, LTV, PR, Gateway, HLS, Payloads

3.10.3 Lunar Surface Dust Tolerance

[EVASC.0156] The Artemis System shall tolerate the liberation of at least 200 grams of lunar surface dust in the cabin environment from the xEVA system after each two-crew lunar surface EVA.

Rationale: The total value per two-crew EVA in this requirement is established to provide a bounding condition for nominal scenarios so that surface assets such as Human Landers/PRs can size Environmental Control/Life Support System (ECLSS) hardware and other mitigation features provided by the vehicle. This requirement only identifies the amount of dust that may be liberated from the xEVA system after an EVA and does not account for deployed utilization payloads

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 117 of 280
Title: EHP xEVA System Compatibility Standards	

returned to the vehicle. This value assumes that EVA crew had adequate time at the end of the EVA, before vehicle ingress, to remove excess dust. The 200 grams value assumes two suits and additional non-suit hardware return to the vehicle, so if more than 2 crewmembers perform a simultaneous EVA, then an average of 100 grams per crewmember can be assumed (reference EX-24-008, Artemis System Lunar Dust Tolerance – Amount of Dust Justification and Calculation). Non-suit hardware could include sample containers, EVA cameras, supplemental lighting, geology tools (if IVA reconfiguration is needed), etc. EVA tools are not nominally returned into the cabin/airlock between EVAs unless needed for battery recharge. Contingency scenarios which lead to the termination or abort of a lunar surface EVA will likely bring in more dust than prescribed in the requirement due to reduced or eliminated time and ability to execute dust mitigation activities.

Gravity Field Applicability: Partial Gravity

Allocation: PR, HLS

3.10.4 Lunar Surface Dust Mitigation

[EVASC.0168] The Artemis System shall limit the amount of dust liberated in the cabin environment to less than 100 grams for two-crew lunar surface EVA.

Rationale: The total value per two-crew EVA in this requirement is established to provide a best-case bounding condition for nominal scenarios so that surface assets such as Human Landers/PR can size ECLSS filters and other mitigation features provided by the vehicle to meet or exceed the vehicle system cleanliness requirements established for lunar dust. This requirement only identifies the amount of dust that may be liberated from the xEVA system after an EVA. Utilization payloads, instruments, or other items returned to the vehicle after an EVA are not part of this 100 g value. This value assumes that EVA crew had adequate time at the end of the EVA, before vehicle ingress, to remove excess dust. 100 grams is based upon an allocation of no more than 50 grams per crew. If more than 2 crewmembers perform a simultaneous EVA, then it is expected to average 50 grams of dust per suit. EVA tools are not nominally returned into the cabin/airlock between EVAs unless needed for battery recharge. It is likely that contingency scenarios will have the potential to bring in more dust than prescribed in the requirement. It is acknowledged that this requirement is for nominal scenarios only; contingency events which lead to the termination or abort of a lunar surface EVA will likely reduce or eliminate the time and ability to execute dust mitigation activities.

Gravity Field Applicability: Partial Gravity

Allocation: EVA Dev

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 118 of 280
Title: EHP xEVA System Compatibility Standards	

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 119 of 280
Title: EHP xEVA System Compatibility Standards	

4.0 VERIFICATION

This section contains the EVA compatibility matrixed requirements that are necessary to show compliance with each “shall” statement in section 3.0 of this document.

Verification methods for “should” statements are not included as they are not mandatory requirements. If a provider chooses to implement a “should” statement, the subsequent verification methods may be negotiated by that Artemis System and EHP. “Should” statements are located in Appendix B.

4.1 RESPONSIBILITIES

Verifications to these requirements will be provided by:

- Artemis Systems, including Gateway Program (modules/external payloads), HLS Program (Integrated Lander and HDL), EVA Development Project Office or xEVA Systems hardware providers, LTV, PR, Payloads, and any other future lunar surface assets that will be utilized for exploration or science that must accommodate suited EVA crewmember.
- EHP, who employs additional verification methods to reduce risk. These are 1) Vehicle Integration Test Office (VITO) demonstration and inspection and 2) Configuration Analysis Modeling and Mass Properties (CAMMP) Model Analysis.

It is the responsibility of each Program or project which is allocated requirements to establish and control the traceability between each of the Requirements defined in this Standard and the development of any associated ICDs.

Verifications will demonstrate that the system is in compliance with the requirements established in the Standard, and that the “as built” status is in compliance with the associated ICD.

In the event stand-alone verification of any requirement is not sufficient and that active participation from the other Partner is necessary to fully verify the compliance with the requirements, interfacing parties are jointly responsible for the successful completion of interface verification. Joint participation activities are defined in the associated Bilateral Integration & Verification Plan (BIVP).

4.2 VERIFICATION APPROACH

This section contains a description of verification methods, roles and responsibilities, and a verification matrix describing the methods recommended for each requirement. For more information on EHP verification and validation, please see EHP-10013, *EHP Verification and Validation Plan*.

Artemis System vehicle providers, including Gateway Program (modules/external payloads), HLS Program (Integrated Lander and HDL), LTV, PR, Payloads, and any other future lunar surface assets that will be utilized for exploration or science that must accommodate suited EVA crewmember, are expected to verify the allocated

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 120 of 280
Title: EHP xEVA System Compatibility Standards	

requirements according to EHP-10013, *EHP Verification and Validation Plan*, processes for interface verifications.

Artemis System verification methods describe those coordinated and performed by Artemis System vehicle program and/or its vendor(s). In some cases, requirement verification is expected to include evidence of multiple methods, which are detailed in Table 4.5-1. Many Artemis-supporting hardware deliverables and mission capabilities are vendor-provided. Accordingly, contract terms often dictate the terms of developing verification methods, arriving at verification agreements, and providing verification compliance evidence. Where interface requirements are described “Hardware provider” in the Verification Matrix (section 4.4), the matrix provides the expected verification methods to demonstrate requirement compliance. Whether verification plans are solely vendor-provided for NASA review and approval, or jointly developed between vendors and NASA project teams, the Verification Matrix provides the expectations for verifying the implementation of the EHP xEVA System and support hardware compatibility. Artemis System verification may be accomplished by inspection, analysis, demonstration and test which are defined in EHP-10013, *EHP Verification and Validation Plan*, Section 6.0. In defining these methods, EHP has applied historical knowledge of accepted ISS verification deliverable. EHP acknowledges that other Artemis Programs and/or their vendors may employ different definitions for inspection, analysis, demonstration, and test. Accordingly, the statements herein describe EHP’s expectations against which Program or vendor evidence will be evaluated. The listed verification methods are considered the threshold, however, alternative methods that meet the intent of the entire verification herein (method, objective, and success criteria) may be adjudicated with EHP’s requirement integrator (defined in EHP-10013, *EHP Verification and Validation Plan*, Section 3.0) and may be found acceptable.

Section 4.5 identifies the joint interface verifications (JIV), defined as a verification that requires multiple programs/entities to sign off on the planning, objectives and risk acceptance associated with the verification and interface requirements closure. JIV events between Artemis System’s vehicles and the xEVA System will be jointly planned the verification event(s) (VE) in a BIVP. Deliverables, dependencies, and other cross-program exchanges noted in the plan should then be incorporated into cross-program Joint-Technical Agreement Memos (JTAMs) and/or Joint Agreement Memorandums (JAMs).

Some additional (VE) will be coordinated and performed by EHP. Those events, described in Section 6.2 of EHP-10013, *EHP Verification and Validation Plan* are intended to reduce risk. In those cases, the Artemis System vehicle may provide access to their vehicle or computer-aided design (CAD) models. Those access agreements or deliverables will also be captured in BIVPs, JTAMs and/or JAMs.

As described previously, these compatibility requirements were built upon the ISS EMU. Accordingly, the parallel development early in the Artemis campaign of both the

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 121 of 280
Title: EHP xEVA System Compatibility Standards	

commercially provided xEVA System and Artemis System vehicles complicates this approach. The ideal case would be that these compatibility standards encompass the known EVAS-provider design. Since xEVA Systems are still in development for early Artemis missions, xEVA Systems hardware providers managed by the EHP EVA Development Project Office will be required to verify that their designs are compliant with the constraints and inherent assumptions contained in this set of compatibility standards and levied on Artemis Systems (i.e., Artemis vehicles).

4.3 VERIFICATION METHODS

The requirements in Section 3.0 are verified by inspection, analysis, demonstration, test, or inspection of records. Additional methods may be employed by EHP as a means to reduce risk. These are VITO demonstration and inspection and CAMMP Model Analysis. All methods for the Program, Project Office, and hardware provider are defined in EHP-10013, *EHP Verification and Validation Plan*, section 6.0. EHP-10013 section 6.2 provides additional Program methods of CAMMP Model Analysis and VITO Crew Equipment Interface Test (CEIT) and Sharp Edge Inspections.

4.4 VERIFICATION MATRIX

Table 4.4-1 *Artemis Systems Verification Matrix* identifies all requirements outlined in this document and identifies the EHP expectation for verification for verification methods.

EHP inspection of records for the submission of requirement compliance evidence is anticipated for every requirement, and not listed in the matrix (e.g., section 4.5).

Verification statements are contained in the following Sections:

- Section 4.5 Artemis System Verifications
- Section 4.6 EHP Verifications

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 122 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.4-1 ARTEMIS SYSTEMS VERIFICATION MATRIX

Requirement Number	Requirement Name	Artemis System Provider Method				EHP Method			JIV	Notes
		I	A	D	T	EVA Model Analysis	VITO CEIT	Sharp Edge Inspection		
[EVASC.0001]	Microgravity Translation Path Hardware Protrusion		X			X				If necessary, a facility demonstration could include Neutral Buoyancy Laboratory (NBL) evaluation and performed to validate translation, work volume, worksites, etc.
[EVASC.0171]	Partial Gravity Translation Path Hardware Protrusion		X							If necessary, a facility demonstration could include NBL evaluation and performed to validate translation, work volume, worksites, etc.
[EVASC.0002]	Tool Interface Clearance	X				X	X			
[EVASC.0007]	Bolt Clearance	X					X			
[EVASC.0003]	Gloved-Hand Clearance		X			X	X			
[EVASC.0004]	Handrail/ Handhold Mounting Clearance	X				X	X			
[EVASC.0005]	Handrail/ Handhold Side Clearance	X				X	X			
[EVASC.0006]	APFR EVA Installation/Removal Clearance	X				X	X			
[EVASC.0152]	Contiguous EVA Translation Path		X			X				If necessary, a facility demonstration could include NBL evaluation and performed to validate translation, work volume, worksites, etc.
[EVASC.0008]	Translation Path		X			X				If necessary, a facility demonstration could include NBL evaluation and

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 123 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.4-1 ARTEMIS SYSTEMS VERIFICATION MATRIX

Requirement Number	Requirement Name	Artemis System Provider Method				EHP Method			JIV	Notes
		I	A	D	T	EVA Model Analysis	VITO CEIT	Sharp Edge Inspection		
										performed to validate translation, work volume, worksites, etc.
[EVASC.0154]	Incapacitated Crewmember Interfaces		X	X					X	<u>Joint Integrated Verification</u> - Demonstration is critical, and NASA teams to ensure that providers work together to produce a high-fidelity demonstration with demonstration report as deliverable.
[EVASC.0010]	EVA Contact Hazards Preclusion Beyond Translation Corridor		X			X				
[EVASC.0159]	Partial Gravity EVA Contact Hazards Preclusion		X			X				
[EVASC.0012]	Microgravity EVA Equipment Handling Features	X								
[EVASC.0160]	Partial Gravity EVA Handling Capability <TBR-EHP-10028-003>		X							
[EVASC.0162]	Partial Gravity Step Design Parameter	X								Can consider verification via demonstration in conjunction with other VEs, such as [EVASC.0153]
[EVASC.0135]	Pressurized xEVA Suit Transfer through Airlock Hatch	X		X						A facility demonstration could include NBL evaluation to provide acceptance rationale for exceptions for translation, work volume, worksites, etc.

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 124 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.4-1 ARTEMIS SYSTEMS VERIFICATION MATRIX

Requirement Number	Requirement Name	Artemis System Provider Method				EHP Method			JIV	Notes
		I	A	D	T	EVA Model Analysis	VITO CEIT	Sharp Edge Inspection		
										Demonstration report provided as deliverable.
[EVASC.0153]	EVA Lunar Hatch Dimensions	X		X						If necessary, a facility demonstration could include NBL evaluation and performed to validate translation, work volume, worksites, etc. Demonstration report provided as deliverable.
[EVASC.0014]	Mounting Alignment Devices	X								
[EVASC.0015]	Handling Alignment Marks	X								
[EVASC.0016]	Locking Device				X					
[EVASC.0124]	Over Center Device Force				X					
[EVASC.0017]	Soft Dock or Soft Capture Mechanisms	X								
[EVASC.0125]	Soft Capture	X								
[EVASC.0126]	Soft Dock Force				X					
[EVASC.0024]	Unique Labeling	X					X			
[EVASC.0025]	Hand Tool Operations			X			X			
[EVASC.0135a]	Hand Ratcheting		X				X			
[EVASC.0026]	Throw Angle		X							
[EVASC.0127]	Ratchet Mechanism	X								
[EVASC.0128]	Back Drive Torque		X		X					
[EVASC.0129]	Running Torque		X		X					
[EVASC.0027]	Pressurized Glove Compatibility			X			X			EHP would expect a test or demo as part of the Verification Closure Notice (VCN) (expect that hardware provider

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 125 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.4-1 ARTEMIS SYSTEMS VERIFICATION MATRIX

Requirement Number	Requirement Name	Artemis System Provider Method				EHP Method			JIV	Notes
		I	A	D	T	EVA Model Analysis	VITO CEIT	Sharp Edge Inspection		
										would submit an analysis supported by test or demonstration).
[EVASC.0087]	Inadvertent Actuation of EVA -Actuated Controls	X								
[EVASC.0029]	Handrail/Handhold Cross-Section	X								
[EVASC.0030]	Handrail/Handhold/Handle Length	X								
[EVASC.0031]	Handrail/Handhold Accessibility		X			X	X			Analyze via EHP Model analysis, with expectation that programs/vendors will provide the appropriate fidelity of CAD models to enable a good assessment of the requirement statement. Once this work is complete, team should assess whether it is necessary to raise a flag to Program Management that the programs aren't positioned to recover from a failed EVA Model Analysis team verification with a facility demonstration due to procurement plans and schedule misalignments.
[EVASC.0032]	Continuous Handrail/Handhold Spacing		X			X				Analyze via EHP Model analysis, with expectation that programs/vendors will provide the appropriate fidelity of CAD models to enable a good assessment of the requirement statement. Once this work is complete, team should assess whether it is necessary to raise

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 126 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.4-1 ARTEMIS SYSTEMS VERIFICATION MATRIX

Requirement Number	Requirement Name	Artemis System Provider Method				EHP Method			JIV	Notes
		I	A	D	T	EVA Model Analysis	VITO CEIT	Sharp Edge Inspection		
										a flag to Program Management that the programs aren't positioned to recover from a failed EVA Model Analysis team verification with a facility demonstration due to procurement plans and schedule misalignments.
[EVASC.0144]	EVA Egress Handrail Spacing	X				X				
[EVASC.0035]	Work Envelope		X			X				A facility demonstration could include NBL evaluation to provide acceptance rationale for exceptions for translation, work volume, worksites, etc.
[EVASC.0170]	Working Volume Operation of Vehicle		X			X				
[EVASC.0157]	EVA Egress/Ingress Methods		X	X		X			X	<u>Joint Integrated Verification</u> - Demonstration is critical, and NASA teams to ensure that providers work together to produce a high-fidelity demonstration.
[EVASC.0036]	Crew Aids for Stability		X			X				
[EVASC.0037]	EVA Worksites Using Robotic Work Platform		X			X				
[EVASC.0038]	Lubricant Use	X								
[EVASC.0040]	Worksite Mobility		X			X				Suit model approach will need to account for newer EVAS designs or otherwise be very conservative.

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 127 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.4-1 ARTEMIS SYSTEMS VERIFICATION MATRIX

Requirement Number	Requirement Name	Artemis System Provider Method				EHP Method			JIV	Notes
		I	A	D	T	EVA Model Analysis	VITO CEIT	Sharp Edge Inspection		
[EVASC.0041]	Worksite Field of View		X			X				Suit model approach will need to account for newer EVAS designs or otherwise be very conservative.
[EVASC.0042]	Working Volume		X			X				Suit model approach will need to account for newer EVAS designs or otherwise be very conservative.
[EVASC.0148]	Dedicated EVA Worksite		X			X				Suit model approach will need to account for newer EVAS designs or otherwise be very conservative.
[EVASC.0149]	Robotic Assisted EVA Worksites		X			X				Suit model approach will need to account for newer EVAS designs or otherwise be very conservative.
[EVASC.0150]	Free-Float Worksites		X			X				Suit model approach will need to account for newer EVAS designs or otherwise be very conservative.
[EVASC.0045]	Installation of Passive WIF Sockets	X				X				
[EVASC.0046]	Foot Restraint Ingress/Egress Aids	X				X				
[EVASC.0047]	Visual Contrast at Worksite		X							Testing or demonstration could be utilized to substantiate the analysis
[EVASC.0033]	Color for EVA Handling and EVA Equipment Tether Points	X								
[EVASC.0048]	EVA Equipment Tether Point	X			X					
[EVASC.0134]	Soft Goods Tether Point Loads	X			X					

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 128 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.4-1 ARTEMIS SYSTEMS VERIFICATION MATRIX

Requirement Number	Requirement Name	Artemis System Provider Method				EHP Method			JIV	Notes
		I	A	D	T	EVA Model Analysis	VITO CEIT	Sharp Edge Inspection		
[EVASC.0050]	Microgravity EVA Crew-Induced Loads		X							Analysis supported by test
[EVASC.0155]	Partial Gravity EVA Crew-Induced Loads		X							Analysis supported by test
[EVASC.0018]	EVA Actuated Loads		X							Analysis supported by test
[EVASC.0164]	EVA Mechanical Feedback		X							
[EVASC.0143]	WIF Structural Loading		X							
[EVASC.0053]	One-Handed EVA Operable Connectors			X		X				
[EVASC.0054]	Connector Accessibility		X			X				
[EVASC.0138]	Gloved-Hand Connector Accessibility		X			X	X			
[EVASC.0055]	Incorrect Connector Mating Prevention	X								
[EVASC.0056]	Connector Coding	X								
[EVASC.0057]	Connector Protective Caps	X								
[EVASC.0058]	Connector Rotational Travel	X	X							
[EVASC.0059]	Connector Locking Feature	X								
[EVASC.0060]	Visible Connector Alignment Markings	X					X			
[EVASC.0061]	Connector Locking Ring			X						
[EVASC.0062]	Connector Mating Status			X						

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 129 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.4-1 ARTEMIS SYSTEMS VERIFICATION MATRIX

Requirement Number	Requirement Name	Artemis System Provider Method				EHP Method			JIV	Notes
		I	A	D	T	EVA Model Analysis	VITO CEIT	Sharp Edge Inspection		
[EVASC.0052]	Mating/Demating of Powered Connectors		X							
[EVASC.0065]	Connectors – Electrical Bonding				X					
[EVASC.0066]	Scoop-Proof Electrical Connector Design	X								
[EVASC.0063]	Connector Grip Point			X						
[EVASC.0131]	Arc Containment		X							Analysis should include EEE expert to verify. For the arc containment evaluation method, demonstration could also be considered since the operational conditions can be applied to demonstrate that there are no arc issues and or any arc produced will be contained.
[EVASC.0132]	Uncontrolled Motion of Flex Hoses	X		X						
[EVASC.0133]	Venting Feature				X					
[EVASC.0145]	Indication of Pressure Flow – Nonbrazed or Nonwelded	X								
[EVASC.0068]	Captive Fastener Design	X					X			
[EVASC.0069]	Prohibited Locking Methods for Fasteners	X								
[EVASC.0067]	One-Handed Fastener Operations			X						

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 130 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.4-1 ARTEMIS SYSTEMS VERIFICATION MATRIX

Requirement Number	Requirement Name	Artemis System Provider Method				EHP Method			JIV	Notes
		I	A	D	T	EVA Model Analysis	VITO CEIT	Sharp Edge Inspection		
[EVASC.0072]	Scalloped Knobs	X								
[EVASC.0073]	Knobs Head Diameter and Height	X								
[EVASC.0074]	EVA Hand-Actuated Fasteners Contingency Operation	X					X			
[EVASC.0075]	EVA PIP Pin Grasping Interface			X			X			
[EVASC.0076]	PIP Pin Restraint Tether	X								
[EVASC.0078]	Driver-Type Push Force Tools				X					
[EVASC.0079]	EVA Fastener Head	X					X			
[EVASC.0084]	Fastener Deep Socket	X								
[EVASC.0151]	Fastener Torque Data									Considered to be a data deliverable (consists of test and analysis)
[EVASC.0080]	Power and Hand Tool Operability			X						
[EVASC.0141]	Fastener Gauge/ Measurement	X					X			
[EVASC.0083]	Fastener Alignment Features	X								
[EVASC.0085]	Battery Replacement		X			X				
[EVASC.0086]	Battery Voltage Monitoring		X							
[EVASC.0028]	Auxiliary Controls		X			X				
[EVASC.0088]	EVA Tools		X							
[EVASC.0089]	Cable Routing/ Restraint	X								

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 131 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.4-1 ARTEMIS SYSTEMS VERIFICATION MATRIX

Requirement Number	Requirement Name	Artemis System Provider Method				EHP Method			JIV	Notes
		I	A	D	T	EVA Model Analysis	VITO CEIT	Sharp Edge Inspection		
[EVASC.0090]	Cable Routing Spacing	X								
[EVASC.0091]	Paint		X							
[EVASC.0092]	Hazardous Material Control		X							
[EVASC.0093]	Venting or Exposure Compatibility		X							
[EVASC.0094]	Deliberate Hazardous Venting		X							
[EVASC.0097]	Electrical and Safety Design		X							
[EVASC.0098]	Shock Protection		X							
[EVASC.0099]	Operation During Pressure Change				X					
[EVASC.0100]	Component Hazardous Energy Provision		X							Test may be necessary based on hazard analysis
[EVASC.0101]	Detachable Parts	X	X							
[EVASC.0103]	Touch Temperatures		X							
[EVASC.0011]	xEVA Suit Glove Palm External Touch-Temperature Compliance for Small, Handheld Hardware		X							
[EVASC.0104]	Incidental and Unlimited Contact Heat Transfer Rates		X							
[EVASC.0105]	Sharp Edge and Protrusions Protection	X						X		

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 132 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.4-1 ARTEMIS SYSTEMS VERIFICATION MATRIX

Requirement Number	Requirement Name	Artemis System Provider Method				EHP Method			JIV	Notes
		I	A	D	T	EVA Model Analysis	VITO CEIT	Sharp Edge Inspection		
[EVASC.0107]	Levers, Cranks, Hooks, and Controls	X	X					X		
[EVASC.0109]	Exposed Holes or Slots	X								
[EVASC.0110]	Pinch Hazard Protection	X	X							Test may be necessary based on hazard analysis
[EVASC.0111]	Equipment Clearance for Entrapment Hazard	X	X							
[EVASC.0112]	Protection from Moving or Rotating Equipment	X	X							Test may be necessary based on hazard analysis
[EVASC.0113]	Emergency Ingress of Pressurized Habitable Environment		X	X					X	<u>Joint Integrated Verification</u> - Demonstration is critical, and NASA teams to ensure that providers work together to produce a high-fidelity demonstration.
[EVASC.0169]	Emergency Ingress/Egress of Unpressurized Mobile Assets		X	X					X	<u>Joint Integrated Verification</u> - Demonstration is critical, and NASA teams to ensure that providers work together to produce a high-fidelity demonstration
[EVASC.0140]	External Surface Cleanliness Level	X								
[EVASC.0115]	Tethering / Restraint	X								
[EVASC.0116]	Crew Safety Tether Attachment	X	X							
[EVASC.0161]	Crew Safety Restraint	X	X							
[EVASC.0117]	Color For Crew Load-Rated Handhold/Handrails,	X								

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 133 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.4-1 ARTEMIS SYSTEMS VERIFICATION MATRIX

Requirement Number	Requirement Name	Artemis System Provider Method				EHP Method			JIV	Notes
		I	A	D	T	EVA Model Analysis	VITO CEIT	Sharp Edge Inspection		
	Crew Safety Tethers And Safety Restraint Attachment Points									
[EVASC.0118]	Manual Overrides	X								
[EVASC.0137]	Entanglement, Cable Clamps, Ducts, or Retractors	X								
[EVASC.0121]	Magnetic Field Limit				X					
[EVASC.0123]	Voltage Potential Discharge		X							
[EVASC.0156]	Lunar Surface Dust Tolerance		X		X					
[EVASC.0168]	Lunar Surface Dust Mitigation		X		X					

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 134 of 280
Title: EHP xEVA System Compatibility Standards	

4.5 ARTEMIS SYSTEM VERIFICATIONS

The following table outlines the verification statements for the Artemis System.

TABLE 4.5-1 ARTEMIS SYSTEM VERIFICATIONS

Requirement ID	Title	Method	Verification Objective	Success Criteria	Notes
[EVASC.0001]	Microgravity Translation Path Hardware Protrusion	Analysis	Verification analysis shall consist of the Artemis System's analytical CAD models that utilize Artemis System hardware design drawings to verify that the translation path consists of an unobstructed minimum of 1.1 m (43 in) diameter path with only the translation aids included within the designated path that meets Figures 3.1.1-1 and 3.1.1-2 for microgravity gravity.	Verification will be considered successful when the Artemis System's CAD analysis that utilize hardware design drawings confirms that all translation paths maintain an unobstructed minimum 1.1 m (43-in) diameter path with only translation aids included and comply with Figures 3.1.1-1 and 3.1.1-2 for microgravity).	If necessary, a facility demonstration could include NBL evaluation and performed to validate translation, work volume, worksites, etc.
[EVASC.0171]	Partial Gravity Translation Path Hardware Protrusion	Analysis	Verification analysis shall consist of the Artemis System's analytical CAD models that utilize Artemis System hardware design drawings to verify that the translation path consists of an unobstructed, other than translation aids, minimum of 1.1 m (43 in) diameter clearance centered on the suited crewmember as the crewmember traverses the designated path that meets Figures 3.1.2-1 and 3.1.2-2 for partial gravity.	Verification will be considered successful when the Artemis System's CAD analysis that utilize hardware design drawings confirms that all translation paths maintain an unobstructed, other than translation aids, minimum of 1.1 m (43 in) diameter clearance centered on the suited crewmember as the crewmember traverses the designated path that meets Figures 3.1.2-1 and 3.1.2-2 for partial gravity.	If necessary, a facility demonstration could include evaluation performed to validate translation, work volume, worksites, etc.
[EVASC.0002]	Tool Interface Clearance	Inspection	Verification shall consist of an inspection of Artemis System hardware design drawings to ensure compliance with the	Verification will be considered successful when inspection of the hardware design drawings shows tool interface clearance is	

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 135 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.5-1 ARTEMIS SYSTEM VERIFICATIONS

Requirement ID	Title	Method	Verification Objective	Success Criteria	Notes
			specified tool interference clearance as defined in Figure 3.1.3-1.	maintained as defined in Figure 3.1.3-1.	
[EVASC.0007]	Bolt Clearance	Inspection	Verification shall consist of an inspection of Artemis System hardware design drawings to ensure specified clearance around the outer bolt edge is provided as defined in Figure 3.1.3-1.	Verification will be considered successful when inspection of the hardware design drawings shows clearance around the bolt head outer edge to ensure tool access as defined in Figure 3.1.3-1.	
[EVASC.0003]	Gloved-Hand Clearance	Analysis	Verification shall consist of the Artemis System's analytical CAD models that utilize the Artemis System hardware design drawings to verify the design maintains clearances for a glove hand as specified in Figure 3.1.5-1.	The verification will be considered successful when the CAD analysis that utilizes hardware design drawings confirms minimum glove clearance outlined in Figure 3.1.5-1 is provided for EVA hand actuation operations on Artemis System hardware within the translation corridors, work areas, and incidental contact zones.	
[EVASC.0004]	Handrail / Handhold Mounting Clearance	Inspection	Verification shall consist of an inspection of Artemis System hardware design drawings to ensure that the minimum clearance distance between the lower surface of the handrail/handhold and the mounting surface is provided as defined in Figure 3.1.6-1.	Verification will be considered successful when inspection shows that the minimum clearance distance between the lower surface of the handrail/handhold and the mounting surface of 5.72 cm (2.25 in) meets Figure 3.1.6-1.	
[EVASC.0005]	Handrail/ Handhold Side Clearance	Inspection	Verification shall consist of an inspection of Artemis System hardware design drawings to ensure that the minimum clearance around all edges of the handrail/handholds has been	Verification will be considered successful when inspection of hardware design drawings confirms that a minimum 10 cm (4 in) clearance around all edges of	

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 136 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.5-1 ARTEMIS SYSTEM VERIFICATIONS

Requirement ID	Title	Method	Verification Objective	Success Criteria	Notes
			provided as defined in Figure 3.1.7-1.	the handrail/handholds is provided as defined in the Figure 3.1.7-1.	
[EVASC.0006]	APFR EVA Installation/ Removal Clearance	Inspection	Verification shall consist of an inspection of Artemis System hardware design drawings to ensure that clearance is maintained around the Passive WIF to support EVA installation/removal of the APFR as defined in Figure 3.1.8-1.	Verification will be considered successful when inspection of hardware design drawings shows that maintains clearance the envelope around the Passive WIF to support EVA installation/removal of the APFR as defined in Figure 3.1.8-1.	
[EVASC.0152]	Contiguous EVA Translation Path	Analysis	Verification analysis shall consist of the Artemis System's analytical CAD models that utilize the Artemis System hardware design drawings to verify that provides EVA compatible accommodations supporting contiguous translation to adjacent berthed/docked vehicles.	Verification will be considered successful when the Artemis System's CAD analysis that utilize hardware design drawings confirms provides EVA compatible accommodations that support contiguous translation to adjacent berthed/docked vehicles.	If necessary, a facility demonstration could include NBL evaluation and performed to validate translation, work volume, worksites, etc.
[EVASC.0008]	Translation Path	Analysis	Verification analysis shall consist of the Artemis System's analytical CAD models that utilize the Artemis System's hardware design drawings to verify provides a translation path to all EVA worksites, with only the allowable interferences (i.e., handrails/handholds in microgravity) and otherwise free from contact hazards.	Verification will be considered successful when CAD analysis that utilize hardware design drawings confirms that Artemis System provides a translation path to all EVA worksites, with only the allowable interferences (i.e., handrails/handholds in microgravity) and otherwise free from contact hazards.	If necessary, a facility demonstration could include NBL evaluation and performed to validate translation, work volume, worksites, etc.
[EVASC.0154]	Incapacitated Crewmember Interfaces	Analysis Demonstration	<u>Analysis:</u> Verification shall consist of an analysis performed to ensure that Artemis System provides hardware	<u>Analysis:</u> Verification will be considered successful when the analysis shows that Artemis	<u>Joint Integrated Verification</u> - Demonstration is

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 137 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.5-1 ARTEMIS SYSTEM VERIFICATIONS

Requirement ID	Title	Method	Verification Objective	Success Criteria	Notes
			that aids a suited crewmember in the rescue of an incapacitated suited crewmember from the surface or microgravity location and return for ingress the vehicle through the EVA hatch, culminating in positioning the suited incapacitated crewmember in an xEVA System for extraction which may require a 1st percentile female to rescue a 99th percentile male. Analysis needs to take into consideration the emergency ingress requirement [EVASC.0113] provides EVA translation path aids for the suited crewmembers to navigate the shortest paths to an airlock, including hardware, aids, and pathway markings or indications for vehicle ingress and egress so that an EVA crewmember can complete an emergency ingress as specified in Table 3.9.17-1 Emergency Ingress Capability. <u>Microgravity Demonstration:</u> Verification demonstration shall include FOD and safety participation and consist of using representative Artemis System hardware to verify that a suited crewmember can support rescue of an incapacitated crewmember by	System provides hardware that aids a suited crewmember in the rescue of an incapacitated suited crewmember from the surface or microgravity location and return for ingress the vehicle through the EVA hatch, culminating in positioning the suited incapacitated crewmember in an xEVA System for extraction which may require a 1st percentile female to rescue a 99th percentile male. <u>Microgravity Demonstration:</u> Verification will be considered successful when demonstration shows that the suited crewmember can ingress an incapacitated crewmember into the vehicle and secure the suited incapacitated crewmember for extraction once inside the vehicle. <TBR-EHP-10028-018> <u>Partial Gravity Demonstration:</u> Verification will be considered successful when demonstration that included FOD and safety participation shows that the Artemis System confirms provision of hardware to aid a 1st-99th% suited crewmember rescue an	critical, and NASA teams to ensure that providers work together to produce a high-fidelity demonstration.

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 138 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.5-1 ARTEMIS SYSTEM VERIFICATIONS

Requirement ID	Title	Method	Verification Objective	Success Criteria	Notes
			ingressing the vehicle through the EVA hatch, culminating in positioning the suited incapacitated crewmember in an xEVA System for extraction from the suit. Demonstration results will inform the verification of the emergency ingress requirement [EVASC.0113] by providing information on ingress time. <TBR-EHP-10028-018> <u>Partial Gravity Demonstration:</u> Verification demonstration shall include FOD and safety participation and consist of using flight-like Artemis System hardware to verify that a 1st-99th% suited crewmember can rescue of an incapacitated crewmember from the surface or microgravity location, returns for ingress into the vehicle, and suited incapacitated crewmember is secured for extraction once inside the vehicle, which may require a 1st percentile female to rescue a 99th percentile male.	incapacitated crewmember from the surface, ingress into the vehicle, and suited incapacitated crewmember is secured for extraction once inside the vehicle, which may require a 1st percentile female to rescue a 99th percentile male.	
[EVASC.0010]	EVA Contact Hazards Preclusion Beyond Translation Corridor	Analysis	Verification analysis shall consist of analytical CAD models that utilize the Artemis System's hardware design drawings to confirm that precludes EVA contact hazards within 0.61 m (24 in) beyond the	Verification will be considered successful when the CAD analysis that utilize hardware design drawings confirms that the translation corridors have no contact hazards within 0.61 m (24	

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 139 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.5-1 ARTEMIS SYSTEM VERIFICATIONS

Requirement ID	Title	Method	Verification Objective	Success Criteria	Notes
			translation corridor dimension as defined in Figure 3.2.4-1.	in) beyond the translation corridor dimension as outlined in Figure 3.2.4-1 EVA Contact Hazard Preclusions Envelope.	
[EVASC.0159]	Partial Gravity EVA Contact Hazards Preclusion	Analysis	Verification analysis shall consist of analytical CAD models that utilize the Artemis System's hardware design drawings to verify that precludes EVA contact hazards within specified constraints outlined in Figure 3.2.5-1 Side View Partial Gravity EVA Contact Hazards Height.	Verification will be considered successful when the CAD analysis that utilize the hardware design drawings confirms that there is no EVA contact hazards within the specified constraints outlined in Figure 3.2.5-1 Side View Partial Gravity EVA Contact Hazards Height.	
[EVASC.0012]	Microgravity EVA Equipment Handling Features	Inspection	Verification shall consist of an inspection of Artemis System hardware design drawings to ensure that items being transferred from one location to another include EVA handling features that are commensurate with the size of the hardware and the expected EVA operations as defined in Table 3.3.1-1.	Verification will be considered successful when the inspection of hardware design drawings shows Artemis System hardware items to be transferred from one location to another include EVA handling features that are commensurate with the size of the hardware and the expected EVA operations as defined in Table 3.3.1-1 Constraints for EVA Transport of Objects.	
[EVASC.0160]	Partial Gravity EVA Handling Capability <TBR-EHP-10028-003>	Analysis	Verification analysis shall consist of analytical CAD models that utilize the Artemis Systems' hardware design drawings to verify that limits properties of items to be handled by suited crewmembers as specified in Tables 3.3.2-1 through 3.3.2-3	Verification will be considered successful when the CAD analysis that utilize hardware design drawings confirms that the Artemis System limits properties of items to be handled by suited crewmembers to the values	

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 140 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.5-1 ARTEMIS SYSTEM VERIFICATIONS

Requirement ID	Title	Method	Verification Objective	Success Criteria	Notes
				outlined in Table 3.3.2-1 through 3.3.2-3.	
[EVASC.0162]	Partial Gravity Design Parameter	Inspection	Verification shall consist of an inspection of Artemis System hardware design drawings to ensure that the height of a crewmember's step is limited to the specified values defined in Table 3.3.3-1.	Verification will be considered successful when the inspection of hardware design drawings shows Artemis System hardware limits the height of a crewmember's step to the values outlined in Table 3.3.3.-1 Partial Gravity Step Design Parameter.	Can consider verification via demonstration in conjunction with other VEs, such as [EVASC.0153]
[EVASC.0135]	Pressurized xEVA Transfer through Airlock Hatch	Inspection Demonstration	<u>Inspection:</u> Verification shall consist of an inspection of Artemis System hardware design drawings to ensure that hatch design provides a minimum hatch diameter of 1000 mm (39.37 in) diameter that allows for EVA suited pressurized crewmember transfer between pressurized and unpressurized environments. <u>Demonstration:</u> Verification demonstration shall consist of evaluation to show that EVA hatch design provides a hatch diameter minimum of 1000 mm (39.37 in) that allows for EVA suited pressurized crewmember transfer between pressurized and unpressurized environments .	<u>Inspection:</u> Verification will be considered successful when inspection of hardware design drawings shows hatch design provides a minimum hatch diameter of 1000 mm (39.37 in)that allows for EVA suited pressurized crewmember transfer between pressurized and unpressurized environments . <u>Demonstration:</u> Verification will be considered successful when demonstration shows hatch design provides a minimum hatch diameter of 1000 mm (39.37 in) that allows for EVA suited pressurized crewmember transfer between pressurized and unpressurized environments.	A facility demonstration could include NBL evaluation to provide acceptance rationale for exceptions for translation, work volume, worksites, etc. Demonstration report provided as deliverable.
[EVASC.0153]	EVA Lunar Hatch Dimensions	Inspection Demonstration	<u>Inspection:</u> Verification shall consist of an inspection of Artemis System hardware design drawings to	<u>Inspection:</u> Verification will be considered successful when inspection of hardware design	If necessary, a facility demonstration could

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 141 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.5-1 ARTEMIS SYSTEM VERIFICATIONS

Requirement ID	Title	Method	Verification Objective	Success Criteria	Notes
			ensure that EVA Lunar hatch design provides a minimum opening of 1.02 m (horizontal dimension) X 1.52 m (vertical dimension) (40 in X 60 in) for EVA egress/ingress. <u>Demonstration:</u> Verification shall consist of a demonstration to show that the Artemis System provides a minimum EVA Lunar hatch opening 1.02 m (horizontal dimension) X 1.52 m (vertical dimension) (40 in X 60 in) for EVA egress/ingress.	drawings shows EVA Lunar hatch design provides a minimum EVA Lunar hatch opening of 1.02 m (horizontal dimension) X 1.52 m (vertical dimension) (40 in X 60 in) for EVA egress/ingress. <u>Demonstration:</u> Verification will be considered successful when the demonstration confirms EVA Lunar hatch design provides a minimum EVA Lunar hatch opening of 1.02 m (horizontal dimension) X 1.52 m (vertical dimension) (40 in X 60 in) for EVA egress/ingress.	include NBL evaluation and performed to validate translation, work volume, worksites, etc. Demonstration report provided as deliverable.
[EVASC.0014]	Mounting Alignment Devices	Inspection	Verification shall consist of an inspection of Artemis System hardware design drawings to ensure that alignment devices are incorporated for any hardware that requires mounting by an EVA crewmember.	Verification will be considered successful when inspection of hardware design drawings shows that alignment devices are incorporated in the design for hardware that requires mounting by EVA crewmembers to preclude incorrect installations. Prior to inspection, Artemis System needs to identify the hardware that requires mounting by an EVA crewmember.	
[EVASC.0015]	Handling Alignment Marks	Inspection	Verification shall consist of an inspection of Artemis System hardware design drawings to ensure that alignment marks are	Verification will be considered successful when inspection of hardware design drawings shows that alignment markings are	

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 142 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.5-1 ARTEMIS SYSTEM VERIFICATIONS

Requirement ID	Title	Method	Verification Objective	Success Criteria	Notes
			provided for hardware requiring alignment or actuation.	incorporated in the design for hardware that requires alignment or actuation by EVA crewmembers. Prior to inspection, Artemis System needs to identify the hardware that requires mounting by an EVA crewmember.	
[EVASC.0016]	Locking Device	Test	Verification test shall utilize Artemis System flight-like hardware and consist of driving the device through the lock and unlock positions and measuring the percentage of the total travel distance required to change state.	Verification will be considered successful when test quantifies that mechanism locking devices (such as latches and latch locks) travel at least 30% of their full travel before changing state (locked to unlocked or vice versa). Prior to test, Artemis System needs to identify the hardware that requires locking device operated by EVA crewmember.	
[EVASC.0124]	Over Center Device Force	Test	Verification test shall utilize Artemis-System flight-like hardware and consist of driving the over-center device through the lock and unlock positions measuring the peak force to ensure does not exceed 44 N (10lbf).	Verification shall be considered successful when test confirms hardware with over center devices have a minimum peak force of twice the force required to initiate travel, that does not exceed 44 N (10 lbf). Thermal testing shows that the measured force required for hardware with over center devices require less than 44 N (10 lbf).	

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 143 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.5-1 ARTEMIS SYSTEM VERIFICATIONS

Requirement ID	Title	Method	Verification Objective	Success Criteria	Notes
				Prior to test, Artemis system needs to identify the hardware that requires over-center device operated by EVA crewmember.	
[EVASC.0017]	Soft Dock or Soft Capture Mechanisms	Inspection	Verification shall consist of an inspection of Artemis System hardware design drawings to ensure that a soft dock or soft capture is provided for any hardware that EVA crewmembers need to manipulate.	Verification will be considered successful when inspection of hardware design drawings shows a soft dock or soft capture feature has been provided for all hardware that cannot be held in place while being permanently installed. Prior to inspection, Artemis system needs to identify the hardware that requires soft dock or soft capture to be operated by EVA crewmember.	
[EVASC.0125]	Soft Capture	Inspection	Verification shall consist of an inspection of Artemis System hardware design drawings to ensure that a soft capture mechanism is provided that meets criteria.	Verification will be considered successful when inspection of the hardware design drawings shows a soft capture mechanism has been provided in each of the following cases for hardware that is not held in place while being permanently installed.1) all hardware 272 kg (600 lbm) or greater. 2) all large hardware that presents crewmember visibility problems when verifying position for mating 3) complex hardware with unusual shapes or centers of gravity which	

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 144 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.5-1 ARTEMIS SYSTEM VERIFICATIONS

Requirement ID	Title	Method	Verification Objective	Success Criteria	Notes
				preclude a stable or positive soft capture 4) Hardware installations where partial gravity vectors can be shown to maintain a stable or positive configuration before a hard capture is engaged. Prior to inspection, Artemis system needs to identify the hardware that requires soft capture to be operated by EVA crewmember.	
[EVASC.0126]	Soft Dock Force	Test	Verification test shall consist of measuring the forces required to operate Artemis System qualification components during a unit thermal cycle test.	Verification will be considered successful when test measurements indicating that the force required to initially and temporarily secure an object (soft dock) are sized proportionally to the mass of the object per the Table 3.3.8.5-1 Soft Dock Force. Prior to test, Artemis system needs to identify the hardware that requires soft dock to be operated by EVA crewmember.	
[EVASC.0024]	Unique Labeling	Inspection	Verification shall consist of an inspection of Artemis System hardware design drawings to ensure provision of cues/indicators/labels are marked or coded for intended use and meet EVA-EXP-0049, EVA Office	Verification will be considered successful when inspection of hardware design drawings confirm that translation paths, worksites, and any other callouts for the EVA crewmember with appropriate cues/indicators/labels are marked or coded for intended use as	

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 145 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.5-1 ARTEMIS SYSTEM VERIFICATIONS

Requirement ID	Title	Method	Verification Objective	Success Criteria	Notes
			Exploration EVA (xEVA) System Labeling Guidelines.	defined in EVA-EXP-0049, EVA Office xEVA System Labeling Guidelines.	
[EVASC.0025]	Hand Tool Operations	Demonstration	Verification shall consist of a demonstration to show that a hand in a pressurized glove can successfully operate the tool with either hand.	Verification will be considered successful when demonstration shows the tool can be operated with each hand (in a pressurized glove) individually.	
[EVASC.0135a]	Hand Ratcheting	Analysis	Verification analysis shall consist of a CAD analysis of the worksite, equipment, and/or tools with any nearby obstructions to ensure that EVA operable fasteners have at least 90° throw angle when ratcheted by an EVA gloved hand.	Verification will be considered successful when CAD analysis of the worksite, equipment, and/or tools with any nearby obstructions show that EVA operable fasteners have at least 90° throw angle when ratcheted by an EVA gloved hand.	
[EVASC.0026]	Throw Angle	Analysis	Verification analysis shall consist of a CAD analysis of the hand ratcheting tool handle coupled with appropriate extensions, a gloved hand on the ratchet, and any obstructions to ensure that provides throw angle of 45°.	Verification will be considered successful when CAD analysis data confirms Artemis System EVA worksites that necessitate hand ratcheting have throw angles for hand tools that incorporate a ratcheting handle are a minimum of 45°.	
[EVASC.0127]	Ratchet Mechanism	Inspection	Verification shall consist of an inspection of Artemis System hardware design drawings to ensure that hardware requiring a ratchet wrench incorporates a fine-tooth mechanism that allows engagement at every 10° maximum.	Verification will be considered successful when inspection of Artemis System hardware design drawings confirms that the ratchet wrench fine-tooth mechanism allows engagement at every 10° maximum.	

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 146 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.5-1 ARTEMIS SYSTEM VERIFICATIONS

Requirement ID	Title	Method	Verification Objective	Success Criteria	Notes
[EVASC.0128]	Back Drive Torque	Analysis Test	<u>Test:</u> Verification shall be by test utilizing flight-like Artemis System hardware consisting of back driving wrench and measuring the torque using a pressurized glove. <u>Analysis:</u> If the test doesn't include pressurized glove, analysis can be used to show it is compatible with pressurized glove.	Verification will be considered successful when test (and analysis) confirms the EVA wrench design to have a maximum back drive torque of 0.5 Nm (7 in-oz) while operated by a pressurized glove.	
[EVASC.0129]	Running Torque	Analysis Test	<u>Test:</u> Verification test shall utilize flight-like Artemis System hardware and consist of driving wrench and measuring the torque. <u>Analysis:</u> If the test doesn't include pressurized glove, analysis can be used to show it is compatible.	Verification will be considered successful when test (and analysis) confirms fastener design used with EVA ratchet wrench to have a running torque that is greater than the maximum back drive torque of 0.5 Nm (0.7 in-oz).	
[EVASC.0027]	Pressurized Glove Compatibility	Demonstration	Verification shall consist of a demonstration utilizing flight-like Artemis System hardware showing that controls can be operated by crewmembers wearing a pressurized glove.	Verification will be considered successful when demonstration (confirms all controls an EVA crewmember might operate are compatible with a pressurized glove.	EHP would expect a test or demo as part of the VCN (expect that hardware provider would submit an analysis supported by test or demonstration). If the demonstration doesn't include pressurized glove, analysis can be used to show it is compatible.

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 147 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.5-1 ARTEMIS SYSTEM VERIFICATIONS

Requirement ID	Title	Method	Verification Objective	Success Criteria	Notes
[EVASC.0087]	Inadvertent Actuation of EVA -Actuated Controls	Inspection	Verification shall consist of an inspection of Artemis System hardware design drawings to ensure that EVA actuated controls prevent inadvertent actuation.	Verification will be considered successful when inspection of hardware design drawings shows design on EVA actuated controls have implemented features to prevent inadvertent actuation.	
[EVASC.0029]	Handrail/ Handhold Cross-Section	Inspection	Verification shall consist of an inspection of Artemis System hardware design drawings to ensure that handrail/handholds meet cross-sectional profile and tolerances defined in Figures 3.3.12.1-1 or 3.3.12.1-2.	Verification will be considered successful when inspection of hardware design drawings shows handholds/handrails meet Figure 3.3.12.1-1 Handrail/Handhold Profile Dimensions, for microgravity/dual gravity field application with tolerances per SDG33107728, via Top Mounted Handrail drawing SEG33106347 latest revision or Figure 3.3.12.1-2 Partial Gravity Handrail/Handhold Profile Dimensions.	
[EVASC.0030]	Handrail/ Handhold/ Handle Length	Inspection	Verification shall consist of an inspection of Artemis System hardware design drawings to ensure that handrail/handholds/handles meet minimum grip length of 0.15 m (6in).	Verification will be considered successful when inspection of hardware design drawings shows handhold/handrail/handle design meets minimum grip length of 0.15 m (6 in) and can be operated by a pressurized glove.	
[EVASC.0031]	Handrail/ Handhold Accessibility	Analysis	Verification analysis shall consist of a CAD analysis on Artemis System hardware containing EVA handrail/handholds and other EVA interfaces installed in the EVA translation path or worksite showing	Verification will be considered successful when the CAD analysis confirms handrail/handholds and other EVA interfaces located in the defined crew translation path or worksite are left exposed and usable to the crew as a translation	

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 148 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.5-1 ARTEMIS SYSTEM VERIFICATIONS

Requirement ID	Title	Method	Verification Objective	Success Criteria	Notes
			that these interfaces are left exposed for crew use.	aid or if covered it is accessible to the crew.	
[EVASC.0032]	Continuous Handrail/Handhold Spacing	Analysis	Verification analysis shall consist of a CAD analysis on Artemis System hardware to ensure EVA handrail/handholds and integral safety tether attach points spacing does not exceed 0.61 m (24 in) as specified in Figure 3.3.12.4-1.	Verification will be considered successful when the CAD analysis confirms Artemis System EVA handrail/handholds and integral safety tether attach points spacing does not exceed 0.61 m (24 in) between the standoff and handrail/handhold as defined in Figure 3.3.12.4-1 Handrail/Handhold Gaps.	
[EVASC.0144]	EVA Egress Handrail Spacing	Inspection	Verification shall consist of an inspection of Artemis System hardware design drawings to ensure that handrails/handholds are within 30.5 cm (12 in) of the EVA egress location.	Verification shall be considered successful when inspection of hardware design drawings shows handrail/handholds spacing for EVA egress are located within 30.5 cm (12 in) of the EVA egress location.	
[EVASC.0035]	Work Envelope	Analysis	Verification analysis shall consist of a CAD analysis on Artemis System hardware for compliance with the one-handed or two-handed gravity field work volumes outlined in Figures 3.3.12.6.1-1 through 3.3.12.6-4 and Table 3.3.12.6.1-1, per delivered task list overview.	Verification will be considered successful when the CAD analysis confirms EVA worksites with either one-handed or two-handed EVA glove operations reside within work envelopes defined in Figures 3.3.12.6.1-1 through Figure 3.3.12.6.1-4 and Table 3.3.12.6.1-1.	Expanded work envelopes may meet the intent of this requirement if task analysis and NASA HITL mobility demonstrations and Crew Consensus show that activities outside the scope of the figures are acceptable

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 149 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.5-1 ARTEMIS SYSTEM VERIFICATIONS

Requirement ID	Title	Method	Verification Objective	Success Criteria	Notes
[EVASC.0170]	Working Volume Operation of Vehicle	Analysis	Verification analysis shall consist of a CAD analysis on Artemis System hardware for ensuring that is within <TBD-EHP-10028-019> crewmember volume for operation of vehicle on the lunar surface.	Verification will be considered successful when the CAD analysis confirms crewmember working volume for operation of vehicle on the lunar surface is within <TBD-EHP-10028-019> .	
[EVASC.0157]	EVA Egress/Ingress Methods	Analysis Demonstration	<u>Analysis:</u> Verification analysis shall consist of analytical CAD models that utilize the Artemis Systems' hardware design drawings to verify the Artemis System provides EVA egress/ingress methods from/to the vehicle that accommodate the 1st to 99th% pressurized crewmembers' mobility. <u>Demonstration:</u> Verification demonstration shall consist of a high-fidelity evaluation that shows the Artemis System provides EVA egress/ingress methods from/to the vehicle that accommodate the 1st-99 % pressurized crewmembers' mobility.	<u>Analysis:</u> Verification will be considered successful when the CAD analysis that utilize hardware design drawings confirms that the Artemis System provides EVA egress/ingress methods from/to the vehicle that accommodate the 1st to 99th% pressurized crewmembers' mobility. <u>Demonstration:</u> Verification will be considered successful when demonstration shows that the Artemis System provides EVA egress/ingress methods from/to the vehicle that accommodate 1st to 99th% pressurized crewmembers' mobility.	<u>Joint Integrated Verification-</u> Demonstration is critical, and NASA teams to ensure that providers work together to produce a high-fidelity demonstration.
[EVASC.0036]	Crew Aids for Stability	Analysis	Verification analysis shall consist of analytical CAD models that utilize the Artemis Systems' hardware design drawings to verify that crew aids for stability or grasp points for dedicated, robotic, or free-float worksite are provided that comply	Verification will be considered successful when the CAD analysis that utilize hardware design drawings confirms crew stability aids or grasp points for dedicated, robotic, or free-float worksite are located within 0.46 m (18 in) above or below the center of the	

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 150 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.5-1 ARTEMIS SYSTEM VERIFICATIONS

Requirement ID	Title	Method	Verification Objective	Success Criteria	Notes
			with Figure 3.3.12.6.4-1, per delivered task list overview.	crewmember's two-handed work envelope, 0.6m (24 in) to the left or right of the body centerline, and between 0.13 m (5 in) and 0.58 m (23 in) from the bolt pattern on the foot restraint (same depth as the one-handed work volume) as defined in Figure 3.3.12.6.4-1 Crew Aids for Stability.	
[EVASC.0037]	EVA Worksites Using Robotic Work Platform	Analysis	Verification analysis shall consist of analytical CAD models that utilize the Artemis Systems' hardware design drawings to verify that a robotic work platform provides handrails/handholds within reach and orientations that support stability or arresting reaction forces.	Verification will be considered successful when the CAD analysis that utilize hardware design drawings confirms worksites requiring a robotic work platform are designed to include handholds/handrails within the EVA crewmember worksite area to react forces or moments and provide within reach and orientations that support stability or arresting reaction forces.	
[EVASC.0038]	Lubricant Use	Inspection	Verification shall consist of an inspection of Artemis System hardware design drawings and corresponding identification of all areas in which lubricants are used to show either mitigations for the transfer of any lubrications onto the xEVA glove/suit or provides labeling designating as "no touch areas." Verification may also include confirmation that the system does not employ lubricants.	Verification will be considered successful when inspection of hardware design drawings shows Artemis System hardware shows either mitigations for the transfer of any lubrications onto the xEVA glove/suit or provides labeling designating as "no touch areas." Verification may also include confirmation that the Artemis System does not employ lubricants.	

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 151 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.5-1 ARTEMIS SYSTEM VERIFICATIONS

Requirement ID	Title	Method	Verification Objective	Success Criteria	Notes
[EVASC.0040]	Worksite Mobility	Analysis	Verification analysis shall consist of analytical CAD models that utilize the Artemis Systems' hardware design drawings to verify that foot restraint enables access and reach according to the work envelope (Figure 3.3.12.6.1-1 Work Volume-Microgravity in requirement [EVASC.0035]) that is aligned with the task objectives for the microgravity EVA.	Verification will be considered successful when CAD analysis that utilize hardware design drawings confirms that foot restraints enable access and reach according to the work envelope that is aligned with the task objectives for the microgravity EVA are as defined in the Figure 3.3.12.6.1-1 Work Volume-Microgravity in requirement [EVASC.0035].	Suit model approach will need to account for newer EVAS designs or otherwise be very conservative.
[EVASC.0041]	Worksite Field of View	Analysis	Verification consists of an analysis that identifies Artemis Systems' planned worksites and tasks. The analysis shall utilize analytical CAD models of the Artemis System worksite and hardware design drawings to verify that EVA worksites and the crewmember's line of sight are not obstructed by surrounding hardware.	Verification will be considered successful when analysis, using the Artemis System worksite and hardware CAD models, confirms that crewmember's line of sight are not obstructed by surrounding hardware.	Suit model approach will need to account for newer EVAS designs or otherwise be very conservative.
[EVASC.0042]	Working Volume	Analysis	Verification consists of an analysis that identifies Artemis Systems' planned worksites and tasks. The analysis shall consist of analytical CAD models that utilize the Artemis Systems' hardware design drawings to verify that any hardware that is manipulated by EVA crewmembers meets the specified field of gravity working	Verification will be considered successful when CAD analysis that utilize hardware design drawings of planned worksites and tasks confirms that Artemis System provides Worksite field of gravity work volumes as specified in Figure 3.4.2.2.-1 and Figure 3.4.2.2-2.	Suit model approach will need to account for newer EVAS designs or otherwise be very conservative.

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 152 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.5-1 ARTEMIS SYSTEM VERIFICATIONS

Requirement ID	Title	Method	Verification Objective	Success Criteria	Notes
			volume as designated in Figure 3.4.2.2-1 and Figure 3.4.2.2-2.		
[EVASC.0148]	Dedicated EVA Worksite	Analysis	Verification consists of an analysis that identifies Artemis Systems' planned worksites and tasks. The analysis shall consist of analytical CAD models that utilize the Artemis Systems' hardware design drawings to verify that dedicated worksites provide access to crew restraint where conditions apply as outlined in Table 3.4.3-1.	Verification will be considered successful when CAD analysis that utilize hardware design drawings of planned worksites and tasks confirm that dedicated EVA worksites provide restraint accommodations and capability for the crewmember at the worksite with either an APFR or BRT for worksite access where conditions apply as outlined in Table 3.4.3-1 Worksite Forces.	Suit model approach will need to account for newer EVAS designs or otherwise be very conservative.
[EVASC.0149]	Robotic Assisted EVA Worksite	Analysis	Verification consists of an analysis that identifies Artemis Systems' planned worksites and tasks. The analysis shall consist of analytical CAD models that utilize the Artemis Systems' hardware design drawings to verify that robotic assisted EVA worksite provide the force reaction mechanism as defined in requirement [EVASC.0036] independent of the robotic stabilization platform for EVA worksite operations requiring crew-generated forces greater than 44N (10 lbf).	Verification will be considered successful when CAD analysis that utilize hardware design drawings of planned worksites and tasks confirm that the identified Robotic assisted EVA worksite(s) on Artemis System(s) provides a force reaction mechanism as defined in requirement [EVASC.0036] independent of the robotic stabilization platform for EVA worksite operations requiring crew-generated forces greater than 44N (10 lbf).	Suit model approach will need to account for newer EVAS designs or otherwise be very conservative.

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 153 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.5-1 ARTEMIS SYSTEM VERIFICATIONS

Requirement ID	Title	Method	Verification Objective	Success Criteria	Notes
[EVASC.0150]	Free-Float Worksites	Analysis	Verification consists of an analysis that identifies Artemis Systems' planned worksites and tasks. The analysis shall consist of analytical CAD models that utilize the Artemis Systems' hardware design drawings to verify that free-float worksites are located with the one-handed envelope defined in Figure 3.3.12.6.1-1.	Verification will be considered successful when CAD analysis that utilize hardware design drawings of planned worksites and tasks confirm all tasks to be performed by free-float crewmember are located within the one-handed envelope defined in Figure 3.3.12.6.1-1. Work Volume - Microgravity Figure in requirement [EVASC.0035].	Suit model approach will need to account for newer EVAS designs or otherwise be very conservative.
[EVASC.0045]	Installation of Passive WIF Sockets	Inspection	Verification shall identify Artemis Systems' planned worksites and tasks. Once EVA worksites are identified, the verification shall consist of an inspection of Artemis System hardware design drawings to ensure that any EVA worksites that include a passive WIF provides handrails/handholds that are located within 0.61 m (24 in) to support foot restraint installation.	Verification will be considered successful when inspection of hardware design drawings for planned worksites and tasks confirm that EVA worksites accommodate passive WIF sockets with handholds/handrails located within 24-in. (0.61 m) of the passive WIF to support foot restraint installation.	
[EVASC.0046]	Foot Restraint Ingress/Egress Aids	Inspection	Verification identifies Artemis Systems' planned worksites and tasks. Once EVA worksites are identified, the verification shall consist of an inspection of Artemis System hardware design drawings to ensure that EVA Worksites containing WIFs include crew mobility aids located within 0.91 to	Verification will be considered successful when inspection of hardware design drawings for planned worksites and tasks confirm that EVA worksites containing WIFs include crew mobility aids located within 0.91 to 1.2 m (36 to 48 in) above the foot restraint/boot interface, 0.61 m (24	

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 154 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.5-1 ARTEMIS SYSTEM VERIFICATIONS

Requirement ID	Title	Method	Verification Objective	Success Criteria	Notes
			1.2 m (36 to 48 in) above the foot restraint/boot interface, 0.61 m (24 in) to the left or right of the foot restraint/boot interface centerline, and 0.46 m (18 in) in front of the centerline axis of the foot restraint/boot interface to aid in ingress/egress of the foot restraint as specified in Figure 3.4.7-1.	in) to the left or right of the foot restraint/boot interface centerline, and 0.46 m (18 in) in front of the centerline axis of the foot restraint/boot interface to aid in ingress and egress of the foot restraint as specified in Figure 3.4.7-1 Ingress and Egress Aids for Foot Restraint.	
[EVASC.0047]	Visual Contrast at Worksite	Analysis	Verification analysis shall consist of analytical models that utilize the Artemis Systems' hardware design drawings to assess visual contrast of Artemis System EVA worksites and all associated equipment, such as labeling, markings, and displays are designed to be clearly readable by crewmembers in extreme lighting conditions with a ratio of 5:1 at worksite as well as include a lighting analysis.	Verification will be considered successful when model and lighting analysis show that the Artemis System EVA worksites and all associated equipment, such as labeling, markings, and displays are designed to be clearly readable by crewmembers in extreme lighting conditions with a contrast ration of 5:1.	Testing or demonstration could be utilized to substantiate the analysis
[EVASC.0033]	Color for EVA Handling and EVA Equipment Tether Points	Inspection	Verification shall consist of an inspection of Artemis Systems' flight drawings to ensure that handholds or tether points intended to be used only for equipment handling are colored according to Table 3.5.1.1-1 EVA Handling and EVA Equipment Tether Point Colors <TBR-EHP-10028-012>.	Verification will be considered successful when the inspection of flight drawings shows that Artemis System provides handholds or tether points intended to be used only for equipment handling are colored according to Table 3.5.1.1-1 EVA Handling and EVA Equipment Tether Point Colors <TBR-EHP-10028-012>.	
[EVASC.0048]	EVA Equipment Tether Point	Inspection	<u>Inspection</u> : Verification shall consist of an inspection of Artemis	<u>Inspection</u> : Verification will be considered successful when the	

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 155 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.5-1 ARTEMIS SYSTEM VERIFICATIONS

Requirement ID	Title	Method	Verification Objective	Success Criteria	Notes
		Test	Systems' flight drawings to ensure that non-soft goods tether point receptacle dimensions are in accordance with Figure 3.5.1.2-1, EVA Equipment Tether Point Dimensions. <u>Test:</u> Verification shall consist of a test measuring loads of Artemis System flight-like hardware to verify that the Artemis System hardware has a minimum load of 334 N (75 lbf) in any direction.	inspection of flight hardware drawings shows that the non-soft goods tether point meets the required dimensions outlined in Figure 3.5.1.2-1, EVA Equipment Tether Point Dimensions. <u>Test:</u> Verification will be considered successful when the test shows that Artemis System structural components meet the minimum load of 334 N (75 lbf) in any direction.	
[EVASC.0134]	Soft Goods Tether Point Loads	Inspection Test	<u>Inspection:</u> Verification shall consist of an inspection of Artemis Systems' flight drawings to ensure that soft goods tether point receptacle dimensions are in accordance with Figure 3.5.1.2-1, EVA Equipment Tether Point Dimensions across its entire length. <u>Test:</u> Verification shall consist of a test measuring loads of Artemis System flight-like hardware to verify that the Artemis System hardware that have a minimum load of 133 N (30 lbf) in any direction.	<u>Inspection:</u> Verification will be considered successful when the inspection of flight drawings shows that the soft goods tether point meets the required dimensions outlined in Figure 3.5.1.2-1, EVA Equipment Tether Point Dimensions across its entire length. <u>Test:</u> Verification will be considered successful when the test shows that Artemis System structural components meet the that have a minimum load of 133 N (30 lbf) in any direction.	
[EVASC.0050]	Microgravity EVA Crew-Induced Loads	Analysis	Verification shall consist of an analysis to verify that the Artemis System hardware withstands the loads identified in Table 3.5.2-1	Verification will be considered successful when the analysis shows that Artemis System hardware withstands specified	Analysis supported by test

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 156 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.5-1 ARTEMIS SYSTEM VERIFICATIONS

Requirement ID	Title	Method	Verification Objective	Success Criteria	Notes
			Microgravity EVA Crew-Induced Loads.	EVA crew-induced loads as defined in Table 3.5.2-1 Microgravity EVA Crew-Induced Loads. This analysis will be substantiated by supporting Test, consisting of a test measuring loads of Artemis System flight-like hardware to verify that withstands the loads identified in Table 3.5.2-1 Microgravity EVA Crew-Induced Loads.	
[EVASC.0155]	Partial Gravity EVA Crew-Induced Loads	Analysis	Verification shall consist of an analysis to verify that the Artemis System hardware, or IVA hardware expected to interface with the suited crewmember withstands the loads identified in Table 3.5.3-1 Partial Gravity EVA Crew-Induced Loads <TBR-EHP-10028-002> .	Verification will be considered successful when the analysis shows that Artemis System hardware, or IVA hardware expected to interface with the suited crewmember withstands specified EVA crew-induced loads as defined in Table 3.5.3-1 Partial Gravity EVA Crew-Induced Loads <TBR-EHP-10028-002> . This analysis will be substantiated by supporting Test, consisting of a test measuring loads of Artemis System flight-like hardware to verify that withstands the loads identified in Table 3.5.3-1 Partial Gravity EVA Crew-Induced Loads <TBR-EHP-10028-002> .	Analysis supported by test
[EVASC.0018]	EVA Actuated Loads	Analysis	Verification consists of an analysis that identifies Artemis Systems' planned worksites, tasks, and required tools for operation. Once	Verification will be considered successful when analysis identifies Artemis Systems' planned worksites, tasks, and tools for	Analysis supported by test

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 157 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.5-1 ARTEMIS SYSTEM VERIFICATIONS

Requirement ID	Title	Method	Verification Objective	Success Criteria	Notes
			EVA worksites are identified, the verification shall consist of an analysis to verify that the Artemis System items to be actuated by EVA crewmembers are operable by the loads defined in Table 3.5.4-1 Miscellaneous Crew Induced Limit Loads.	operation to confirm that Artemis System hardware items to be actuated by EVA crewmembers are operable by the loads defined in Table 3.5.4-1 Miscellaneous Crew Induced Limit Loads. This analysis will be substantiated by supporting Test, consisting of a test measuring loads of Artemis System flight-like hardware to verify that withstands the loads identified in Table 3.5.4-1 Miscellaneous Crew Induced Loads.	
[EVASC.0164]	EVA Mechanical Feedback	Analysis	Verification consists of an analysis that identifies Artemis Systems' planned worksites, tasks, and required tools for operation. Once EVA worksites are identified, the verification shall consist of an analysis to verify that the Artemis System EVA mechanical feedback actuation force is greater than 15.5 N (3.5 lbf).	Verification will be considered successful when analysis identifies Artemis Systems' planned worksites, tasks, and tools for operation to confirm that Artemis System shows that EVA mechanical feedback actuation force is greater than 15.5 N (3.5 lbf).	Analysis supported by test
[EVASC.0143]	WIF Structural Loading	Analysis	Verification shall consist of an analysis to verify that the Artemis System maintains positive margins of safety for loads at the WIF to structure interface of 1220 N (274 lbf) in any direction and a moment of 474 Nm (4200 lbf-in) in any direction, simultaneously, as defined in Figure 3.5.6-1 Top	Verification will be considered successful when the analysis shows that Artemis System maintains positive margins of safety for loads at the WIF to structure interface of 1220 N (274 lbf) in any direction and a moment of 474 Nm (4200 lbf-in) in any direction, simultaneously, as	

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 158 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.5-1 ARTEMIS SYSTEM VERIFICATIONS

Requirement ID	Title	Method	Verification Objective	Success Criteria	Notes
			Mounted WIF Worksite Half Loading Criteria and to be used only in conjunction with a limiting APFR.	defined in Figure 3.5.6-1 Top Mounted WIF Worksite Half Loading Criteria and to be used only in conjunction with a limiting APFR.	
[EVASC.0053]	One-Handed EVA Operable Connectors	Demonstration	Verification shall consist of a demonstration to show that a hand in a pressurized glove can successfully mated/demated with either hand.	Verification will be considered successful when demonstration shows the connector can be aligned and mated with each hand (in a pressurized glove) individually. This demonstration is performed after the Artemis System has identified planned worksites, tasks, and need for EVA operable connectors.	
[EVASC.0054]	Connector Accessibility	Analysis	Verification consists of an analysis that identifies Artemis Systems' planned worksites, tasks, and EVA operable connectors. Once identified, the verification shall consist of an analysis to verify that the Artemis System provides individual EVA connectors that can be operated by EVA crewmembers without having to remove/replace another connector.	Verification will be considered successful when analysis identifies Artemis Systems' planned worksites, tasks and EVA operable connectors and shows that individual EVA connectors can be operated by EVA crewmembers without having to remove/replace another connector. Verification should consider connector spacing outlined in Appendix B EVABP-27, Figure B-1 EVABP-27 Connector Spacing.	

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 159 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.5-1 ARTEMIS SYSTEM VERIFICATIONS

Requirement ID	Title	Method	Verification Objective	Success Criteria	Notes
[EVASC.0138]	Gloved-Hand Connector Accessibility	Analysis	Verification consists of an analysis that identifies Artemis Systems' planned worksites, tasks, and EVA operable connectors. Once identified, the verification shall consist of an analysis to verify that the Artemis System provides gloved-hand clearance around entire EVA connector as specified in Figure 3.1.5-1 Gloved-Hand Clearance (reference [EVASC.0003]).	Verification will be considered successful when analysis identifies Artemis Systems' planned worksites, tasks and EVA operable connectors and shows that Artemis System provides gloved-hand clearance around entire EVA connector as specified in Figure 3.1.5-1 Gloved-Hand Clearance (reference [EVASC.0003]) to enable EVA crew to actuate (including the actuating mechanism, if applicable). Verification should consider connector spacing outlined in Appendix B EVABP-27, Figure B-1EVABP-27 Connector Spacing. Note: Because spacing requirements are highly dependent on design and size of the connector, all connector spacing designs not meeting gloved hand access requirements in every direction may need to be demonstrated to be acceptable by test with pressurized EVA gloves with flight-like hardware.	
[EVASC.0055]	Incorrect Connector Mating Prevention	Inspection	Verification shall consist of an inspection of Artemis Systems' flight hardware drawings and shows EVA connectors on the vehicle or	Verification will be considered successful when the inspection shows that the Artemis System flight hardware was built to	

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 160 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.5-1 ARTEMIS SYSTEM VERIFICATIONS

Requirement ID	Title	Method	Verification Objective	Success Criteria	Notes
			EVA worksite to verify that each connector contains features to preclude incorrect mating.	drawings. In addition, inspection of the qualification hardware reveals that each EVA connector contains features to physically prevent improper mating. Note: If there are findings that indicate that an EVA connector is in potential violation of the requirement, a demonstration may need to be performed to ensure that the EVA connector can be mated as intended and not inadvertently mated to any other EVA connector in the vicinity.	
[EVASC.0056]	Connector Coding	Inspection	Verification shall consist of an inspection of Artemis Systems' flight hardware to verify it meets approved drawings and shows that both sides of the EVA operable connectors include a label, code, or other identifier that is unique to that mate/demate operation and visible to the EVA crewmembers.	Verification will be considered successful when the inspection shows Artemis System hardware locate unique identifier labels or codes on both sides of mating EVA connectors and associated items are visible when both connected and disconnected.	
[EVASC.0057]	Connector Protective Caps	Inspection	Verification shall consist of an inspection of Artemis Systems' flight hardware drawings to ensure that individual EVA connectors contain a protective cap that contain tethering capability (e.g., can be integrated) for all connector plugs and jack when not connecting to the mated part and that it	Verification will be considered successful when the inspection of flight hardware drawings shows that Artemis System provides individual EVA connectors that have protective caps containing tethering capability (e.g., can be integrated) for all connector plugs and jack when not connecting to	

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 161 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.5-1 ARTEMIS SYSTEM VERIFICATIONS

Requirement ID	Title	Method	Verification Objective	Success Criteria	Notes
			accounts for the items outlined in [EVASC.0057] to prevent any physical damage or contamination.	the mated part and that it accounts for the items outlined in [EVASC.0057] to prevent any physical damage or contamination.	
[EVASC.0058]	Connector Rotational Travel	Inspection Analysis	<u>Inspection:</u> Verification shall consist of an inspection of Artemis Systems' flight hardware drawings to ensure that EVA twist-on connectors have the limited maximum of 180° for installation or removal. <u>Analysis:</u> Verification shall consist of an analysis to verify that the Artemis System hardware containing EVA twist-on connectors have the limited maximum of 180° for installation or removal.	<u>Inspection:</u> Verification will be considered successful when the inspection of flight hardware drawings shows that Artemis System EVA connector rotational travel is limited to a maximum of 180° for installation or removal of twist-on connectors. <u>Analysis:</u> Verification will be considered successful when the analysis shows that Artemis System provides EVA connector rotational travel that is limited to a maximum of 180° for installation or removal of twist-on connectors. Note: The connector keying and orientation of the EVA connector relative to intended operation should provide initial assessment regarding the amount of rotation. A demonstration/or test may need to be performed to ensure that the EVA connector meets the rotational travel under pressure.	
[EVASC.0059]	Connector Locking Feature	Inspection	Verification shall consist of an inspection of Artemis Systems' flight hardware drawings to ensure	Verification will be considered successful when the inspection of flight hardware drawings shows	

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 162 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.5-1 ARTEMIS SYSTEM VERIFICATIONS

Requirement ID	Title	Method	Verification Objective	Success Criteria	Notes
			that EVA connector plugs include a locking feature to prevent inadvertent demate.	that Artemis System EVA connector plugs include a locking feature to prevent inadvertent demate.	
[EVASC.0060]	Visible Connector Alignment Markings	Inspection	Verification shall consist of an inspection of Artemis Systems' flight hardware drawings to verify that visible alignment markings or other feature are provided on EVA connectors to ensure proper alignment during mating/demating.	Verification will be considered successful when the inspection of flight hardware drawings shows that Artemis System provides visible alignment markings or other feature on EVA connectors to ensure proper alignment during mating/demating.	
[EVASC.0061]	Connector Locking Ring	Demonstration	Verification shall consist of a demonstration of Artemis System EVA connectors to ensure that the locking ring does not turn independent from the connector body.	Verification will be considered successful when the demonstration of Artemis System EVA connectors shows that the locking ring does not turn independent from the connector body.	
[EVASC.0062]	Connector Mating Status	Demonstration	Verification shall consist of a demonstration of EVA connector operation to verify that the Artemis System EVA connectors provide status indication of full mate/demate.	Verification will be considered successful when the demonstration of EVA connector operation shows that Artemis System provides EVA connectors that indicate full mate/demate status.	
[EVASC.0052]	Mating/ Demating of Powered Connectors	Analysis	Verification shall consist of review of hazard analysis to identify any potential for hazardous currents that may be produced by the voltages and impedances at the contacts of the EVA electrical connectors.	Verification will be considered successful when the hazard analysis shows that Artemis System mitigates potential for shock hazards to the EVA crewmember during mate/demate operation of EVA connectors.	

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 163 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.5-1 ARTEMIS SYSTEM VERIFICATIONS

Requirement ID	Title	Method	Verification Objective	Success Criteria	Notes
[EVASC.0065]	Connectors - Electrical Bonding	Test	Verification shall consist of a test on flight-like Artemis System hardware to measure the electrical resistance between the conductive outer surface of each assembly and the disconnected EVA electrical connector at the far end of each attached cable (measure both ways, from one end of the assembly at the other end and vice versa).	Verification will be considered successful when the test shows that Artemis System all EVA electrical connectors provide a Class H (or NASA approved alternate electrical bonding standard with Cross-Program Chief Engineer approval) per NASA-STD-4003, Electrical Bonding for NASA Launch Vehicles, Spacecraft, Payloads, and Flight Equipment to the wiring harness shielding of cables and to conductive outer surfaces of electrical and electronic assemblies.	
[EVASC.0066]	Scoop-Proof Electrical Connector Design	Inspection	Verification shall consist of an inspection of Artemis Systems' flight hardware drawings of EVA connectors to ensure that electrical pins cannot contact the opposing connector half during mating/demating.	Verification will be considered successful when the inspection of flight hardware drawings shows that all Artemis System EVA electrical connectors prevent electrical pins from contacting the opposing connector half during mating/demating.	
[EVASC.0063]	Connector Grip Point	Demonstration	Verification shall consist of a demonstration to verify that the Artemis System EVA connector backshells provide grip point(s) for engagement/disengagement of electrical connectors with pressurized gloved hand in a glove box simulating EVA suit pressures by pressurized gloved.	Verification will be considered successful when the demonstration shows that the Artemis System EVA connector backshells' grip point(s) can be used for engagement/disengagement electrical connectors with by	

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 164 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.5-1 ARTEMIS SYSTEM VERIFICATIONS

Requirement ID	Title	Method	Verification Objective	Success Criteria	Notes
				pressurized gloved hand in a glove box simulating EVA suit pressures.	
[EVASC.0131]	Arc Containment	Inspection Analysis	<u>Inspection:</u> Verification shall consist of an inspection of Artemis Systems' flight hardware drawings to ensure that EVA electrical connectors confine and isolate electrical arc and sparks during mating/demating. <u>Analysis:</u> Verification shall consist of an analysis to verify that the Artemis System flight EVA connectors confine and isolate electrical arc and sparks during mating/demating.	<u>Inspection:</u> Verification will be considered successful when the inspection of flight hardware drawings shows that Artemis System provides EVA electrical connectors that confine and isolate electrical arcs and sparks during mating/demating. <u>Analysis:</u> Verification will be considered successful when the analysis shows that Artemis System provides EVA electrical connectors that confine and isolate electrical arcs and sparks during mating/demating that are verified by EEE expert.	Analysis should include EEE expert to verify. For the arc containment evaluation method, demonstration could also be considered since the operational conditions can be applied to demonstrate that there are no arc issues and or any arc produced will be contained.
[EVASC.0132]	Uncontrolled Motion of Flex Hoses	Inspection Demonstration	<u>Inspection:</u> Verification shall consist of an inspection of Artemis Systems' flight hardware drawings to ensure that restraint features are provided on the EVA serviceable fluid flex hoses and lines with delta pressure or under pressure (with exception of xEVA suit umbilical) to prevent injury to the EVA crewmembers. <u>Demonstration:</u> Verification shall consist of a demonstration to verify that the Artemis System flight EVA	<u>Inspection:</u> Verification will be considered successful when the inspection of flight hardware drawings shows that Artemis System restraint features are provided on the EVA serviceable fluid flex and lines with delta pressure or under pressure (with exception of xEVA suit umbilical) to prevent injury to the EVA crewmembers. <u>Demonstration:</u> Verification will be considered successful when the	

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 165 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.5-1 ARTEMIS SYSTEM VERIFICATIONS

Requirement ID	Title	Method	Verification Objective	Success Criteria	Notes
			serviceable fluid flex hoses and lines with delta pressure or under pressure (with exception of xEVA suit umbilical) can be restrained.	demonstration shows that Artemis System provides flight EVA serviceable fluid flex hoses and lines with delta pressure or under pressure (with exception of xEVA suit umbilical) can be restrained.	
[EVASC.0133]	Venting Feature	Test	Verification shall consist of a test to verify that each Artemis System flight EVA operable connectors can be connected, pressurized, depressurized, and disconnected during a unit thermal cycle test. Thermal analysis or instrument readings shall be used to show that the component has reached thermal equilibrium.	Verification will be considered successful when the test confirms that the Artemis System incorporated mechanism operates as intended in the specified thermal environment for each flight EVA operable fluid connectors. Test Note: Direct testing of pressure removal in the thermal environment is necessary since the pressure relief may be impacted by the cold and heat of space. Wherever possible, qualification hardware verifications should be performed by test.	
[EVASC.0145]	Indication of Pressure Flow - Nonbrazed or Nonwelded	Inspection	Verification shall consist of an inspection of Artemis System flight-like hardware drawings of nonbrazed or nonwelded gas and liquid lines provide a positive indication of the gas pressure/fluid flow to verify that the line is passive before disconnection of non-self-sealing connectors.	Verification will be considered successful when the inspection of flight-like hardware drawings shows that nonbrazed or nonwelded gas and liquid lines provide a positive indication of the gas pressure/fluid flow to verify that the line is passive before disconnection of non-self-sealing connectors.	

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 166 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.5-1 ARTEMIS SYSTEM VERIFICATIONS

Requirement ID	Title	Method	Verification Objective	Success Criteria	Notes
[EVASC.0068]	Captive Fastener Design	Inspection	Verification identifies Artemis Systems' planned worksites, tasks, and EVA operable fasteners. Verification shall consist of an inspection of Artemis Systems' flight hardware drawings to ensure that all EVA operable fasteners are captive.	Verification will be considered successful when the inspection of flight hardware drawings shows that Artemis System provides external EVA operable fasteners are captive or restraint features.	
[EVASC.0069]	Prohibited Locking Methods for Fasteners	Inspection	Verification identifies Artemis Systems' planned worksites, tasks, and EVA operable fasteners. Verification shall consist of an inspection of Artemis Systems' flight hardware drawings to ensure that none of the EVA operable fasteners have included any of the prohibited specified methods for locking.	Verification will be considered successful when the inspection of flight hardware drawings shows that Artemis System does not use safety wire, cotter pins, or jam nuts as a method for locking EVA operable fasteners.	
[EVASC.0067]	One-Handed Fastener Operation	Demonstration	Verification identifies Artemis Systems' planned worksites, tasks, and EVA operable fasteners. Verification shall consist of a demonstration utilizing a glove box/1g evaluation to verify that the Artemis System-provided EVA fasteners intended to be manipulated without the use of a tool can be operated one-handed while wearing a pressurized glove.	Verification will be considered successful when the glove box/1g demonstration of EVA operable fasteners shows that Artemis System provides EVA fasteners to be manipulated without a tool can be operated one-handed while wearing a pressurized glove.	

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 167 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.5-1 ARTEMIS SYSTEM VERIFICATIONS

Requirement ID	Title	Method	Verification Objective	Success Criteria	Notes
[EVASC.0072]	Scalloped Knobs	Inspection	Verification shall consist of an inspection of Artemis Systems' flight hardware drawings to ensure that scalloped knobs for actuation by pressurized EVA gloved hand contain no fewer than four (4) and no more than eight (8) scallops.	Verification will be considered successful when the inspection of flight hardware drawings show that the Artemis System provides scalloped knobs for actuation by pressurized EVA gloved hand that contain no fewer than four (4) and no more than eight (8) scallops.	
[EVASC.0073]	Knobs Head Diameter and Height	Inspection	Verification shall consist of an inspection of Artemis Systems' flight hardware drawings to ensure that knobs operated by pressurized EVA glove contain head diameters between the ranges of 3.8 cm - 5.1 cm (1.5 in. – 2.0 in) with a minimum head height of 1.9 cm (0.75 in).	Verification will be considered successful when the inspection of flight hardware drawings shows that Artemis System provides knobs for EVA suited pressurized gloved hand operation that contain head diameters between the ranges of 3.8 cm - 5.1 cm (1.5 in. – 2.0 in) with a minimum head height of 1.9 cm (0.75 in).	
[EVASC.0074]	EVA Hand-Actuated Fasteners Contingency Operation	Inspection	Verification shall consist of an inspection of the Artemis Systems' flight hardware drawings to ensure that an external hexagonal drive interface as specified in Table 3.7.3.4-1 is provided for rotational fasteners per section 3.7.5 for contingency operation with a hand tool.	Verification will be considered successful when the inspection of flight hardware drawings shows that the Artemis System provides an external hexagonal drive interface as specified in Table 3.7.3.4-1 for rotational fasteners per section 3.7.5 for contingency operation with a hand tool.	
[EVASC.0075]	EVA PIP Pin Grasping Interface	Demonstration	Verification shall consist of a Demonstration of Artemis Systems' flight hardware drawings to ensure that EVA PIP pin grasping interface provides a 3.8 cm (1.5 in) minimum diameter ring or T-handle to the pin	Verification will be considered successful when the demonstration shows that Artemis System provides a 3.8 cm (1.5 in) minimum diameter ring or T-	

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 168 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.5-1 ARTEMIS SYSTEM VERIFICATIONS

Requirement ID	Title	Method	Verification Objective	Success Criteria	Notes
			that is compatible with a pressurized glove.	handle to the pin that is compatible with a pressurized glove.	
[EVASC.0076]	PIP Pin Restraint Tether	Inspection	Verification shall consist of an inspection of Artemis Systems' flight hardware drawings to ensure that a restraint tether is attached to the body (handle) of the PIP pin and not to the activation mechanism.	Verification will be considered successful when the inspection of flight hardware design drawings shows that Artemis System provides a restraint tether attached to the body (handle) of the pin and not to the activation mechanism.	
[EVASC.0078]	Driver-Type Push Force Tools	Test	Verification shall consist of a test to verify that Artemis System EVA fasteners do not include a push force greater than 9N (2 lbf) to maintain tool engagement while providing torque.	Verification will be considered successful when the test confirms that the Artemis System provides EVA fasteners that do not require a driver-type hand tool with a push force greater than 9N (2 lbf) to maintain tool engagement while providing torque.	
[EVASC.0079]	EVA Fastener Head	Inspection	Verification shall consist of an inspection of the Artemis Systems' flight hardware drawings to ensure that all EVA-actuated bolt heads conform to the specifications outlined in Figures 3.7.5.2-1 and 3.7.5.2-2 at the minimum to maximum operational temperatures to enable interface with the standard EVA hand and power tools.	Verification will be considered successful when the inspection of flight hardware drawings shows that all Artemis System EVA actuated bolts conform to specifications outlined in Figures 3.7.5.2-1 and 3.7.5.2-2 at the minimum to maximum operational temperatures to enable interface with the standard EVA hand and power tools.	
[EVASC.0084]	Fastener Deep Socket	Inspection	Verification shall consist of an inspection of Artemis Systems' flight hardware drawings to ensure EVA fasteners have an extra deep well on the socket that is at	Verification will be considered successful when the inspection of flight hardware drawings shows that Artemis System EVA fasteners incorporates an extra	

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 169 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.5-1 ARTEMIS SYSTEM VERIFICATIONS

Requirement ID	Title	Method	Verification Objective	Success Criteria	Notes
			minimum equivalent to the double hex-head height.	deep well on the socket that is at minimum equivalent to the double hex-head height.	
[EVASC.0151]	Fastener Torque Data	N/A	N/A	N/A	Considered to be a data deliverable
[EVASC.0080]	Power and Hand Tool Operability	Demonstration	Verification shall consist of a demonstration of Artemis Systems' flight-like EVA fasteners requiring use of a power tool are removable and/or installable with a hand-operated tool.	Verification will be considered successful when the demonstration shows that Artemis System EVA fasteners installed with power tools are removable and/or installable with a hand-operated tool.	
[EVASC.0141]	Fastener Gauge/ Measurement	Inspection	Verification shall consist of an inspection of Artemis Systems' flight hardware drawings to ensure that EVA bolt heads are gauged/measured and meet dimensional requirements at the minimum to maximum operational temperatures prior to installation into assembly.	Verification will be considered successful when the inspection of flight hardware drawings shows that Artemis System EVA bolt heads are gauged/measured and meet dimensional requirements at the minimum to maximum operational temperatures prior to installation into assembly.	
[EVASC.0083]	Fastener Alignment Features	Inspection	Verification shall consist of an inspection of Artemis Systems' flight hardware drawings to ensure that EVA fastener alignment features are provided for initial engagement.	Verification will be considered successful when the inspection of flight hardware drawings shows that Artemis System provides alignment features to ensure initial engagement and preclude cross-threading.	
[EVASC.0085]	Battery Replacement	Analysis	Verification shall consist of an analysis to verify that the Artemis System provides batteries can be changed out at the worksite.	Verification will be considered successful when the analysis shows that any Artemis System batteries that necessitate change	

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 170 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.5-1 ARTEMIS SYSTEM VERIFICATIONS

Requirement ID	Title	Method	Verification Objective	Success Criteria	Notes
				out via EVA will be replaceable at the worksite. Ensure that any hardware containing the battery is compatible with pressurized EVA glove, that the operation and hardware design is compliant with the environment (e.g., thermal, dust, restraints), that the battery MI meets the applicable tools and access requirements in this document, and evaluation of the EVA worksite to comply with EVA worksite requirements. This analysis may include a HITL for some aspects of the overall verification analysis.	
[EVASC.0086]	Battery Voltage Monitoring	Analysis	Verification shall consist of an analysis of Artemis Systems' battery architecture to verify that battery-powered devices include battery monitoring capability before/during EVA usage.	Verification will be considered successful when the analysis battery architecture shows that Artemis System battery-powered devices include battery monitoring capability before/during EVA usage to protect against catastrophic hazards and insufficient capacity to support mission time.	
[EVASC.0028]	Auxiliary Controls	Analysis	Verification shall consist of an analysis of Artemis Systems' flight hardware drawings to verify that auxiliary controls are positioned for EVA crewmembers to operate and	Verification will be considered successful when the analysis of flight hardware drawings shows that Artemis System auxiliary controls are positioned for	

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 171 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.5-1 ARTEMIS SYSTEM VERIFICATIONS

Requirement ID	Title	Method	Verification Objective	Success Criteria	Notes
			do not interfere with the tool/fastener position.	operation by an EVA-gloved hand to activate the control without disturbing the tool/fastener position.	
[EVASC.0088]	EVA Tools	Analysis	Verification shall consist of an analysis of flight hardware to identify EVA operations with associated EVA tools and to verify that the system uses EVA tools and equipment in accordance with EHP-10074, EHP EVA Tool List and Process for EVA Operations <TBR-EHP-10028-008>.	Verification will be considered successful when the analysis of flight hardware shows that Artemis System has identified tools needed for EVA operations and the EVA tools are listed in EHP-10074, EHP EVA Tool List and Process for EVA Operations <TBR-EHP-10028-008>.	
[EVASC.0089]	Cable Routing/ Restraint	Inspection	Verification shall consist of an inspection of Artemis Systems' flight hardware drawings and flight hardware to verify that translation paths that include cable routing/restraint for hardware also include handrails, thereby precluding the potential for EVA crew to use cable routing/restraint hardware as EVA handling and translation aids.	Verification will be considered successful when the inspection of flight hardware drawings shows that Artemis System translation paths that include cable routing/restraint for hardware also include handrails, thereby precluding the potential for EVA crew to use cable routing/restraint hardware as EVA handling and translation aids.	
[EVASC.0090]	Cable Routing Spacing	Inspection	Verification shall consist of an inspection of Artemis Systems' flight hardware drawings and flight hardware to verify EVA-installed cable routing includes restraints spaced at a maximum of 0.61m (24 in).	Verification will be considered successful when the inspection of flight hardware drawings and flight hardware shows that Artemis System EVA-installed cable routing includes restraints spaced at a maximum of 0.61m (24 in).	
[EVASC.0091]	Paint	Analysis	Verification shall consist of an analysis of painted surfaces or	Verification will be considered successful when the analysis	

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 172 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.5-1 ARTEMIS SYSTEM VERIFICATIONS

Requirement ID	Title	Method	Verification Objective	Success Criteria	Notes
			features to verify that they are able to withstand environmental conditions observed by the Artemis System during transit and while on the lunar surface along with the wear-and-tear due to EVA contact (planned and inadvertent) on EVA worksites and within EVA translation paths/traverses, or areas on the vehicle which the EVA crewmember will access without flaking or contaminating the EVA crewmember.	shows that Artemis System provides evidence that painted surfaces or features are able to withstand environmental conditions observed by the Artemis System during transit and while on the lunar surface along with the wear-and-tear due to EVA contact (planned and inadvertent) on EVA worksites and within EVA translation paths/traverses, or areas on the vehicle which the EVA crewmember will access without flaking or contaminating the EVA crewmember. Artemis System will provide this assessment as part of Phase 3 Safety Review Package approval.	
[EVASC.0092]	Hazardous Material Control	Analysis	Verification shall consist of a hazard analysis to verify that the Artemis System reviews materials and chemicals selected for construction, propulsion, and maneuver capabilities (including propellants) and their intended use during operation.	Verification will be considered successful when the hazard analysis shows that Artemis System provides means of containment commensurate with the hazard level for toxic or hazardous chemicals/materials to preclude contaminating an EVA crewmember or xEVA suit (hazardous to EVA crewmember, xEVA suit performance, or to IVA environment if brought back inside habitable volume).	

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 173 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.5-1 ARTEMIS SYSTEM VERIFICATIONS

Requirement ID	Title	Method	Verification Objective	Success Criteria	Notes
				Artemis System will provide this assessment as part of Phase 3 Safety Review Package approval.	
[EVASC.0093]	Venting or Exposure Compatibility	Analysis	Verification shall consist of a hazard analysis to verify that the Artemis System hardware precludes venting or exposure of hazardous materials to an EVA crewmember.	Verification will be considered successful when the hazard analysis shows that Artemis System provides hardware that precludes venting or exposure of hazardous materials to an EVA crewmember (hazardous to EVA crewmember or to IVA environment if brought back inside habitable volume). Artemis System will provide this assessment as part of Phase 3 Safety Review Package approval.	
[EVASC.0094]	Deliberate Hazardous Venting	Analysis	Verification shall consist of a hazard analysis to verify that the Artemis System directs deliberate hazardous venting away from the EVA crewmember and translation path/worksites.	Verification will be considered successful when the hazard analysis shows that Artemis System provides methods for directing deliberate hazardous venting away from the EVA crewmember and translation path/worksites. Artemis System will provide this assessment as part of Phase 3 Safety Review Package approval.	
[EVASC.0097]	Electrical and Safety Design	Analysis	Verification shall consist of a hazard analysis performed to mitigate Artemis System hazardous	Verification will be considered successful when the hazard analysis shows that Artemis System precludes exposure of	

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 174 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.5-1 ARTEMIS SYSTEM VERIFICATIONS

Requirement ID	Title	Method	Verification Objective	Success Criteria	Notes
			electrical potential to the EVA crewmember.	hazardous electrical potential to the EVA crewmember. Artemis System will provide this assessment as part of Phase 3 Safety Review Package approval.	
[EVASC.0098]	Shock Protection	Analysis	Verification shall consist of a hazard analysis performed to ensure the Artemis System hardware contains no shock hazards present on any exposed conducting surfaces that are accessible to contact by EVA crewmembers.	Verification will be considered successful when the hazard analysis shows that Artemis System hardware does not contain shock hazards on any exposed conducting surfaces that are accessible to contact by EVA crewmembers (e.g., translation corridors, work areas, and incidental contact zones). Artemis System will provide this assessment as part of Phase 3 Safety Review Package approval.	
[EVASC.0099]	Operation During Pressure Change	Test	Verification shall consist of a test performed to ensure Artemis System hardware functions during depressurization and repressurization per EHP-10029, Exploration EVA (xEVA) System to Artemis Systems Interface Requirements Document (IRD) <TBR-EHP-10028-016>, and Table 3.9.5-1 withstands the differential pressure without resulting in a hazard to the EVA crewmember.	Verification will be considered successful when the test performed shows that Artemis System hardware functions during depressurization and repressurization per EHP-10029, xEVA System to Artemis Systems IRD <TBR-EHP-10028-016>, and Table 3.9.5-1 Depress and Repress Rates without resulting in a hazard to the EVA crewmember.	

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 175 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.5-1 ARTEMIS SYSTEM VERIFICATIONS

Requirement ID	Title	Method	Verification Objective	Success Criteria	Notes
[EVASC.0100]	Component Hazardous Energy Provision	Analysis	Verification shall consist of a hazard analysis performed to ensure that the Artemis System hardware does not expose the EVA crewmember to the release of hazardous energy.	Verification will be considered successful when the hazard analysis shows that Artemis System hardware precludes the potential for exposing EVA crewmembers to the release of hazardous energy. Artemis System will provide this assessment as part of Phase 3 Safety Review Package approval.	Test may be necessary based on hazard analysis
[EVASC.0101]	Detachable Parts	Inspection Analysis	<u>Inspection:</u> Verification shall consist of an inspection of Artemis Systems' flight hardware drawings to verify that the hardware provides a method for securing detachable parts. <u>Analysis:</u> Verification shall consist of analysis performed to ensure that Artemis System hardware provides a method for securing detachable parts.	<u>Inspection:</u> Verification will be considered successful when the inspection of flight hardware drawings shows that Artemis System hardware provides methods for securing detachable parts. <u>Analysis:</u> Verification will be considered successful when the analysis shows that Artemis System hardware provides a method for securing detachable parts. Artemis System will provide this assessment as part of Phase 3 Safety Review Package approval.	
[EVASC.0103]	Touch Temperatures	Analysis	Verification shall consist of a thermal analysis performed to ensure that all Artemis System hardware protects the xEVA suit	Verification will be considered successful when the thermal analysis shows that all Artemis System hardware protects the	

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 176 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.5-1 ARTEMIS SYSTEM VERIFICATIONS

Requirement ID	Title	Method	Verification Objective	Success Criteria	Notes
			from specified high- and low-touch-temperature extremes in the translation corridors, work areas, and incidental contact zones as defined in Table 3.9.8-1 < TBR-EHP-10028-011 >.	xEVA suit from specified high- and low-touch-temperature extremes as defined in Table 3.9.8-1 Touch Temperatures < TBR-EHP-10028-011 >, in the translation corridors, work areas, and incidental contact zones. Artemis System will provide this assessment as part of Phase 3 Safety Review Package approval.	
[EVASC.0011]	xEVA Glove Palm External Touch-Temperature Compliance for Small, Handheld Hardware	Analysis	Verification shall consist of a thermal analysis performed to ensure that all Artemis System hardware complies with the xEVA suit glove palm touch temperature limits for crewmember protection from high and low skin touch temperatures specified in Tables 3.9.9-1 < TBR-EHP-10028-011 > and 3.9.9-2 < TBR-EHP-10028-011 >.	Verification will be considered successful when the thermal analysis shows that all Artemis System hardware complies with the 30-minute xEVA suit glove palm touch temperature limits for crewmember protection from high and low skin touch temperatures as defined in Table 3.9.9-1 30-Minute Touch Temperature Compliance For Small, Handheld Hardware < TBR-EHP-10028-011 >, and Table 3.9.9-2 30-Minute Touch Temperature Heat Rate Compliance < TBR-EHP-10028-011 >. Artemis System will provide this assessment as part of Phase 3 Safety Review Package approval.	
[EVASC.0104]	Incidental and Unlimited Touch	Analysis	Verification shall consist of a thermal analysis performed to	Verification will be considered successful when the thermal	

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 177 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.5-1 ARTEMIS SYSTEM VERIFICATIONS

Requirement ID	Title	Method	Verification Objective	Success Criteria	Notes
	Heat Transfer Rates		ensure that all Artemis System hardware limits heat transfer rates for incidental unlimited contact as specified in Table 3.9.10-1 < TBR-EHP-10028-011 >.	analysis shows that all Artemis System hardware limits heat transfer rates for incidental and unlimited contact to the values defined in Table 3.9.10-1 Heat Transfer Rates < TBR-EHP-10028-011 >. Artemis System will provide this assessment as part of Phase 3 Safety Review Package approval.	
[EVASC.0105]	Sharp Edge and Protrusions Protection	Inspection	Verification shall consist of an inspection of Artemis Systems' flight hardware drawings to verify that the EVA crewmember is not subjected to harm from end item edges and protrusions as specified in Tables 3.9.11-1 / 3.9.11-2 and Figures 3.9.11-1 / 3.9.11-2.	Verification will be considered successful when the inspection of flight hardware drawings shows that Artemis System hardware protects the EVA crewmember from end item edges and protrusions on all hardware located within 0.61 m (24 in) of a translation path (reference requirements [EVASC.0010] and [EVASC.0159]), within a worksite envelope, and on each equipment item needing an EVA interface through compliance with Table 3.9.11-1 Edge, Corner, and Protrusion Criteria – Edge and In-Plane Corner Radii, Figure 3.9.11-1 Exposed Corner and Edges, and Table 3.9.11-2 Edge, Corner, and Protrusions Criteria – Protrusions and Outside Corners.	

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 178 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.5-1 ARTEMIS SYSTEM VERIFICATIONS

Requirement ID	Title	Method	Verification Objective	Success Criteria	Notes
				Artemis System will provide this assessment as part of Phase 3 Safety Review Package approval.	
[EVASC.0107]	Levers, Cranks, Hooks, and Controls	Inspection Analysis	<u>Inspection:</u> Verification shall consist of an inspection of Artemis Systems' flight hardware drawings to verify that levers, cranks, hooks, and controls are located to preclude a safety/snag hazard. <u>Analysis:</u> Verification shall consist of a hazard analysis performed to ensure that all Artemis System hardware levers, cranks, hooks, and controls are located to preclude a safety/snag hazard.	<u>Inspection:</u> Verification will be considered successful when the inspection of flight hardware drawings shows that Artemis System hardware containing levers, cranks, hooks, and controls are located where they cannot pinch, snag, cut, abrade, or provide kickback to the EVA crewmember appendages or equipment such as tethers. <u>Analysis:</u> Verification will be considered successful when the hazard analysis shows that all Artemis System hardware containing levers, cranks, hooks, and controls are located where they cannot pinch, snag, cut, abrade, or provide kickback to the EVA crewmember appendages or equipment such as tethers. Artemis System will provide this assessment as part of Phase 3 Safety Review Package approval.	
[EVASC.0109]	Exposed Holes or Slots	Inspection	Verification shall consist of an inspection of Artemis Systems' flight hardware drawings to verify that round or slotted exposed holes,	Verification will be considered successful when the inspection of flight hardware drawings shows that Artemis System hardware that	

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 179 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.5-1 ARTEMIS SYSTEM VERIFICATIONS

Requirement ID	Title	Method	Verification Objective	Success Criteria	Notes
			within the range of 13 mm (0.5 in) to 36 mm (1.4 in) in diameter are covered to preclude entrapment of an xEVA-gloved finger.	contains exposed holes that are round or slotted, within the range of 13 mm (0.5 in) to 36 mm (1.4 in) in diameter are covered to preclude entrapment of an xEVA-gloved finger. Artemis System will provide this assessment as part of Phase 3 Safety Review Package approval.	
[EVASC.0110]	Pinch Hazard Protection	Inspection Analysis	<u>Inspection:</u> Verification shall consist of an inspection of Artemis Systems' flight hardware drawings to verify EVA crewmembers are protected from pinch hazards. <u>Analysis:</u> Verification shall consist of an analysis performed to ensure that all Artemis System hardware protects the EVA crewmember from pinch hazards.	<u>Inspection:</u> Verification will be considered successful when the inspection of flight hardware drawings shows that Artemis System hardware protects the EVA crewmember from pinch hazards. <u>Analysis:</u> Verification will be considered successful when the analysis shows that all Artemis System hardware protects the EVA crewmember from pinch hazards. Artemis System will provide this assessment as part of Phase 3 Safety Review Package approval.	Test may be necessary based on hazard analysis (prior to Phase 3)
[EVASC.0111]	Equipment Clearance for Entrapment Hazard	Inspection Analysis	<u>Inspection:</u> Verification shall consist of an inspection of Artemis Systems' flight hardware drawings to verify provides clearance for equipment removal and	<u>Inspection:</u> Verification will be considered successful when the inspection of flight hardware drawings shows that Artemis System hardware provides	

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 180 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.5-1 ARTEMIS SYSTEM VERIFICATIONS

Requirement ID	Title	Method	Verification Objective	Success Criteria	Notes
			replacement that meets clearances identified in any or all of the requirements [EVASC.0109], [EVASC.0003], and [EVASC.0001]. <u>Analysis:</u> Verification shall consist of a hazard analysis performed to ensure that all Artemis System hardware provides clearance for equipment removal and replacement that meets clearances identified in any or all of the requirements [EVASC.0109], [EVASC.0003], and [EVASC.0001]/[EVASC.0171].	clearance for equipment removal and replacement to prevent the creating of a crew entrapment hazard that can be caused by insufficient clearance that results in not meeting any or all of the following requirements: [EVASC.0109], [EVASC.0003], and [EVASC.0001]. <u>Analysis:</u> Verification will be considered successful when the hazard analysis shows that all Artemis System hardware provides clearance for equipment removal and replacement to prevent the creating of a crew entrapment hazard that can be caused by insufficient clearance that results in not meeting any or all of the following requirements: [EVASC.0109], [EVASC.0003], and [EVASC.0001]/[EVASC.0171]. Artemis System will provide this assessment as part of Phase 3 Safety Review Package approval.	
[EVASC.0112]	Protection from Moving or Rotating Equipment	Inspection Analysis	<u>Inspection:</u> Verification shall consist of an inspection of Artemis Systems' flight hardware drawings to verify that protects EVA crewmembers from impacts from moving, rotating equipment, or	<u>Inspection:</u> Verification will be considered successful when the inspection of flight hardware drawings shows that Artemis System hardware protects EVA crewmembers from impacts from	Test may be necessary based on hazard analysis (prior to Phase 3)

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 181 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.5-1 ARTEMIS SYSTEM VERIFICATIONS

Requirement ID	Title	Method	Verification Objective	Success Criteria	Notes
			inadvertent deployment, separation, or jettison of hardware. <u>Analysis:</u> Verification shall consist of a hazard analysis performed to ensure that all Artemis System hardware protect EVA crewmembers from impacts from moving, rotating equipment, or inadvertent deployment, separation, or jettison of hardware.	moving, rotating equipment, or inadvertent deployment, separation, or jettison of hardware. <u>Analysis:</u> Verification will be considered successful when the hazard analysis shows that all Artemis System hardware protects EVA crewmembers from impacts from moving, rotating equipment, or inadvertent deployment, separation, or jettison of hardware Artemis System will provide this assessment as part of Phase 3 Safety Review Package approval.	
[EVASC.0113]	Emergency Ingress of Pressurized Habitable Environment	Analysis Demonstration	<u>Analysis:</u> Verification shall consist of a hazard analysis performed to ensure that Artemis System provides EVA translation path that aids the suited crewmembers in navigating the shortest paths to an airlock, including hardware, aids, and pathway markings or indications for vehicle ingress and egress so that an EVA crewmember can complete an emergency ingress as specified in Table 3.9.17-1 Emergency Ingress Capability. <u>Demonstration:</u> Verification shall consist of a demonstration of	<u>Analysis:</u> Verification will be considered successful when the hazard analysis shows that Artemis System provides EVA translation path that aids the suited crewmembers in navigating the shortest paths to an airlock, including hardware, aids, and pathway markings or indications for vehicle ingress and egress so that an EVA crewmember can complete an emergency ingress as specified in Table 3.9.17-1 Emergency Ingress Capability. Artemis System will provide this assessment as part of Phase 3	<u>Joint Integrated Verification:</u> Demonstration is critical, and NASA teams to ensure that providers work together to produce a high-fidelity demonstration.

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 182 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.5-1 ARTEMIS SYSTEM VERIFICATIONS

Requirement ID	Title	Method	Verification Objective	Success Criteria	Notes
			Artemis System EVA translation path aids the suited crewmembers in navigating the shortest paths to an airlock, including hardware, aids, and pathway markings or indications for vehicle ingress and egress so that an EVA crewmember can complete an emergency ingress as specified in Table 3.9.17-1 Emergency Ingress Capability.	Safety Review Package approval. <u>Demonstration:</u> Verification will be considered successful when the demonstration shows that Artemis System EVA translation path that aids the suited crewmembers in navigating the shortest paths to an airlock, including hardware, aids, and pathway markings or indications for vehicle ingress and egress so that an EVA crewmember can complete an emergency ingress as specified in Table 3.9.17-1 Emergency Ingress Capability.	
[EVASC.0169]	Emergency Ingress/ Egress of Unpressurized Mobile Asset	Analysis Demonstration	<u>Analysis:</u> Verification shall consist of a hazard analysis performed to ensure that Artemis System provides EVA crew translation paths, aids, accommodations, and restraints so that an EVA crewmember can complete an emergency ingress/egress with the total time of five minutes (threshold)/three minutes (goal). <u>Demonstration:</u> Verification shall consist of a demonstration of Artemis System vehicle to ensure that provides EVA crew translation paths, aids, accommodations, and restraints so that an EVA	<u>Analysis:</u> Verification will be considered successful when the hazard analysis shows that Artemis System provides EVA crew translation paths, aids, accommodations, and restraints so that an EVA crewmember can complete an emergency ingress/egress with the total time of five minutes (threshold)/three minutes (goal). <u>Demonstration:</u> Verification will be considered successful when the demonstration shows that Artemis System provides EVA crew translation paths, aids,	<u>Joint Integrated Verification:</u> Demonstration is critical, and NASA teams to ensure that providers work together to produce a high-fidelity demonstration.

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 183 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.5-1 ARTEMIS SYSTEM VERIFICATIONS

Requirement ID	Title	Method	Verification Objective	Success Criteria	Notes
			crewmember can complete an emergency ingress/egress with the total time of five minutes (threshold)/three minutes (goal).	accommodations, and restraints so that an EVA crewmember can complete an emergency ingress/egress with the total time of five minutes (threshold)/three minutes (goal).	
[EVASC.0140]	External Surface Cleanliness Level	Inspection	Verification shall consist of an inspection of Artemis Systems' flight hardware drawings to verify that hardware that interfaces with external surfaces of xEVA System hardware meet a VC-S level upon delivery as identified in Table 3.9.19-1 External Surface Cleanliness Level (or NASA approved cleanliness alternate standard with Cross-Program Chief Engineer approval), or higher.	Verification will be considered successful when the inspection of flight hardware drawings shows that Artemis System hardware that interfaces with external surfaces of xEVA System hardware meet a VC-S level upon delivery as identified in Table 3.9.19-1 External Surface Cleanliness Level (or NASA approved cleanliness alternate standard with Cross-Program Chief Engineer approval), or higher	
[EVASC.0115]	Tethering / Restraint	Inspection	Verification shall consist of an inspection of Artemis Systems' flight hardware drawings to verify that hardware items provide at least one means of positive restraint independent of EVA grasp (e.g., tether points, handrails, etc.).	Verification will be considered successful when the inspection of flight hardware drawings shows that Artemis System hardware items provide at least one means of positive restraint independent of EVA grasp (e.g., tether points, handrails, etc.).	
[EVASC.0116]	Crew Safety Tether Attachment	Inspection Analysis	<u>Inspection</u> : Verification shall consist of an inspection of Artemis Systems' flight hardware drawings to verify that handholds/handrails on microgravity translation paths intended for crewmember restraint	<u>Inspection</u> : Verification will be considered successful when the inspection of flight hardware drawings shows that Artemis System handholds/handrails on microgravity translation paths	

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 184 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.5-1 ARTEMIS SYSTEM VERIFICATIONS

Requirement ID	Title	Method	Verification Objective	Success Criteria	Notes
			are incorporated a safety tether attachment point into its support structure. <u>Analysis:</u> Verification shall consist of a load analysis performed to ensure that all Artemis System safety tether attachment points incorporated on support structure withstand the specified EVA crewmember safety tether loads identified in [EVASC.0050] Table 3.5.2-1 Microgravity EVA Crew-Induced Loads, Design Limit Load Type “Crew Safety Tether Attachment”, and only to be used in conjunction with the load-limiting crew safety tethers.	intended for crewmember restraint are incorporated a safety tether attachment point into its support structure. <u>Analysis:</u> Verification will be considered successful when the load analysis shows that all Artemis System safety tether attachment points on support structure withstand the specified EVA crewmember safety tether loads identified in [EVASC.0050] Table 3.5.2-1 Microgravity EVA Crew-Induced Loads, Design Limit Load Type “Crew Safety Tether Attachment”, and only to be used in conjunction with the load-limiting crew safety tethers.	
[EVASC.0161]	Crew Safety Restraint	Inspection Analysis	<u>Inspection:</u> Verification shall consist of an inspection of Artemis Systems' flight hardware drawings to verify that attachment points for Crew Safety Restraints, employed in areas of EVA operations where reaction to Crew Slips or Falls is necessary for safety of crewmember are incorporated into support structure. <u>Analysis:</u> Verification shall consist of a load analysis performed to ensure that all Artemis System	<u>Inspection:</u> Verification will be considered successful when the inspection of flight hardware drawings shows that Artemis System attachment points for Crew Safety Restraints, employed in areas of EVA operations where reaction to Crew Slips or Falls is necessary for safety of crewmember are incorporated into support structure. <u>Analysis:</u> Verification will be considered successful when the	

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 185 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.5-1 ARTEMIS SYSTEM VERIFICATIONS

Requirement ID	Title	Method	Verification Objective	Success Criteria	Notes
			attachment points for Crew Safety Restraints are incorporated into support structure that withstands the specified EVA crewmember loads identified in requirement [EVASC.0155], Table 3.5.3-1 Partial Gravity EVA Crew-Induced Loads, under Load/Energy Category “L12, Swinging/ or Falling”.	load analysis shows that all Artemis System attachment points for Crew Safety Restraints are incorporated into support structure that withstands the specified EVA crewmember loads identified in requirement [EVASC.0155], Table 3.5.3-1 Partial Gravity EVA Crew-Induced Loads, under Load/Energy Category “L12, Swinging/ or Falling”.	
[EVASC.0117]	Color For Crew Load-Rated Handhold/Handrails, Crew Safety Tethers And Safety Restraint Attachment Points	Inspection	Verification shall consist of an inspection of Artemis Systems' flight hardware drawings to verify that handholds/handrail, Crew Safety Tethers and Crew Safety Restraint attachment points intended for use by the crew are colored according to Table 3.9.23-1 Crew Load-Rated Colors < TBR-EHP-10028-012 >, where the color value has a contrast ratio of 5:1 or greater than the background.	Verification will be considered successful when the inspection of flight hardware drawings shows that Artemis System handholds/handrail, Crew Safety Tethers and Crew Safety Restraint attachment points intended for use by the crew are colored according to Table 3.9.23-1 Crew Load-Rated Colors < TBR-EHP-10028-012 >, where the color value has a contrast ratio of 5:1 or greater than the background.	
[EVASC.0118]	Manual Overrides	Inspection	Verification shall consist of an inspection of Artemis Systems' flight hardware drawings to verify includes a manual override that accommodates EVA operation with a hand tool for remotely commanded mechanisms requiring EVA as a backup.	Verification will be considered successful when the inspection shows that Artemis System hardware includes a manual override that accommodates EVA operation with a hand tool for remotely commanded mechanisms requiring EVA as a backup.	

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 186 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.5-1 ARTEMIS SYSTEM VERIFICATIONS

Requirement ID	Title	Method	Verification Objective	Success Criteria	Notes
[EVASC.0137]	Entanglement, Cable Clamps, Ducts, or Retractors	Inspection	Verification shall consist of an inspection of Artemis Systems' flight hardware drawings to verify cables, conductors, bundles and hoses within an EVA worksite or translation path are secured by means of restraints (e.g., clamps, wire ties, etc.) unless they are contained in a cable management system (wiring ducts or cable retractors).	Verification will be considered successful when the inspection of flight hardware drawings shows that Artemis System cables, conductors, bundles and hoses within an EVA worksite or translation path are secured by means of restraints (e.g., clamps, wire ties, etc.) unless they are contained in a cable management system (wiring ducts or cable retractors).	
[EVASC.0121]	Magnetic Field Limit	Test	Verification shall consist of a test of Artemis System associated with RE 101A and RE 101D values do not exceed those outlined in Table 3.10.1-1 Magnetic Field Limit for crew translation corridors, work areas, and incidental contact zones during EVA operations internal and external to the pressurized volume.	Verification will be considered successful when the test shows that Artemis System hardware RE 101A and RE 101D values do not exceed those outlined in Table 3.10.1-1 Magnetic Field Limit for crew translation corridors, work areas, and incidental contact zones during EVA operations internal and external to the pressurized volume.	
[EVASC.0123]	Voltage Potential Discharge	Analysis	Verification shall consist of a hazard analysis performed to ensure that all Artemis System meets performance requirements during and after exposure to voltage/current transients of <TBD-EHP-10028-014> that occur during contact with xEVA operations when the vehicle and xEVA suit or payloads come to equilibrium.	Verification will be considered successful when the hazard analysis shows that the Artemis System meets performance requirements during and after exposure to voltage/current transients of <TBD-EHP-10028-014> that occur during contact with xEVA operations when the vehicle and xEVA suit or payloads	

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 187 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.5-1 ARTEMIS SYSTEM VERIFICATIONS

Requirement ID	Title	Method	Verification Objective	Success Criteria	Notes
				come to equilibrium. Artemis System will provide this assessment as part of Phase 3 Safety Review Package approval.	
[EVASC.0156]	Lunar Surface Dust Tolerance	Analysis Test	<u>Analysis:</u> Verification shall consist of a hazard analysis performed to ensure that all Artemis Systems that need to tolerate dust (e.g., ECLSS, etc.) can survive the liberation of at least 200 grams of lunar surface dust in the cabin environment from the xEVA system after each two-crew lunar surface EVA. <u>Test:</u> Verification shall consist of a test performed to ensure that all Artemis Systems that need to tolerate dust (e.g., ECLSS, etc.) survives the liberation at least 200 grams of lunar surface dust in the cabin environment from the xEVA system after each two-crew lunar surface EVA.	<u>Analysis:</u> Verification will be considered successful when the hazard analysis shows that the Artemis System that need to tolerate dust (e.g., ECLSS, etc.) survives the liberation at least 200 grams of lunar surface dust in the cabin environment from the xEVA system after each two-crew lunar surface EVA. <u>Test:</u> Verification will be considered successful when the test shows that all Artemis Systems that need to tolerate dust (e.g., ECLSS, etc.) survives the liberation at least 200 grams of lunar surface dust in the cabin environment from the xEVA system after each two-crew lunar surface EVA.	
[EVASC.0168]	Lunar Surface Dust Mitigation	Analysis Test	<u>Analysis:</u> Verification shall consist of a hazard analysis performed to ensure that the Artemis System (xEVA System) employs hardware or other means to limit the amount of dust liberated in the cabin environment to less than 100 grams	<u>Analysis:</u> Verification will be considered successful when the hazard analysis shows that the Artemis System employs hardware or other means to limit the amount of dust liberated in the cabin environment to less than 100	Applicable to EVA Dev

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 188 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.5-1 ARTEMIS SYSTEM VERIFICATIONS

Requirement ID	Title	Method	Verification Objective	Success Criteria	Notes
			of lunar surface dust after each two-crew lunar surface EVA. <u>Test:</u> Verification shall consist of a test performed to ensure that Artemis System (xEVA System) employs hardware or other means to limit the amount of dust liberated in the cabin environment to less than 100 grams of lunar surface dust after each two-crew lunar surface EVA.	grams of lunar surface dust after each two-crew lunar surface EVA. <u>Test:</u> Verification will be considered successful when the test shows that the Artemis System employs hardware or other means to limit the amount of dust liberated in the cabin environment to less than 100 grams of lunar surface dust after each two-crew lunar surface EVA.	

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 189 of 280
Title: EHP xEVA System Compatibility Standards	

4.6 EHP VERIFICATIONS

The following table outlines the verification statements for EHP and includes an Inspection of Record for each requirement.

TABLE 4.6-1 EHP VERIFICATIONS

Requirement ID	Title	Method	Inspection of Records for every requirement	Verification Objective	Success Criteria
[EVASC.0001]	Microgravity Translation Path Hardware Protrusion	EVA Model Analysis	Inspection of Records will need to be performed by EHP to verify that the Artemis System provides a minimum of 1.1 m (43 in) diameter path with only the translation aids included within the designated path that complies with Figures 3.1.1-1 and 3.1.1-2. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	EHP CAMMP Team shall utilize the Artemis System's analytical CAD models to perform an integrated analysis to confirm that the Artemis System provides a translation path that consists of a minimum 1.1 m (43 in) diameter path with only the translation aids included within the microgravity translation path.	Verification will be considered successful when the EHP CAMMP Team analysis report confirms that the Artemis System is in compliance with the requirement.
[EVASC.0171]	Partial Gravity Translation Path Hardware Protrusion	EVA Model Analysis	Inspection of Records will need to be performed by EHP to verify that the Artemis System provides an unobstructed, other than translation aids, minimum of 1.1 m (43 in) diameter clearance centered on the suited crewmember as the crewmember traverses the designated path complies with Figures 3.1.2-1 and 3.1.2-2 for partial gravity. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	EHP CAMMP Team shall utilize the Artemis System's analytical CAD models to perform an integrated analysis to confirm that the Artemis System provides a translation path that consists of an unobstructed, other than translation aids, minimum of 1.1 m (43 in) diameter clearance centered on the suited crewmember as the crewmember traverses the designated path complies with Figures 3.1.2-1 and 3.1.2-2 for partial gravity.	Verification will be considered successful when the EHP CAMMP Team analysis report confirms that the Artemis System is in compliance with the requirement.

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 190 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.6-1 EHP VERIFICATIONS

Requirement ID	Title	Method	Inspection of Records for every requirement	Verification Objective	Success Criteria
[EVASC.0002]	Tool Interface Clearance	EVA Model Analysis VITO CEIT	Inspection of Records will need to be performed by EHP to verify that the Artemis System maintains a 7.6 cm (3.0 in) clearance between the gripping portions of a tool interface and any surrounding hardware or obstructions with a tool handle offset maximum of 35.5 cm (14.0 in) as defined in Figure 3.1.3-1. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	<u>EVA Model Analysis</u> - EHP CAMMP Team shall utilize the Artemis System's analytical CAD models to perform an integrated analysis to confirm that the Artemis System maintains the required tool clearance. <u>VITO CEIT</u> - VITO Team shall perform tool to interface fit checks on flight-like Artemis System hardware and issues final report. Any identified issues need to be fixed per approved procedure.	<u>EVA Model Analysis</u> - Verification will be considered successful when the EHP CAMMP Team analysis report confirms that the Artemis System is in compliance with the requirement. <u>VITO CEIT</u> - Verification will be considered successful when the VITO final report shows that the Artemis System hardware complies with the requirement and meets Figure 3.1.3-1.
[EVASC.0007]	Bolt Clearance	VITO CEIT	Inspection of Records will need to be performed by EHP to verify that the Artemis System provides 2.5 cm (1 in) clearance around the bolt head outer edge with a minimum tool head engagement height of 0.3 in. (0.7 cm) as defined in Figure 3.1.3-1. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	VITO Team shall perform tool to interface fit checks on flight-like Artemis System hardware and issues final report. Any identified issues need to be fixed per approved procedure.	Verification will be considered successful when the VITO final report shows that the Artemis System hardware complies with the requirement and meets Figure 3.1.3-1.
[EVASC.0003]	Gloved-Hand Clearance	EVA Model Analysis	Inspection of Records will need to be performed by EHP to verify that the Artemis System provides 2.5 cm (1 in) clearance around the bolt head outer edge with a minimum tool head	<u>EHP Model Analysis</u> - EHP CAMMP Team shall utilize the Artemis System's analytical CAD models to perform an integrated analysis to confirm	<u>EVA Model Analysis</u> - Verification will be considered successful when the EHP CAMMP Team analysis report confirms that the Artemis

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 191 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.6-1 EHP VERIFICATIONS

Requirement ID	Title	Method	Inspection of Records for every requirement	Verification Objective	Success Criteria
		VITO CEIT	engagement height of 0.3 in. (0.7 cm) as defined in Figure 3.1.5-1. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	that the Artemis System maintains the required clearances to accommodate an EVA gloved-hand. <u>VITO CEIT</u> - VITO Team shall perform xEVA glove to interface fit checks on flight-like Artemis System hardware and issues final report. Any identified issues need to be fixed per approved procedure.	System is in compliance with the requirement. <u>VITO CEIT</u> - Verification will be considered successful when the VITO final report shows that the Artemis System hardware complies with the requirement and meets Figure 3.1.5-1.
[EVASC.0004]	Handrail / Handhold Mounting Clearance	EVA Model Analysis VITO CEIT	Inspection of Records will need to be performed by EHP to verify that the Artemis System provides a minimum clearance distance between the lower surface of the handrail/handhold and the mounting surface of 5.72 cm (2.25 in) as specified in Figure 3.1.6-1. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	<u>EHP Model Analysis</u> - EHP CAMMP Team shall utilize the Artemis System's analytical CAD models to perform an integrated analysis to confirm that the Artemis System maintains the required handrail/handhold mounting clearances. <u>VITO CEIT</u> - VITO Team shall perform xEVA glove to interface fit checks on flight-like Artemis System hardware and issues final report. Any identified issues need to be fixed per approved procedure.	<u>EVA Model Analysis</u> - Verification will considered successful when the EHP CAMMP Team analysis report confirms that the Artemis System is in compliance with the requirement. <u>VITO CEIT</u> - Verification will be considered successful when the VITO final report shows that the Artemis System hardware complies with the requirement and meets Figure 3.1.6-1.

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 192 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.6-1 EHP VERIFICATIONS

Requirement ID	Title	Method	Inspection of Records for every requirement	Verification Objective	Success Criteria
[EVASC.0005]	Handrail/ Handhold Side Clearance	EVA Model Analysis VITO CEIT	Inspection of Records will need to be performed by EHP to verify that the Artemis System provides minimum 10 cm (4 in) clearance around all edges of the handrail/handholds as defined in Figure 3.1.7-1. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	<u>EHP Model Analysis</u> - EHP CAMMP Team shall utilize the Artemis System's analytical CAD models to perform an integrated analysis to confirm that the Artemis System maintains the required handrail/handhold mounting clearances. <u>VITO CEIT</u> - VITO Team shall perform xEVA glove to interface fit checks on flight-like Artemis System hardware and issues final report. Any identified issues need to be fixed per approved procedure.	<u>EVA Model Analysis</u> - Verification will be considered successful when the EHP CAMMP Team analysis report confirms that the Artemis System is in compliance with the requirement. <u>VITO CEIT</u> - Verification will be considered successful when the VITO final report shows that the Artemis System hardware complies with the requirement and meets Figure 3.1.7-1.

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 193 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.6-1 EHP VERIFICATIONS

Requirement ID	Title	Method	Inspection of Records for every requirement	Verification Objective	Success Criteria
[EVASC.0006]	APFR EVA Installation/ Removal Clearance	EVA Model Analysis VITO CEIT	Inspection of Records will need to be performed by EHP to verify that the Artemis System provides clearance around the passive WIF as defined in Figure 3.1.8-1. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	<u>EHP Model Analysis</u> - EHP CAMMP Team shall utilize the Artemis System's analytical CAD models to perform an integrated analysis to confirm that the Artemis System maintains the required WIF clearances for installation/removal of an APFR. <u>VITO CEIT</u> - VITO Team shall perform tool to interface fit checks on flight-like Artemis System hardware and issues final report. Any identified issues need to be fixed per approved procedure.	<u>EVA Model Analysis</u> - Verification will considered successful when the EHP CAMMP Team analysis report confirms that the Artemis System is in compliance with the requirement. <u>VITO CEIT</u> - Verification will be considered successful when the VITO final report shows that the Artemis System hardware complies with the requirement and meets Figure 3.1.8-1.
[EVASC.0152]	Contiguous EVA Translation Path	EVA Model Analysis	Inspection of Records will need to be performed by EHP to verify that the Artemis System provides EVA compatible accommodations supporting contiguous translation to adjacent berthed/docked vehicles. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	EHP Model Analysis - EHP CAMMP Team shall utilize the Artemis System's analytical CAD models to perform an integrated analysis to confirm that the Artemis System provides a contiguous translation path, including cases where the translation path crosses Artemis System boundaries, such as while one vehicle is berthed/docked to another.	EVA Model Analysis - Verification will considered successful when the EHP CAMMP Team analysis report confirms that the Artemis System is in compliance with the requirement.

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 194 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.6-1 EHP VERIFICATIONS

Requirement ID	Title	Method	Inspection of Records for every requirement	Verification Objective	Success Criteria
[EVASC.0008]	Translation Path	EVA Model Analysis	Inspection of Records will need to be performed by EHP to verify that the Artemis System provides an EVA translation path to all EVA worksites. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	EHP Model Analysis - EHP CAMMP Team shall utilize the Artemis System's analytical CAD models to perform an integrated analysis to confirm that the Artemis System provides a translation path to all EVA worksites.	EVA Model Analysis - Verification will considered successful when the EHP CAMMP Team analysis report confirms that the Artemis System is in compliance with the requirement.
[EVASC.0154]	Incapacitated Crewmember Interfaces		Inspection of Records will need to be performed by EHP to verify that the Artemis System provides hardware that aids a suited crewmember in the rescue of an incapacitated crewmember. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A
[EVASC.0010]	EVA Contact Hazards Preclusion Beyond Translation Corridor	EVA Model Analysis	Inspection of Records will need to be performed by EHP to verify that the Artemis System provides an EVA translation path that has no contact hazards within 0.61 m (24 in) beyond the translation corridor dimension. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	EHP Model Analysis - EHP CAMMP Team shall utilize the Artemis System's analytical CAD models to perform an integrated analysis to confirm that the Artemis System precludes EVA contact hazards within 0.61 m (24 in) beyond the translation corridor dimension.	EVA Model Analysis - Verification will considered successful when the EHP CAMMP Team analysis report confirms that the Artemis System is in compliance with the requirement.

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 195 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.6-1 EHP VERIFICATIONS

Requirement ID	Title	Method	Inspection of Records for every requirement	Verification Objective	Success Criteria
[EVASC.0159]	Partial Gravity EVA Contact Hazards Preclusion	EVA Model Analysis	Inspection of Records will need to be performed by EHP to verify that the Artemis System provides an EVA translation path that has no contact hazards within specified constraints. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	EHP Model Analysis - EHP CAMMP Team shall utilize the Artemis System's analytical CAD models to perform an integrated analysis to confirm that the Artemis System precludes EVA contact hazards within specified 3.2.5-1 Side View Partial Gravity EVA Contact Hazards Height.	EVA Model Analysis - Verification will considered successful when the EHP CAMMP Team analysis report confirms that the Artemis System is in compliance with the requirement.
[EVASC.0012]	Microgravity EVA Equipment Handling Features		Inspection of Records will need to be performed by EHP to verify that the Artemis System provides handling features for items to be transferred from one location to another. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A
[EVASC.0160]	Partial Gravity EVA Handling Capability <TBR-EHP-10028-003>		Inspection of Records will need to be performed by EHP to verify that the Artemis System limits properties of items to be transferred from one location to another. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 196 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.6-1 EHP VERIFICATIONS

Requirement ID	Title	Method	Inspection of Records for every requirement	Verification Objective	Success Criteria
[EVASC.0162]	Partial Gravity Design Parameter		Inspection of Records will need to be performed by EHP to verify that the Artemis System hardware limits the height a crewmember needs to step. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A
[EVASC.0135]	Pressurized xEVA Transfer through Airlock Hatch		Inspection of Records will need to be performed by EHP to verify that the Artemis System designs hatches that complies with the xEVA pressurized suited crewmembers. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A
[EVASC.0153]	EVA Lunar Hatch Dimensions		Inspection of Records will need to be performed by EHP to verify that the Artemis System limits dimensions of hardware that needs to be transferred in/out of the airlock. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 197 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.6-1 EHP VERIFICATIONS

Requirement ID	Title	Method	Inspection of Records for every requirement	Verification Objective	Success Criteria
[EVASC.0014]	Mounting Alignment Devices		Inspection of Records will need to be performed by EHP to verify that the Artemis System provides alignment devices for hardware that needs to be mounted during EVA. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A
[EVASC.0015]	Handling Alignment Marks		Inspection of Records will need to be performed by EHP to verify that the Artemis System provides alignment marks are provided for hardware requiring alignment or actuation. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A
[EVASC.0016]	Locking Device		Inspection of Records will need to be performed by EHP to verify that the Artemis System locking devices meet travel specification. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 198 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.6-1 EHP VERIFICATIONS

Requirement ID	Title	Method	Inspection of Records for every requirement	Verification Objective	Success Criteria
[EVASC.0124]	Over Center Device Force		Inspection of Records will need to be performed by EHP to verify that the Artemis System over center device meets minimum peak force. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A
[EVASC.0017]	Soft Dock or Soft Capture Mechanisms		Inspection of Records will need to be performed by EHP to verify that the Artemis System soft dock or soft capture is provided for hardware not held in place when crew installs it. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A
[EVASC.0125]	Soft Capture		Inspection of Records will need to be performed by EHP to verify that the Artemis System soft capture is used in the identified cases for crew safety. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 199 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.6-1 EHP VERIFICATIONS

Requirement ID	Title	Method	Inspection of Records for every requirement	Verification Objective	Success Criteria
[EVASC.0126]	Soft Dock Force		Inspection of Records will need to be performed by EHP to verify that the Artemis System hardware that requires soft dock force meets specification for respective hardware masses. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A
[EVASC.0024]	Unique Labeling	VITO CEIT	Inspection of Records will need to be performed by EHP to verify that the Artemis System hardware incorporates labels for translation paths, worksites, and other callouts that complies with EVA-EXP-0049, EVA Office Exploration EVA (xEVA) System Labeling Guidelines. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	VITO Team shall perform tool to interface fit checks on flight-like Artemis System hardware and issues final report. Any identified issues need to be fixed per approved procedure.	Verification will be considered successful when the VITO final report shows that the Artemis System hardware complies with the requirement.
[EVASC.0025]	Hand Tool Operations	VITO CEIT	Inspection of Records will need to be performed by EHP to verify that the Artemis System hardware that requires hand tool operations can be operated as indicated in requirement. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	VITO Team shall perform tool to interface fit checks on flight-like Artemis System hardware and issues final report. Any identified issues need to be fixed per approved procedure.	Verification will be considered successful when the VITO final report shows that the Artemis System hardware complies with the requirement.

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 200 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.6-1 EHP VERIFICATIONS

Requirement ID	Title	Method	Inspection of Records for every requirement	Verification Objective	Success Criteria
[EVASC.0135a]	Hand Ratcheting	VITO CEIT	Inspection of Records will need to be performed by EHP to verify that the Artemis System hardware that requires use of hand ratcheting provides necessary sweep angle. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	VITO Team shall perform tool to interface fit checks on flight-like Artemis System hardware to validate manual ratcheting angle of at least 90 ° on all EVA fasteners and issues final report. Any identified issues need to be fixed per approved procedure.	Verification will be considered successful when the VITO final report shows that the Artemis System hardware complies with the requirement.
[EVASC.0026]	Throw Angle		Inspection of Records will need to be performed by EHP to verify that the Artemis System hardware that requires use of hand ratcheting provides necessary sweep angle. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A
[EVASC.0127]	Ratchet Mechanism		Inspection of Records will need to be performed by EHP to verify that the Artemis System hardware that requires a ratchet wrench has a fine-tooth mechanism that allows engagement at every 10° maximum. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 201 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.6-1 EHP VERIFICATIONS

Requirement ID	Title	Method	Inspection of Records for every requirement	Verification Objective	Success Criteria
[EVASC.0128]	Back Drive Torque		Inspection of Records will need to be performed by EHP to verify that the Artemis System hardware (EVA wrenches has a back drive torque while operating with a pressurized glove. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A
[EVASC.0129]	Running Torque		Inspection of Records will need to be performed by EHP to verify that the Artemis System hardware EVA fasteners requiring use of EVA ratchet wrench has a running torque greater than the maximum back drive torque while operating with a pressurized glove. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 202 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.6-1 EHP VERIFICATIONS

Requirement ID	Title	Method	Inspection of Records for every requirement	Verification Objective	Success Criteria
[EVASC.0027]	Pressurized Glove Compatibility	VITO CEIT	Inspection of Records will need to be performed by EHP to verify that the Artemis System hardware EVA fasteners requiring use of EVA ratchet wrench has a running torque greater than the maximum back drive torque while operating with a pressurized glove. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	VITO Team shall perform tool to interface fit checks on flight-like Artemis System hardware and issues final report. Any identified issues need to be fixed per approved procedure.	Verification will be considered successful when the VITO final report shows that the Artemis System hardware complies with the requirement.
[EVASC.0087]	Inadvertent Actuation of EVA -Actuated Controls		Inspection of Records will need to be performed by EHP to verify that the Artemis System hardware prevents inadvertent actuation on all EVA-actuated controls. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A
[EVASC.0029]	Handrail/ Handhold Cross-Section		Inspection of Records will need to be performed by EHP to verify that the Artemis System hardware contains handrails/handholds that meet EVA cross-sections. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 203 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.6-1 EHP VERIFICATIONS

Requirement ID	Title	Method	Inspection of Records for every requirement	Verification Objective	Success Criteria
[EVASC.0030]	Handrail/ Handhold/ Handle Length		Inspection of Records will need to be performed by EHP to verify that the Artemis System hardware contains handrails/handholds/handles that meet EVA handrail/handhold/handles lengths. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A
[EVASC.0031]	Handrail/ Handhold Accessibility	EVA Model Analysis VITO CEIT	Inspection of Records will need to be performed by EHP to verify that the Artemis System leaves EVA handrails/handholds installed in EVA translation and at EVA worksites unexposed for use by crew. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	<u>EHP Model Analysis</u> - EHP CAMMP Team shall utilize the Artemis System's analytical CAD models to perform an integrated analysis to confirm that the Artemis System maintains the required handrail/handhold mounting clearances. <u>VITO CEIT</u> - VITO Team shall perform tool to interface fit checks on flight-like Artemis System hardware and issues final report. Any identified issues need to be fixed per approved procedure.	<u>EVA Model Analysis</u> - Verification will be considered successful when the EHP CAMMP Team analysis report confirms that the Artemis System is in compliance with the requirement. <u>VITO CEIT</u> - Verification will be considered successful when the VITO final report shows that the Artemis System hardware complies with the requirement.

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 204 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.6-1 EHP VERIFICATIONS

Requirement ID	Title	Method	Inspection of Records for every requirement	Verification Objective	Success Criteria
[EVASC.0032]	Continuous Handrail/ Handhold Spacing	EVA Model Analysis	Inspection of Records will need to be performed by EHP to verify that the Artemis System spaces EVA handrails/handholds installed in EVA translation paths and the integral safety tether attach points do not exceed specifications. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	EHP CAMMP Team shall utilize the Artemis System's analytical CAD models to perform an integrated analysis to confirm that the Artemis System maintains the required handrail/handhold mounting clearances.	Verification will be considered successful when the EHP CAMMP Team analysis report confirms that the Artemis System is in compliance with the requirement and meets Figure 3.3.12.4-1.
[EVASC.0144]	EVA Egress Handrail Spacing	EVA Model Analysis	Inspection of Records will need to be performed by EHP to verify that the Artemis System spaces EVA handrails/handholds required for EVA egress to be within specifications. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	EHP CAMMP Team shall utilize the Artemis System's analytical CAD models to perform an integrated analysis to confirm that the Artemis System maintains the required handrail/handhold mounting clearances.	Verification will be considered successful when the EHP CAMMP Team analysis report confirms that the Artemis System is in compliance with the requirement.
[EVASC.0035]	Work Envelope	EVA Model Analysis	Inspection of Records will need to be performed by EHP to verify that the Artemis System provides one-handed or two-handed glove operations at an EVA worksite that meets the specified gravity field volumes. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	EHP CAMMP Team shall utilize the Artemis System's analytical CAD models to perform an integrated analysis to confirm that the Artemis System maintains the required work envelope for both gravity fields.	Verification will be considered successful when the EHP CAMMP Team analysis report confirms that the Artemis System is in compliance with the requirement.

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 205 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.6-1 EHP VERIFICATIONS

Requirement ID	Title	Method	Inspection of Records for every requirement	Verification Objective	Success Criteria
[EVASC.0170]	Working Volume Operation of Vehicle	EVA Model Analysis	Inspection of Records will need to be performed by EHP to verify that the Artemis System provides crewmember working volume for operation of vehicle on the lunar surface within <TBD-EHP-10028-019> . This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	EHP CAMMP Team shall utilize the Artemis System's analytical CAD models to perform an integrated analysis to confirm that the Artemis System provides crewmember working volume for operation of vehicle on the lunar surface within <TBD-EHP-10028-019> .	Verification will be considered successful when the EHP CAMMP Team analysis report confirms that the Artemis System is in compliance with the requirement.
[EVASC.0157]	EVA Egress/ Ingress Methods	EVA Model Analysis	Inspection of Records will need to be performed by EHP to verify that the Artemis System provides EVA egress/ingress methods from/to the vehicle that accommodate the pressurized crewmembers' mobility. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	EHP CAMMP Team shall utilize the Artemis System's analytical CAD models to perform an integrated analysis to confirm that the Artemis System provides EVA egress/ingress methods from/to the vehicle that accommodate the pressurized crewmembers' mobility.	EVA Model Analysis - Verification will considered successful when the EHP CAMMP Team analysis report confirms that the Artemis System is in compliance with the requirement.
[EVASC.0036]	Crew Aids for Stability	EVA Model Analysis	Inspection of Records will need to be performed by EHP to verify that the Artemis System provides crew stability aids that enable performance of EVA tasks. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	EHP CAMMP Team shall utilize the Artemis System's analytical CAD models to perform an integrated analysis to confirm that the Artemis System provides crew stability aids that enable performance of EVA tasks within the specified values.	Verification will be considered successful when the EHP CAMMP Team analysis report confirms that the Artemis System is in compliance with the requirement.

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 206 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.6-1 EHP VERIFICATIONS

Requirement ID	Title	Method	Inspection of Records for every requirement	Verification Objective	Success Criteria
[EVASC.0037]	EVA Worksites Using Robotic Work Platform	EVA Model Analysis	Inspection of Records will need to be performed by EHP to verify that the Artemis System provides a robotic work platform that reacts forces/moments for stability of the EVA crewmember. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	EHP CAMMP Team shall utilize the Artemis System's analytical CAD models to perform an integrated analysis to confirm that the Artemis System an EVA worksite that requires a robotic work platform for crew stability meets specifications.	Verification will be considered successful when the EHP CAMMP Team analysis report confirms that the Artemis System is in compliance with the requirement.
[EVASC.0038]	Lubricant Use		Inspection of Records will need to be performed by EHP to verify that the Artemis System uses lubricants that will not transfer to the Artemis System xEVA glove/suit or provides labels designating as a "not touch" area. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A
[EVASC.0040]	Worksite Mobility	EVA Model Analysis	Inspection of Records will need to be performed by EHP to verify that the Artemis System provides foot restraints that accommodate the EVA crewmember. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	EHP CAMMP Team shall utilize the Artemis System's analytical CAD models to perform an integrated analysis to confirm that the Artemis System provides foot restraints that accommodate the crewmember's range of motion.	Verification will be considered successful when the EHP CAMMP Team analysis report confirms that the Artemis System is in compliance with the requirement.

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 207 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.6-1 EHP VERIFICATIONS

Requirement ID	Title	Method	Inspection of Records for every requirement	Verification Objective	Success Criteria
[EVASC.0041]	Worksite Field of View	EVA Model Analysis	Inspection of Records will need to be performed by EHP to verify that the Artemis System provides EVA worksites that have an unobstructed view. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	EHP CAMMP Team shall utilize the Artemis System's analytical CAD models to perform an integrated analysis to confirm that the Artemis System provides EVA worksites that are not obstructed by surrounding hardware.	Verification will be considered successful when the EHP CAMMP Team analysis report confirms that the Artemis System is in compliance with the requirement.
[EVASC.0042]	Working Volume	EVA Model Analysis	Inspection of Records will need to be performed by EHP to verify that the Artemis System provides EVA worksites that have the specified field of gravity work volume. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	EHP CAMMP Team shall utilize the Artemis System's analytical CAD models to perform an integrated analysis to confirm that the Artemis System provides a work volume that meets Figure 3.4.2.2-1 and Figure 3.4.2.2-2.	Verification will be considered successful when the EHP CAMMP Team analysis report confirms that the Artemis System is in compliance with the requirement.
[EVASC.0148]	Dedicated EVA Worksite	EVA Model Analysis	Inspection of Records will need to be performed by EHP to verify that the Artemis System provides dedicated EVA worksites that are within the specified conditions. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	EHP CAMMP Team shall utilize the Artemis System's analytical CAD models to perform an integrated analysis to confirm that the Artemis System provides dedicated EVA worksites that meet the specified conditions.	Verification will be considered successful when the EHP CAMMP Team analysis report confirms that the Artemis System is in compliance with the requirement.

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 208 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.6-1 EHP VERIFICATIONS

Requirement ID	Title	Method	Inspection of Records for every requirement	Verification Objective	Success Criteria
[EVASC.0149]	Robotic Assisted EVA Worksite	EVA Model Analysis	Inspection of Records will need to be performed by EHP to verify that the Artemis System provides robotic assisted EVA worksites that are within the specified conditions. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	EHP CAMMP Team shall utilize the Artemis System's analytical CAD models to perform an integrated analysis to confirm that the Artemis System provides robotic assisted EVA worksites that meet the specified conditions.	Verification will be considered successful when the EHP CAMMP Team analysis report confirms that the Artemis System is in compliance with the requirement.
[EVASC.0150]	Free-Float Worksites	EVA Model Analysis	Inspection of Records will need to be performed by EHP to verify that the Artemis System provides free float worksites that are located within the one-handed envelope defined in Figure 3.3.12.6.1-1. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	EHP CAMMP Team shall utilize the Artemis System's analytical CAD models to perform an integrated analysis to confirm that the Artemis System provides free float worksites that are located within Figure 3.3.12.6.1-1 utilizing the one-handed envelope.	Verification will be considered successful when the EHP CAMMP Team analysis report confirms that the Artemis System is in compliance with the requirement.
[EVASC.0045]	Installation of Passive WIF Sockets	EVA Model Analysis	Inspection of Records will need to be performed by EHP to verify that the Artemis System provides handrails/handholds with 0.61 m (24 in) of the passive WIF for all EVA worksites that necessitate a WIF for restraining a crewmember. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	EHP CAMMP Team shall utilize the Artemis System's analytical CAD models to perform an integrated analysis to confirm that the Artemis System provides passive WIFs that includes handholds/handrails within 0.61 m (24 in) of the passive WIF.	Verification will be considered successful when the EHP CAMMP Team analysis report confirms that the Artemis System is in compliance with the requirement.

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 209 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.6-1 EHP VERIFICATIONS

Requirement ID	Title	Method	Inspection of Records for every requirement	Verification Objective	Success Criteria
[EVASC.0046]	Foot Restraint Ingress/Egress Aids	EVA Model Analysis	Inspection of Records will need to be performed by EHP to verify that the Artemis System provides EVA Worksites that contain WIFs includes crew mobility aids with specified distances that supports ingress/egress of the foot restraint. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	EHP CAMMP Team shall utilize the Artemis System's analytical CAD models to perform an integrated analysis to confirm that the Artemis System provides EVA Worksites that contain WIFs includes crew mobility aids with specified distances defined in Figure 3.4.7-1 Ingress and Egress Aids for Foot Restraint.	Verification will be considered successful when the EHP CAMMP Team analysis report confirms that the Artemis System is in compliance with the requirement.
[EVASC.0047]	Visual Contrast at Worksite		Inspection of Records will need to be performed by EHP to verify that the Artemis System provides visual contrast at the EVA worksite such that crewmembers are able to read in extreme lighting conditions with a contrast ratio of 5:1. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 210 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.6-1 EHP VERIFICATIONS

Requirement ID	Title	Method	Inspection of Records for every requirement	Verification Objective	Success Criteria
[EVASC.0033]	Color for EVA Handling and EVA Equipment Tether Points		Inspection of Records will need to be performed by EHP to verify that the Artemis System provides handholds or EVA equipment tether points are colored according to Table 3.5.1.1-1 EVA Handling and EVA Equipment Tether Point Colors <TBR-EHP-10028-012> . These types of handholds/tethers do not require additional loading constraints as it only is applicable for hardware to be manipulated by crew. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A
[EVASC.0048]	EVA Equipment Tether Point		Inspection of Records will need to be performed by EHP to verify that the Artemis System provides non-soft goods EVA equipment tether points included on hardware that meet specifications. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 211 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.6-1 EHP VERIFICATIONS

Requirement ID	Title	Method	Inspection of Records for every requirement	Verification Objective	Success Criteria
[EVASC.0134]	Soft Goods Tether Point Loads		Inspection of Records will need to be performed by EHP to verify that the Artemis System provides soft goods tether points included on hardware that meet specifications. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A
[EVASC.0050]	Microgravity EVA Crew-Induced Loads		Inspection of Records will need to be performed by EHP to verify that the Artemis System withstands specified EVA crew-induced loads as defined in Table 3.5.2-1 Microgravity EVA Crew-Induced Loads. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A
[EVASC.0155]	Partial Gravity EVA Crew-Induced Loads		Inspection of Records will need to be performed by EHP to verify that the Artemis System hardware withstands specified EVA crew-induced loads as defined in Table 3.5.3-1 Partial Gravity EVA Crew-Induced Loads <TBR-EHP-10028-002> . This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 212 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.6-1 EHP VERIFICATIONS

Requirement ID	Title	Method	Inspection of Records for every requirement	Verification Objective	Success Criteria
[EVASC.0018]	EVA Actuated Loads		Inspection of Records will need to be performed by EHP to verify that the Artemis System items to be actuated by EVA crewmembers are operable by the loads defined in Table 3.5.4-1 Miscellaneous Crew Induced Limit Loads. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A
[EVASC.0164]	EVA Mechanical Feedback		Inspection of Records will need to be performed by EHP to verify that the Artemis System EVA mechanical feedback actuation force meets specifications. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 213 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.6-1 EHP VERIFICATIONS

Requirement ID	Title	Method	Inspection of Records for every requirement	Verification Objective	Success Criteria
[EVASC.0143]	WIF Structural Loading		Inspection of Records will need to be performed by EHP to verify that the Artemis System maintain positive margins of safety for loads at the WIF to structure interface of 1220 N (274 lbf) in any direction and a moment of 474 Nm (4200 lbf-in) in any direction, simultaneously, as defined in Figure 3.5.6-1 Top Mounted WIF Worksite Half Loading Criteria and to be used only in conjunction with a limiting APFR. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A
[EVASC.0053]	One-Handed EVA Operable Connectors	EVA Model Analysis	Inspection of Records will need to be performed by EHP to verify that the Artemis System provides EVA connectors that meet specifications one-handed operation. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	EHP CAMMP Team shall utilize the Artemis System's analytical CAD models to perform an integrated analysis to confirm that the Artemis System provides ability for EVA crewmembers to operate EVA connectors one-handed as specified.	Verification will be considered successful when the EHP CAMMP Team analysis report confirms that the Artemis System is in compliance with the requirement.

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 214 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.6-1 EHP VERIFICATIONS

Requirement ID	Title	Method	Inspection of Records for every requirement	Verification Objective	Success Criteria
[EVASC.0054]	Connector Accessibility	EVA Model Analysis	Inspection of Records will need to be performed by EHP to verify that the Artemis System provides individual EVA connectors that can be operated without having to remove/replace another connector. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	EHP CAMMP Team shall utilize the Artemis System's analytical CAD models to perform an integrated analysis to confirm that the Artemis System provides ability for EVA crewmembers to operate individual EVA connectors without having to remove/replace another connector.	Verification will be considered successful when the EHP CAMMP Team analysis report confirms that the Artemis System is in compliance with the requirement.
[EVASC.0138]	Gloved-Hand Connector Accessibility	EVA Model Analysis VITO CEIT	Inspection of Records will need to be performed by EHP to verify that the Artemis System provides gloved-hand access around any point of the EVA connector as specified in Figure 3.1.5-1. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	<u>EHP Model Analysis</u> - EHP CAMMP Team shall utilize the Artemis System's analytical CAD models to perform an integrated analysis to confirm that the Artemis System provides gloved-hand access around any point of the EVA connector as specified in Figure 3.1.5-1. <u>VITO CEIT</u> - VITO Team shall perform tool to interface fit checks on flight-like Artemis System hardware and issues final report. Any identified issues need to be fixed per approved procedure.	<u>EVA Model Analysis</u> - Verification will be considered successful when the EHP CAMMP Team analysis report confirms that the Artemis System is in compliance with the requirement. <u>VITO CEIT</u> - Verification will be considered successful when the VITO final report shows that the Artemis System hardware complies with the requirement.
[EVASC.0055]	Incorrect Connector		Inspection of Records will need to be performed by EHP to verify that the Artemis System provides EVA	N/A	N/A

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 215 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.6-1 EHP VERIFICATIONS

Requirement ID	Title	Method	Inspection of Records for every requirement	Verification Objective	Success Criteria
	Mating Prevention		connectors on a vehicle or at a worksite can't be incorrectly mated or cross-connected. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.		
[EVASC.0056]	Connector Coding		Inspection of Records will need to be performed by EHP to verify that the Artemis System provides unique labels or codes on both sides of EVA connectors that are to be mated that are visible to the crew when both mated and demated. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A
[EVASC.0057]	Connector Protective Caps		Inspection of Records will need to be performed by EHP to verify that the Artemis System provides protective caps for EVA connectors that prevent physical damage and contamination from the specified conditions. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 216 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.6-1 EHP VERIFICATIONS

Requirement ID	Title	Method	Inspection of Records for every requirement	Verification Objective	Success Criteria
[EVASC.0058]	Connector Rotational Travel		Inspection of Records will need to be performed by EHP to verify that the Artemis System provides EVA connectors with a rotational travel that has a limited maximum of 180° for installation or removal of twist-on connectors. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A
[EVASC.0059]	Connector Locking Feature		Inspection of Records will need to be performed by EHP to verify that the Artemis System provides EVA connector plugs that include a locking feature to prevent inadvertent demate. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A
[EVASC.0060]	Visible Connector Alignment Markings	VITO CEIT	Inspection of Records will need to be performed by EHP to verify that the Artemis System provides visible alignment markings or other feature on EVA connectors to ensure proper alignment during mating/demating. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	VITO Team shall perform tool to interface fit checks on flight-like Artemis System hardware and issues final report. Any identified issues need to be fixed per approved procedure.	Verification will be considered successful when the VITO final report shows that the Artemis System hardware complies with the requirement.

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 217 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.6-1 EHP VERIFICATIONS

Requirement ID	Title	Method	Inspection of Records for every requirement	Verification Objective	Success Criteria
[EVASC.0061]	Connector Locking Ring		Inspection of Records will need to be performed by EHP to verify that the Artemis System provides EVA connectors with locking rings that do not turn independently from the connector body. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A
[EVASC.0062]	Connector Mating Status		Inspection of Records will need to be performed by EHP to verify that the Artemis System provides EVA connectors that indicate full mate/demate status. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A
[EVASC.0052]	Mating/ Demating of Powered Connectors		Inspection of Records will need to be performed by EHP to verify that the Artemis System EVA connectors do not provide shock hazards to the EVA crewmember. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 218 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.6-1 EHP VERIFICATIONS

Requirement ID	Title	Method	Inspection of Records for every requirement	Verification Objective	Success Criteria
[EVASC.0065]	Connectors - Electrical Bonding		Inspection of Records will need to be performed by EHP to verify that the Artemis System provides Proper bonding ensures the majority of harmful electrical current under hazardous fault conditions is shunted away from the EVA crewmember, preventing a shock hazard. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A
[EVASC.0066]	Scoop-Proof Electrical Connector Design		Inspection of Records will need to be performed by EHP to verify that the Artemis System provides EVA connectors that prevents the electrical pins from inadvertently contacting the opposing connector half during mating/demating. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 219 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.6-1 EHP VERIFICATIONS

Requirement ID	Title	Method	Inspection of Records for every requirement	Verification Objective	Success Criteria
[EVASC.0063]	Connector Grip Point		Inspection of Records will need to be performed by EHP to verify that the Artemis System provides EVA connectors with backshells that can be engaged/disengaged by a pressurized gloved hand. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A
[EVASC.0131]	Arc Containment		Inspection of Records will need to be performed by EHP to verify that the Artemis System provides EVA electrical connectors that confine and isolate electrical arcs and sparks during mating/demating. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A
[EVASC.0132]	Uncontrolled Motion of Flex Hoses		Inspection of Records will need to be performed by EHP to verify that the Artemis System provides EVA fluid hoses that are restrained so as to not provide injury to the EVA crew. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A
[EVASC.0133]	Venting Feature		Inspection of Records will need to be performed by EHP to verify that the Artemis System provides method for	N/A	N/A

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 220 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.6-1 EHP VERIFICATIONS

Requirement ID	Title	Method	Inspection of Records for every requirement	Verification Objective	Success Criteria
			removing pressure before demating EVA operable fluid connectors and lines. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.		
[EVASC.0145]	Indication of Pressure Flow - Nonbrazed or Nonwelded		Inspection of Records will need to be performed by EHP to verify that the Artemis System provides positive indication of the gas pressure/fluid flow to verify that the line is passive before disconnection of non-self-sealing connectors. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A
[EVASC.0068]	Captive Fastener Design	VITO CEIT	Inspection of Records will need to be performed by EHP to verify that the Artemis System provides external EVA operable fasteners are captive or provide restraint features to mitigate impact hazards to the crew and vehicle. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	VITO Team shall perform tool to interface fit checks on flight-like Artemis System hardware and issues final report. Any identified issues need to be fixed per approved procedure.	Verification will be considered successful when the VITO final report shows that the Artemis System hardware complies with the requirement.

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 221 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.6-1 EHP VERIFICATIONS

Requirement ID	Title	Method	Inspection of Records for every requirement	Verification Objective	Success Criteria
[EVASC.0069]	Prohibited Locking Methods for Fasteners		Inspection of Records will need to be performed by EHP to verify that the Artemis System does not include any of the specified prohibited fasteners as a method for locking EVA operable fasteners. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A
[EVASC.0067]	One-Handed Fastener Operation		Inspection of Records will need to be performed by EHP to verify that the Artemis System provides EVA operable fasteners (those intended for manipulation without the use of a tool) that can be operated one handed. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A
[EVASC.0072]	Scalloped Knobs		Inspection of Records will need to be performed by EHP to verify that the Artemis System provides scalloped knobs for actuation by the pressurized EVA glove with no fewer than four (4) and no more than eight (8) scallops. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 222 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.6-1 EHP VERIFICATIONS

Requirement ID	Title	Method	Inspection of Records for every requirement	Verification Objective	Success Criteria
[EVASC.0073]	Knobs Head Diameter and Height		Inspection of Records will need to be performed by EHP to verify that the Artemis System provides knobs for EVA suited pressurized gloved hand operation to have head diameters between the ranges of 3.8 cm - 5.1 cm (1.5 in. – 2.0 in) with a minimum head height of 1.9 cm 0.75 in). This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A
[EVASC.0074]	EVA Hand-Actuated Fasteners Contingency Operation	VITO CEIT	Inspection of Records will need to be performed by EHP to verify that the Artemis System external hexagonal drive interface is provided for rotational fasteners to be operated with a hand tool as specified in Table 3.7.3.4-1, EVA External Hexagonal Drive Interface. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	VITO Team shall perform tool to interface fit checks on flight-like Artemis System hardware and issues final report. Any identified issues need to be fixed per approved procedure.	Verification will be considered successful when the VITO final report shows that the Artemis System hardware complies with the requirement.
[EVASC.0075]	EVA PIP Pin Grasping Interface	VITO CEIT	Inspection of Records will need to be performed by EHP to verify that the Artemis System provides EVA PIP Pin grasping interface as specified. This inspection of records shall include a review of the Artemis Systems and Artemis System	VITO Team shall perform tool to interface fit checks on flight-like Artemis System hardware and issues final report. Any identified issues need to be fixed per approved procedure.	Verification will be considered successful when the VITO final report shows that the Artemis System hardware complies with the requirement.

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 223 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.6-1 EHP VERIFICATIONS

Requirement ID	Title	Method	Inspection of Records for every requirement	Verification Objective	Success Criteria
			vendors VCNs for compliance with the requirement.		
[EVASC.0076]	PIP Pin Restraint Tether		Inspection of Records will need to be performed by EHP to verify that the Artemis System provides a restraint tether is attached to the body (handle) of the pin and not to the activation mechanism. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A
[EVASC.0078]	Driver-Type Push Force Tools		Inspection of Records will need to be performed by EHP to verify that the Artemis System provides EVA fasteners that do not require a driver-type hand tool with a push force greater than 9N (2 lbf) to maintain tool engagement while providing torque. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 224 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.6-1 EHP VERIFICATIONS

Requirement ID	Title	Method	Inspection of Records for every requirement	Verification Objective	Success Criteria
[EVASC.0079]	EVA Fastener Head	VITO CEIT	Inspection of Records will need to be performed by EHP to verify that the Artemis System provides EVA-actuated bolts that meet specifications. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	VITO Team shall perform tool to interface fit checks on flight-like Artemis System hardware and issues final report. Any identified issues need to be fixed per approved procedure.	Verification will be considered successful when the VITO final report shows that the Artemis System hardware complies with the requirement.
[EVASC.0084]	Fastener Deep Socket		Inspection of Records will need to be performed by EHP to verify that the Artemis System provides an extra-deep well in the EVA socket and/or a wobble feature helps mitigate stripping concerns. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A
[EVASC.0151]	Fastener Torque Data		N/A	N/A	N/A
[EVASC.0080]	Power and Hand Tool Operability		Inspection of Records will need to be performed by EHP to verify that the Artemis System provides EVA fasteners installed with power tools are removable and/or installable with a hand-operated tool. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 225 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.6-1 EHP VERIFICATIONS

Requirement ID	Title	Method	Inspection of Records for every requirement	Verification Objective	Success Criteria
[EVASC.0141]	Fastener Gauge/ Measurement	VITO CEIT	Inspection of Records will need to be performed by EHP to verify that the Artemis System provides EVA bolt heads that are gauged/measured to ensure they meet dimensional requirements at the minimum to maximum operational temperatures prior to installation into assembly. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	VITO Team shall perform tool to interface fit checks on flight-like Artemis System hardware and issues final report. Any identified issues need to be fixed per approved procedure.	Verification will be considered successful when the VITO final report shows that the Artemis System hardware complies with the requirement.
[EVASC.0083]	Fastener Alignment Features		Inspection of Records will need to be performed by EHP to verify that the Artemis System provides EVA fastener alignment features for initial engagement to preclude cross-threading. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 226 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.6-1 EHP VERIFICATIONS

Requirement ID	Title	Method	Inspection of Records for every requirement	Verification Objective	Success Criteria
[EVASC.0085]	Battery Replacement	EVA Model Analysis	Inspection of Records will need to be performed by EHP to verify that the Artemis System provides batteries that can be changeout at the EVA and not need to be brought back into the vehicle. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	EHP CAMMP Team shall utilize the Artemis System's analytical CAD models to perform an integrated analysis to confirm that the Artemis System provides batteries that can be changeout at the EVA worksite.	Verification will be considered successful when the EHP CAMMP Team analysis report confirms that the Artemis System is in compliance with the requirement.
[EVASC.0086]	Battery Voltage Monitoring		Inspection of Records will need to be performed by EHP to verify that the Artemis System provides battery-powered devices shall include battery monitoring capability before/during EVA usage to protect against catastrophic hazards and insufficient capacity to support mission time. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 227 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.6-1 EHP VERIFICATIONS

Requirement ID	Title	Method	Inspection of Records for every requirement	Verification Objective	Success Criteria
[EVASC.0028]	Auxiliary Controls	EVA Model Analysis	Inspection of Records will need to be performed by EHP to verify that the Artemis System provides auxiliary controls are positioned for operation by an EVA-gloved hand to activate the control without disturbing the tool/fastener position. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	EHP CAMMP Team shall utilize the Artemis System's analytical CAD models to perform an integrated analysis to confirm that the Artemis System provides auxiliary controls are positioned for EVA crewmembers to operate and do not interfere with the tool/fastener position	Verification will be considered successful when the EHP CAMMP Team analysis report confirms that the Artemis System is in compliance with the requirement.
[EVASC.0088]	EVA Tools		Inspection of Records will need to be performed by EHP to verify that the Artemis System utilizes EVA tools listed in EHP-10074, EHP EVA Tool List and Process for EVA Operations <TBR-EHP-10028-008. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A
[EVASC.0089]	Cable Routing/ Restraint		Inspection of Records will need to be performed by EHP to verify that the Artemis System cable routing/restraint for hardware precludes the use of EVA handling and translation aids as routing paths or restraints. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 228 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.6-1 EHP VERIFICATIONS

Requirement ID	Title	Method	Inspection of Records for every requirement	Verification Objective	Success Criteria
[EVASC.0090]	Cable Routing Spacing		Inspection of Records will need to be performed by EHP to verify that the Artemis System EVA-installed cable routing includes restraints spaced at a maximum of 0.61m (24 in). This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A
[EVASC.0091]	Paint		Inspection of Records will need to be performed by EHP to verify that the Artemis System provides evidence is provided that painted surfaces or features are able to withstand environmental conditions and wear-and-tear due to EVA contact (planned and inadvertent) on EVA worksites and within EVA translation paths/traverses, or areas on the vehicle which the EVA crewmember will access without flaking or contaminating the EVA crewmember. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A
[EVASC.0092]	Hazardous Material Control		Inspection of Records will need to be performed by EHP to verify that the Artemis System provide means of containment commensurate with the hazard level for toxic or hazardous	N/A	N/A

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 229 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.6-1 EHP VERIFICATIONS

Requirement ID	Title	Method	Inspection of Records for every requirement	Verification Objective	Success Criteria
			chemicals/materials to preclude contaminating an EVA crewmember or xEVA suit (hazardous to EVA crewmember, xEVA suit performance, or to IVA environment if brought back inside habitable volume). This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.		
[EVASC.0093]	Venting or Exposure Compatibility		Inspection of Records will need to be performed by EHP to verify that the Artemis System hardware to preclude venting or exposure of hazardous materials to an EVA crewmember (hazardous to EVA crewmember or to IVA environment if brought back inside habitable volume). This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 230 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.6-1 EHP VERIFICATIONS

Requirement ID	Title	Method	Inspection of Records for every requirement	Verification Objective	Success Criteria
[EVASC.0094]	Deliberate Hazardous Venting		Inspection of Records will need to be performed by EHP to verify that the Artemis System directs deliberate hazardous venting away from the EVA crewmember and translation path/worksites. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A
[EVASC.0097]	Electrical and Safety Design		Inspection of Records will need to be performed by EHP to verify that the Artemis System hardware precludes exposure of any hazardous electrical potential to the EVA crewmember. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A
[EVASC.0098]	Shock Protection		Inspection of Records will need to be performed by EHP to verify that the Artemis System hardware does not contain shock hazards on any exposed conducting surfaces that are accessible to contact by EVA crewmembers (e.g., translation corridors, work areas, and incidental contact zones). This inspection of records shall include a review of the Artemis Systems and Artemis System	N/A	N/A

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 231 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.6-1 EHP VERIFICATIONS

Requirement ID	Title	Method	Inspection of Records for every requirement	Verification Objective	Success Criteria
			vendors VCNs for compliance with the requirement.		
[EVASC.0099]	Operation During Pressure Change		Inspection of Records will need to be performed by EHP to verify that the Artemis System hardware withstands the differential pressure of depressurization, repressurization, and the depressurized condition without resulting in a hazard to the EVA crewmember as defined in EHP-10029, xEVA System to Artemis System Interface Requirements Documents (IRD) <TBR-EHP-10028-016> . This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A
[EVASC.0100]	Component Hazardous Energy Provision		Inspection of Records will need to be performed by EHP to verify that the Artemis System hardware does not expose the EVA crewmember to the release of hazardous energy This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 232 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.6-1 EHP VERIFICATIONS

Requirement ID	Title	Method	Inspection of Records for every requirement	Verification Objective	Success Criteria
[EVASC.0101]	Detachable Parts		Inspection of Records will need to be performed by EHP to verify that the Artemis System hardware provides methods of securing detachable parts. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A
[EVASC.0103]	Touch Temperatures		Inspection of Records will need to be performed by EHP to verify that the Artemis System hardware protects the xEVA suit from specified high- and low-touch-temperature extremes in the translation corridors, work areas, and incidental contact zones. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 233 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.6-1 EHP VERIFICATIONS

Requirement ID	Title	Method	Inspection of Records for every requirement	Verification Objective	Success Criteria
[EVASC.0011]	xEVA Glove Palm External Touch-Temperature Compliance for Small, Handheld Hardware		Inspection of Records will need to be performed by EHP to verify that the Artemis System hardware complies with the 30-minute xEVA suit glove palm touch temperature limits for crewmember protection from high and low skin touch temperatures as defined in Table 3.9.9-1 30-Minute Touch Temperature Compliance For Small, Handheld Hardware <TBR-EHP-10028-011> , and Table 3.9.9-2 30-Minute Touch Temperature Heat Rate Compliance <TBR-EHP-10028-011> . This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A
[EVASC.0104]	Incidental and Unlimited Touch Heat Transfer Rates		Inspection of Records will need to be performed by EHP to verify that the Artemis System hardware limits heat transfer rates for incidental unlimited contact. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 234 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.6-1 EHP VERIFICATIONS

Requirement ID	Title	Method	Inspection of Records for every requirement	Verification Objective	Success Criteria
[EVASC.0105]	Sharp Edge and Protrusions Protection	Sharp Edge Inspection	Inspection of Records will need to be performed by EHP to verify that the Artemis System hardware does not expose EVA crewmembers to harm from end item edges and protrusions. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	VITO Team shall perform independent final flight configuration sharp edge inspection and issues final report. Labels and alignment marks/devices are also inspected. The sharp edge inspection will be negotiated with the vendor either at vendor site or launch site, dependent on hardware access and the flight hardware configuration. Only trained personnel can perform a swatch test. Any identified issues need to be fixed per approved procedure.	Verification will be considered successful when the VITO final report shows that the Artemis System hardware complies with the requirement.

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 235 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.6-1 EHP VERIFICATIONS

Requirement ID	Title	Method	Inspection of Records for every requirement	Verification Objective	Success Criteria
[EVASC.0107]	Levers, Cranks, Hooks, and Controls	Sharp Edge Inspection	Inspection of Records will need to be performed by EHP to verify that the Artemis System hardware locates levers, cranks, hooks, and controls to preclude a safety/snag hazard. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	VITO Team shall perform independent final flight configuration sharp edge inspection and issues final report. Labels and alignment marks/devices are also inspected. The sharp edge inspection will be negotiated with the vendor either at vendor site or launch site, dependent on hardware access and the flight hardware configuration. Only trained personnel can perform a swatch test. Any identified issues need to be fixed per approved procedure.	Verification will be considered successful when the VITO final report shows that the Artemis System hardware complies with the requirement.
[EVASC.0109]	Exposed Holes or Slots		Inspection of Records will need to be performed by EHP to verify that the Artemis System hardware exposed holes that are round or slotted, within the range of 13 mm (0.5 in) to 36 mm (1.4 in) in diameter are covered to preclude entrapment of an xEVA-gloved finger. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 236 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.6-1 EHP VERIFICATIONS

Requirement ID	Title	Method	Inspection of Records for every requirement	Verification Objective	Success Criteria
[EVASC.0110]	Pinch Hazard Protection		Inspection of Records will need to be performed by EHP to verify that the Artemis System hardware protects the EVA crewmember from pinch hazards. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A
[EVASC.0111]	Equipment Clearance for Entrapment Hazard		Inspection of Records will need to be performed by EHP to verify that the Artemis System hardware provides clearance for equipment removal and replacement that meets clearances identified in any or all of the requirements [EVASC.0109], [EVASC.0003], and [EVASC.0001]/[EVASC.0171]. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 237 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.6-1 EHP VERIFICATIONS

Requirement ID	Title	Method	Inspection of Records for every requirement	Verification Objective	Success Criteria
[EVASC.0112]	Protection from Moving or Rotating Equipment		Inspection of Records will need to be performed by EHP to verify that the Artemis System hardware protects EVA crewmembers from impacts from moving, rotating equipment, or inadvertent deployment, separation, or jettison of hardware. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A
[EVASC.0113]	Emergency Ingress of Pressurized Habitable Environment		Inspection of Records will need to be performed by EHP to verify that the Artemis System EVA translation path aids the suited crewmembers in navigating the shortest paths to an airlock, including hardware, aids, and pathway markings or indications for vehicle ingress and egress so that an EVA crewmember can complete an emergency ingress as specified in Table 3.9.17-1 Emergency Ingress Capability. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 238 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.6-1 EHP VERIFICATIONS

Requirement ID	Title	Method	Inspection of Records for every requirement	Verification Objective	Success Criteria
[EVASC.0169]	Emergency Ingress/ Egress of Unpressurized Mobile Asset		Inspection of Records will need to be performed by EHP to verify that the Artemis System provides EVA crew translation paths, aids, accommodations, and restraints so that an EVA crewmember can complete an emergency ingress/egress with the total time of five minutes (threshold)/three minutes (goal).	N/A	N/A
[EVASC.0140]	External Surface Cleanliness Level		Inspection of Records will need to be performed by EHP to verify that the Artemis System hardware that interfaces with external surfaces of xEVA System hardware shall meet a VC-S level upon delivery as identified in Table 3.9.19-1 External Surface Cleanliness Level (or NASA approved cleanliness alternate standard), or higher. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A
[EVASC.0115]	Tethering / Restraint		Inspection of Records will need to be performed by EHP to verify that the Artemis System hardware items shall provide at least one means of positive restraint independent of EVA grasp. This inspection of records shall include a review of the Artemis Systems and Artemis System	N/A	N/A

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 239 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.6-1 EHP VERIFICATIONS

Requirement ID	Title	Method	Inspection of Records for every requirement	Verification Objective	Success Criteria
			vendors VCNs for compliance with the requirement.		
[EVASC.0116]	Crew Safety Tether Attachment		Inspection of Records will need to be performed by EHP to verify that the Artemis System handholds/handrails on microgravity translation paths intended for crewmember restraint incorporate a safety tether attachment point into its support structure that withstand the specified EVA crewmember safety tether loads identified in [EVASC.0050] Table 3.5.2-1 Microgravity EVA Crew-Induced Loads, Design Limit Load Type "Crew Safety Tether Attachment", and only to be used in conjunction with the load-limiting crew safety tethers.. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 240 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.6-1 EHP VERIFICATIONS

Requirement ID	Title	Method	Inspection of Records for every requirement	Verification Objective	Success Criteria
[EVASC.0161]	Crew Safety Restraint		Inspection of Records will need to be performed by EHP to verify that the Artemis System attachment points for Crew Safety Restraints, employed in areas of EVA operations where reaction to Crew Slips or Falls is necessary for safety of crewmember are incorporated into support structure that withstands the specified EVA crewmember loads identified in requirement [EVASC.0155], Table 3.5.3-1 Partial Gravity EVA Crew-Induced Loads, under Load/Energy Category "L12, Swinging/ or Falling". This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A
[EVASC.0117]	Color For Crew Load-Rated Handhold/ Handrails, Crew Safety Tethers And Safety Restraint Attachment Points		Inspection of Records will need to be performed by EHP to verify that the Artemis System handholds/handrail, Crew Safety Tethers and Crew Safety Restraint attachment points intended for use by the crew shall be colored according to Table 3.9.23-1 Crew Load-Rated Colors <TBR-EHP-10028-012> , where the color value has a contrast ratio of 5:1 or greater than the background. This inspection of records shall include a review of the Artemis Systems and	N/A	N/A

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 241 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.6-1 EHP VERIFICATIONS

Requirement ID	Title	Method	Inspection of Records for every requirement	Verification Objective	Success Criteria
			Artemis System vendors VCNs for compliance with the requirement.		
[EVASC.0118]	Manual Overrides		Inspection of Records will need to be performed by EHP to verify that the Artemis System includes a manual override that accommodates EVA operation with a hand tool for remotely commanded mechanisms that require EVA as a backup. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A
[EVASC.0137]	Entanglement, Cable Clamps, Ducts, or Retractors		Inspection of Records will need to be performed by EHP to verify that the Artemis System cables, conductors, bundles and hoses within an EVA worksite or translation path shall be secured by means of restraints (e.g., clamps, wire ties, etc.) unless they are contained in a cable management system (wiring ducts or cable retractors). This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 242 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.6-1 EHP VERIFICATIONS

Requirement ID	Title	Method	Inspection of Records for every requirement	Verification Objective	Success Criteria
[EVASC.0121]	Magnetic Field Limit		Inspection of Records will need to be performed by EHP to verify that the Artemis System does not generate static or time varying magnetic fields that exceed values outlined in Table 3.10.1-1 Magnetic Field Limit in crew translation corridors, work areas, and incidental contact zones during EVA operations internal and external to the pressurized volume. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A
[EVASC.0123]	Voltage Potential Discharge		Inspection of Records will need to be performed by EHP to verify that the Artemis System meets performance requirements during and after exposure to voltage/current transients of <TBD-EHP-10028-014> that occur during contact with xEVA operations when the vehicle and xEVA suit or payloads come to equilibrium. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 243 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE 4.6-1 EHP VERIFICATIONS

Requirement ID	Title	Method	Inspection of Records for every requirement	Verification Objective	Success Criteria
[EVASC.0156]	Lunar Surface Dust Tolerance		Inspection of Records will need to be performed by EHP to verify that the Artemis System tolerate the liberation of at least 200 grams of lunar surface dust in the cabin environment from the xEVA system after each two-crew lunar surface EVA. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A
[EVASC.0168]	Lunar Surface Dust Mitigation		Inspection of Records will need to be performed by EHP to verify that the Artemis System limits the amount of dust liberated in the cabin environment to less than 100 grams of lunar surface dust after each two-crew lunar surface EVA. This inspection of records shall include a review of the Artemis Systems and Artemis System vendors VCNs for compliance with the requirement.	N/A	N/A

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 244 of 280
Title: EHP xEVA System Compatibility Standards	

APPENDIX A ACRONYMS AND ABBREVIATIONS AND GLOSSARY OF TERMS

APPENDIX A-1 ACRONYMS AND ABBREVIATIONS

Acronym	Definition/Meaning
ACD	Artemis Campaign Development
AIT	Analysis Integration Team
AML	Advanced Materials Laboratory
APFR	Articulating Portable Foot Restraint
AU	Astronomical Unit
AVT	Acceptance Vibration Test
BIVP	Bilateral Integration and Verification Plan
BOL	Beginning of Life
BP	Best Practices
BRT	Body Restraint Tether
BTU	British Thermal Unit
C	Centigrade
CAD	Computer-Aided Design
CB	Control Board
CCW	Counterclockwise
CEIT	Crew Equipment Interface Test
CG	Center of Gravity
cm	centimeter
COTS	Commercial-Off-the Shelf
CR	Change Request
CW	Clockwise
CxP	Constellation Program
DCU	Display and Control Unit
DI	Systems Engineering & Integration mailcode
dia/diam	diameter
DRM	Design Reference Mission
DSNE	Design Specification for Natural Environments

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 245 of 280
Title: EHP xEVA System Compatibility Standards	

Acronym	Definition/Meaning
ECLSS	Environmental Control/Life Support System
EEE	Electrical, Electronic, and Electromechanical
EHP	Extravehicular Activity and Human Surface Mobility Program
EMI	Electromagnetic Interference
EMU	Extravehicular Mobility Unit
EOL	End of Life
ESPRIT	European System Providing Refueling Infrastructure and Telecommunications- Refueler Module
EVA	Extravehicular Activity
EVR	Extravehicular Robotics
F	Fahrenheit
FOD	Flight Operations Directorate, Foreign Object Debris
FPU	Fluid Pumping Unit
FS	Factor of Safety
ft	feet
ft ³	cubic foot
g	gravity
GDRD	Generic Design Requirements Document
GERS	Gateway External Robotics Systems
GFE	Government Furnished Equipment
HDL	Human-class Delivery Lander
HFE	Hydrofluoroether
Hg	Mercury
HIDH	Human Integration Design Handbook
HITL	Human-in-the-Loop
HLS	Human Landing System
HTV	Human Thermal Vacuum
HUT	Hard Upper Torso
Hz	Hertz
ICD	Interface Control Document

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 246 of 280
Title: EHP xEVA System Compatibility Standards	

Acronym	Definition/Meaning
in	inch
in-oz	inch-ounce
IRCD	Interface Requirements Control Document
IRD	Interface Requirements Document
ISS	International Space Station
IVA	Intravehicular Activity
J	Joule
JAM	Joint Agreement Memorandum
JANTX	Joint Army-Navy Technical Exchange
JIV	Joint Interface Verification
JSC	Johnson Space Center
JTAM	Joint Technical Agreement Memo
K	Kelvin
kg	kilogram
kPa	Kilo Pascal
lb	pound
lbf	pound force
lbf-in	pound force inch
lbm	pound mass
LOS	Loss of Signal
LTA	Lower Torso Assembly
LTV	Lunar Terrain Vehicle
m	meter
MDRD	Mockup Development Requirements Document
MEK	Methyl Ethyl Ketone
MHz	Mega Hertz
MI	Maintenance Item
MLI	Multi-Layer Insulation
mm	millimeter
MMOD	Micrometeoroid Orbital Debris

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 247 of 280
Title: EHP xEVA System Compatibility Standards	

Acronym	Definition/Meaning
N	Newton
NASA	National Aeronautics and Space Administration
NBL	Neutral Buoyancy Laboratory
NDS	NASA Docking System
Nm	Newton meter
NZGL	NASA Zero Gravity Lever
OCAD	Operational Control Agreement Database
OPR	Office of Primary Responsibility
ORU	Orbital Replacement Unit
PaSRD	Partner System Requirements Document
PDA	Pre-Delivery Acceptance
PGT	Pistol Grip Tool
PIA	Pre-Installation Acceptance
PIO2	Inspired Partial Pressure of Oxygen
PIP	Push-in-Pull
PLSS	Portable Life Support Subsystem
PR	Pressurized Rover
psi	Pounds per square inch
QD	Quick Disconnect
REEA	Retracting End Effector Assembly
RF	Radio Frequency
RIVA	Requirements, Interfaces, Verification, Analysis
RMS	Remote Manipulator System
RPM	Revolution per Minute
RTV	Room Temperature Vulcanizing (silicone rubber)
SE&I	Systems Engineering & Integration
sec	second
SEMP	Systems Engineering Management Plan
SI	International Systems of Units
SPEC	Specification

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 248 of 280
Title: EHP xEVA System Compatibility Standards	

Acronym	Definition/Meaning
SRD	System Requirements Document
SSA	Space Suit Assembly
SSP	Space Station Program
TA	Technical Authority
TBD	To Be Determined
TBR	To Be Resolved
TMG	Thermal Micrometeoroid Garment
US	United States
V	Volts
VC	Visibly Clean
VCN	Verification Closure Notice
VC-S	Visibly Clean – Sensitive
VE	Verification Event
VITO	Vehicle Integrated Test Office
W	Watt
WIF	Worksite Interface
xEMU	Exploration EMU
xEVA	Exploration EVA

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 249 of 280
Title: EHP xEVA System Compatibility Standards	

APPENDIX A-2 GLOSSARY OF TERMS

Term	Description
Cargo	Logistics items that are transported from one location to another for intended use beyond the system providing the transportation.
Contingency EVA	An EVA performed to deal with critical failures or circumstances, which are not adequately protected by redundancy or other means. An EVA not scheduled in the pre-mission timeline required to affect the safety of the crew, outpost, and/or safe return of the vehicle.
Contractor Mission Systems	Mission Systems provided by the contractor.
Crew	Human onboard the spacecraft or space system during a mission.
Dedicated Worksite	Any vehicle-specific location in which an EVA suited crewmember is expected to perform planned or unplanned tasks, whether for vehicle systems operability, maintenance, or payload interactions. The dedicated worksite ensures that volumetric clearances, visual cues, stability aids and restraints are properly located to allow EVA crewmember safe and efficient performance.
Demate	The act of isolating umbilical services by physical disconnection and removal of one compatible umbilical connector from another.
Design Reference Mission (DRM)	Typical mission scenario encompassing tasks that are most likely to drive the architecture system design requirements. The DRMs are analyzed for all mission aspects from failure tolerance to hardware layout, software functionality and design suitability.
EVA crewmember	An astronaut wearing an EVA space suit.
Fastener	Any device or mechanism that is operable by an EVA crewmember with intent to engage or release structural integrity between two parts. Examples include, but not limited to, bolts or screws driven by hand or power tools; PIP-Pins, shear-pins, latches, and levers which require a change of position or removal/installation. In limited circumstances (driven by design and safety considerations), Velcro and magnets may also be considered as a fastening means.
Flight	This is the sequence of events that takes place between liftoff and landing of a transportation vehicle.
Flight-like	Non-flight component built, inspected, and tested to flight component specifications, used in flight operating conditions, and built with manufacturing processes that are identical to those used for flight equipment.
Logistics Items	Goods, equipment, and supplies (such as utilization equipment, vehicle equipment, crew consumables, and

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 250 of 280
Title: EHP xEVA System Compatibility Standards	

Term	Description
	trash/waste, but not including propellant or vehicle dry mass), that are needed to support mission activities.
Logistics / Logistics Management / Logistics Systems	As used within NASA, encompasses the functions associated with planning for and implementation of program life-cycle logistics support (i.e., Integrated Logistics Support), transportation, supply support, supply chain management related to logistics support functions, property management, and property disposition (from NPD 7500.1D). Logistics capabilities are associated with packaging, handling, storage, transportation, and tracking of logistics items needed to support mission operations.
Maintenance Item	An MI is defined as a hardware component or software component that can be removed and replaced, and/or repaired. Note: On-Orbit Replacement Unit (ORU) was the terminology used for ISS EVA heritage to cover on-orbit replacement for maintenance.
Mate	The process of physically joining one compatible umbilical connector to another such that the connectors will, thereafter, remain joined without continued crew effort. This includes the opening of any isolation devices and joining of electrical and data pins and sockets necessary to provide full umbilical services.
Microgravity	An environment where gravity has little or no measurable effect, commonly referred to as "weightlessness" or "zero-g".
Mission Systems	The end-to-end terrestrial-based hardware and software infrastructure, whether owned or operated by NASA or the contractor, supporting mission operations from the point of Radio Frequency (RF) receipt/transmit at RF communications ground sites, through associated data processing and distribution infrastructure, transport networks, mission control center systems and facilities, interfaces between operations facilities, to the end user display/control systems and support infrastructure. Mission Systems include all associated hardware and software infrastructure supporting crew and flight controller training for operations, including but not limited to simulations, networks, mockups, and interfaces to support infrastructure.
NASA Mission Systems	Mission Systems provided by NASA.
Work Envelope	Defines the volume that an EVA Crewmember can manipulate hardware by hand from either a dedicated Foot Restraint (Microgravity) or Standing (Partial Gravity) position. One-handed Envelope illustrates volume for tasks which only require a single hand (left or right) to perform the

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 251 of 280
Title: EHP xEVA System Compatibility Standards	

Term	Description
	operation, such as switches and connectors. The Two-handed Envelope is much smaller in volume for tasks needing simultaneous use of both hands.
Over the Center Device	An over center mechanism is a bistable mechanical system that has two stable equilibrium (i.e., resting) states. Work must be performed on the system to move the mechanism from one stable position to just past the "peak" system instability, at which point the mechanism goes "over center" to its secondary stable position. [Examples are an electric wall switch, EVA Bayonet Receptacle slide lock (SEG33106697), or the door latch on the EVA Tool Stowage Device (ETSD) (SEG33106600).]
Payload	Hardware inclusive of scientific or technological instruments for utilization purposes.
Pressurized	The suit is considered pressurized at any point that it is sealed and a pressure delta over the ambient is incurred. When pressurized, the softgoods portion of the pressure bladder will inflate, impacting the stiffness of the suit, thereby, impacting the suited crewmember dimensions and mobility. The stiffening may cause the natural position of extremities to shift, thereby, impacting the neutral stance of the crewmember and the field of view.
Reach Envelope	This is defined and illustrated as the range of reach that the EVA Crew can reasonably access from a given position (e.g., such as a Foot Restraint in microgravity, or standing in one place in Partial Gravity). Intended to allow Vehicle or Payload designers to place EVA operable interfaces within reach at defined EVA Worksites.
Requirement	A necessary, quantifiable, and verifiable capability, function, property, characteristic, or behavior that a product must exhibit to solve a real-world problem, or a constraint that it must satisfy or be satisfied.
Safe Haven	A specifically designated area, location or facility that protects crew or ground team from unplanned hazardous or dangerous events. A safe haven provides essential life support functions to keep personnel alive and healthy until the dangerous condition has been mitigated or rescue is performed.
Spacecraft	A spacecraft is considered to be any host vehicles such as habitats, rovers, Landers, as well as orbiting and transit vehicles.
Supporting Spacecraft	A spacecraft capable of operating independently of other spacecraft that is required for the contractor to execute its responsibilities pursuant to the terms of this contract, but that is not an Integrated Lander Element, an HDL Element, nor a

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 252 of 280
Title: EHP xEVA System Compatibility Standards	

Term	Description
	launch vehicle. Supporting Spacecraft functions could include, but are not limited to, propellant transfer, propellant storage, or orbital maneuvering and transfer.
Unpressurized	The suit is considered unpressurized at any point that it is not sealed, e.g., visors up. Some minimal pressure may build up in the umbilical and certain areas of the suit due to active inlet breathing gas flow; this has typically been up to approximately 2kPa-d (0.3 psid). However, the suit is not inflated at this pressure. Since this pressure is not associated with sealed suit loop performance, leakage and pressure drop requirements do not apply to this pressure mode.
Utilization	Inclusive term for Artemis science, research, and technology payloads portfolio, and associated equipment.
Utilization Items / Equipment	A subcategory of logistics items with the purpose of supporting utilization activities (e.g., payloads, geology tools, sample collection containers, payload support equipment).
Vehicle	Often synonymous with the term spacecraft.
Working Volume	This is defined and illustrated as the region/volume around the entire EVA crewmember, as warranted by Vehicle or Payloads (dedicated or contingency) EVA Worksites locations. This volume ensures that there is enough physical clearance to allow the suited crewmember to perform body movements at the torso to aid in execution of the task. In Microgravity, this is applied at the foot restraint location, and for Partial Gravity, at the intended stationary location for the intended worksites. The working volume is intended to be free of obstacles or hazards which may preclude the crew's movement, sight, or access at the worksite to complete the designated worksite task(s).
Worksite – Microgravity	Microgravity EVA Worksites are locations which have been intentionally designed to be compatible with EVA Crew. EVA Worksites come in various forms, including Dedicated EVA Worksites which provide the capability to position the EVA Crew for performing a specified task. A Robotic-Assisted EVA Worksite provides the capability for robotically positioned EVA Crew to conduct tasks. A Free-Float EVA Worksite provides the capability for an unrestrained crewmember to perform specified tasks (such as single-hand operations). EVA, particularly when facilitated by mobility elements such as robotics, can access many areas of a spacecraft which were not formally designed to be EVA Worksites. Although conceptually possible, that should not be used as an automatic reason to apply EVA Worksite Outfitting requirements to the entire vehicle exterior.

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 253 of 280
Title: EHP xEVA System Compatibility Standards	

Term	Description
Worksite – Partial Gravity	Location at vehicle or payload interface where a suited EVA Crewmember is required to perform a physical task or operation, whether planned or for Pre-defined Contingencies. This may include, but not limited to, tasks such as external vehicle reconfiguration, maintenance/repair/replacement, stowage of Scientific Samples, and deployment/re-stowage of external Payloads.

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 254 of 280
Title: EHP xEVA System Compatibility Standards	

APPENDIX B EVA BEST PRACTICES (BP)

This appendix is a culmination of lessons learned/BP for performing EVAs.

EVABP-1 Multiple fasteners should be used in cases where single fastener usage would result in excessive torque specification for a single 7/16-inch or 11 mm fastener head. Currently, only one size of EVA driver is planned for numerous reasons (availability, commonality, EVA overhead, etc.). If required by hardware design, alternate EVA fasteners must be worked with EHP, since this will drive the need for additional sizes of EVA drivers.

EVABP-2 JSC EVA community has found that PIP pins can be particularly problematic since most are Commercial off the Shelf (COTS). The following acceptance cleaning/testing have been implemented at JSC and has resulted in successful screening and use of COTS PIP pins on orbit without failure.

1. Cleaning Provisions

Flight PIP pins should be cleaned to the following procedure to remove any commercial lubricants and/or contamination that may prevent the PIP pin from operating properly:

- a) Soak in a Hydrofluoroether (HFE) 7100, Methyl Ethyl Ketone (MEK), or other NASA M&P approved solvent bath for 45 minutes.
- b) Sonic soak in a HFE 7100, MEK, or other approved solvent bath for 120 minutes.
- c) Internal flush for 60 minutes.
- d) Dry and verify exterior cleanliness to Visibly Clean (VC) – Sensitive as identified by JPR.5322.1/GSDO-RQMT-1080, Cross-Program Contamination Control Requirements.
- e) Section one sample PIP pin from the lot and inspect removed components for grease residue or contamination.

Rationale: Various types of grease are used on COTS PIP pins. Destructive evaluation of one item from lot post-cleaning is necessary to verify the solvent removes the COTS lubricant.

2. Vibration Testing

Flight PIP pins should be subjected to an Acceptance Vibration Test (AVT) per Table B-1 *PIP Pin Acceptance Vibration Test Levels*. The AVT should be performed in both the longitudinal and transverse axes.

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 255 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE B-1 PIP PIN ACCEPTANCE VIBRATION TEST (AVT) LEVELS

Load Factor	Frequency Range	Amplitude
	20 - 80 Hz	+3.0 dB/Octave
6.1 G _{rms}	80 - 350 Hz	0.040 g ² /Hz
	350 - 2000 Hz	-3.0 dB/Octave

Test Duration - one minute per axis

3. Thermal Testing

Flight PIP pins should be actuated at -73° C (-100° F).

4. Destructive evaluations

Destructive evaluation must be performed on representative samples from the flight lot to verify workmanship.

5. Proven Design

A design that has proven to perform well is the AviBank Part #56789.

PIP pins assembled by welding are not recommended as repeated thermal cycles may lead to propagation of weld flaws.

EVABP-3 Double Acting PIP Pin

Equipment should provide double acting PIP pins with ease of actuation of the mechanism both during insertion and removal.

Rationale: This assumes that the actuation shaft is pushed in with the insertion of the PIP pin and pulled out with the removal of the PIP pin.

EVABP-3a PIP Pin Drive-Out Feature

Equipment should be designed to provide access for a drive-out feature.

Rationale: All PIP Pins are subject to jamming. Having a drive-out feature helps limit risk associated with this known failure mode common to all PIP Pins.

EVABP-4 External hardware should always be assessed for EVA contact hazards since planned locations can change and worksites can be created in unexpected places.

EVABP-5 Hardware should be designed to be removed/replaced via EVA in a contingency situation.

EVABP-6 Contact the EHP at NASA JSC early in the hardware development to facilitate design for EVA access and use with minimal cost and schedule impacts. Consider adding an EVA approval signature on all hardware drawings with EVA interfaces.

EVABP-7 Many EVA tools are developed for unique purposes that should not be used for generic purposes. Do not design hardware to use these unique tools without specific needs and discussions with the EHP at NASA JSC. Example: fasteners should use standard bolt heads (reference [EVASC.0088]).

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 256 of 280
Title: EHP xEVA System Compatibility Standards	

EVABP-8 Design hardware to accept torque values with as wide a tolerance as possible. Thermal extremes can adversely affect the ability of a tool to provide a specific torque.

EVABP-9 Using established hardware, including COTS hardware, might be more cost effective than designing new hardware if the established hardware meets the requirements and interfaces. However, COTS hardware may have limitations that are not readily apparent, that make them inappropriate for the harsh environment. Several COTS tools flown have encountered problems, so make the decision with care and as much information as possible. Do not let cost alone decide as there is nothing cost effective in a tool that does not work. Among the problems encountered before:

- a) Commercial hardware fabricated from tool steel may be subject to brittle fracture below -29 to -46°C (-20 to -50° F). Cold case material properties must be evaluated for all off the shelf equipment.
- b) Most commercial lubricants fail at “cold” temperatures that are actually somewhat benign compared to many spaceflight extremes, leading to the surprise that an otherwise acceptable COTS solution must be reworked to utilize a significantly different lubrication approach. Even small amounts of residue lubricant can cause mechanism failure. In order to assure hardware operation at cold case temperature extremes, remove all grease and replace with baked-on dry film coatings. Excess dry film must be removed in order to prevent buildup that can cause contamination and mechanical interference.
- c) COTS equipment often exhibits wide variation in end item dimensions and material properties. Fit checks, measurements and lot testing of materials should be implemented to assure that hardware will meet use requirements. Fit checks at ambient environment will not demonstrate on-orbit “thermal fit”. Gauges need to take into consideration thermal fit.
- d) Vendors have changed hardware but not part numbers and did not notify the customer. Crucial product lines (for spares/replacements) may become unavailable. For inexpensive, long shelf-life COTS items, a surplus should be purchased to ensure sufficient spares.

EVABP-10 Multi-Layer Insulation (MLI)/Softgoods Good Practices [Design for thermal blankets]

- a) Tether loops should be plentiful, concentrated near the mating seams. Use of contrasting color/stitching can improve effectivity and visibility.
- b) MLI seams to be mated EVA should have alignment marks both horizontally and vertically (forming a T at the edge) of a contrasting color to aid in EVA installation.
- c) Velcro should have either a metal backing or a glove pocket behind the strips to aid in proper mating of hook and pile.
- d) Restraint straps should be provided to hold MLI out of the way once opened. Straps should be long enough to reach a handrail or provide a self-restraint method. Velcro should also be used on the strap for stowage. All tether straps should include a tether loop.

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 257 of 280
Title: EHP xEVA System Compatibility Standards	

- e) Velcro that may be exposed for a length of time should be covered with a flap to prevent degrading. Flaps should also be added to MLI for EVA interfaces to allow access without removing entire MLI cover wherever feasible. MLI covers/blankets with multiple flaps should be numbered/labeled sequentially to aid in proper identification during EVA operations.
- f) All MLI should be designed to be EVA removable (contingency) with ¼ turn fasteners preferred (no tool required).
- g) Stiffeners should be provided on large blankets (more than 1.5 m² (16 ft²) to help in EVA handling).
- h) No staples should be left on exposed MLI for sharp edge considerations.

EVABP-11 Connectors exposed to sunlight should be provided with some means of thermal protection to preclude exceeding touch temperature requirements.

EVABP-12 A full complement of EVA testing should be performed on EVA concepts (glove box, 1-g, zero-g aircraft, Human Thermal Vacuum (HTV)) as necessary. Testing environment depends on the complexity of the EVA concept. Testing that involves EVA mockups should be of the appropriate fidelity to perform thorough testing (e.g., functional, volumetric, etc.). If testing at JSC, requirements for mock-up hardware and training simulators should be coordinated with the EHP at NASA JSC. A MDRD should be developed in conjunction with EVA Mission Operations Directorate personnel. Instructions for preparing an MDRD are found in JSC 28528, *Preparation and Revision of CX Mockup Development Requirements Document (MDRD)*. EVA testing is always preferable to analysis and takes precedence over analytical results.

EVABP-13 Wherever possible, perform verification of flight hardware by test. If another method is used, documented rationale should substantiate why testing cannot be accomplished.

EVABP-14 Mechanism/robotics commanding and movement should be limited and, if possible, eliminated during EVA to prevent placing the EVA crew at risk.

EVABP-15 Battery power indication should only be active when the battery is integrated into the power device, e.g., by being incorporated as part of a power device.

EVABP-16 Do not cover or interfere with handrails/handholds, tether points, or foot restraint receptacles with MLI or other objects.

EVABP-17 EEE Parts Sources

1. Approved parts are listed in the following locations except for Kapton wire (MIL-DTL-81381, *Wire, Electric, Polyimide-Insulated, Copper, or Copper Alloy, Detail Specification*), which requires written program approval before use.
2. SSP 30423, *Space Station Approved EEE Parts List*

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 258 of 280
Title: EHP xEVA System Compatibility Standards	

3. EVA-EXP-0034, *Extravehicular Activity (EVA) Office Exploration EVA System Technical Standards*
4. SLS-SPEC-159, *Cross-Program Design Specification for Natural Environments (DSNE)* – EEE Parts used on the lunar surface may not function properly in the higher ionization radiation environment if designed for use only in the station environment.
5. Semiconductors should be Joint Army-Navy Technical Exchange (JANTX) in accordance with MIL-PRF-19500, *Semiconductor Devices, General Specification*.
6. Evaluate any part not selected by any of the above per the *Engineering Directorate Certified EEE Parts Approval Process*, detailed in JSC-61360, and have it approved by its controlling authority.

EVABP-18 Wire ties should be used only when the use has been analyzed effectively and the wire ties have been shown to safely perform the function proposed. Wire ties are not certified to be used more than once or as launch restraints. Refer questions or concerns with the EVA use of wire ties to the EHP.

EVABP-19 Bolt torques should be limited to less than 34 Nm (25 ft-lbs) for the ratchet wrench and 20.3 Nm (15 ft-lbs) for the power tool, which provides for one-handed operation. For overall EVA efficiency, the Artemis System should design all bolted interfaces such that the maximum torque required is no more than 20.3 Nm (15 ft-lbs).

EVABP-20 Techniques for lightening material should not include through holes that can act as an entrapment hazard. A back plate or pocket holes can be used to lighten the material without creating an entrapment hazard.

EVABP-21 Hardware should be designed with as much versatility and resiliency as possible to protect for future or unforeseen activities.

EVABP-22 Rolled Threads and Head Height

1. Use of rolled threads for EVA fasteners - Rolled threads are stronger and less likely to yield, gall, or seize. Utilizing a double height head fastener can often lead to machining of fasteners (more expensive than using high quality aerospace grade fasteners with tight tolerances). The rolled fastener heads need to be confirmed with/concurred with ES/JSC Structural Engineering Division.
2. Head Height - EVA fasteners are based on “double height” bolt heads – it is how good engagement is obtained to deliver high torque to fasteners. Even if COTS bolts are used, the heads should still be “double height” to ensure adequate transfer of load between the socket and the fastener bolt head.

EVABP-23 Some crew-applied dynamically loads (oscillatory in nature) may need some restrictions on their frequency of cyclic application and durations in order to avoid or limit structural dynamic amplification and to lower the likelihood of an overload. Vulnerable

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 259 of 280
Title: EHP xEVA System Compatibility Standards	

structures or mechanisms are those that may possess low-frequency natural modes of vibrations (approximately 2 hertz and lower). Examples are mechanisms and cantilevered masses, whether in temporary or permanent configurations.

EVABP-24 Design hardware to be assessed for EVA contact hazards.

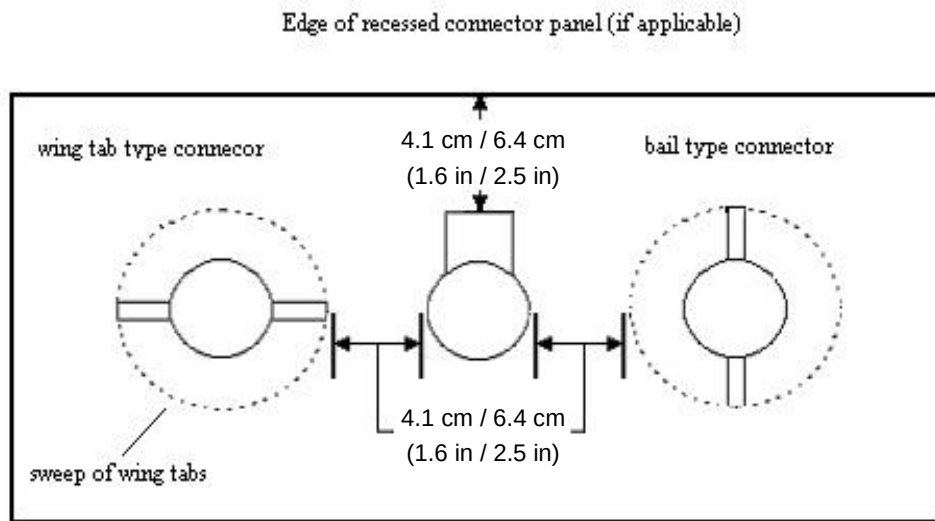
EVABP-25 Programs should seek to obtain commonly implemented interfaces as Government Furnished Equipment (GFE). Use of GFE for common interfaces is beneficial in that it provides consistency across the vehicle. The NASA EHP has standard items available as GFE; these include handrails, tether hooks and other piece parts, foot restraint interfaces, quarter turn fasteners, and brackets.

EVABP-26 Threaded fasteners that perform a safety-critical function should comply with NASA-STD-5020, *Requirements for Threaded Fastening Systems in Spaceflight Hardware*. This does not apply to fasteners that are considered to be any of the following: not safety critical, are internal to components, or shown to be contained.

EVABP-27 Connector spacing for small to medium bail and wingtab type connectors should be designed to approach the minimum spacing requirements shown in Figure B-1 EVABP-27 *Connector Spacing*.

Rationale: When utilizing the spacing outlined in Figure B-1 EVABP-27 Connector Spacing, it may result in greater/or less than the gloved-hand access requirement [EVASC.0003] in certain directions. If there is not sufficient real estate to meet the gloved-hand requirement [EVASC.0003], Figure B-1 EVABP-27 Connector Spacing illustrates what was successful on past Programs and can be considered with the appropriate paperwork and EVA evaluations.

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 260 of 280
Title: EHP xEVA System Compatibility Standards	



*Spacing requirement increases 6.4 cm (2.5 in) for multiple rows of connectors

These are minimum dimensions and should be used only where packaging issues exist.

FIGURE B-1 EVABP-27 CONNECTOR SPACING

EVABP-28 Connector Ultimate Load

Connector backshells should withstand ultimate load as defined in JSC 65828, *Structural Design Requirements and Factors of Safety for Spaceflight Hardware*.

Rationale: Strain relief helps to prevent breakage due to induced loading. Connector backshells designed to provide strain relief, must still preclude sharp edges, pinch points and other hazards as defined in section 3.9.

EVABP-29 EVA Secondary Ingress Method

The Artemis System should provide a secondary ingress method or equivalent for EVA contingency ingress.

Rationale: Heritage airlock operations have shown the need to allow for a secondary ingress in the event of a failure of the primary method, thereby taking the burden off of a module to be designed to go to vacuum in case of a contingency. Secondary ingress/egress does not necessarily mean separate hatches on the same volume. It could, for example, be met by two separate volumes linearly aligned, allowing the inner volume to act as the redundant ingress/egress method if the outer volume were to fail. See EVA-EXP-0031, EVA Airlocks and Alternative Ingress/Egress Methods for reference.

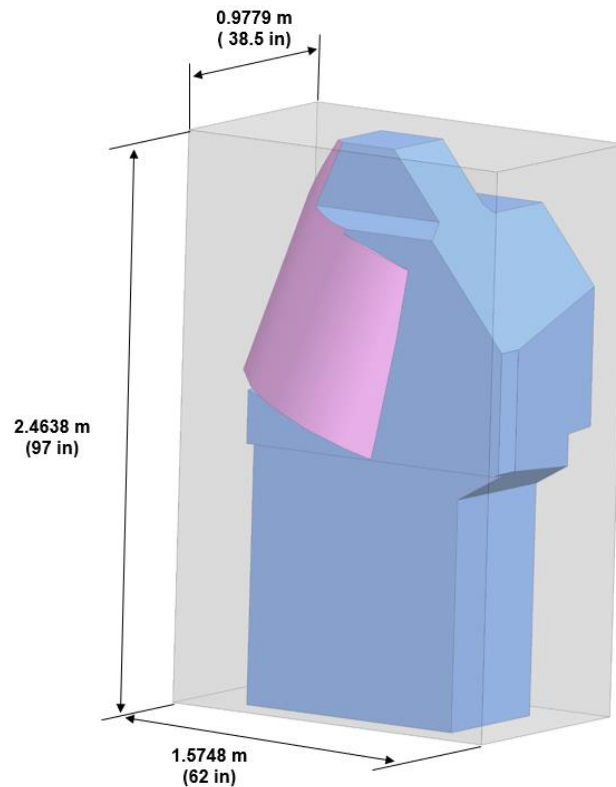
Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 261 of 280
Title: EHP xEVA System Compatibility Standards	

EVABP-30 Hatch/Volumetric Transfer Constraint

The Artemis System should limit dimensions of any hardware to be transferred between pressurized and unpressurized environments not to exceed the free volume of the airlock minus the volume of the two crewmembers (per Figure B-2 *EVABP-30 Minimum Don/Doff Volume Dimensions for a Single xEVA Suit*) with access to airlock systems, controls and procedures for ingress, egress, or emergency operations.

*Rationale: Crew may be expected to transfer hardware as part of normal or contingency operations (e.g., items for maintenance that need to be brought back into the Airlock). MI designs or mission scenarios requiring hardware to be transferred from the pressurized to the unpressurized environment need to be sized such that they can fit within the free volume of their intended vehicle airlock while not impeding the ability of the two EVA crew to perform all ingress and egress operations. These dimensions are vehicle dependent and not universal, and so it falls to hardware providers to account for their specific mission parameters when levying specific requirements that limit the volume of MIs. The free space within the vehicle airlock must accommodate the crew, the to-be-transferred hardware and crew's access to airlock systems. Using the volumetric dimensions of two suited crewmembers, vehicles shall define the volumetric limits of their MIs. Once defined, those limits should be provided to EHP for inclusion in interface definition documents for tool and payload providers. Hatch opening (Height/Width) is the first consideration for assessing this transfer constraint, but Depth of object needs to be analyzed to ensure the object can fit through without uncontrolled handling. Due to the nature of nominal single crew person egress/ingress techniques, it is expected that all transferred hardware will require two-crew members to perform the hand-off at the Airlock Hatch opening/closing threshold. Additional consideration needs to be given to swing or rotational envelopes of the Hatch to open/close, and not restrict access to critical interfaces internal to the airlock. Recommended that this assessment to be analyzed by Test with NASA participation. Figure B-2 EVABP-30 is for reference, see EHP-10029, xEVA System to Artemis System IRD <**TBR-EHP-10028-016**> for latest don/doff volumes.*

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 262 of 280
Title: EHP xEVA System Compatibility Standards	



Volume of Bounding Box Containing xEMU Don/Doff Volume (Using Unassisted PLSS Swing Entry Method)
3516154.30 cm ³ / 214568.90 in ³
3.52 m ³ / 124.17 ft ³

FIGURE B-2 EVABP-30 MINIMUM DON/DOFF VOLUME DIMENSIONS FOR A SINGLE XEVA SUIT

EVABP-31 Hand Grip

The Artemis System should incorporate a non-abrasive, non-slip surface on hand gripping surfaces that have a repetitive use potential to prevent EVA glove slippage and abrasion.

Rationale: A non-abrasive non-slip surface can be achieved through surface finish (e.g., prevents abrasion of EVA glove material) or shape of the object (e.g., knob) for microgravity applications). Partial gravity application includes areas where crew need to grasp with the pressurized EVA glove but adds considerations for

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 263 of 280
Title: EHP xEVA System Compatibility Standards	

crew traversing on the surface given that there may be a layer of dust between the surface and the boot.

EVABP-32 Partial Gravity Tethering

Partial gravity tools should be designed with tether points to provide positive restraint when working on MIs above the surface (e.g., working on a ladder and on top of PR, remove/replace high antenna / camera on LTV, etc.). Tethering tools will reduce the risk to nearby crew of falling items, reduce potential to damage hardware, reduce introduction of FOD to MI interfaces, and improve efficiency for crew not having to retrieve dropped tools.

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 264 of 280
Title: EHP xEVA System Compatibility Standards	

APPENDIX C NON-EVA BPS FOR CONSIDERATIONS

This appendix contains those BPs that are considered to be non-EVA, but worthy of consideration when designing EVA interface.

Fasteners - Orientation/Retention

FAS-1 Orient and/or retain PIP pins so failure of the mechanism during launch or landing precludes release.

Protruding Fasteners

PFAS-1 Provide sharp edge protection for protruding fastener threads.

Rationale: If structurally sound, acorn nuts can provide this sort of protection. Room Temperature Vulcanizing (RTV) does not always stay on exposed threads over a long duration.

Handling – Standard Interface Temperature Differential

HAN-1 Use a standard hardware interface temperature differential per Table C-1 *Standard Interface Temperature Differential*, when designing all EVA hardware to ensure fit and function of interfacing hardware. Use an interface temperature differential of 38°C (100°F) when using material surface coatings with a low EOL optical property ratio (0.9 maximum). Use an interface temperature differential of 93°C (200°F) as a design goal for any interface hardware with a high EOL surface coating (greater than 0.9). A reduced temperature differential, determined by analysis, may be used if required for fit/function, and should be documented in the Safety Data Package.

TABLE C-1 STANDARD INTERFACE TEMPERATURE DIFFERENTIAL

EOL Surface Optical Property Ratio	Standard Interface Temperature Differential
≤ 0.9	38° C (100° F)
> 0.9	93° C (200° F)

Materials and Process Best Practice

MP-1 All currently available soft goods, including Velcro®, Kevlar®, Vectran™, nylon, fiberglass and Nomex® are affected by prolonged exposure to the EVA environment. The combined effects of atomic oxygen, ultraviolet light, ionizing radiation, and abrasion during use can reduce the load capability of soft goods. Soft goods should not be used in load bearing applications that are exposed to the EVA environment without high safety factors, backup load paths and/or protection from atomic oxygen, ionizing radiation, and ultraviolet effects.

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 265 of 280
Title: EHP xEVA System Compatibility Standards	

Non-EVA Best Practices

NEBP-1 Real-time access to high fidelity, flight-like hardware is very beneficial for procedure development and troubleshooting in contingency cases.

NEBP-2 Avoid the use of welded hardware. Repeated thermal cycles may lead to propagation of weld flaws.

Note: this refers to ground welding; on-orbit welding is not available.

NEBP-3 Some locking compounds are difficult to apply without contamination of adjacent surfaces and mechanisms. Results of the contamination may not be apparent until the hardware reaches cold case temperatures.

NEBP-4 Jam nuts can be used on non-EVA operable fasteners; however, neither jam nuts nor anaerobic locking compounds alone provide positive locking and must be used in concert with safety wire and tab washers. The preferred method of locking is preload combined with other conventional aerospace locking features. Self-locking features that retain full running torque after preload is relieved or the threads move can also provide positive locking.

NEBP-5 Some fasteners are difficult (spiral cam fasteners) to remove and need to be demonstrated to work with EVA gloved operations. These types of fasteners can be “splayed” if improperly operated during shield installation (ex. MMOD shields on ISS) and are not recommended for EVA operations. Fasteners that require operations by EVA crewmembers should meet the requirements outlined in section 3.7. If a fastener is splayed during installation, it will not release again nominally, and will not re-engage later. Contingent reapplication should evaluate the minimum number of required fasteners. This is a lesson learned from ISS, which replaced the spiral cam fastener for EVA use.

NEBP-6 A lesson learned from past projects on IVA fluid connections is that if they do not have enough flex in the lines to stay a shut-off valve when connected, they can sustain a hydraulic shock and it can cause problems. The hydraulic shock is a pressure surge that is caused when a fluid experiences a momentum change (also referred to as a water hammer). There was a related experience with an ISS Fluid Pumping Unit (FPU) fluid connector - reference the PRACA document at:

https://part.iss.nasa.gov/record/24868?list_id=281286&list_index=364.

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 266 of 280
Title: EHP xEVA System Compatibility Standards	

APPENDIX D OPEN WORK

APPENDIX D-1 TO BE DETERMINED

Table D1-1 To Be Determined Items lists the specific To Be Determined (TBD) items in the document that are not yet known. The TBD is inserted as a placeholder wherever the required data is needed and is formatted in bold type within carets. The TBD item is numbered based on the document number (i.e., **<TBD-EHP-xxx-001>** is the first undetermined item assigned in the document). As each TBD is resolved, the updated text is inserted in each place that the TBD appears in the document and the item is removed from this table. As new TBD items are assigned, they will be added to this list in accordance with the above-described numbering scheme. Original TBDs will not be renumbered.

TABLE D1-1 TO BE DETERMINED

TBD	Section/ Requirement	Description
<TBD-EHP-10028-014>	3.10.2 [EVASC.0123]	Define the magnitude of voltage potential or current transfer.
<TBD-EHP-10028-019>	3.3.12.7 [EVASC.0170]	Need to define the work volume for seated operation of vehicle through testing. Need to determine if this is applicable to PR based on contingency scenarios. Data will be provided through testing and may consist of both figures and tables like [EVASC.0035].

APPENDIX D-2 TO BE RESOLVED

Table D2-1 To Be Resolved Items lists the specific To Be Resolved (TBR) items in the document that are not yet known. The TBR is inserted as a placeholder wherever the required data is needed and is formatted in bold type within angle brackets. The TBR issue is numbered based on the document number, including the annex, volume, and book number, as applicable (i.e., **<TBR-EHP-xxx-001>** is the first unresolved item assigned in the document). As each TBR is resolved, the updated text is inserted in each place that the TBR appears in the document and the issue is removed from this table. As new TBR issues are assigned, they will be added to this list in accordance with the above-described numbering scheme. Original TBRs will not be renumbered.

TABLE D2-1 TO BE RESOLVED

TBR	Section/ Requirement	Description
<TBR-EHP-10028-002>	3.5.3 [EVASC.0155]	Loads provided by ES is conservative and may need to be refined as designs mature. Requires integrated team

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 267 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE D2-1 TO BE RESOLVED

TBR	Section/ Requirement	Description
		discussion for design parameters for the L12 force limiting device. In addition, need to determine what is the fall height that would require a restraint system.
<TBR-EHP-10028-003>	3.3.2 [EVASC.0160]	Based on discussion at the 2/5/24 Special RIVA Panel TCM it was agreed to keep <TBR-EHP-10028-003> on the entire requirement. Requirement requires future testing to define the operational scenarios and use cases for single and two-handed lift and carry such as repetitive tasks and various heights (e.g., Lifting and manipulating objects while kneeling or stepping up or over something is considered nominal operation/situation along with refining the new suit capabilities). Any updates to requirement require participation by key stakeholders.
<TBR-EHP-10028-008>	3.8.2 [EVASC.0088]	Received new document number and title for, EHP EVA Tool List and Processes for EVA Operations <TBR-EHP-10028-008> . Approval process will be worked via the RIVA/xEVA AIT. <u>Document to include:</u> - Reference and certified tools for: - EVA Tools for Artemis Mission Objectives (surface science) - EVA Tools for Vehicle Maintenance/Contingencies - Processes for: - Requesting a new tool be added to the list - Approving a tool by the RIVA Panel TBR will be closed once document has been Baseline.
<TBR-EHP-10028-011>	3.9.9 [EVASC.0103] 3.9.10 [EVASC.0011] 3.9.11 [EVASC.0104]	Assess updates to xEVAS Glove and PGS Thermal Models needed for EVA Touch Temperature Requirement (reference 20220623-EVA-xEVA AIT-0001). Provided thermal values derived from the current ISS EMU are updated. In addition, a new suit will most likely have new touch temperature limits that need to be included in the thermal requirements.
<TBR-EHP-10028-012>	3.5.1.1 [EVASC.0033] 3.9.24 [EVASC.0117]	Determine viable paths to colored identification of load bearing items: - EHP to engage with M&P to identify viable paint solutions (including cost and schedule for testing as required). Additional testing is being considered at other NASA centers.

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 268 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE D2-1 TO BE RESOLVED

TBR	Section/ Requirement	Description
		- Work Cross-Program on issue resolution.
<TBR-EHP-10028-016>	1.2, 2.1, [EVASC.0099] in Section 3.9.6, 4.5, 4.6, EVABP-30	Will be closed once EHP-10029, xEVA System to Artemis Systems IRD is formally Baselined.
<TBR-EHP-10028-018>	4.5 [EVASC.0154]	Identify future facilities and/or capabilities to perform this verification demonstration.

APPENDIX D-3 FORWARD WORK

TABLE D3-1 FORWARD WORK

FWD	Section/ Requirement	Description

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 269 of 280
Title: EHP xEVA System Compatibility Standards	

APPENDIX E ARTEMIS SYSTEM ALLOCATIONS

The allocations shown in Table E-1 identify the Artemis System obligations to the requirements which are invoked in Section 3. The EHP Projects are individually responsible for traceability and any decomposition or further system allocation.

Requirements applicability to Artemis Missions begin with Artemis III and apply to all subsequent Artemis Missions.

TABLE E-1: ARTEMIS SYSTEM ALLOCATION MATRIX

Requirements		Gravity Field Application		Artemis System Allocations					
Requirement Title	RQT Tag Number	Partial Gravity	Microgravity	EVA Dev	LTV	PR	Gateway	HLS	Payloads
Microgravity Translation Path Hardware Protrusion	[EVASC.0001]	N/A	A	A	N/A	N/A	A	A	N/A
Partial Gravity Translation Path Hardware Protrusion	[EVASC.0171]	A	N/A	A	A	A	N/A	A	N/A
Tool Interface Clearance	[EVASC.0002]	A	A	A	A	A	A	A	A
Bolt Clearance	[EVASC.0007]	A	A	A	A	A	A	A	A
Gloved-Hand Clearance	[EVASC.0003]	A	A	A	A	A	A	A	A
Handrail / Handhold Mounting Clearance	[EVASC.0004]	A	A	A	A	A	A	A	A
Handrail/Handhold Side Clearance	[EVASC.0005]	A	A	A	A	A	A	A	A
APFR EVA Installation/ Removal Clearance	[EVASC.0006]	N/A	A	N/A	N/A	N/A	A	A	N/A

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 270 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE E-1: ARTEMIS SYSTEM ALLOCATION MATRIX

Requirements		Gravity Field Application		Artemis System Allocations					
Requirement Title	RQT Tag Number	Partial Gravity	Microgravity	EVA Dev	LTV	PR	Gateway	HLS	Payloads
Contiguous EVA Translation Path	[EVASC.0152]	N/A	A	N/A	N/A	N/A	A	A	N/A
Translation Path	[EVASC.0008]	A	A	N/A	N/A	N/A	A	A	N/A
Incapacitated Crewmember Interfaces	[EVASC.0154]	A	A	A	A	A	A	A	A
EVA Contact Hazards Preclusion Beyond Translation Corridor	[EVASC.0010]	N/A	A	N/A	N/A	N/A	A	A	N/A
Partial Gravity EVA Contact Hazards Preclusion	[EVASC.0159]	A	N/A	N/A	A	A	N/A	A	A
Microgravity EVA Equipment Handling Features	[EVASC.0012]	N/A	A	A	N/A	N/A	A	A	A
Partial Gravity EVA Handling Capability <TBR-EHP-10028-003>	[EVASC.0160]	A	N/A	A	A	A	N/A	A	A
Partial Gravity Step Design Parameter	[EVASC.0162]	A	N/A	A	A	A	N/A	A	N/A
Pressurized xEVA Transfer through Airlock Hatch	[EVASC.0135]	N/A	A	A	N/A	N/A	A	A	N/A

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 271 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE E-1: ARTEMIS SYSTEM ALLOCATION MATRIX

Requirements		Gravity Field Application		Artemis System Allocations					
Requirement Title	RQT Tag Number	Partial Gravity	Microgravity	EVA Dev	LTV	PR	Gateway	HLS	Payloads
EVA Lunar Hatch Dimensions	[EVASC.0153]	A	N/A	A	N/A	A	N/A	A	N/A
Mounting Alignment Devices	[EVASC.0014]	A	A	A	A	A	A	A	A
Handling Alignment Marks	[EVASC.0015]	A	A	A	A	A	A	A	A
Locking Device	[EVASC.0016]	A	A	A	A	A	A	A	A
Over Center Device Force	[EVASC.0124]	A	A	A	A	A	A	A	A
Soft Dock or Soft Capture Mechanisms	[EVASC.0017]	N/A	A	A	N/A	N/A	A	A	A
Soft Capture	[EVASC.0125]	A	A	N/A	A	A	A	A	A
Soft Dock Force	[EVASC.0126]	N/A	A	A	N/A	N/A	A	A	A
Unique Labeling	[EVASC.0024]	A	A	A	A	A	A	A	A
Hand Tool Operations	[EVASC.0025]	A	A	A	A	A	A	A	A
Hand Ratcheting	[EVASC.0135a]	A	A	A	A	A	A	A	A
Throw Angle	[EVASC.0026]	A	A	A	A	A	A	A	A
Ratchet Mechanism	[EVASC.0127]	A	A	A	A	A	A	A	A
Back Drive Torque	[EVASC.0128]	A	A	A	A	A	A	A	A
Running Torque	[EVASC.0129]	A	A	A	A	A	A	A	A
Pressurized Glove Compatibility	[EVASC.0027]	A	A	A	A	A	A	A	A

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 272 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE E-1: ARTEMIS SYSTEM ALLOCATION MATRIX

Requirements		Gravity Field Application		Artemis System Allocations					
Requirement Title	RQT Tag Number	Partial Gravity	Microgravity	EVA Dev	LTV	PR	Gateway	HLS	Payloads
Inadvertent Actuation of EVA -Actuated Controls	[EVASC.0087]	A	A	A	A	A	A	A	A
Handrail/Handhold Cross-Section	[EVASC.0029]	A	A	A	A	A	A	A	A
Handrail/Handhold/ Handle Length	[EVASC.0030]	A	A	A	A	A	A	A	A
Handrail/Handhold Accessibility	[EVASC.0031]	A	A	N/A	A	A	A	A	N/A
Continuous Handrail/Handhold Spacing	[EVASC.0032]	A	A	N/A	A	A	A	A	N/A
EVA Egress Handrail Spacing	[EVASC.0144]	A	A	N/A	N/A	A	A	A	N/A
Work Envelope	[EVASC.0035]	A	A	A	A	A	A	A	A
Working Volume Operation of Vehicle	[EVASC.0170]	A	N/A	N/A	A	N/A	N/A	N/A	N/A
EVA Egress/Ingress Methods	[EVASC.0157]	A	N/A	A	A	A	N/A	A	N/A
Crew Aids for Stability	[EVASC.0036]	N/A	A	N/A	N/A	N/A	A	A	N/A
EVA Worksites Using Robotic Work Platform	[EVASC.0037]	N/A	A	N/A	N/A	N/A	A	A	N/A
Lubricant Use	[EVASC.0038]	A	A	A	A	A	A	A	A
Worksite Mobility	[EVASC.0040]	N/A	A	N/A	N/A	N/A	A	A	N/A

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 273 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE E-1: ARTEMIS SYSTEM ALLOCATION MATRIX

Requirements		Gravity Field Application		Artemis System Allocations					
Requirement Title	RQT Tag Number	Partial Gravity	Microgravity	EVA Dev	LTV	PR	Gateway	HLS	Payloads
Worksite Field of View	[EVASC.0041]	A	A	N/A	A	A	A	A	A
Working Volume	[EVASC.0042]	A	A	N/A	A	A	A	A	A
Dedicated EVA Worksite	[EVASC.0148]	N/A	A	N/A	N/A	N/A	A	A	N/A
Robotic Assisted EVA Worksite	[EVASC.0149]	N/A	A	N/A	N/A	N/A	A	A	N/A
Free-Float Worksites	[EVASC.0150]	N/A	A	N/A	N/A	N/A	A	A	N/A
Installation of Passive Worksite Interface Sockets	[EVASC.0045]	N/A	A	N/A	N/A	N/A	A	A	N/A
Foot Restraint Ingress/Egress Aids	[EVASC.0046]	N/A	A	N/A	N/A	N/A	A	A	N/A
Visual Contrast at Worksite	[EVASC.0047]	A	A	A	A	A	A	A	A
Color for EVA Handling and EVA Equipment Tether Points	[EVASC.0033]	A	A	A	A	A	A	A	A
EVA Equipment Tether Point	[EVASC.0048]	A	A	A	A	A	A	A	A
Soft Goods Tether Point Loads	[EVASC.0134]	A	A	A	A	A	A	A	A
Microgravity EVA Crew-Induced Loads	[EVASC.0050]	N/A	A	A	N/A	N/A	A	A	A

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 274 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE E-1: ARTEMIS SYSTEM ALLOCATION MATRIX

Requirements		Gravity Field Application		Artemis System Allocations					
Requirement Title	RQT Tag Number	Partial Gravity	Microgravity	EVA Dev	LTV	PR	Gateway	HLS	Payloads
Partial Gravity EVA Crew-Induced Loads	[EVASC.0155]	A	N/A	A	A	A	N/A	A	A
EVA Actuated Loads	[EVASC.0018]	A	A	A	A	A	A	A	A
EVA Mechanical Feedback	[EVASC.0164]	A	A	A	A	A	A	A	A
WIF Structural Loading	[EVASC.0143]	N/A	A	A	N/A	N/A	A	A	N/A
One-Handed EVA Operable Connectors	[EVASC.0053]	A	A	A	A	A	A	A	A
Connector Accessibility	[EVASC.0054]	A	A	A	A	A	A	A	A
Gloved-Hand Connector Accessibility	[EVASC.0138]	A	A	A	A	A	A	A	A
Incorrect Connector Mating Prevention	[EVASC.0055]	A	A	A	A	A	A	A	A
Connector Coding	[EVASC.0056]	A	A	A	A	A	A	A	A
Connector Protective Caps	[EVASC.0057]	A	A	A	A	A	A	A	A
Connector Rotational Travel	[EVASC.0058]	A	A	A	A	A	A	A	A
Connector Locking Feature	[EVASC.0059]	A	A	A	A	A	A	A	A
Visible Connector Alignment Markings	[EVASC.0060]	A	A	A	A	A	A	A	A

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 275 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE E-1: ARTEMIS SYSTEM ALLOCATION MATRIX

Requirements		Gravity Field Application		Artemis System Allocations					
Requirement Title	RQT Tag Number	Partial Gravity	Microgravity	EVA Dev	LTV	PR	Gateway	HLS	Payloads
Connector Locking Ring	[EVASC.0061]	A	A	A	A	A	A	A	A
Connector Mating Status	[EVASC.0062]	A	A	A	A	A	A	A	A
Mating/Demating of Powered Connectors	[EVASC.0052]	A	A	A	A	A	A	A	A
Connectors - Electrical Bonding	[EVASC.0065]	A	A	A	A	A	A	A	A
Scoop-Proof Electrical Connector Design	[EVASC.0066]	A	A	A	A	A	A	A	A
Connector Grip Point	[EVASC.0063]	A	A	A	A	A	A	A	A
Arc Containment	[EVASC.0131]	A	A	A	A	A	A	A	A
Uncontrolled Motion of Flex Hoses	[EVASC.0132]	A	A	A	A	A	A	A	A
Venting Feature	[EVASC.0133]	A	A	A	A	A	A	A	A
Indication of Pressure Flow - Nonbrazed or Nonwelded	[EVASC.0145]	A	A	A	A	A	A	A	A
Captive Fastener Design	[EVASC.0068]	A	A	A	A	A	A	A	A
Prohibited Locking Methods for Fasteners	[EVASC.0069]	A	A	A	A	A	A	A	A
One-Handed Fastener Operation	[EVASC.0067]	A	A	A	A	A	A	A	A

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 276 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE E-1: ARTEMIS SYSTEM ALLOCATION MATRIX

Requirements		Gravity Field Application		Artemis System Allocations					
Requirement Title	RQT Tag Number	Partial Gravity	Microgravity	EVA Dev	LTV	PR	Gateway	HLS	Payloads
Scalloped Knobs	[EVASC.0072]	A	A	A	A	A	A	A	A
Knobs Head Diameter and Height	[EVASC.0073]	A	A	A	A	A	A	A	A
EVA Hand-Actuated Fasteners Contingency Operation	[EVASC.0074]	A	A	A	A	A	A	A	A
EVA PIP Pin Grasping Interface	[EVASC.0075]	A	A	A	A	A	A	A	A
PIP Pin Restraint Tether	[EVASC.0076]	A	A	A	A	A	A	A	A
Driver-Type Push Force Tools	[EVASC.0078]	A	A	A	A	A	A	A	A
EVA Fastener Head	[EVASC.0079]	A	A	A	A	A	A	A	A
Fastener Deep Socket	[EVASC.0084]	A	A	A	A	A	A	A	A
Fastener Torque Data	[EVASC.0151]	A	A	A	A	A	A	A	A
Power and Hand Tool Operability	[EVASC.0080]	A	A	A	A	A	A	A	A
Fastener Gauge/ Measurement	[EVASC.0141]	A	A	A	A	A	A	A	A
Fastener Alignment Features	[EVASC.0083]	A	A	A	A	A	A	A	A
Battery Replacement	[EVASC.0085]	A	A	A	A	A	A	A	A
Battery Voltage Monitoring	[EVASC.0086]	A	A	A	A	A	A	A	A

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 277 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE E-1: ARTEMIS SYSTEM ALLOCATION MATRIX

Requirements		Gravity Field Application		Artemis System Allocations					
Requirement Title	RQT Tag Number	Partial Gravity	Microgravity	EVA Dev	LTV	PR	Gateway	HLS	Payloads
Auxiliary Controls	[EVASC.0028]	A	A	A	A	A	A	A	A
EVA Tools	[EVASC.0088]	A	A	N/A	A	A	A	A	A
Cable Routing/Restraint	[EVASC.0089]	A	A	N/A	A	A	A	A	A
Cable Routing Spacing	[EVASC.0090]	A	A	N/A	A	A	A	A	N/A
Paint	[EVASC.0091]	A	A	N/A	A	A	A	A	A
Hazardous Material Control	[EVASC.0092]	A	A	A	A	A	A	A	A
Venting or Exposure Compatibility	[EVASC.0093]	A	A	A	A	A	A	A	A
Deliberate Hazardous Venting	[EVASC.0094]	A	A	A	A	A	A	A	A
Electrical and Safety Design	[EVASC.0097]	A	A	A	A	A	A	A	A
Shock Protection	[EVASC.0098]	A	A	A	A	A	A	A	A
Operation During Pressure Change	[EVASC.0099]	A	A	A	A	A	A	A	A
Component Hazardous Energy Provision	[EVASC.0100]	A	A	A	A	A	A	A	A
Detachable Parts	[EVASC.0101]	A	A	A	A	A	A	A	A
Touch Temperatures	[EVASC.0103]	A	A	A	A	A	A	A	A

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 278 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE E-1: ARTEMIS SYSTEM ALLOCATION MATRIX

Requirements		Gravity Field Application		Artemis System Allocations					
Requirement Title	RQT Tag Number	Partial Gravity	Microgravity	EVA Dev	LTV	PR	Gateway	HLS	Payloads
xEVA Glove Palm External Touch-Temperature Compliance for Small, Handheld Hardware	[EVASC.0011]	A	A	A	A	A	A	A	A
Incidental and Unlimited Contact Heat Transfer Rates	[EVASC.0104]	A	A	A	A	A	A	A	A
Sharp Edge and Protrusions Protection	[EVASC.0105]	A	A	A	A	A	A	A	A
Levers, Cranks, Hooks, and Controls	[EVASC.0107]	A	A	A	A	A	A	A	A
Exposed Holes or Slots	[EVASC.0109]	A	A	A	A	A	A	A	A
Pinch Hazard Protection	[EVASC.0110]	A	A	A	A	A	A	A	A
Equipment Clearance for Entrapment Hazard	[EVASC.0111]	A	A	A	A	A	A	A	A
Protection from Moving or Rotating Equipment	[EVASC.0112]	A	A	A	A	A	A	A	A
Emergency Ingress of Pressurized Habitable Environment	[EVASC.0113]	A	A	A	N/A	A	A	A	N/A

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 279 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE E-1: ARTEMIS SYSTEM ALLOCATION MATRIX

Requirements		Gravity Field Application		Artemis System Allocations					
Requirement Title	RQT Tag Number	Partial Gravity	Microgravity	EVA Dev	LTV	PR	Gateway	HLS	Payloads
Emergency Ingress/Egress of Unpressurized Mobile Asset	[EVASC.0169]	A	N/A	N/A	A	N/A	N/A	N/A	N/A
External Surface Cleanliness Level	[EVASC.0140]	A	A	A	A	A	A	A	A
Tethering / Restraint	[EVASC.0115]	N/A	A	A	N/A	N/A	A	A	A
Crew Safety Tether Attachment	[EVASC.0116]	N/A	A	A	N/A	N/A	A	A	A
Crew Safety Restraint	[EVASC.0161]	A	N/A	A	A	A	N/A	A	A
Color For Crew Load-Rated Handhold/Handrails, Crew Safety Tethers And Safety Restraint Attachment Points	[EVASC.0117]	A	A	A	A	A	A	A	A
Manual Overrides	[EVASC.0118]	A	A	N/A	A	A	A	A	A
Entanglement, Cable Clamps, Ducts, or Retractors	[EVASC.0137]	A	A	N/A	A	A	A	A	A
Magnetic Field Limit	[EVASC.0121]	A	A	A	A	A	A	A	A
Voltage Potential Discharge	[EVASC.0123]	A	A	A	A	A	A	A	A
Lunar Surface Dust Tolerance	[EVASC.0156]	A	N/A	N/A	N/A	A	N/A	A	N/A

Revision: Baseline	Document No: EHP-10028
Release Date: 09 MAY 2024	Page: 280 of 280
Title: EHP xEVA System Compatibility Standards	

TABLE E-1: ARTEMIS SYSTEM ALLOCATION MATRIX

Requirements		Gravity Field Application		Artemis System Allocations					
Requirement Title	RQT Tag Number	Partial Gravity	Microgravity	EVA Dev	LTV	PR	Gateway	HLS	Payloads
Lunar Surface Dust Mitigation	[EVASC.0168]	A	N/A	A	N/A	N/A	N/A	N/A	N/A